

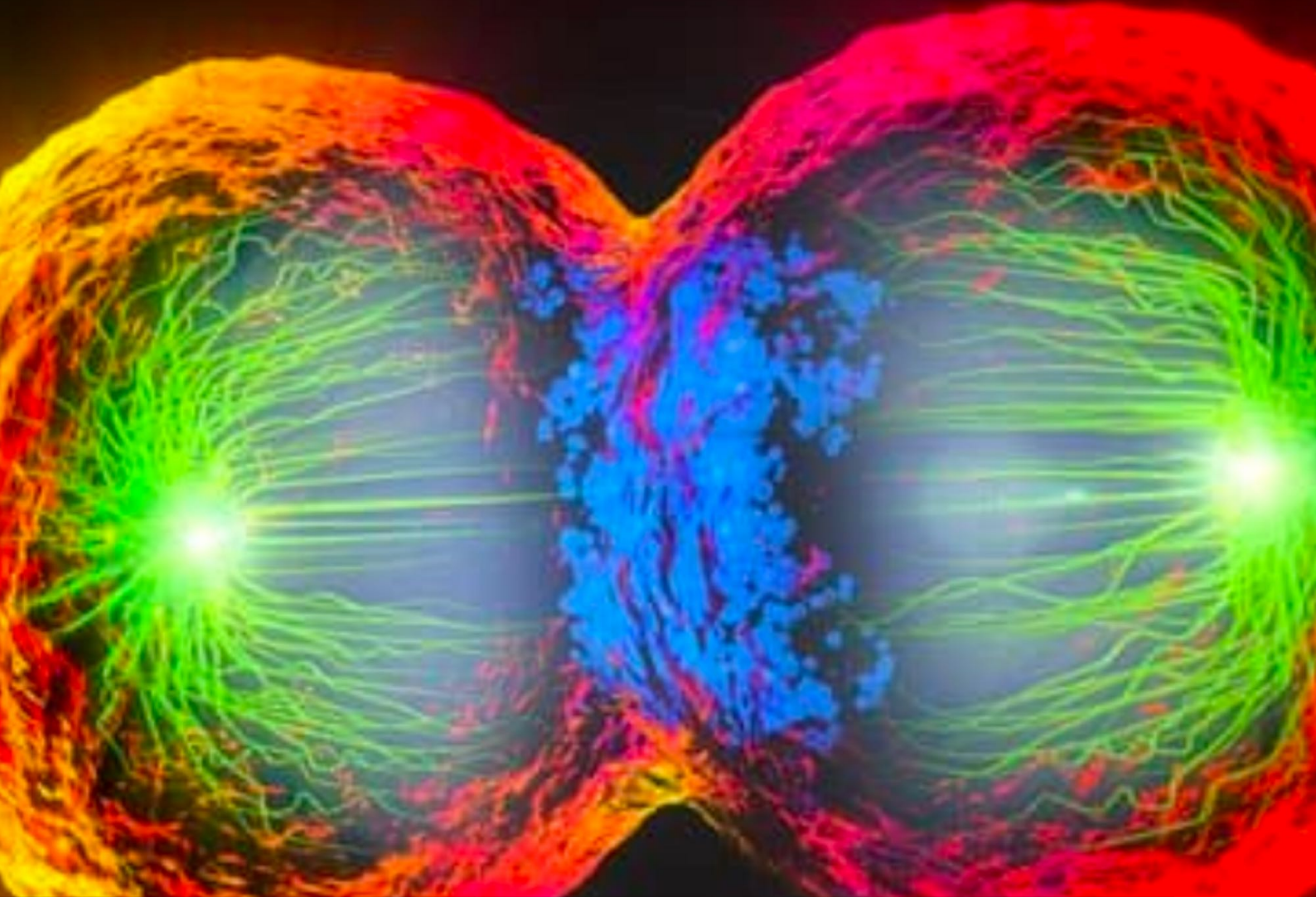
CELL BIOLOGY & BIOCHEMISTRY

NOTES

FOURTH EDITION

PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES

FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT



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A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

If you are new to us, here are a few things to help get the most out of your notes:

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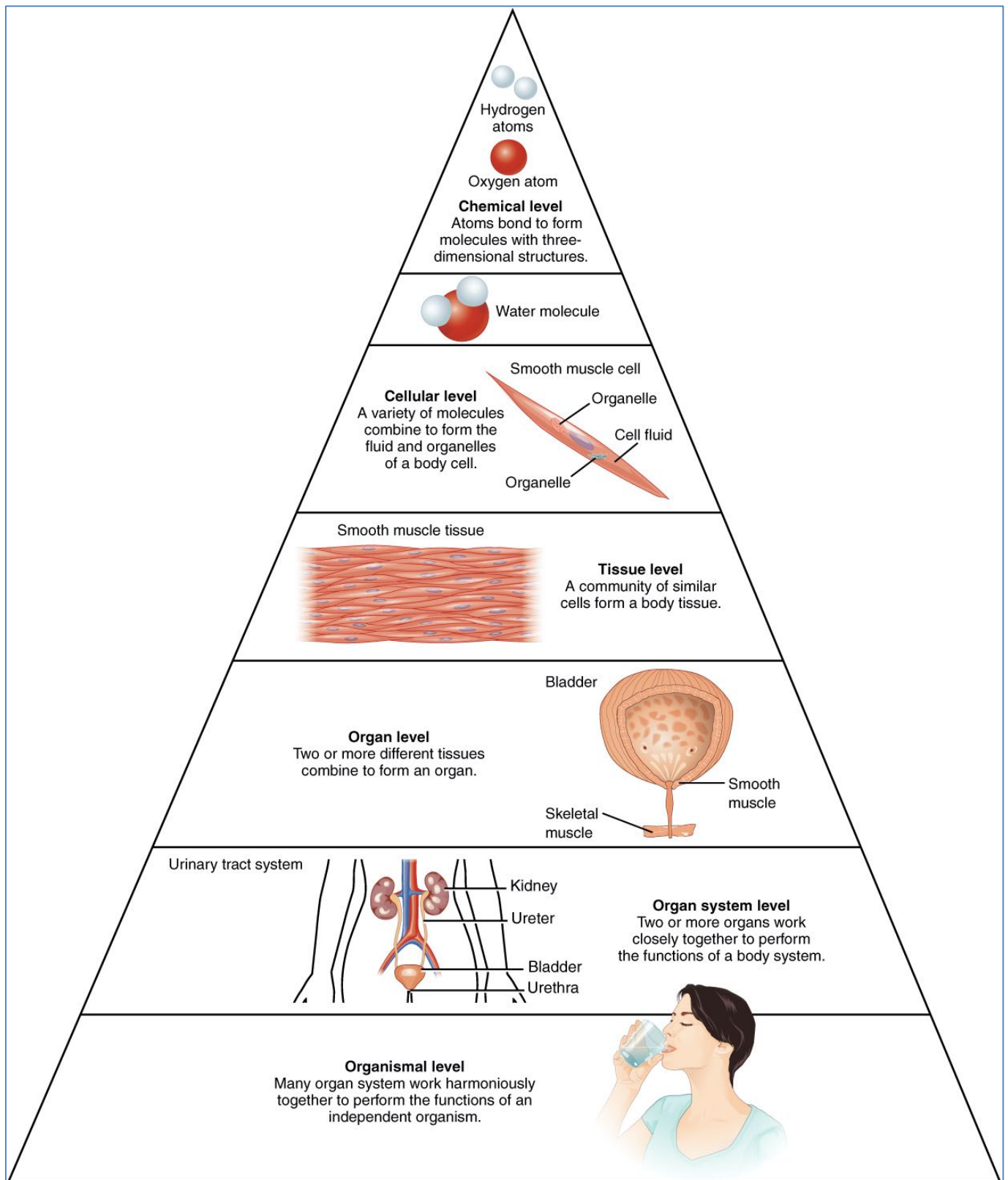
What's included: Ready-to-study summaries of a broad range of cellular physiology and biochemical topics, presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

Cellular Physiology & Biochemistry Topics:

- 5 LEVELS OF ORGANIZATION IN THE BODY
- MOLECULAR BUILDING BLOCKS OF LIFE:
 - o CARBOHYDRATES:
 - o LIPIDS / FATS:
 - o PROTEINS
 - o NUCLEIC ACIDS
- CELL STRUCTURE:
- CELL MEMBRANE & TRANSPORT
- CELLULAR METABOLISM
 - o CARBOHYDRATE METABOLISM
 - o THE CITRIC ACID CYCLE (TCA) / 'KREBS CYCLE':
 - o ELECTRON TRANSPORT CHAINS / OXIDATIVE PHOSPHORYLATION:
 - o AMINO ACID METABOLISM & THE UREA CYCLE
 - o THE UREA CYCLE:
 - o FATTY ACID METABOLISM
- CELLULAR SIGNALLING
- THE 'CELL CYCLE' & CELLULAR REPLICATION
- REGULATION OF CELL FATE
- CANCER, CELL DEATH & CELLULAR AGEING
- EPITHELIAL TISSUES (MEMBRANES & GLANDS)
- CONNECTIVE TISSUE:
- MUSCLE TISSUE:
- NERVOUS TISSUE
- MEMBRANE POTENTIAL & EXCITABLE TISSUES
- TISSUE INJURY & CELLULAR ADAPTATIONS
- MICROBIAL DIVERSITY

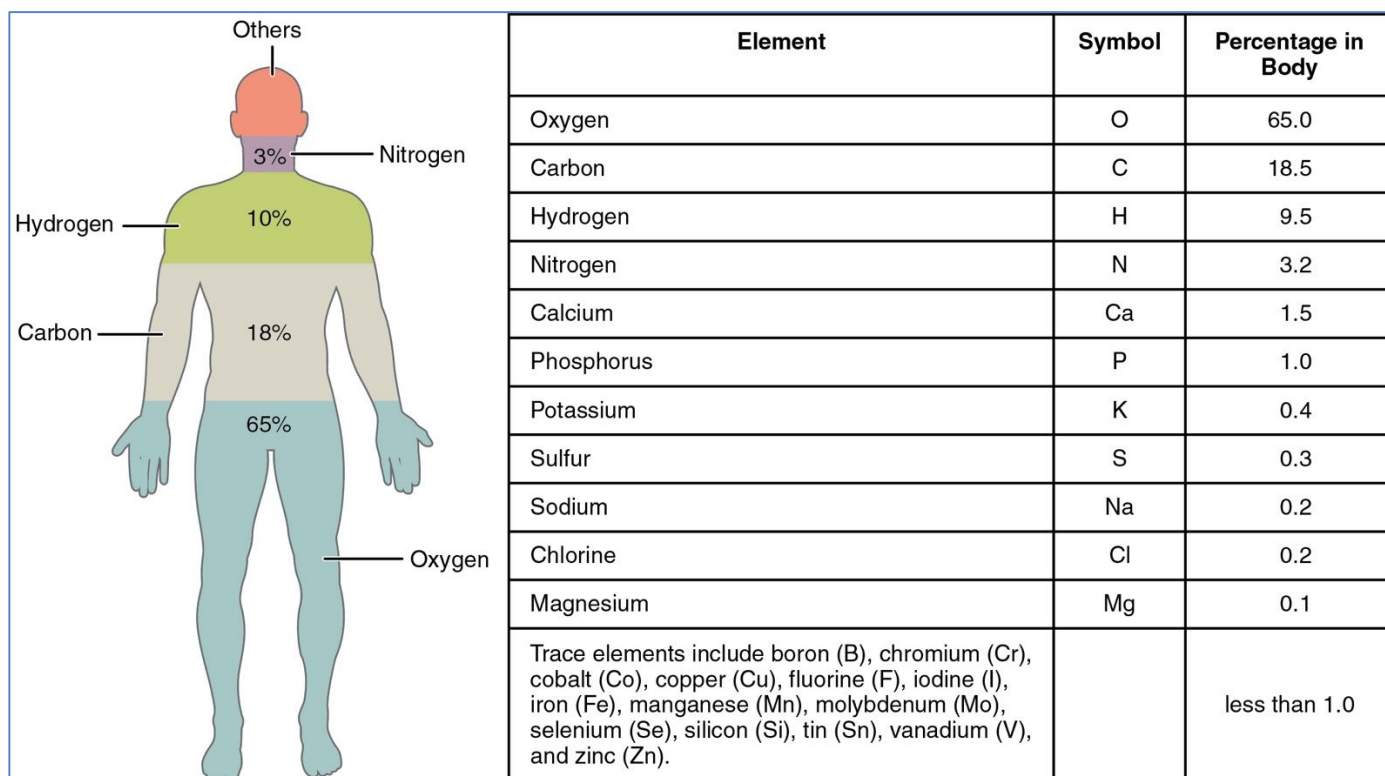
5 LEVELS OF ORGANIZATION IN THE BODY

5 LEVELS OF ORGANIZATION IN THE BODY:



1: The Chemical Level:

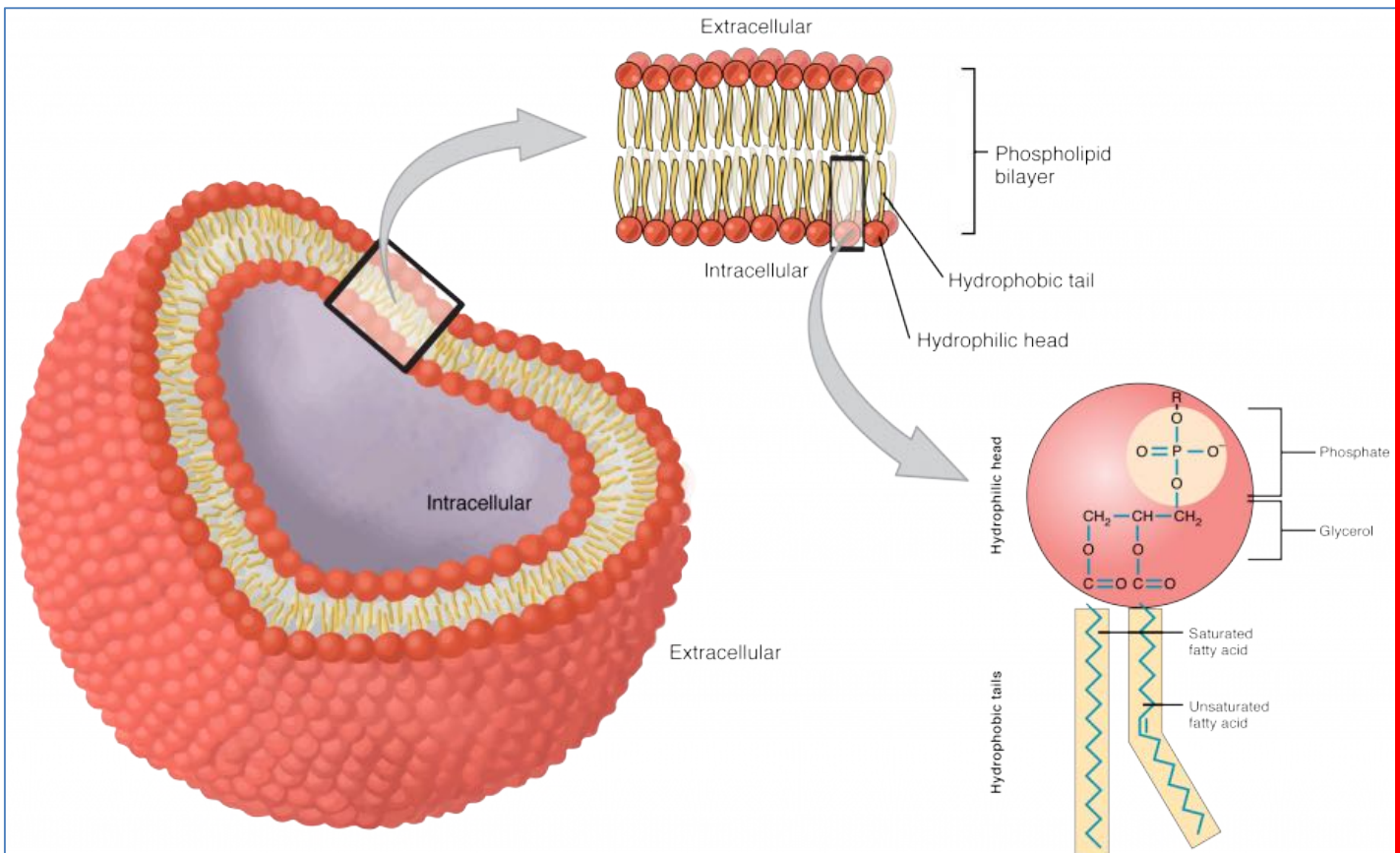
- All matter is made up of atoms & molecules
- 4 Biological Elements account for 96% of living matter:
 - o Carbon
 - o Oxygen
 - o Hydrogen
 - o Nitrogen
- These 4 elements combine to form the 4 major Macromolecules of life:
 - o Proteins
 - o Carbohydrates
 - o Fats
 - o Nucleic Acids



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2: The Cellular Level:

- **Cells are the basic functional units of life**
 - o I.e: There are no *sub*-cellular 'living things'
 - o The cell is the smallest unit capable of carrying out the processes associated with life
 - o Single-celled organisms (eg: Bacteria & amoebas) are the simplest forms of life
 - o Multi-cellular organisms (eg: Humans) are structural aggregates of trillions of cells
- **All cells are bound by a *Plasma Membrane*:**
 - o A bilipid membrane (hydrophilic on outer & inner surfaces; hydrophobic tails facing inwards)
 - o Encapsulates all internal cellular machinery
 - o Contains many proteins necessary for all types of functions:
 - Eg: Transporter proteins (control movement of materials into & out of the cell)
 - Eg: Antigen proteins (allows the body's immune cells to identify 'self' vs 'foreign' cells)
 - Eg: Cell membrane receptors (allows the cell to respond to outside chemical signals)
- **All cells perform certain basic functions:**
 - o Obtain food/nutrients from the environment
 - o Obtain oxygen from the environment
 - o Extract useful energy from food via respiration
 - o Eliminate its own waste products
 - o Synthesize macromolecules necessary for its own maintenance & functions
 - Eg: Proteins for growth
 - Eg: Enzymes for functions
 - Eg: Fats for energy & membrane repair/maintenance
 - Eg: Carbohydrates for energy
 - o Control the exchange of materials between itself and its surroundings
 - o Move materials internally from one part of the cell to another
 - o Sense and respond to changes in surrounding environment
 - o Self-replication (Except nerve and muscle cells)

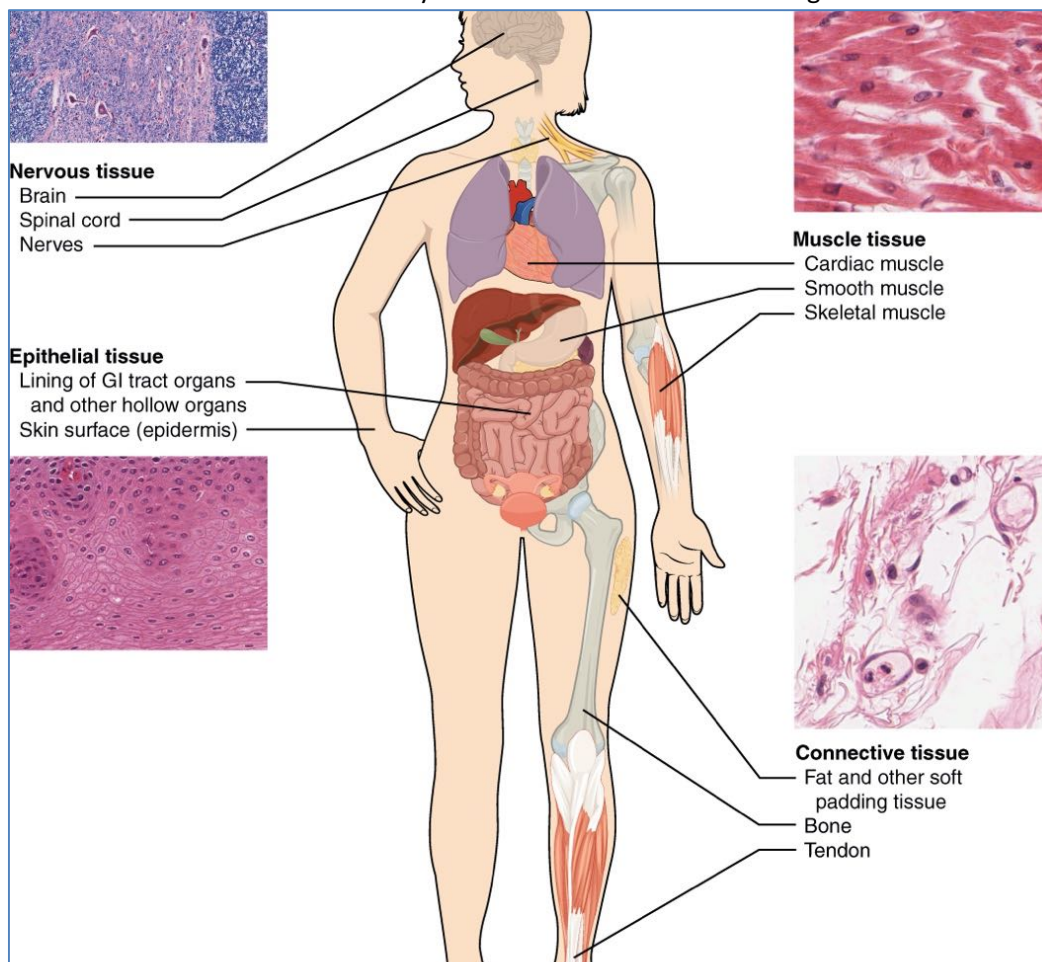


3: The Tissue Level:

- **Cells of similar structure and/or function, combine to form tissues**

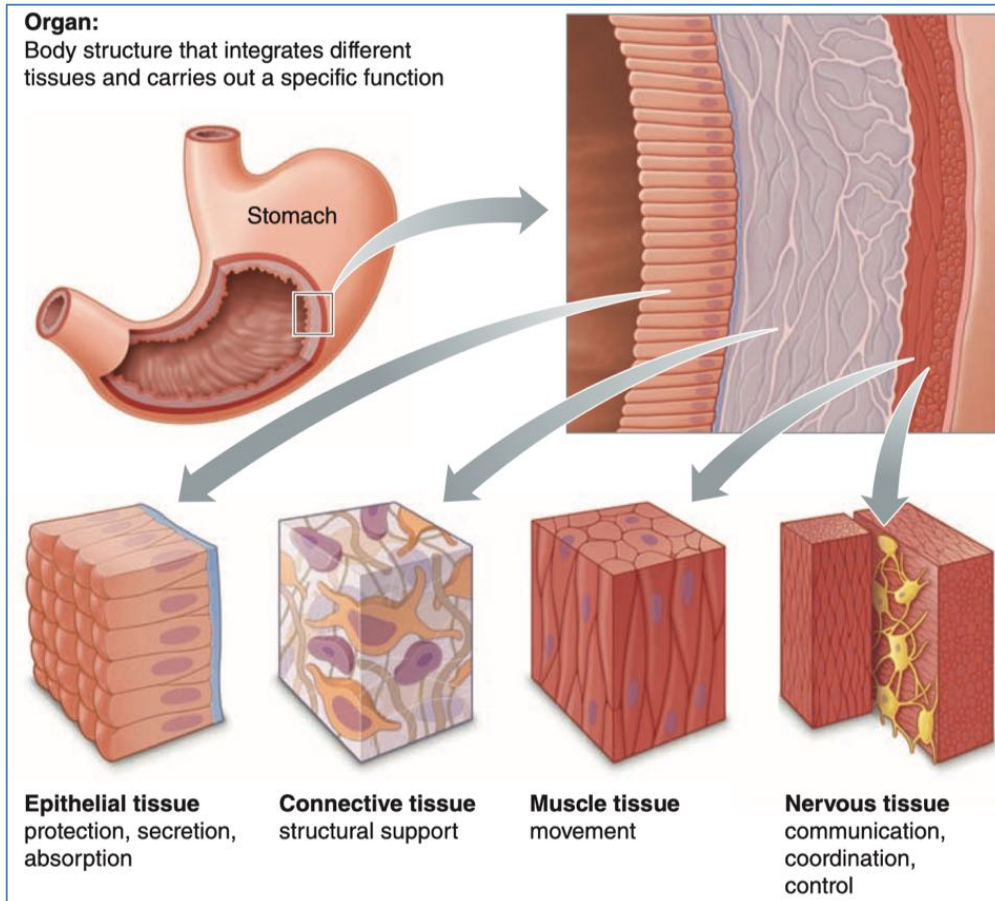
- **4x Primary Tissue Types:**

- **Muscle Tissue:**
 - Consist of contractile cells capable of generating tension & movement
 - 3 Types of Muscle Tissue:
 - Skeletal Muscle
 - Cardiac Muscle
 - Smooth Muscle
- **Nervous Tissue:**
 - Consist of cells specialized for initiating and transmitting electrical impulses
 - Electrical impulses = 'Action Potentials'
 - Signals relay information from one part of the body to another
- **Epithelial Tissue:**
 - Consist of cells specialized for exchanging materials between the cell & its environment
 - ANY substances that ENTERS or LEAVES the body, does so via an Epithelia Membrane
 - 2 Types of Epithelial Tissue:
 - Epithelial Sheets (Membrane)
 - Secretory Glands (Exocrine or Endocrine)
- **Connective Tissue:**
 - Consist of relatively few cells dispersed in an abundant extracellular matrix
 - Role: Connects/supports/anchors various body parts
 - 4 Types of Connective Tissue:
 - Connective Tissue Proper
 - Loose connective tissue
 - Dense connective tissue
 - Cartilage
 - Bone
 - Blood - technically a 'connective tissue' even though it's a fluid



4: The Organ Level:

- **Organ = 2 or more types of primary tissue organized to perform specific functions**
- **Eg: The Stomach:**
 - **Overall function** – To store food, digest food, and move it down the digestive tract
 - **Tissue Types:**
 - **Epithelial** – secrete digesting juices into the lumen
 - **Connective** – Binds together all other tissues
 - **Muscle** – Smooth muscle contractions mix ingested food
 - **Nervous** – controls muscle contraction & gland secretion



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5: The Body System Level:

- **Body System = 2 or more organs organized to perform related functions**
- **Eg: Digestive System; contains:**
 - Stomach
 - Small intestine
 - Large intestine
 - Salivary glands
 - Exocrine pancreas
 - Liver & Gallbladder
- **Body has 11 Systems:**
 - Cardiovascular (Circulatory)
 - Digestive (Gastrointestinal)
 - Respiratory
 - Urinary
 - Skeletal
 - Muscular
 - Integumentary (Skin)
 - Immune
 - Nervous
 - Endocrine
 - Reproductive

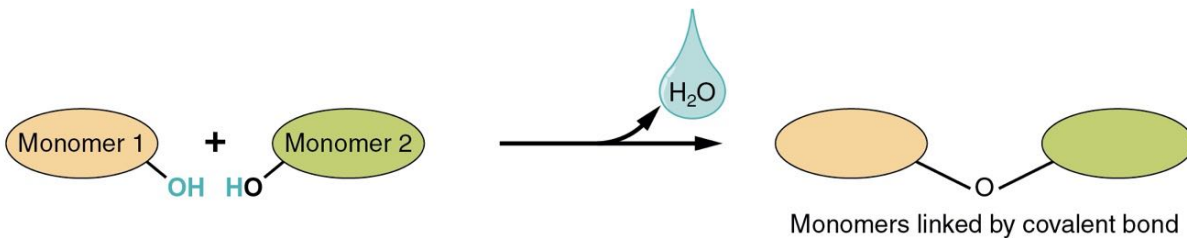
MOLECULAR BUILDING BLOCKS OF LIFE:

MOLECULAR BUILDING BLOCKS OF LIFE:

- All matter is made up of atoms & molecules
- 4 Biological Elements account for 96% of living matter:
 - o Carbon
 - o Oxygen
 - o Hydrogen
 - o Nitrogen
- These 4 elements combine to form the 4 major Macromolecules of life:
 - o Carbohydrates (Sugars)
 - o Fats (Polymers of Fatty Acids)
 - o Proteins (Polymers of Amino Acids)
 - o Nucleic Acids (Polymerizes to form DNA & RNA)
- Most of them are polymers
 - o Made by stringing together many smaller molecules (monomers)
 - o Monomers bond (polymerise) by dehydration reactions and break down by hydrolysis:

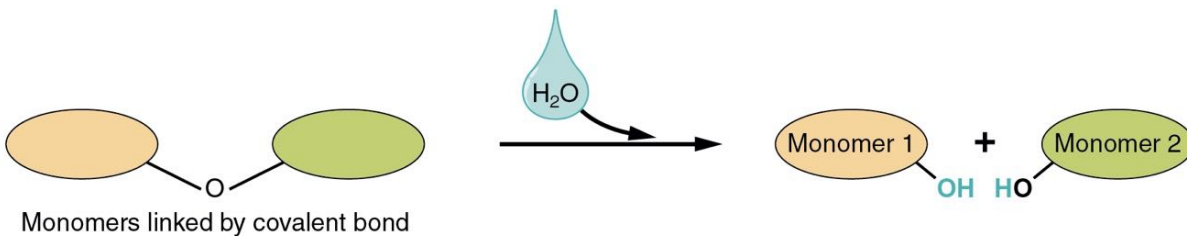
(a) Dehydration synthesis

Monomers are joined by removal of OH from one monomer and removal of H from the other at the site of bond formation.

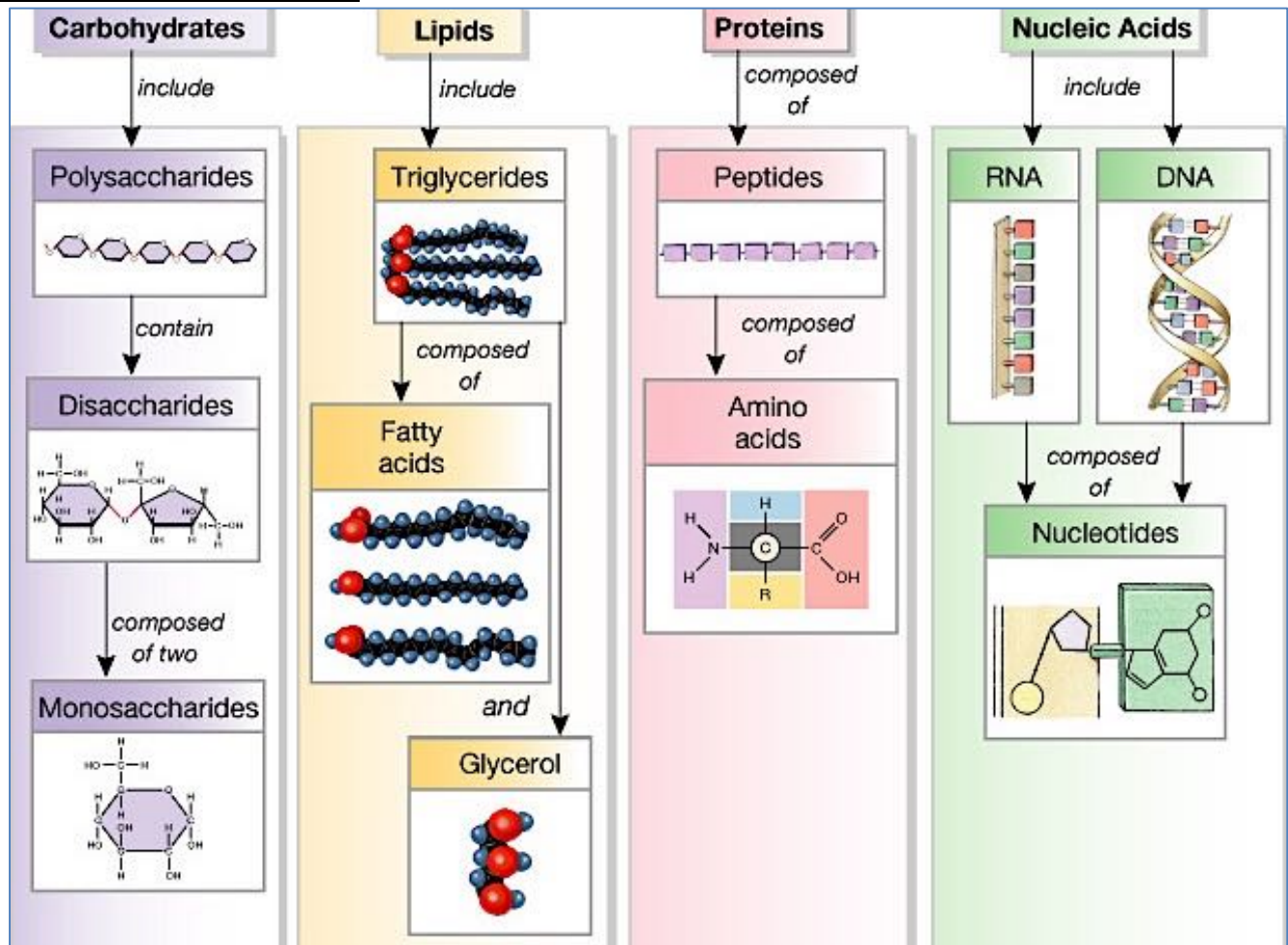


(b) Hydrolysis

Monomers are released by the addition of a water molecule, adding OH to one monomer and H to the other.



The Biological MACROmolecules:



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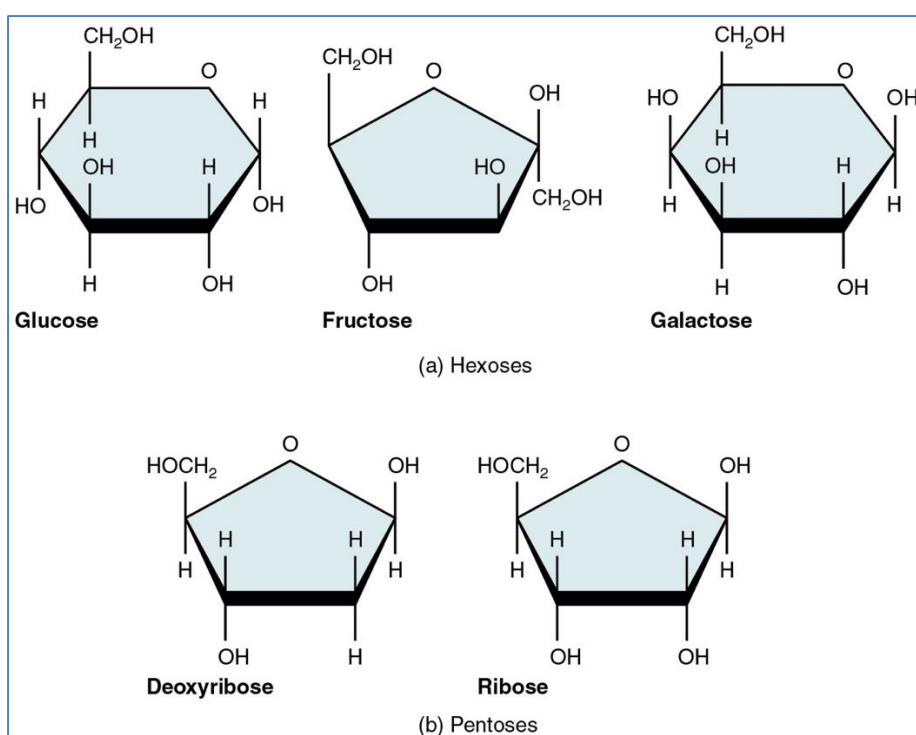
CARBOHYDRATES:

General Info About Carbohydrates:

- **What Are They?**
 - o Biological compounds containing covalently bonded **carbon, hydrogen, and oxygen** (in a 1:2:1 ratio)
- **Importance:**
 - o Monosaccharides = important cellular nutrients
 - o Metabolised by cells to produce usable energy
 - o Important store of energy reserves
- **Structural Classifications:**
 - o **Monosaccharides** = Single-Sugar units:
 - Glucose
 - Fructose
 - Galactose
 - o **Disaccharides** = Double-Sugar units:
 - Sucrose
 - Lactose
 - Maltose
 - o **Polysaccharides** = Multi-Sugar Polymers:
 - Glycogen
 - Starch
 - Cellulose (Plants)

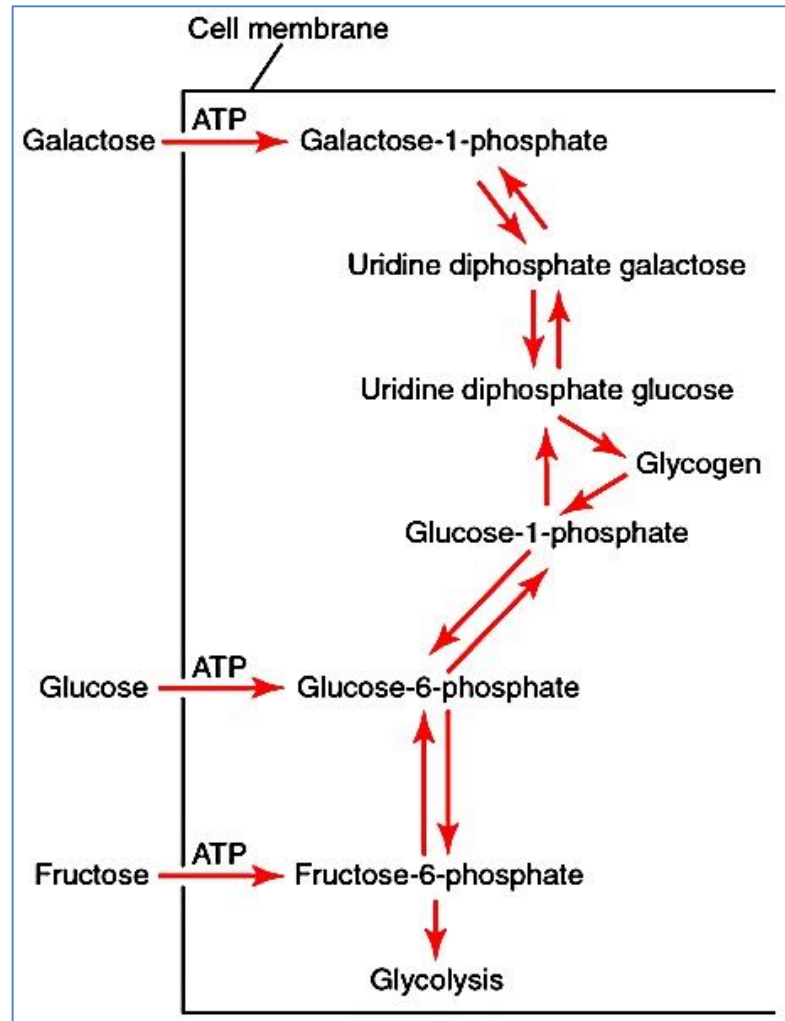
Monosaccharides (Simple Sugars):

- **Monosaccharide Literally Means: “single” “Sugar-unit”**
 - o Cannot be broken down into simpler sugars
- **Examples:**
 - o Glucose
 - o Fructose
 - o Galactose
- **Monosaccharides are Isomers:**
 - o Ie: Have the same chemical formulae but different structural arrangements
 - o Still contain exactly the same amount of energy
- **In aqueous solutions, monosaccharides form rings**
 - o Are the main fuel used by cells



- **Each Monosaccharide Has Its Own Metabolic Pathway:**

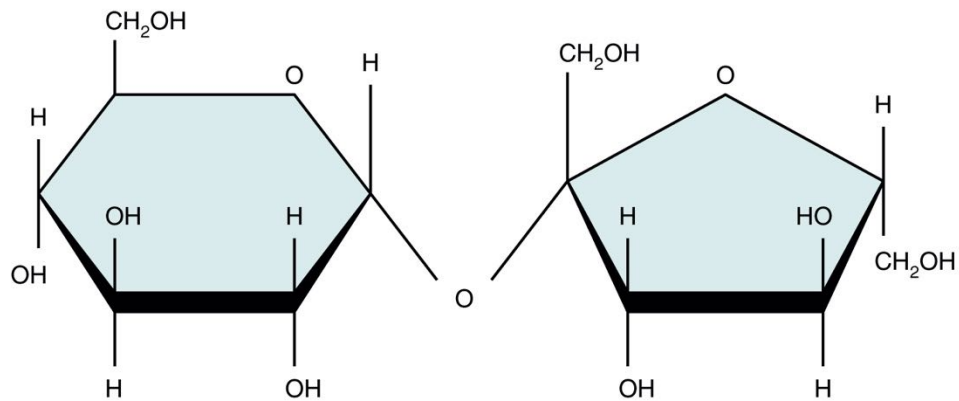
- Glucose → **Glycolysis** → ATP
- Fructose → **Fructolysis** → Glycolysis → ATP
- Galactose → **Leloir Pathway** → Glycolysis → ATP
- (Note how ALL 3 eventually feed into Glycolysis)



Disaccharides:

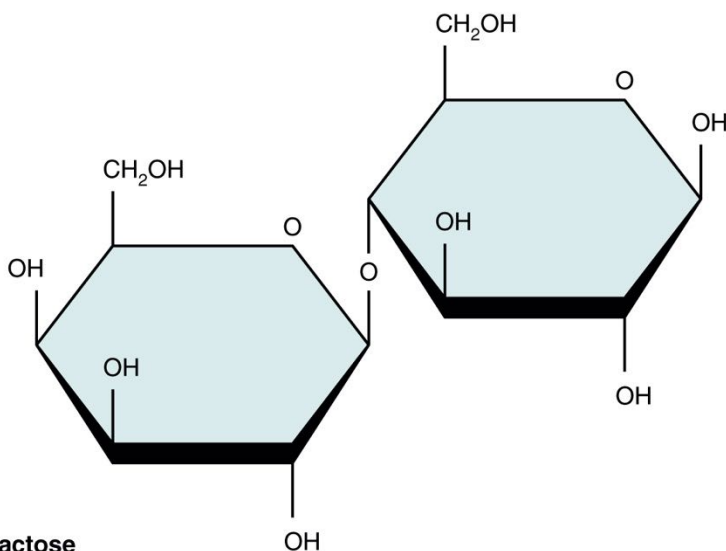
- “Double-Sugar Units”
 - I.e: Consist of 2 monosaccharides
- 3x Digestible Disaccharides:
 - **Maltose**: Glucose + Glucose
 - **Lactose**: Glucose + Galactose
 - **Sucrose (table sugar)**: Glucose + Fructose

(a) The monosaccharides glucose and fructose bond to form sucrose



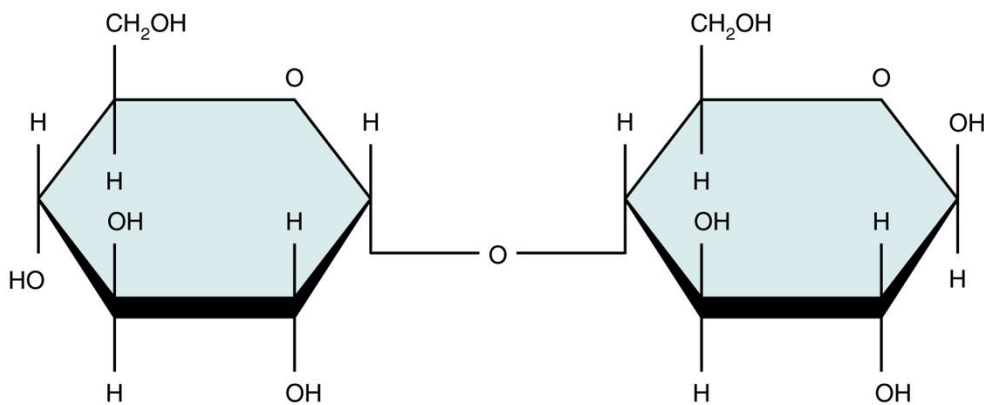
Sucrose

(b) The monosaccharides galactose and glucose bond to form lactose.



Lactose

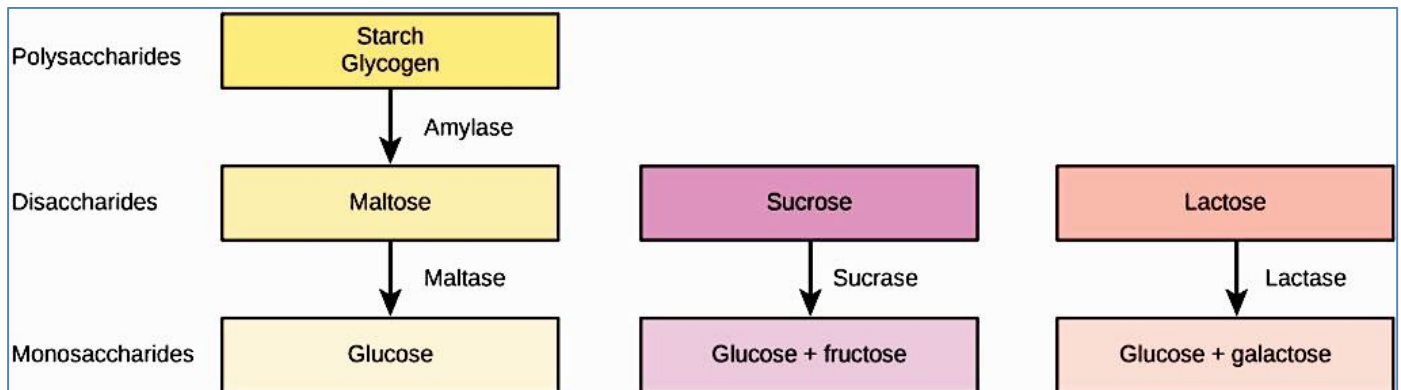
(c) Two glucose monosaccharides bond to form maltose.



Maltose

- **Disaccharide Metabolism:**

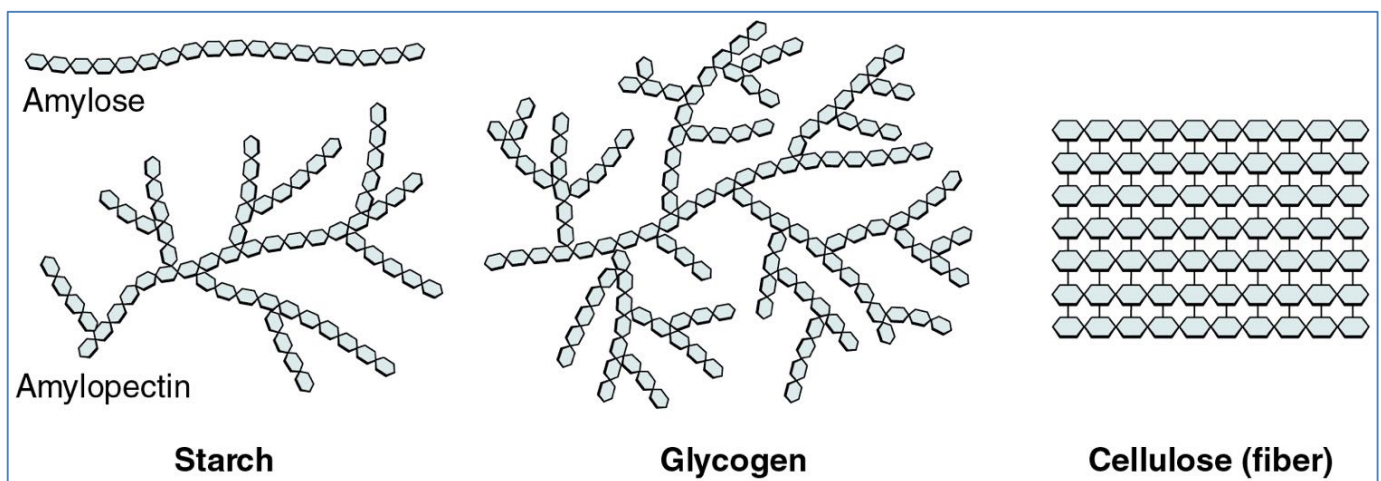
- First requires breakdown into constituent Monosaccharides In order for the body to utilise
- **Requires Specific Enzymes**
 - Sucrase → Hydrolyzes Sucrose into **Glucose + Fructose**
 - Lactase → Hydrolyzes Lactose into **Glucose + Galactose**
 - Maltase → Hydrolyzes Maltose into **Glucose + Glucose**



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Polysaccharides:

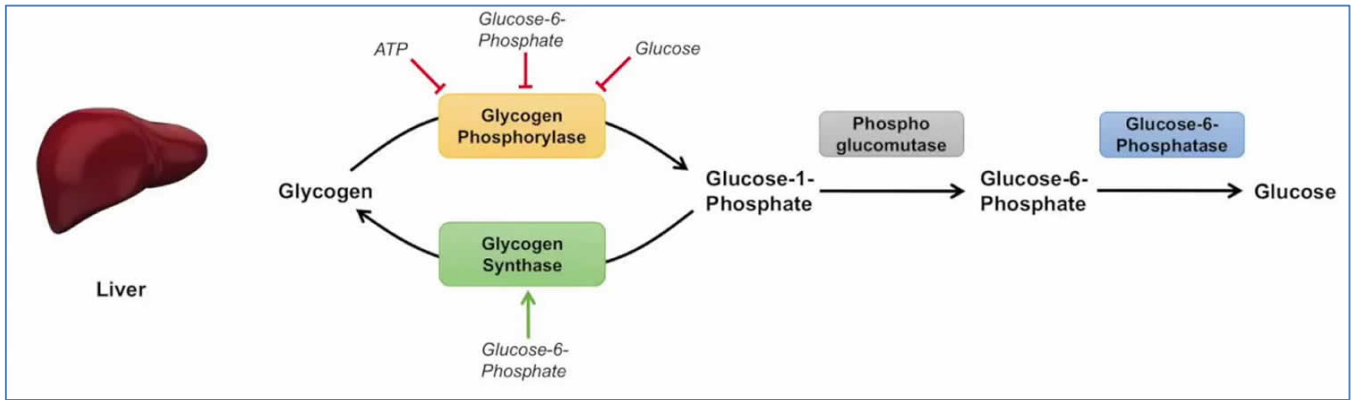
- **AKA: Complex Carbohydrates**
 - Long Polymers of Monosaccharides (Single Sugar Units)
- **3x Polysaccharides:**
 - **Starch**
 - Stores energy in plant cells (potatoes/grains)
 - Made of many Glucose monomers
 - **Glycogen (animal starch)**
 - Animals store excess sugar as glycogen
 - Made of many Glucose monomers
 - Contains many branches
 - **Cellulose**
 - Makes up the structure of plant-cell walls
 - Major component of wood
 - Is a dietary fibre
 - Can only be broken down by grazing animals due to prokaryotes in their digestive system



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- **Glycogen Metabolism & Storage In The Body:**

- **Glycogenesis** = Creating glycogen from excess glucose following a meal
- **Glycogenolysis** = Tapping into glycogen to liberate glucose in times of fasting



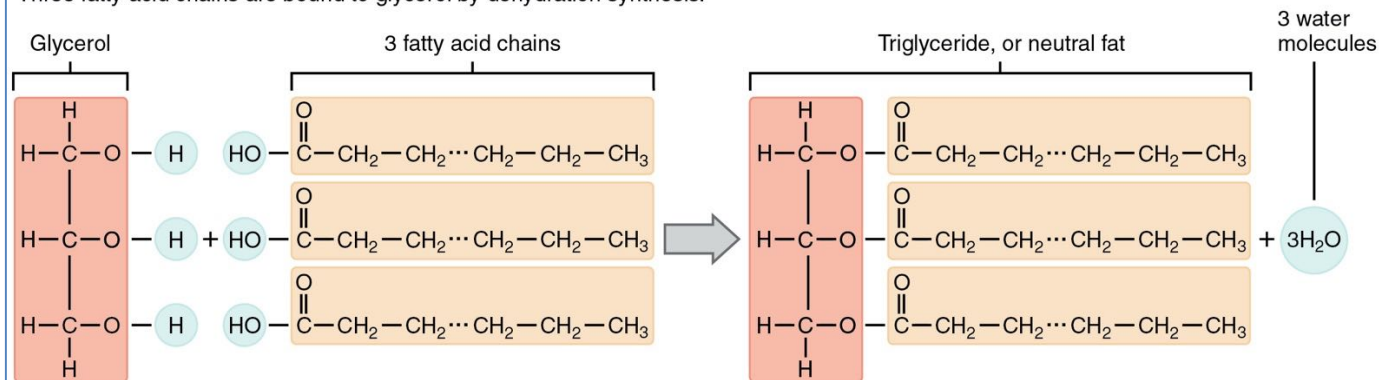
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LIPIDS / FATS:

General Info About Lipids/Fats:

- **What are they?**
 - o Biological compounds containing **hydrocarbons**
 - o Not soluble in water (**hydrophobic**)
 - o Eg: Fats/waxes/oils/sterols/triglycerides/phospholipids
- **Importance:**
 - o Major structural component of cell membranes (lipid bilayer)
 - o Major class of chemical messenger (Eg: Steroid hormones)
 - o Major store of energy (triglycerides)
 - o Major source of energy (fatty acids)
 - o Major solvent for certain vitamins (Vitamins A, D, E & K)
 - o Major functional barriers (Eg: Skin oils, ear wax, cerumen)
 - o Major source of insulation/cushioning of vital organs (Eg: Kidneys and heart)
- **3 Relevant Types:**
 - o **Fatty Acid:** A long hydrocarbon chain with a carboxyl group end
 - **Saturated** fatty acids are **straight**
 - Pack tightly together
 - Solid @ Room Temperature
 - **Unsaturated** fatty acids are **kinked**
 - Pack loosely together
 - Liquid @ Room Temperature
 - o **Triglycerides:** 3 x fatty acids bonded to a glycerol through dehydration (ester linkage)

Three fatty acid chains are bound to glycerol by dehydration synthesis.



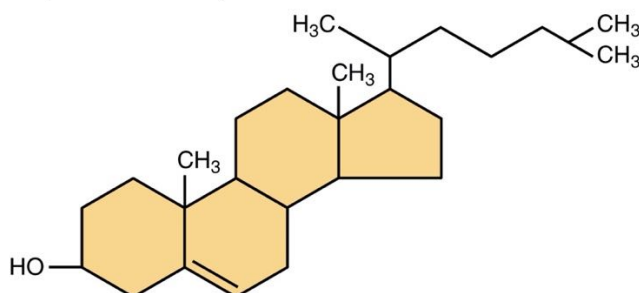
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- o **Steroids:**
 - = any fatty substances containing four carbon rings
 - cholesterol is the “base steroid” from which your body produces other steroids (sex hormones)

Sterols

Four interlocking hydrocarbon rings from a steroid.

Example: Cholesterol (cholesterol is the basis for all steroids formed in the body)



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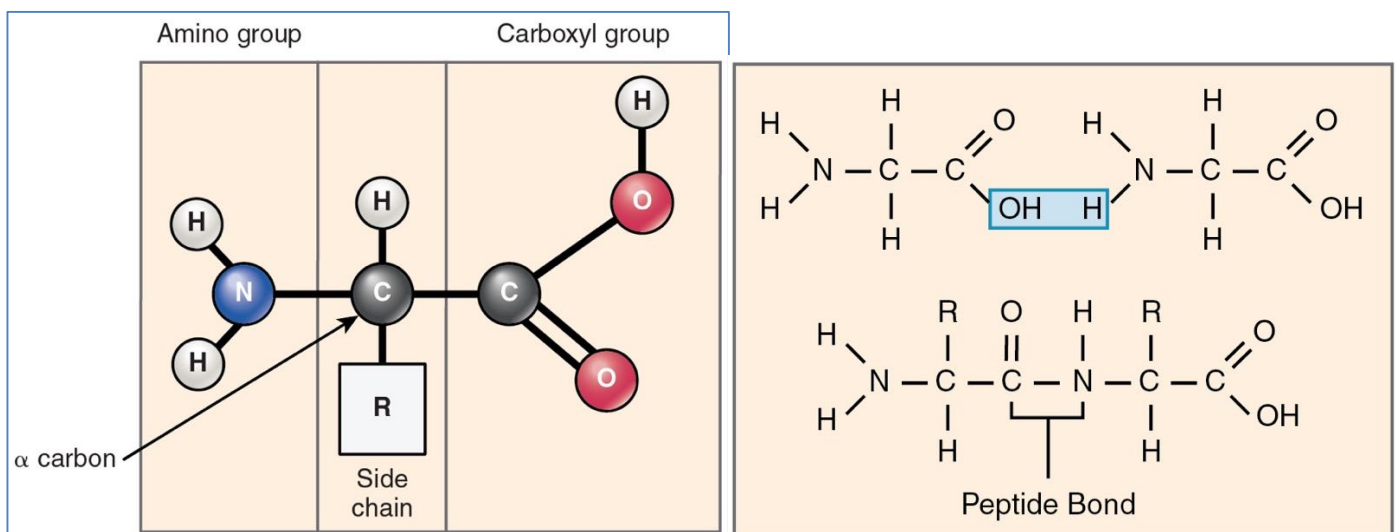
PROTEINS

General Info About Proteins:

- **What are they?**
 - o Biological polymers of linked **amino acid** monomers (via a peptide bond)
 - o The most complex and functionally diverse molecules of living organisms
- **Important Roles Of Proteins In The Body:**
 - o Protein Enzymes (Eg: Digestion, metabolism, cellular repair)
 - o Protein Hormones
 - o Carrier Proteins (Eg: Albumin)
 - o Cellular Receptor Proteins
 - o Membrane Transporter Proteins (Eg: Na/K/ATP-ase)
 - o Contractile Proteins (muscle tissue)
 - o Structural Proteins
 - o Storage Proteins
 - o Defensive Proteins
 - o Sensory Proteins
 - o Gene Regulatory Proteins
 - o Etc etc
- **Think about this:**
 - o The ONLY reason DNA exists, is to encode the creation of all the proteins necessary for you to exist
 - o Proteins are created from DNA Transcription and Translation
- **Base elements:**
 - o **Carbon**
 - o **Hydrogen**
 - o **Oxygen**
 - o **Nitrogen**

Amino Acids:

- **Each amino acid consists of:**
 - o **A central carbon** covalently bonded to 4 partners
 - o **An amino group**
 - o **A carboxyl group**
 - o **A side group** (Variable among all 20 amino acid types)
- **ALL proteins are constructed from Amino Acids**
 - o Amino acids join together by dehydration reactions forming peptide bonds:



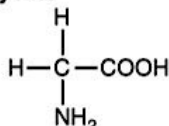
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- **There are 20 relevant amino acids**

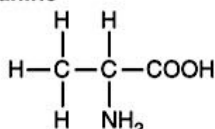
- **Some are Essential** (Either cannot be synthesized at all or cannot be synthesized in sufficient quantities in the body; Therefore, must be consumed in food)
- **Some are Non-Essential** (Body can synthesize them & in sufficient quantity)
- **Note:** There is broad disagreement among textbooks of exactly which are essential or not, some even further classifying certain amino acids as 'conditionally non-essential', so below is Guyton's list:

AMINO ACIDS

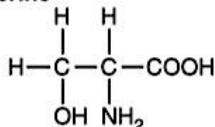
Glycine



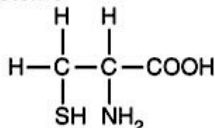
Alanine



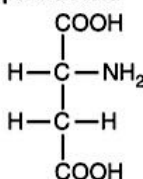
Serine



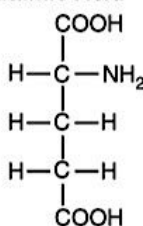
Cysteine



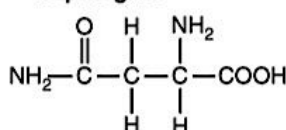
Aspartic Acid



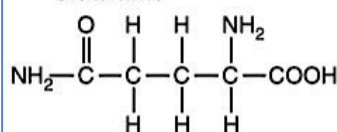
Glutamic Acid



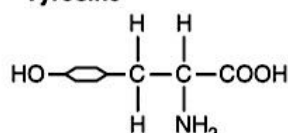
Asparagine



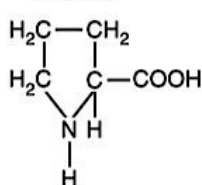
Glutamine



Tyrosine

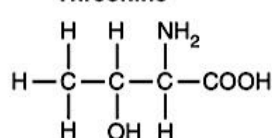


Proline

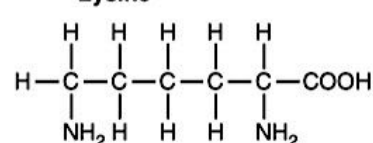


ESSENTIAL AMINO ACIDS

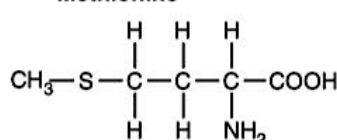
Threonine



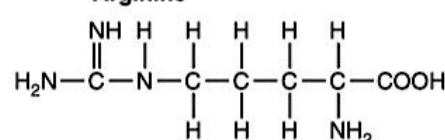
Lysine



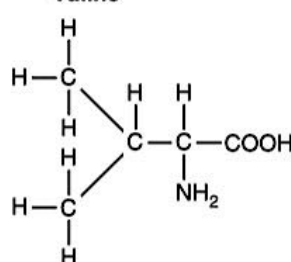
Methionine



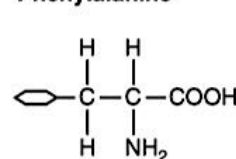
Arginine



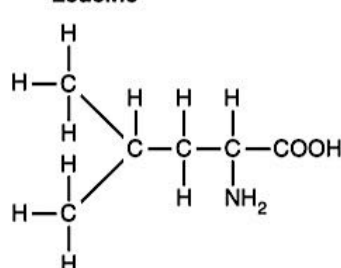
Valine



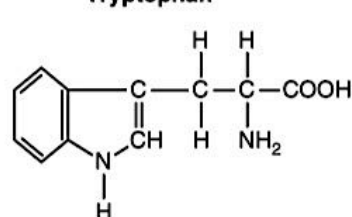
Phenylalanine



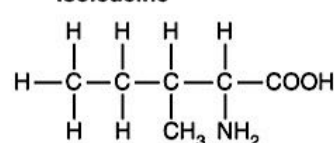
Leucine



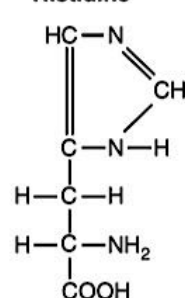
Tryptophan



Isoleucine

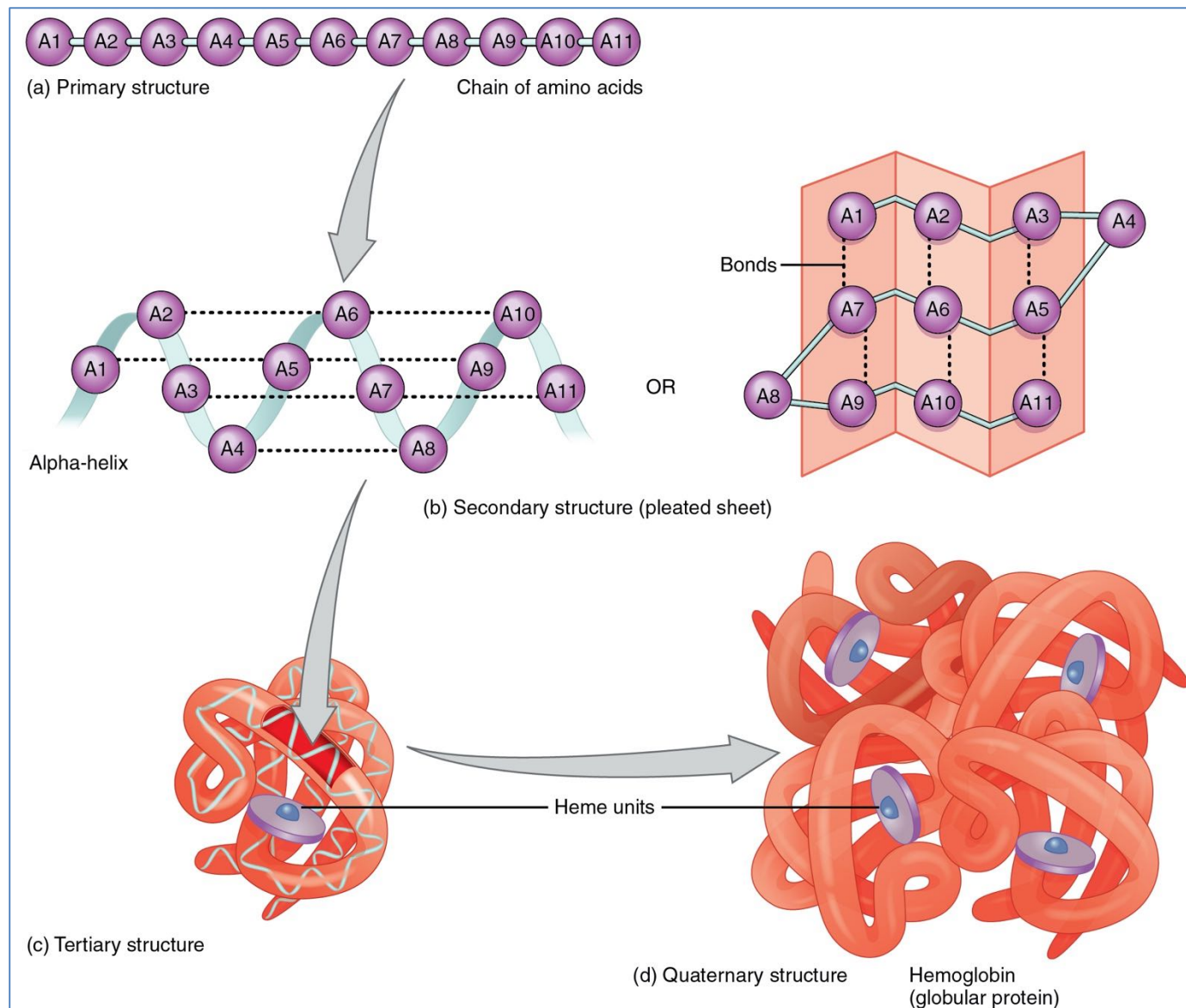


Histidine



Protein Shape/Structures:

- **Primary Protein Structure** (multiple peptide bonds = **polypeptide** *chains*)
 - Written L→R from amino end to carboxylic acid end
- **Secondary Protein Structure**
- **Tertiary Protein Structure**
- **Quaternary Protein Structure**
 - Complete functional protein



The Shape of Proteins (a) The primary structure is the sequence of amino acids that make up the polypeptide chain. (b) The secondary structure, which can take the form of an alpha-helix or a beta-pleated sheet, is maintained by hydrogen bonds between amino acids in different regions of the original polypeptide strand. (c) The tertiary structure occurs as a result of further folding and bonding of the secondary structure. (d) The quaternary structure occurs as a result of interactions between two or more tertiary subunits. The example shown here is hemoglobin, a protein in red blood cells which transports oxygen to body tissues.

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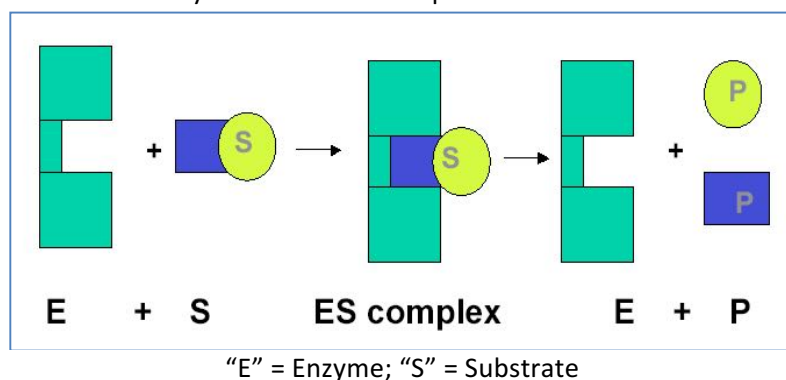
Note: A protein's shape is sensitive to the environment and can be denatured by change in temperature and pH

Proteins as Enzymes:

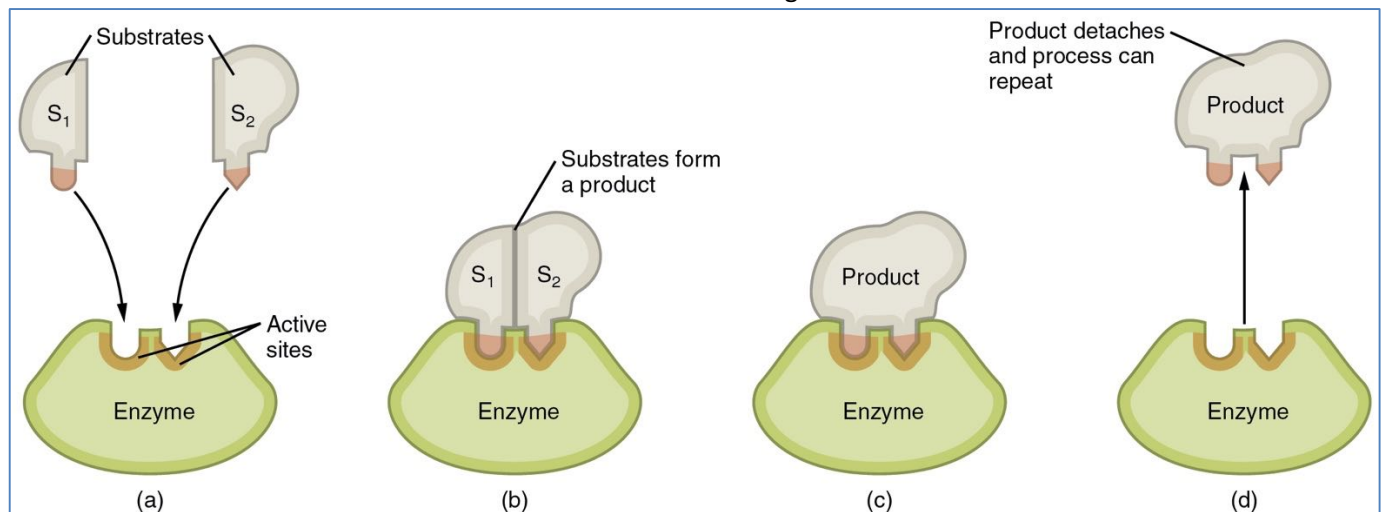
- **What are Enzymes:**
 - Compounds that **Catalyse biological reactions**
 - Almost all enzymes are proteins
 - Act to Lower the activation energy of a reaction
 - May contain cofactors (metal ions for vitamins)
- **Most Enzyme Names end in “-ase”**
 - Is specific for the chemical that it reacts
 - Eg: Sucrase – reacts sucrose
 - Eg: Lipase – reacts lipids
 - Describes the function of that enzyme
 - Eg: Oxidase – catalyses oxidation
 - Eg: Hydrolase – catalyses hydrolysis
 - NOTE: some don't conform: (pepsin, trypsin)

Enzyme Mechanisms:

- **Lock & Key Model:**
 - The active site of the enzyme is the same shape as the substrate that it reacts



- **Induced Fit Model:**
 - The enzyme and active site adjust shape to bind to the substrate
 - Therefore is more versatile in terms of range of substrates



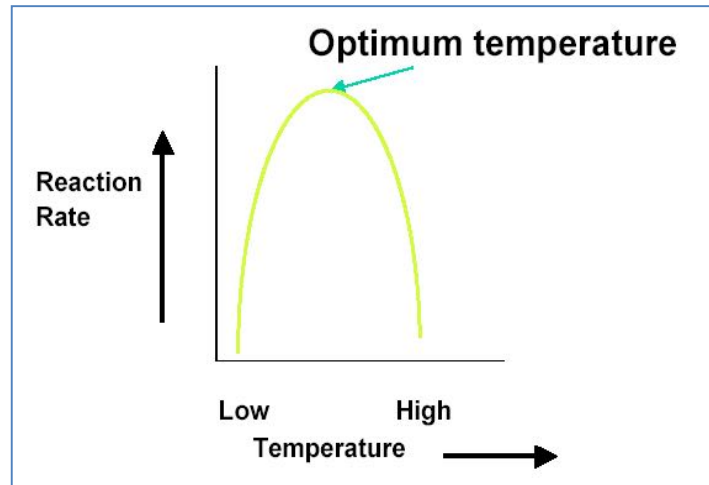
Steps in an Enzymatic Reaction According to the induced-fit model, the active site of the enzyme undergoes conformational changes upon binding with the substrate. (a) Substrates approach active sites on enzyme. (b) Substrates bind to active sites, producing an enzyme-substrate complex. (c) Changes internal to the enzyme-substrate complex facilitate interaction of the substrates. (d) Products are released and the enzyme returns to its original form, ready to facilitate another enzymatic reaction.

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Factors Affecting Enzyme Action:

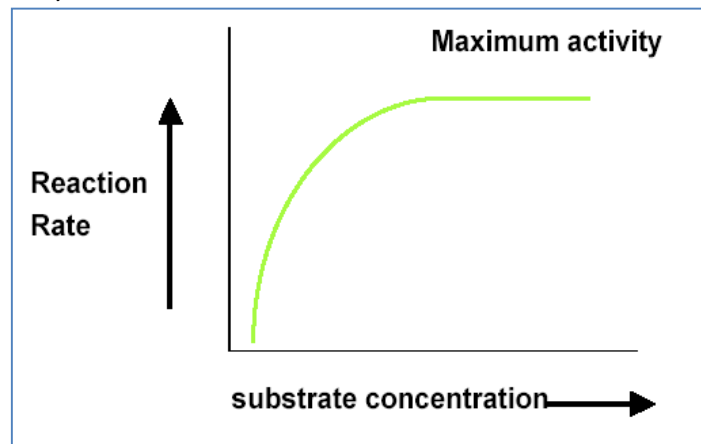
- Temperature:

- Little activity at low temp
- Rate increases with temp increases
- Reach optimum temp (37°C in humans)
- Activity lost @ high temps due to denaturation



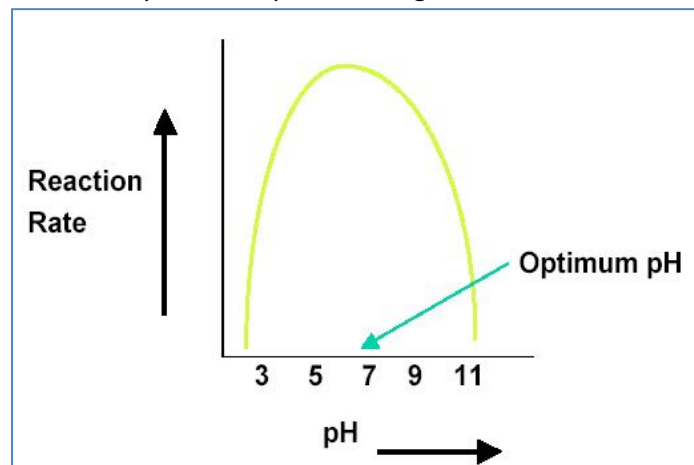
- Substrate Concentration:

- Activity increases with increasing substrate concentration
- Maximum activity reached when concentration of substrate = concentration of enzyme



- pH:

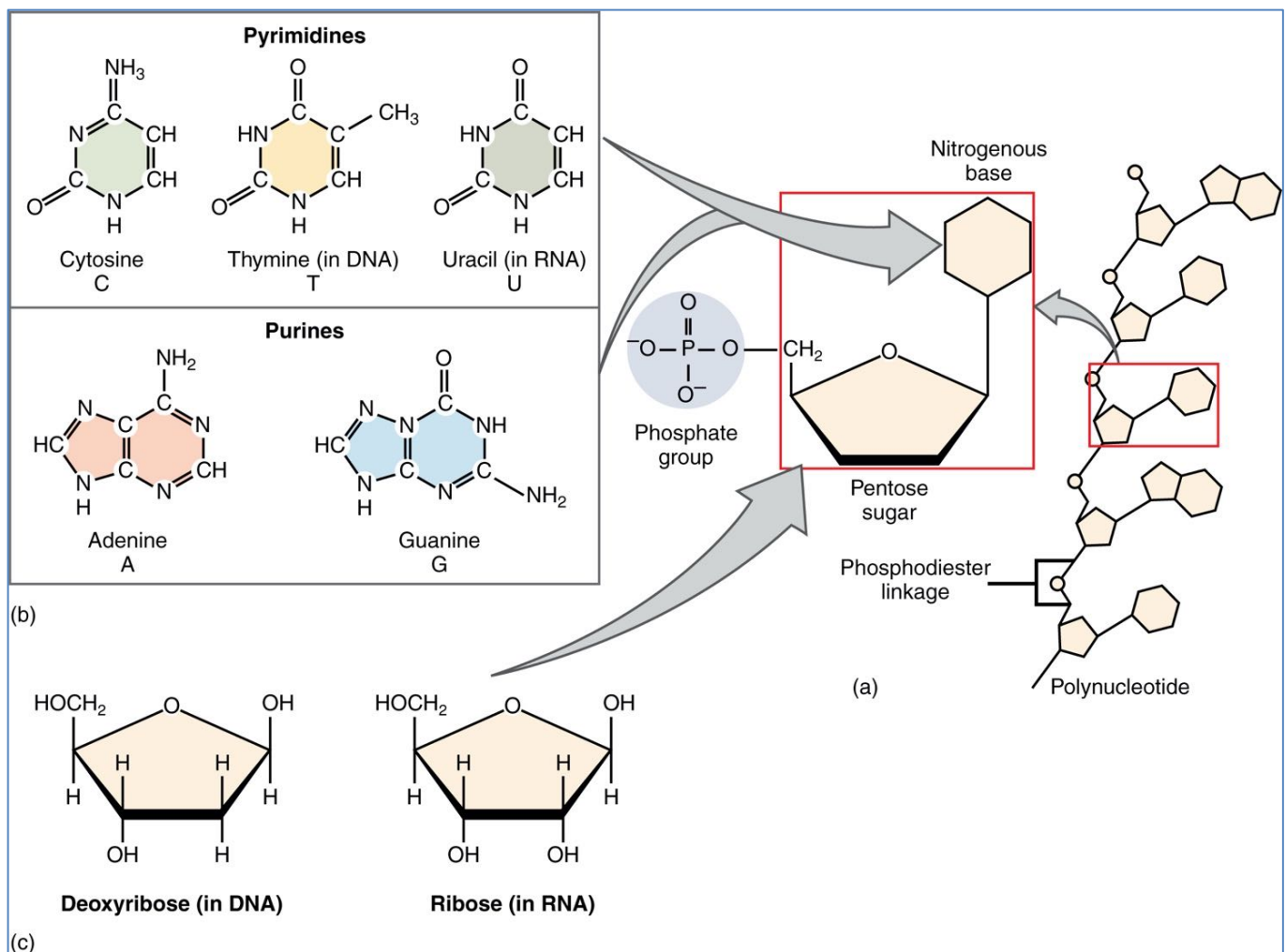
- maximum activity at optimum pH (narrow range)
 - R-groups have proper charge
 - Tertiary structure of enzyme is correct
 - Most lose activity outside optimum range

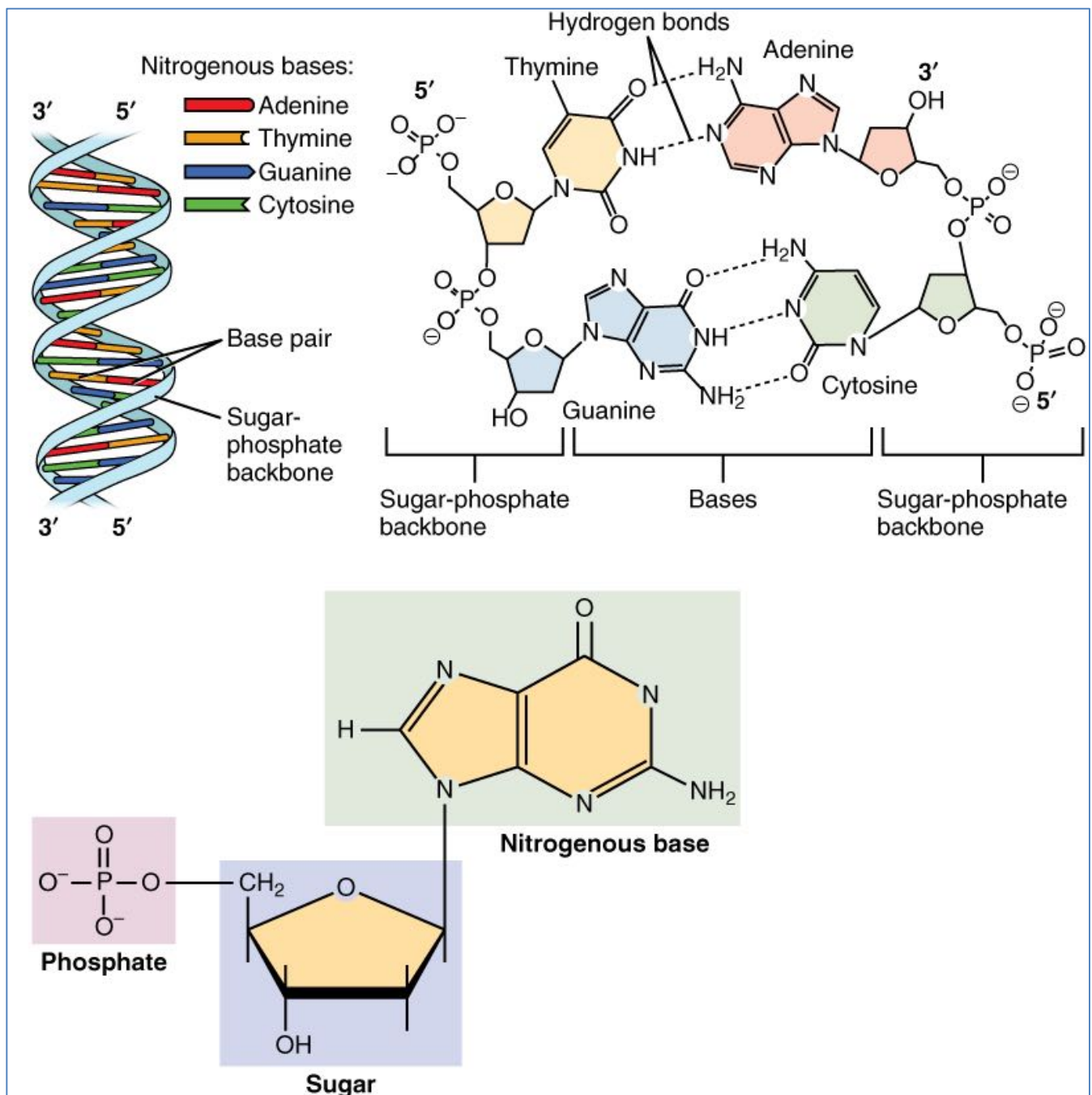


NUCLEIC ACIDS

General Info About Nucleic Acids:

- **What Are They?**
 - "Nucleic acids" = a family of biopolymers, named for their role in the cell nucleus
 - **Composed of chains of monomeric nucleotides ('Bases')**
 - **Adenine (A)** (Only bonds to T)
 - **Guanine (G)** (Only bonds to C)
 - **Thymine (T)** (Only bonds to A)
 - **Cytosine (C)** (Only bonds to G)
 - **(Note: Uracil replaces Thymine in RNA)**
 - Form chains called **polynucleotides** or just **DNA strands**
 - **Joined by a sugar-phosphate backbone**
- **Importance:**
 - Provide the directions for building of all proteins necessary for life
 - Encodes Phenotypes/Traits in all animals
 - Central to the success of evolution (The genes that encode fitness get passed on)

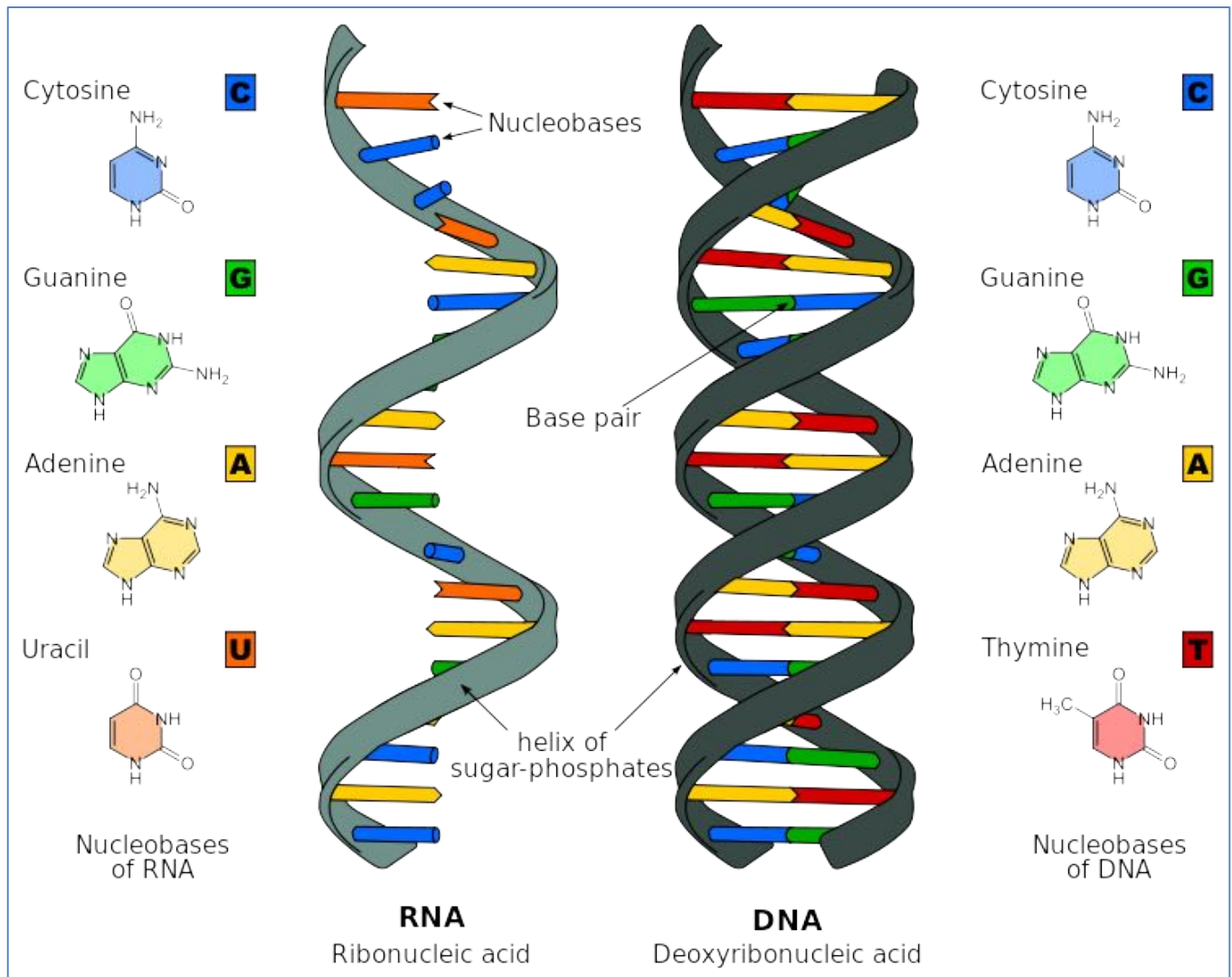




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2 Types Of Nucleic Acids:

- **DNA:** Deoxyribonucleic Acid
 - The stable genetic code stored in the nucleus of cells
- **RNA:** Ribonucleic Acid
 - Translates genetic information from DNA into proteins
 - Acts as a messenger between DNA and the ribosomes (protein synthesis organelles)
 - Has the base **Uracil instead of Thymine**



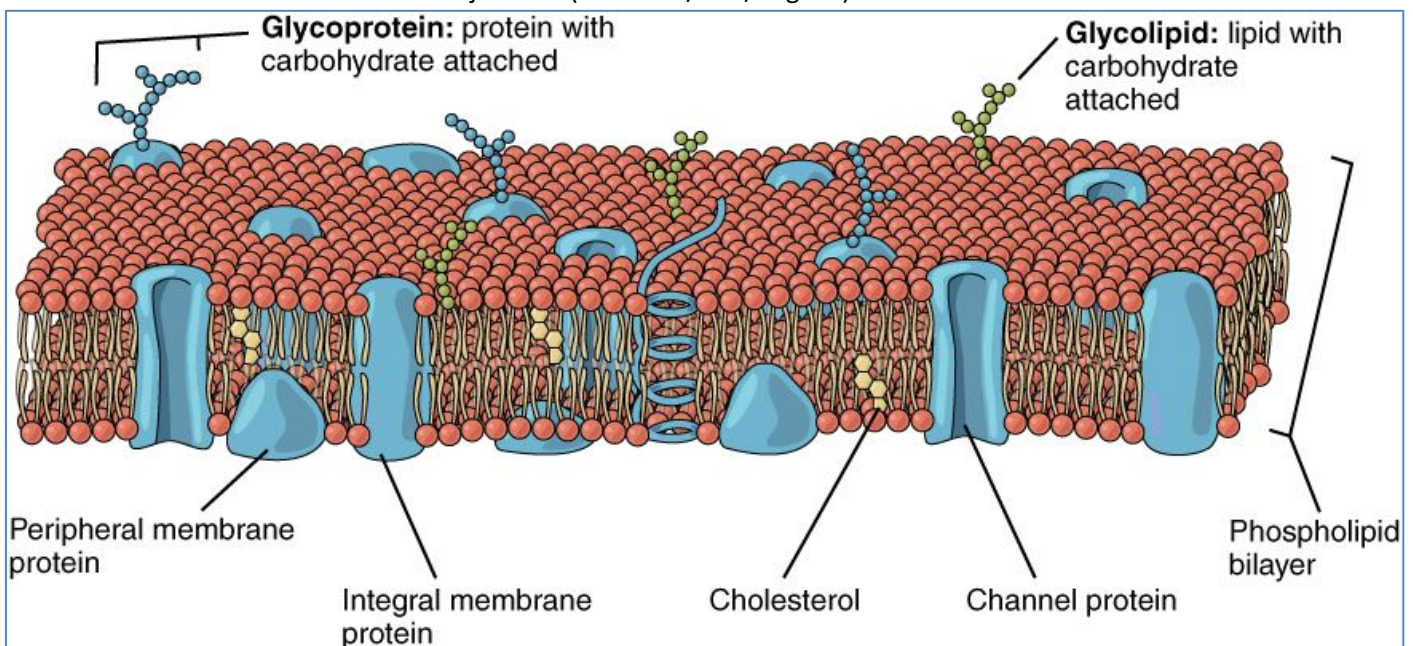
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CELL STRUCTURE:

CELL STRUCTURE:

Structure of a Typical Cell:

- **Plasma membrane**
 - Consists of a bi-lipid layer (diglycerides)
 - Is molten (has properties of both solid and liquid)
 - Contains Cholesterols
 - Contains **proteins**
 - Transportation
 - Catalysis
 - Reception of chemical signals
 - Intercellular joining (2 cells bonding)
 - Cell-Cell Recognition
 - Attachment to extracellular matrix
 - **Membrane Specialisations:**
 - Membrane Junctions (desmosomes/tight/gap)
 - Membrane Projections (microvilli/cilia/flagella)



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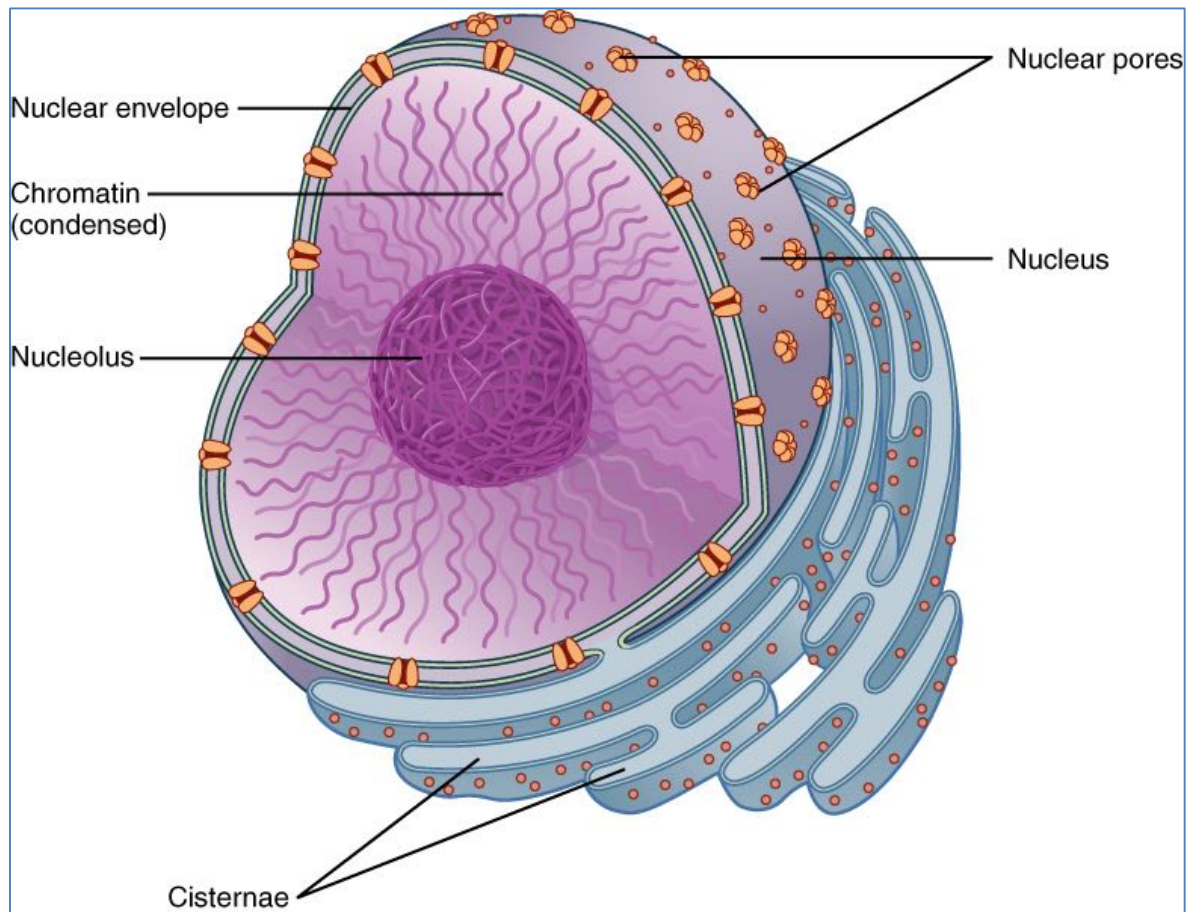
- **Cytoplasm**
 - Everything inside a cell bar the membrane and the nucleus
 - Includes all organelles + cytosol
- **Cytosol**
 - The fluid found within the membrane but outside the organelles
 - Largely water with dissolved protein, salts, sugars & other solutes
- **Inclusions**
 - Chemical substances
 - Glycosomes
 - Glycogen granules
 - Pigment

- **Cytoplasmic Organelles:**

- **Membranous Organelles:**

- **Nucleus**

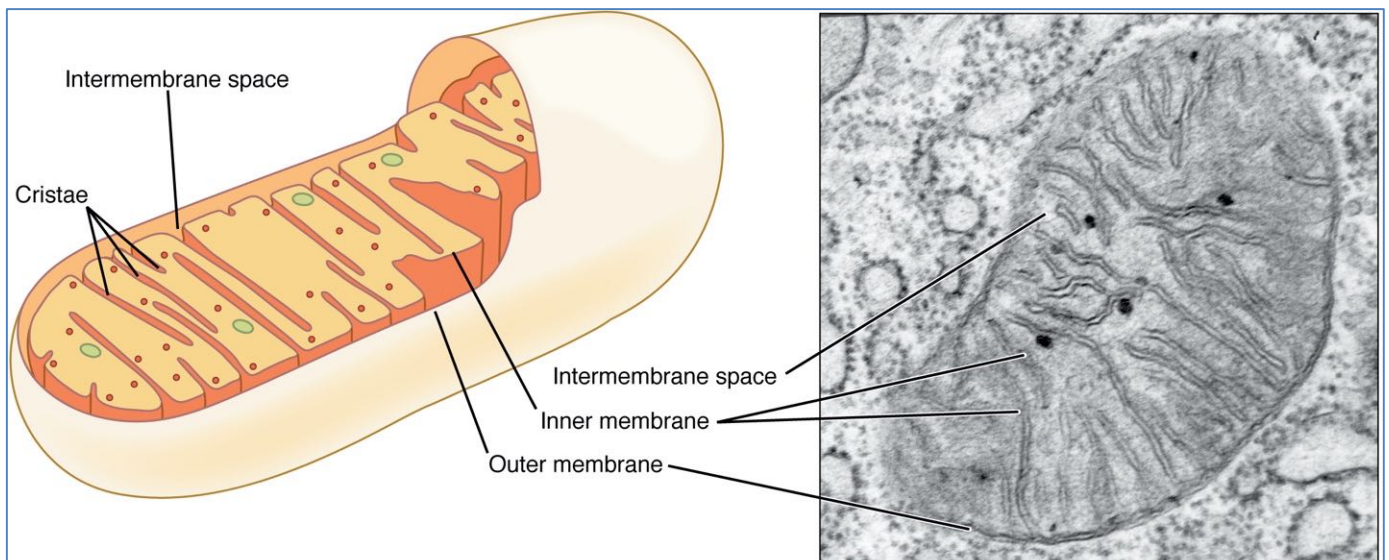
- Nuclear envelope, Nucleoli, Chromatin
 - Contains the genetic library for nearly all cellular proteins
 - Is the place where mitosis begins



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- **Mitochondria**

- Cell power station
 - Double Membrane
 - Synthesise ATP for energy



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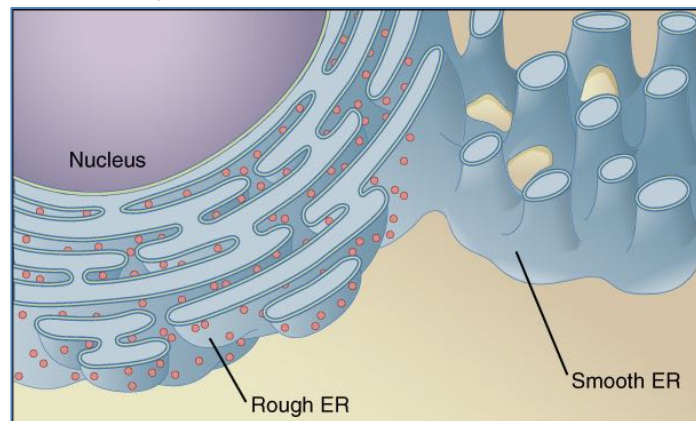
▪ **Endoplasmic reticulum (Rough/Smooth)**

• **Rough:**

- Covered with **ribosomes** (hence rough)
- Synthesis of all proteins secreted from cell + membrane proteins + protein hormones
- Proteins synthesised by ribosomes are then packaged in the Rough ER for export from the cell
- Assist in making cellular membranes

• **Smooth:**

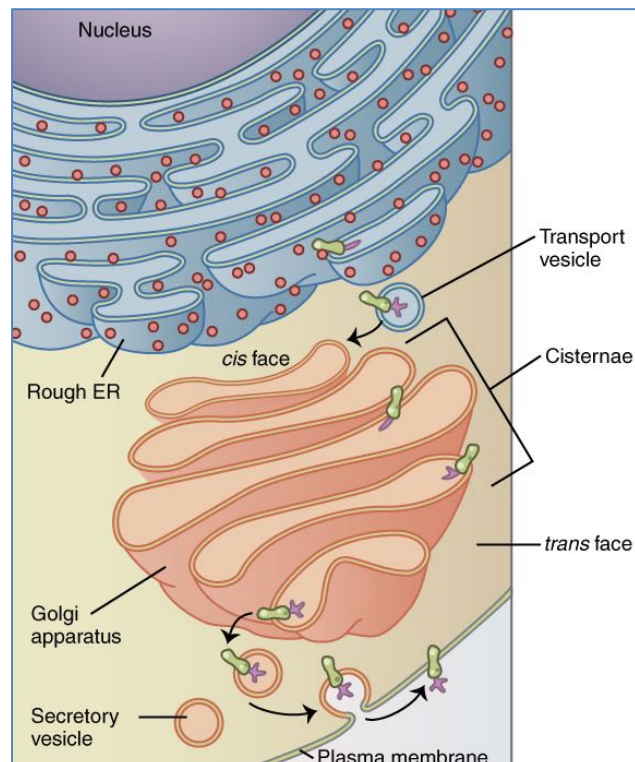
- Not covered with ribosomes (hence smooth)
- Doesn't synthesise proteins
- Metabolises lipids
- Synthesises steroid-based hormones (testosterone/oestrogen)
- Detox of drugs/xenobiotic chemicals
- Storage site of calcium ions in skeletal/cardiac muscle



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▪ **Golgi apparatus**

- The cellular courier
- Modifies, Concentrates and packages proteins and membrane synthesised in the Rough ER for intracellular transport or excretion
- Packaged proteins/membranes are released from the 'shipping face' in a transport vesicle for either excretion or cellular functions



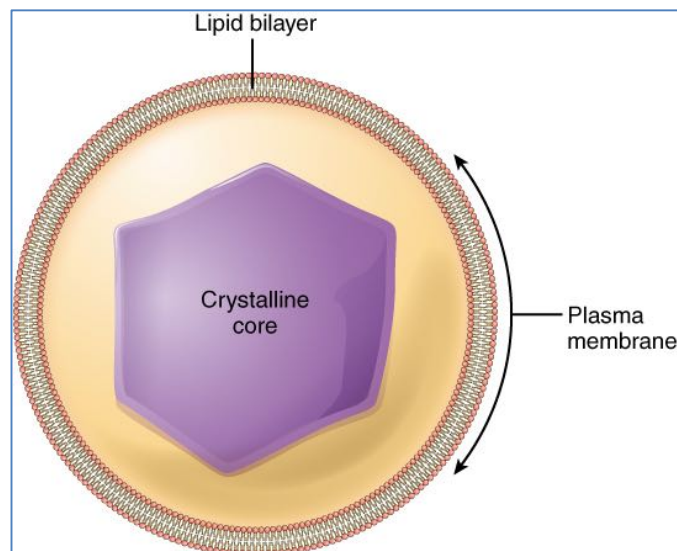
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▪ **Lysosomes:**

- Membranous sacs **created by the Golgi**
- Contain concentrated enzymes
- Inside is acidic for max enzyme function
- Destroy 'old' cellular material
- Destroy bacteria/viruses engulfed by white blood cell

▪ **Peroxisomes**

- Membranous sacs
- Contain enzymes
- Detoxify harmful xenobiotic substances (alcohol)
- Neutralises highly reactive free radicals (by-products of biochemical processes)



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○ **Non-Membranous Organelles**

▪ **Cytoskeleton**

- Elaborate network of large filamentous rod-like proteins
- Provide structural support
- Provide the central mechanism for movement
- Ensures the distribution of organelles throughout cell

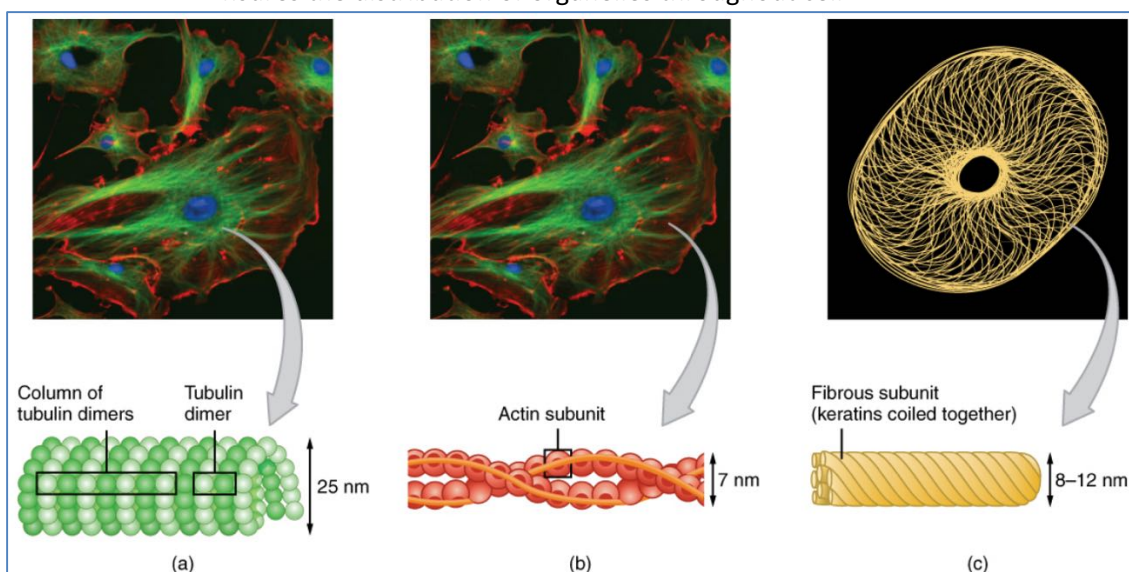
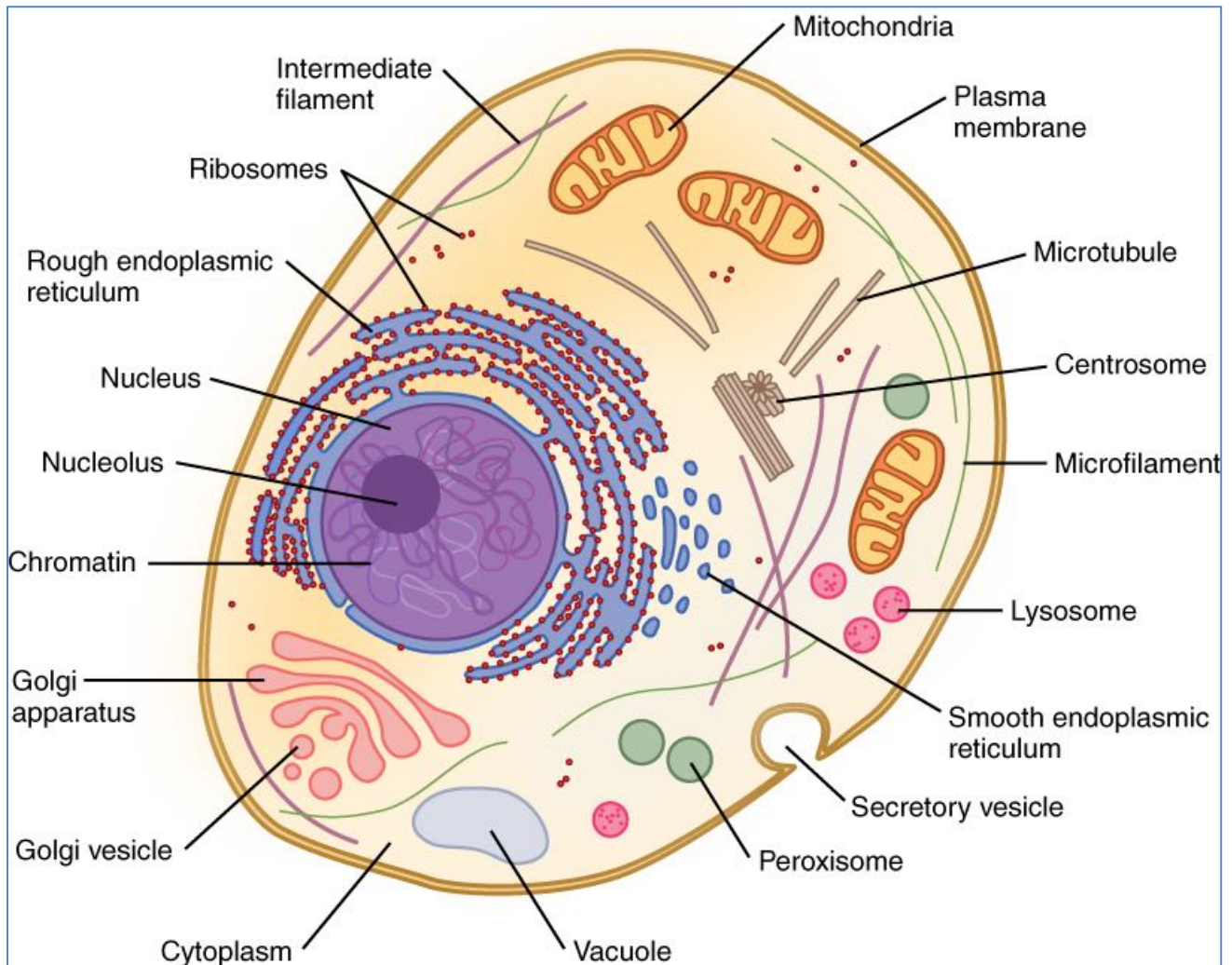


Figure 3.18 The Three Components of the Cytoskeleton The cytoskeleton consists of (a) microtubules, (b) microfilaments, and (c) intermediate filaments. The cytoskeleton plays an important role in maintaining cell shape and structure, promoting cellular movement, and aiding cell division.

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- Centrioles
- Ribosomes
 - Composed of protein & ribosomal RNA (rRNA)
 - Are the site of **protein synthesis**
 - Found either on the Endoplasmic Reticulum or free in the cytosol
 - **ERs with ribosomes** are called '**Rough**' ER



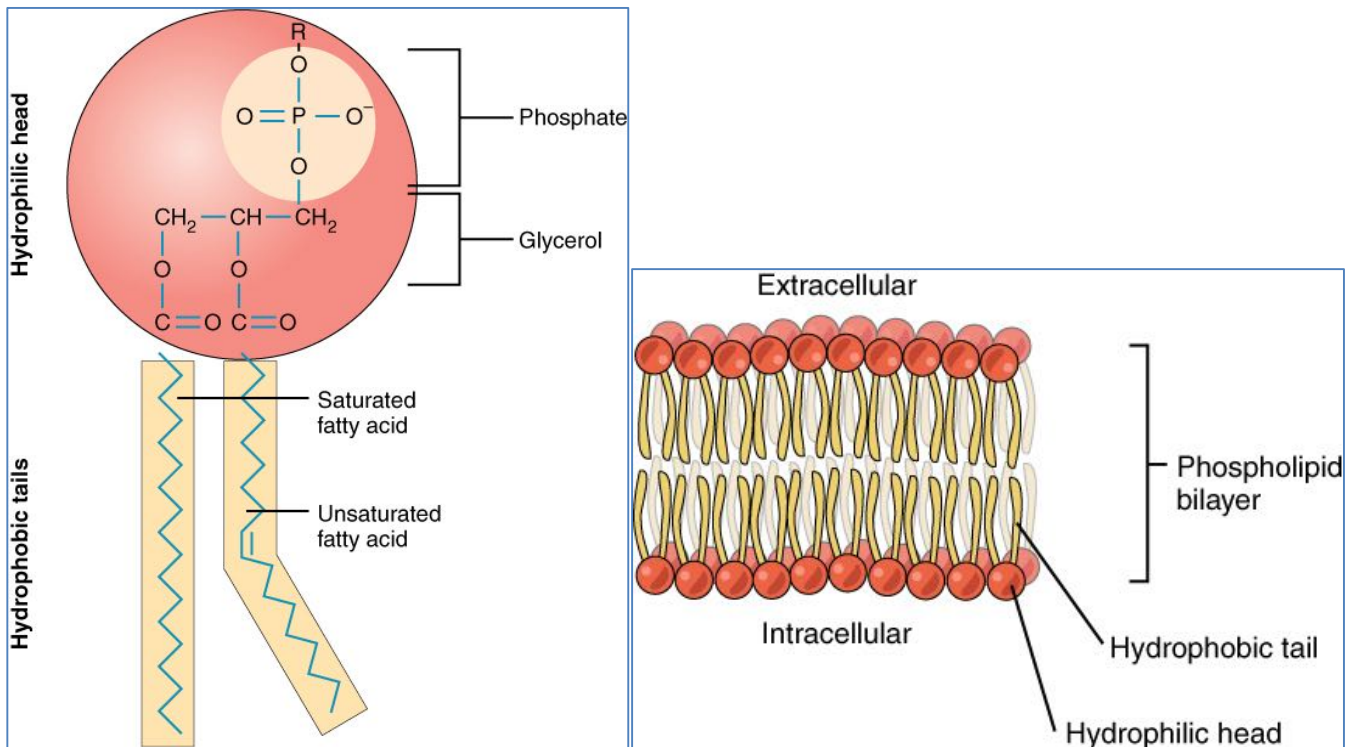
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CELL MEMBRANE & TRANSPORT

CELL MEMBRANE & TRANSPORT

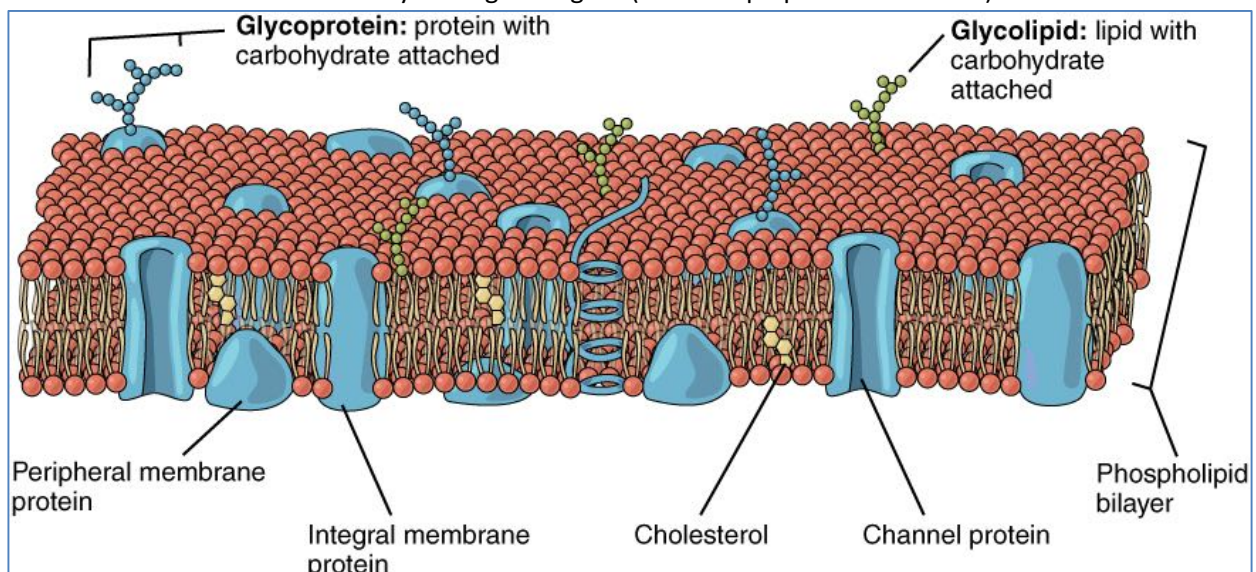
Cell Membranes:

- **Lipid Bilayer** of diglycerides (phospholipids) held together by **hydrophobic forces**
 - **Hydrophilic head group (glycerol)**
 - **Hydrophobic fatty acid tails**; Some saturated + unsaturated



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- **Imbedded Proteins:**
 - **Peripheral:**
 - Associated with the polar head groups
 - Easily removed from the membrane by Δ pH or Δ [salt]
 - **Integral:**
 - Embedded in the membrane
 - Span the width of the membrane
 - Membrane must be destroyed to remove it
 - Done by adding detergent (small amphipathic molecules)



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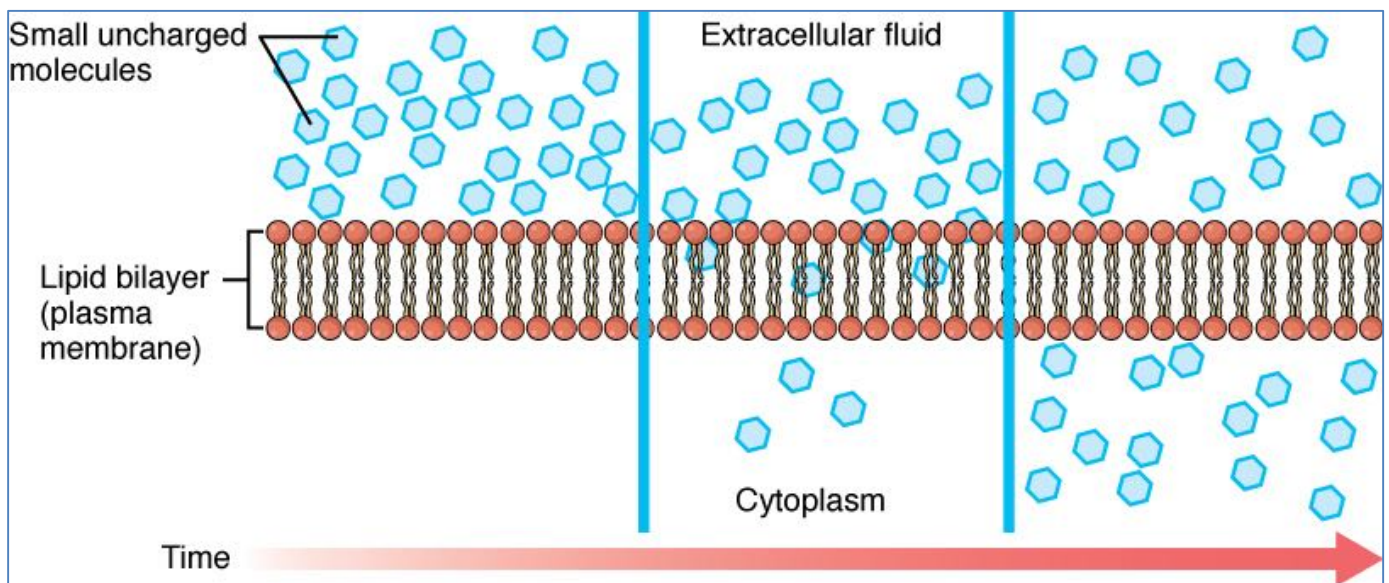
How Substances Cross Cell Membranes:

- Membrane controls the flow of materials in/out of the cell
- Either **passive** or **active processes**:

Passive:

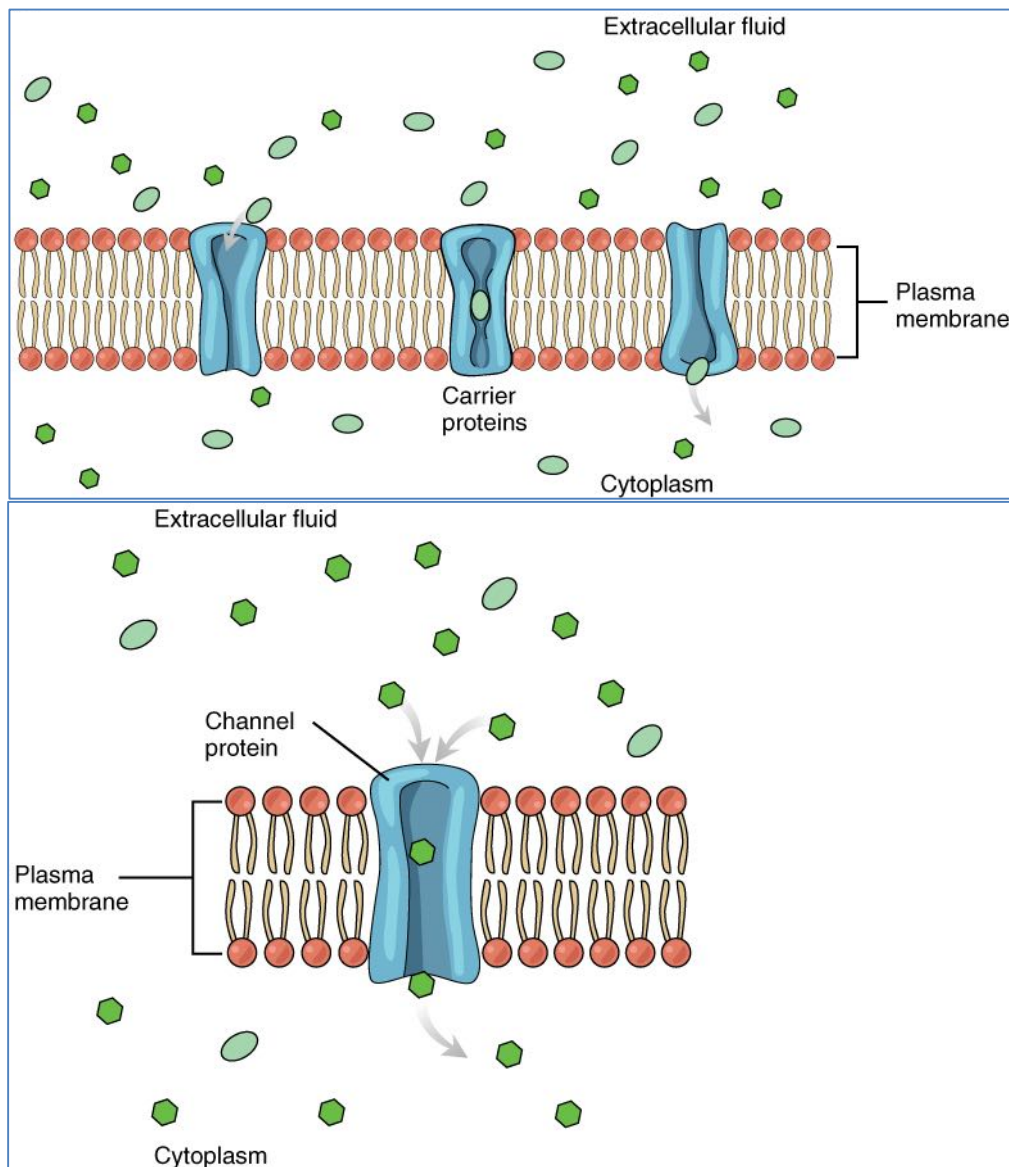
- **Diffusion**

- **Simple Diffusion** – movement of small, uncharged, non-polar and lipid-soluble substances directly through the lipid bilayer (O₂, CO₂, N, Ethanol, Glycerol, Steroids, fat soluble vitamins)
- **Facilitated Diffusion** – where specific molecules diffuse across membranes, through specific transport proteins (Carrier/channel)
- **Factors Affecting Rate of Diffusion:**
 - Concentration gradient
 - Molecular size
 - Temperature (faster @ higher temps)
 - Electric or Pressure gradient

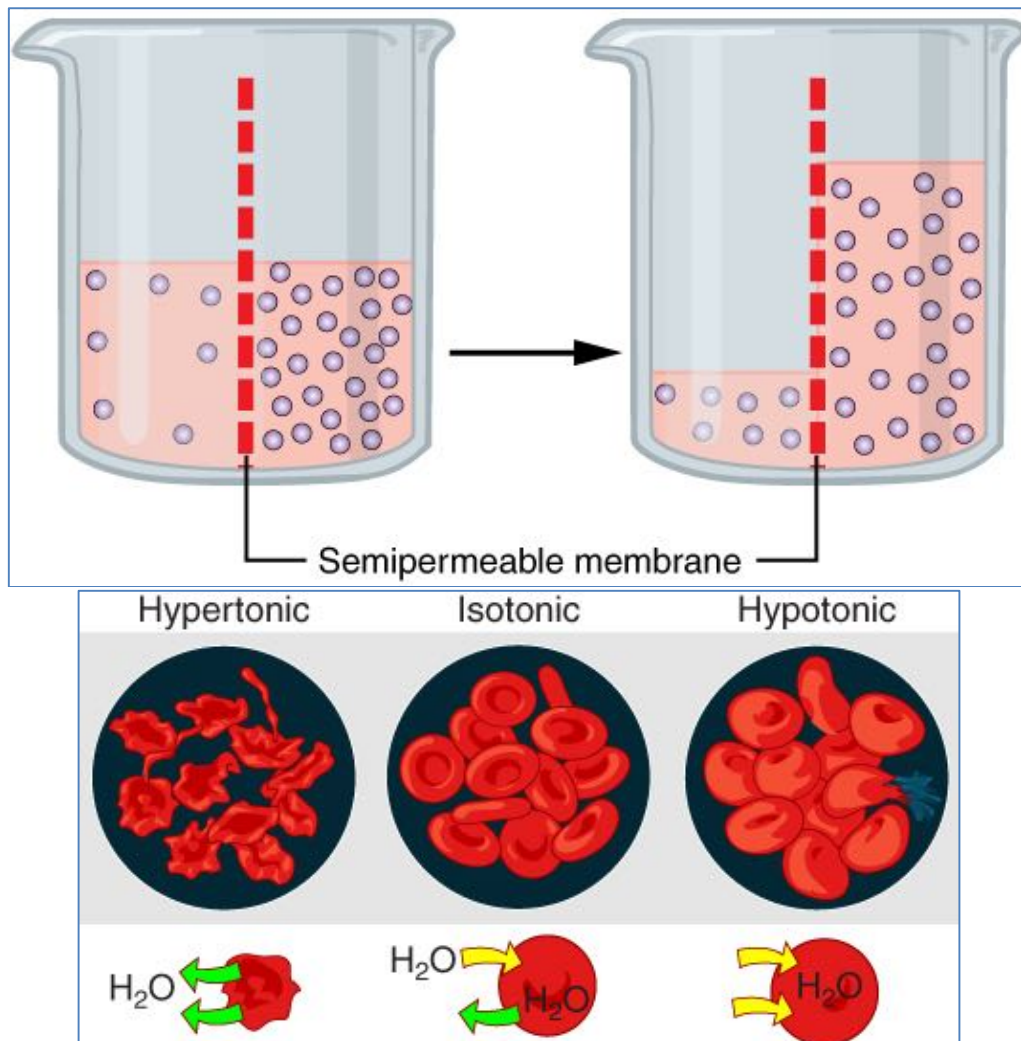


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- **Passive Transport Proteins (Aka: *Facilitated Diffusion*):** – facilitate the diffusion of specific chemicals (glucose/amino acids/nucleotides/ions) through the membrane that would otherwise not pass through the bi-lipid layer
 - **2 Types:**
 - **#1 Passive Carrier Proteins:**
 - Discriminates between solutes based on the shape of the protein's **binding site**
 - - then transfers single molecules across the membrane by **changing its conformation** (similar to a **turnstile**)
 - Has a high affinity for its substrate
 - - are therefore very effective at low substrate concentration
 - Transfer rate is inhibited by temperature
 - **Uniporters:** single solute → down the concentration gradient
 - **#2 Passive Channel Proteins:**
 - Discriminates between solutes mainly on **size and electric charge (usually transports ions)**
 - Act like a **tube** that is **either opened/closed**
 - Has no affinity for its substrate (substrate flow is determined by the concentration gradient of that substrate)
 - Once opened, **ion flow** is very **rapid**
 - Not affected by temperature



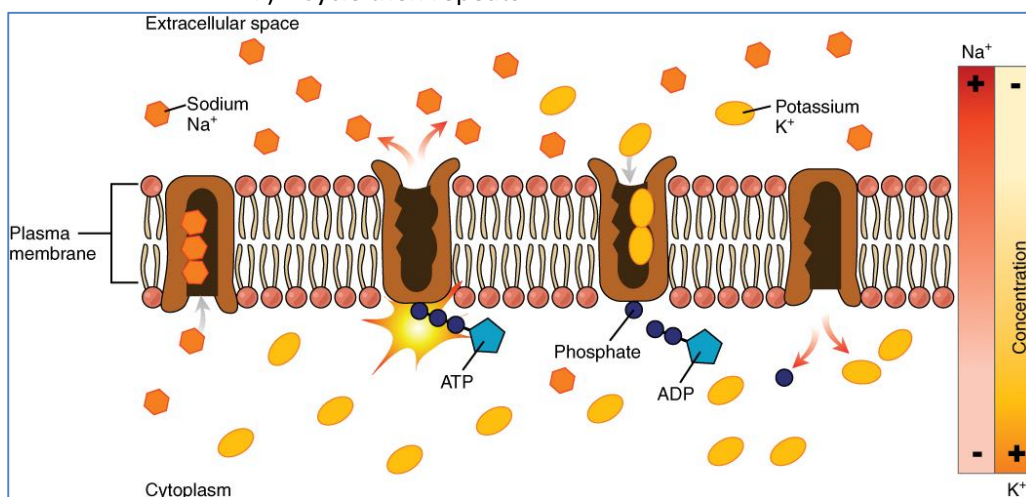
- **Osmosis** – The passive transport of water across a selectively permeable membrane
 - Survival of the cell is dependent on osmoregulation
 - **Water will flow from the hypotonic solution to the hypertonic solution through the lipid bilayer to form an isotonic solution**



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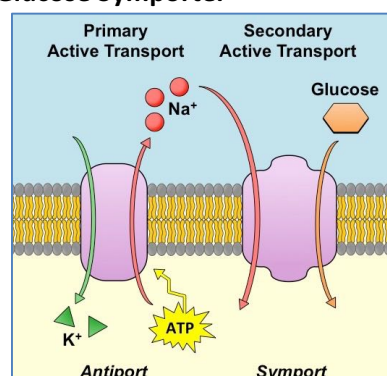
Active Processes:

- Transports substances against their concentration gradient
- Transports substances that would otherwise be too large for channel proteins
- **Active Transport (via carrier proteins):**
 - o (using energy –ATP- to move molecules across a membrane)
 - o Similar to passive facilitated diffusion in that it requires carrier proteins
 - o Active transporters (solute pumps) differ from facilitated diffusion in that they move solutes (mostly ions – Na^+ , K^+ , and Ca^{2+}) uphill **against their concentration gradients**
 - o In so doing, ATP is expended
 - o **2 Classes: Primary & Secondary Active Transport**
 - Distinguished according to their source of energy
 - **Primary Active Transporters:**
 - Energy comes directly from the hydrolysis of ATP
 - Solute binds to the active site – then the protein is phosphorylated, causing it to change its shape and release the solute onto the other side of the membrane
 - **Eg: The Sodium Potassium Pump (The Na^+/K^+ - ATP ase Enzyme)**
 - o **An Antiporter:** 2 solutes $\leftarrow$ opposite directions $\rightarrow$ both against concentration gradients
 - 1) Cytoplasmic Na^+ binds to the protein, stimulating phosphorylation by ATP
 - 2) Phosphorylation causes protein shape to change
 - 3) Change in shape releases Na^+ to the outside
 - 4) K^+ then binds to the protein, triggering the release of the phosphate group
 - 5) Loss of phosphate restores protein to original shape
 - 6) K^+ ions are then released into the cell
 - 7) Cycle then repeats



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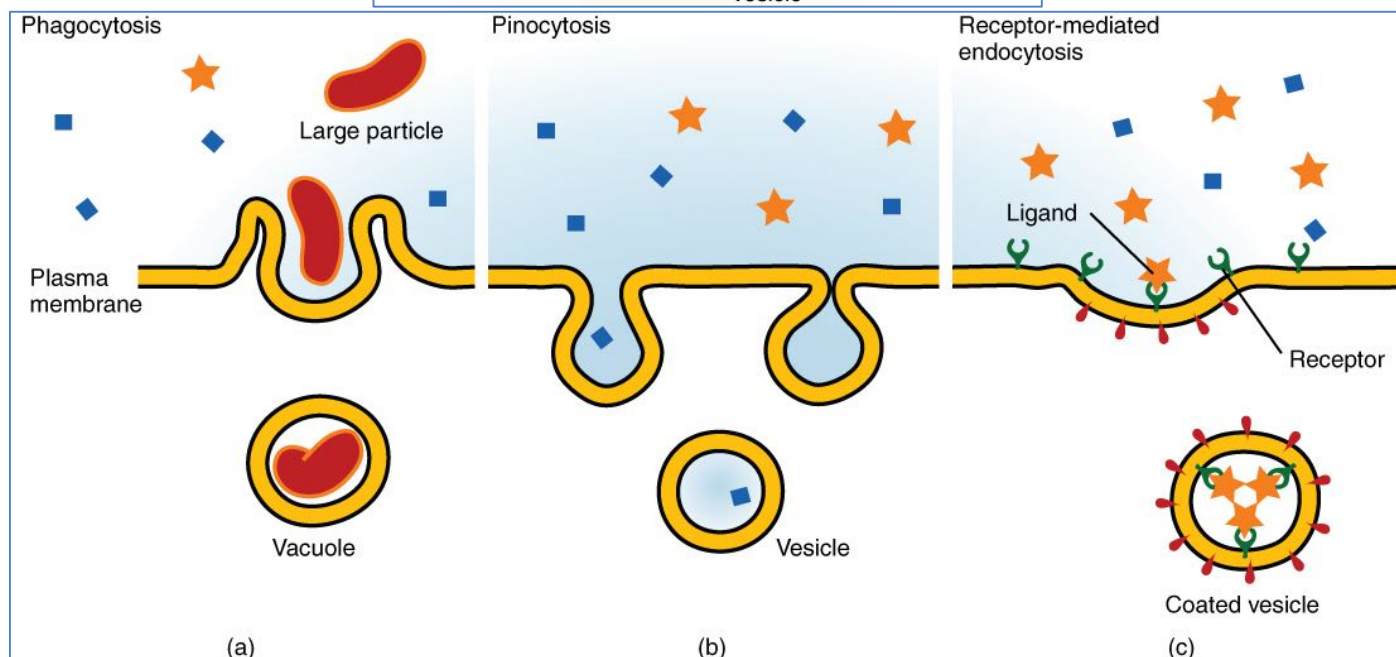
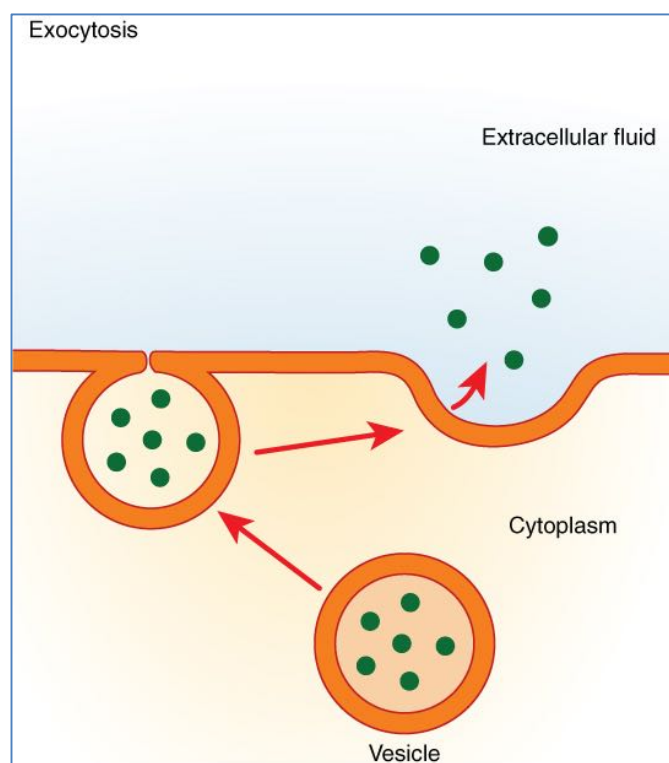
- **Secondary Active Transporters:**
 - **Symporters:** Using the potential energy of the concentration gradient created by a **primary transporter**, the high concentration solute flows downhill, dragging with it another chemical
 - **Eg: Na^+ - Glucose Symporter**



Unattributable

- **Active Transport Via Vesicles:**

- Transport of **large particles, macromolecules and fluids through cell membranes**
- **Exocytosis:** Vesicular transport of substances out of a cell (secretion)
- **Endocytosis:** Vesicular transport of substances into a cell
 - **Phagocytosis:** a large external particle is engulfed and enclosed in a vesicle (**eg:** in white blood cells)
 - **Pinocytosis:** external fluid droplet (containing small solutes) is engulfed and enclosed in a vesicle (absorptive cells – eg: kidney & intestine)
 - **Receptor Mediated:** selective endocytosis – substance binds to membrane receptors & then enclosed in a vesicle



Three Forms of Endocytosis Endocytosis is a form of active transport in which a cell envelopes extracellular materials using its cell membrane. (a) In phagocytosis, which is relatively nonselective, the cell takes in a large particle. (b) In pinocytosis, the cell takes in small particles in fluid. (c) In contrast, receptor-mediated endocytosis is quite selective.

When external receptors bind a specific ligand, the cell responds by endocytosing the ligand.

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CELLULAR METABOLISM

CELLULAR METABOLISM

What is Metabolism?

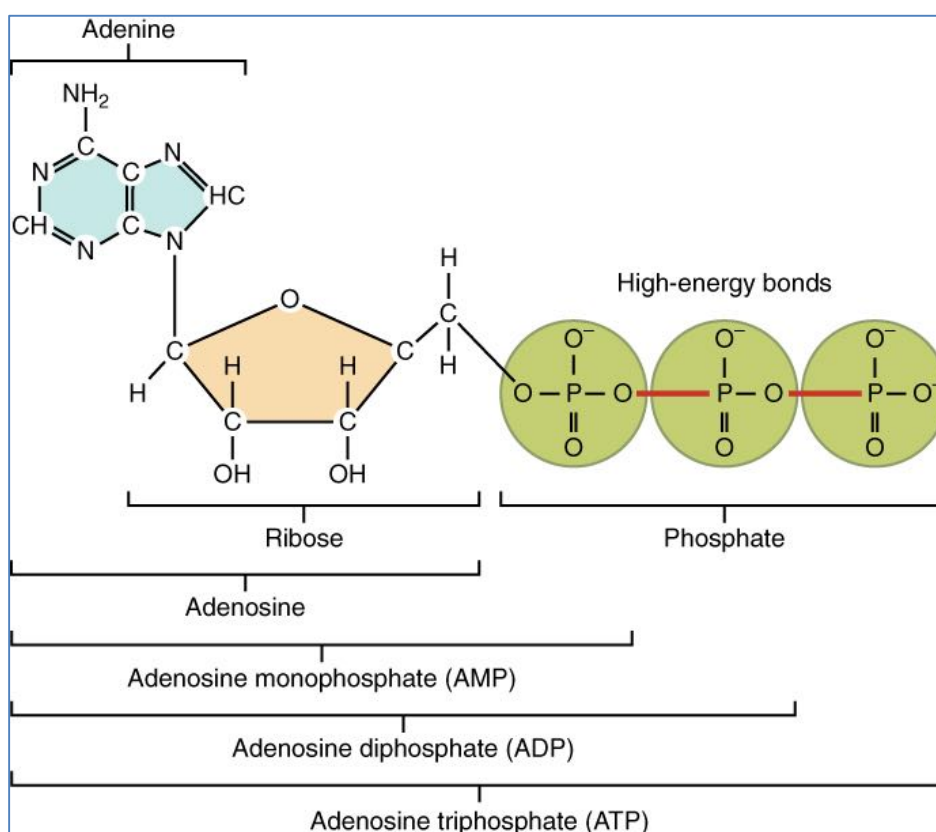
- **Metabolism** is a cell's capacity to acquire energy, build, break apart & release substances
 - Including extracting chemical energy from food through metabolic processes
 - Metabolic reactions lead to some energy loss to the environment (generally heat)
 - Metabolic reactions can release or require energy
- **Why?** Cells engage in metabolism to survive, repair, replicate and carry out their functions
- **Did you know:** All the energy in all the food you eat can be traced back to sunlight
 - The chemical energy stored in food is in the form of sugars and other organic molecules
- **Note: Cyanide & Carbon Monoxide** kill you by disrupting cellular respiration

Autotrophs and Heterotrophs:

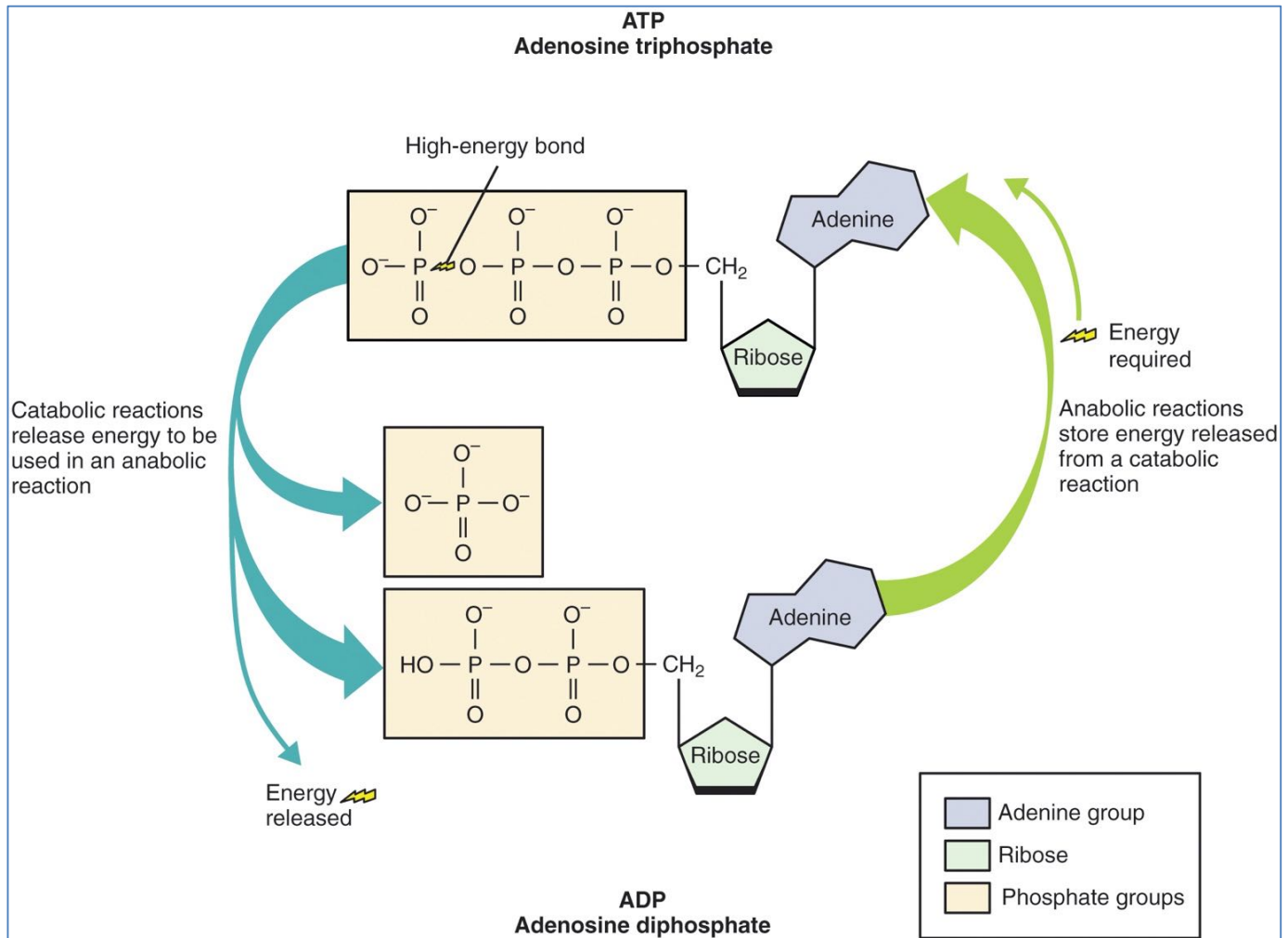
- **Autotrophic organisms ("self-feeders" or "producers")** can convert inorganic substrates (CO_2 and H_2O) into complex organic molecules such as **sugars (to make ATP)** needed for the catabolic reactions in the cell
 - **Eg: Plants and algae that photosynthesize**
- **Heterotrophic organisms ("Consumers")** can only extract energy by breaking down complex molecules (sugars/fats/etc) through respiration
 - **Eg: Most animals including humans**

ATP (Adenosine Triphosphate):

- ATP is the main energy carrier in the cell (Others include NAD, NADH)
- ATP is generated in the cell by **oxidation of nutrients** (Including carbohydrates, amino acids, lipids)
- ATP is consumed by any active/constructive process undertaken by the cell

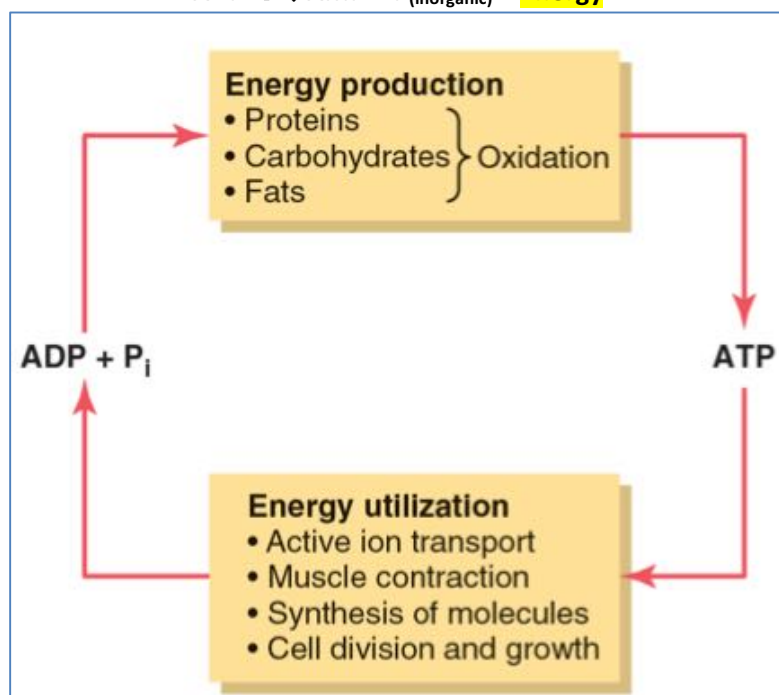
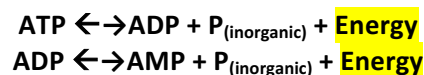


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- **ATP forms when a phosphate is donated to ADP (Adenosine Diphosphate)**



Source: <https://doctorlib.info/physiology/textbook-medical-physiology/67.html>

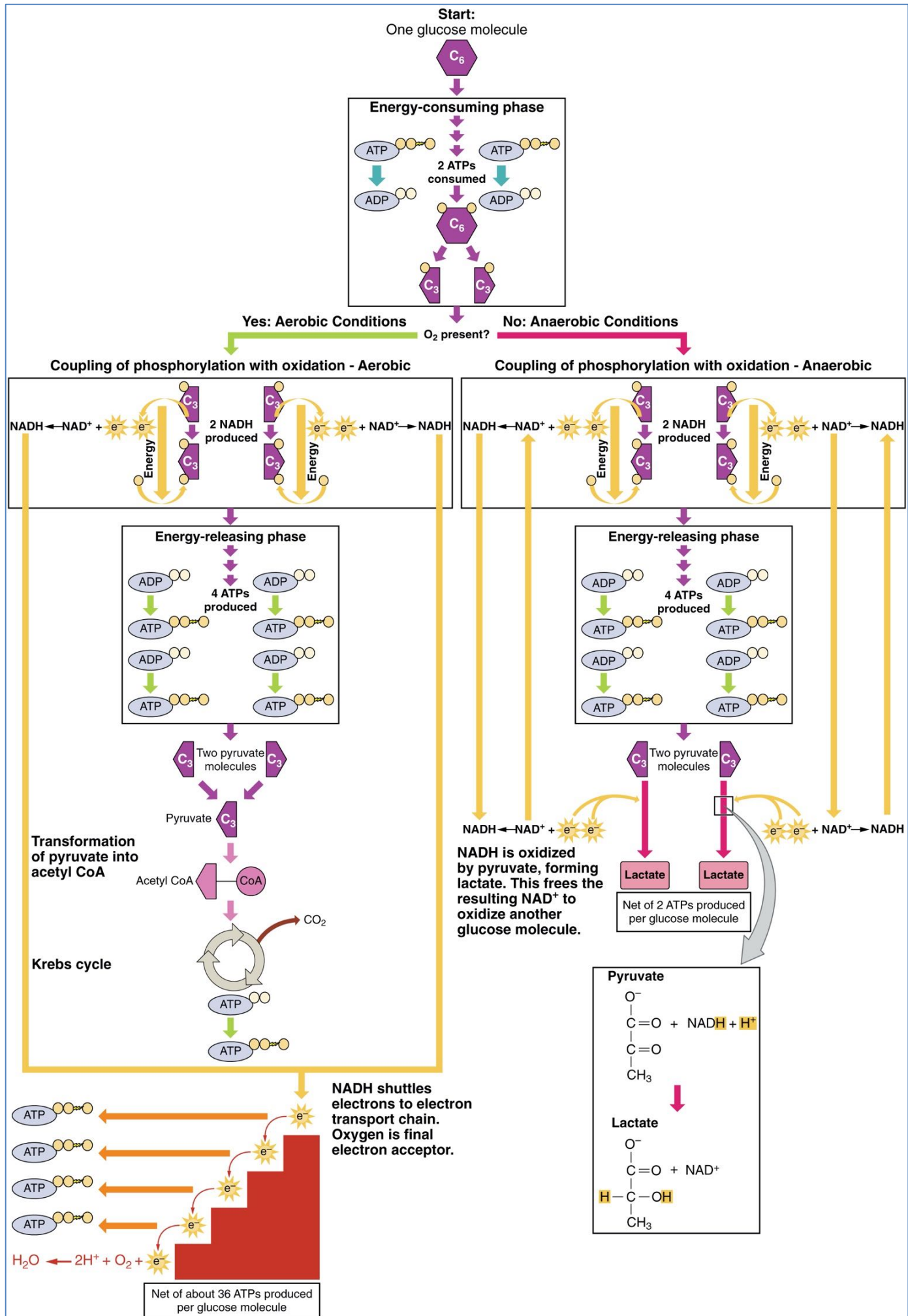
Aerobic Metabolism (Aka: Respiration):

- Oxidation of nutrients generally requires oxygen
- Hence why the majority of cells require oxygen for the majority of the time in order to survive
- Aerobic metabolism is the most efficient way to release energy from nutrients
- Can only occur in the presence of oxygen
- Requires the cell to be able to exchange gases with its surroundings

Anaerobic Metabolism:

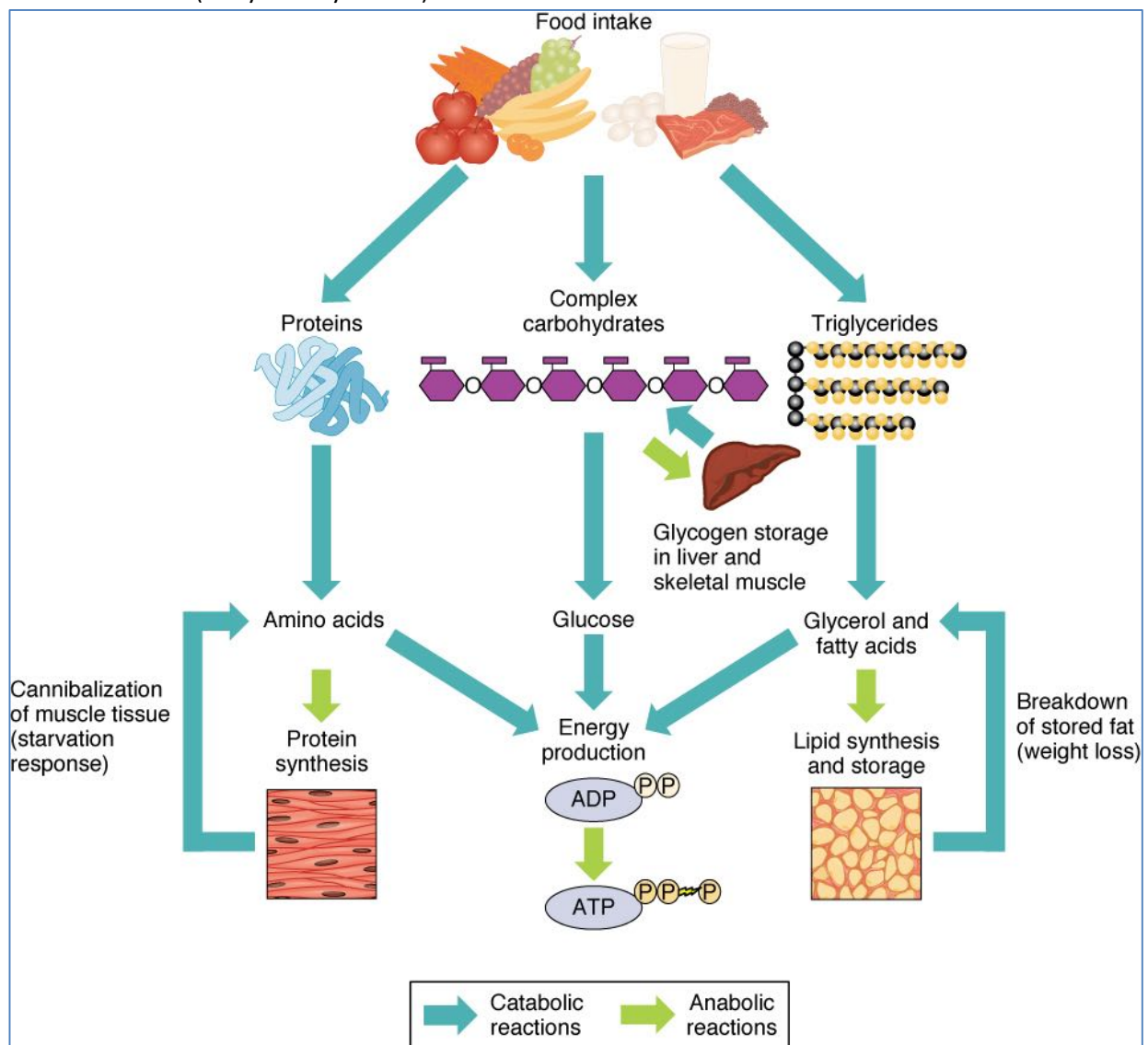
- Certain tissues are capable of surviving via anaerobic metabolism in certain circumstances
- Extracting energy from food without oxygen is not as efficient (lower ATP yield) and produces additional metabolites which need to be processed later or excreted (Eg: Lactic acid)
- Occurs when the demand for oxygen outstrips the body's ability to deliver it

Diagram over the page:



Metabolic Pathways

- **Burning multiple fuels (nutrients) requires multiple metabolic pathways:**
 - **Carbohydrates:**
 - Glycolysis pathway
 - Pentose phosphate pathway
 - The Citric Acid cycle (TCA), aka: 'The Krebs Cycle'
 - The Electron Transport Chain ('Oxidative phosphorylation')
 - (Glycogenesis / Glycogenolysis) – Glucose storage & retrieval
 - (Gluconeogenesis) – Glucose synthesis from other substrates
 - **Amino Acids:**
 - Amino Acid Metabolism
 - The Urea Cycle
 - **Lipids:**
 - Fatty Acid Oxidation
 - Ketogenesis/Ketolysis
 - (Fatty Acid Synthesis)



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- **Each Metabolic pathway is an orderly series of reactions driven by enzymes**
 - Enzymes are catalysts – they lower the activation energy of a reaction + bind substrates
- **Different tissue-types have varying metabolic capabilities:**
 - Eg: Muscles preferentially burn glucose
 - Eg: Liver cells can handle/transform multiple nutrients
 - Eg: Heart muscle can burn glucose or ketone bodies
 - Eg: Brain can burn glucose or ketone bodies

Hormonal Regulation of Metabolism:

Catabolic Hormones

Hormone	Function
Cortisol	Released from the adrenal gland in response to stress; its main role is to increase blood glucose levels by gluconeogenesis (breaking down fats and proteins)
Glucagon	Released from alpha cells in the pancreas either when starving or when the body needs to generate additional energy; it stimulates the breakdown of glycogen in the liver to increase blood glucose levels; its effect is the opposite of insulin; glucagon and insulin are a part of a negative-feedback system that stabilizes blood glucose levels
Adrenaline/epinephrine	Released in response to the activation of the sympathetic nervous system; increases heart rate and heart contractility, constricts blood vessels, is a bronchodilator that opens (dilates) the bronchi of the lungs to increase air volume in the lungs, and stimulates gluconeogenesis

Anabolic Hormones

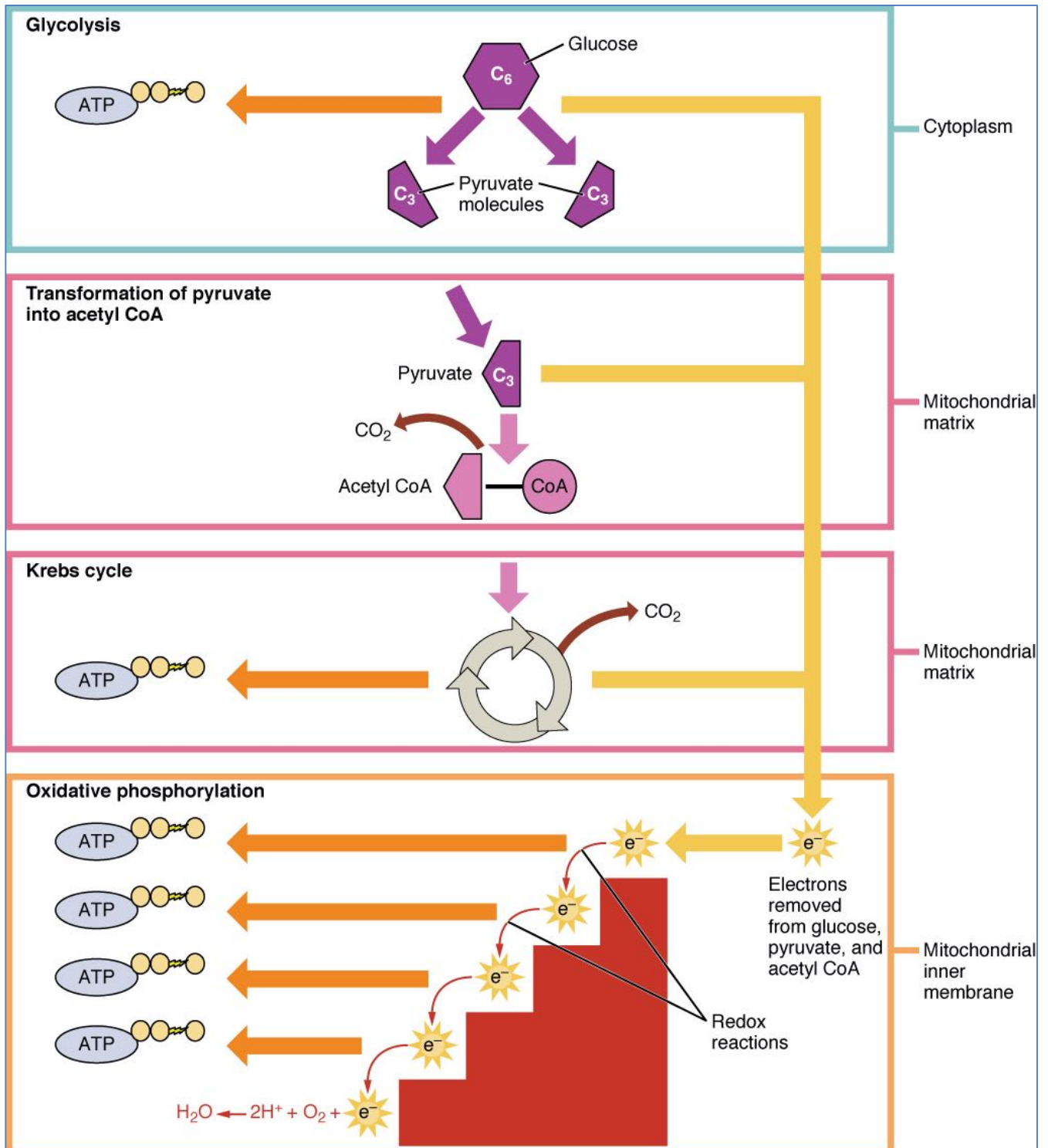
Hormone	Function
Growth hormone (GH)	Synthesized and released from the pituitary gland; stimulates the growth of cells, tissues, and bones
Insulin-like growth factor (IGF)	Stimulates the growth of muscle and bone while also inhibiting cell death (apoptosis)
Insulin	Produced by the beta cells of the pancreas; plays an essential role in carbohydrate and fat metabolism, controls blood glucose levels, and promotes the uptake of glucose into body cells; causes cells in muscle, adipose tissue, and liver to take up glucose from the blood and store it in the liver and muscle as glycogen; its effect is the opposite of glucagon; glucagon and insulin are a part of a negative-feedback system that stabilizes blood glucose levels
Testosterone	Produced by the testes in males and the ovaries in females; stimulates an increase in muscle mass and strength as well as the growth and strengthening of bone
Estrogen	Produced primarily by the ovaries, it is also produced by the liver and adrenal glands; its anabolic functions include increasing metabolism and fat deposition

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CARBOHYDRATE METABOLISM

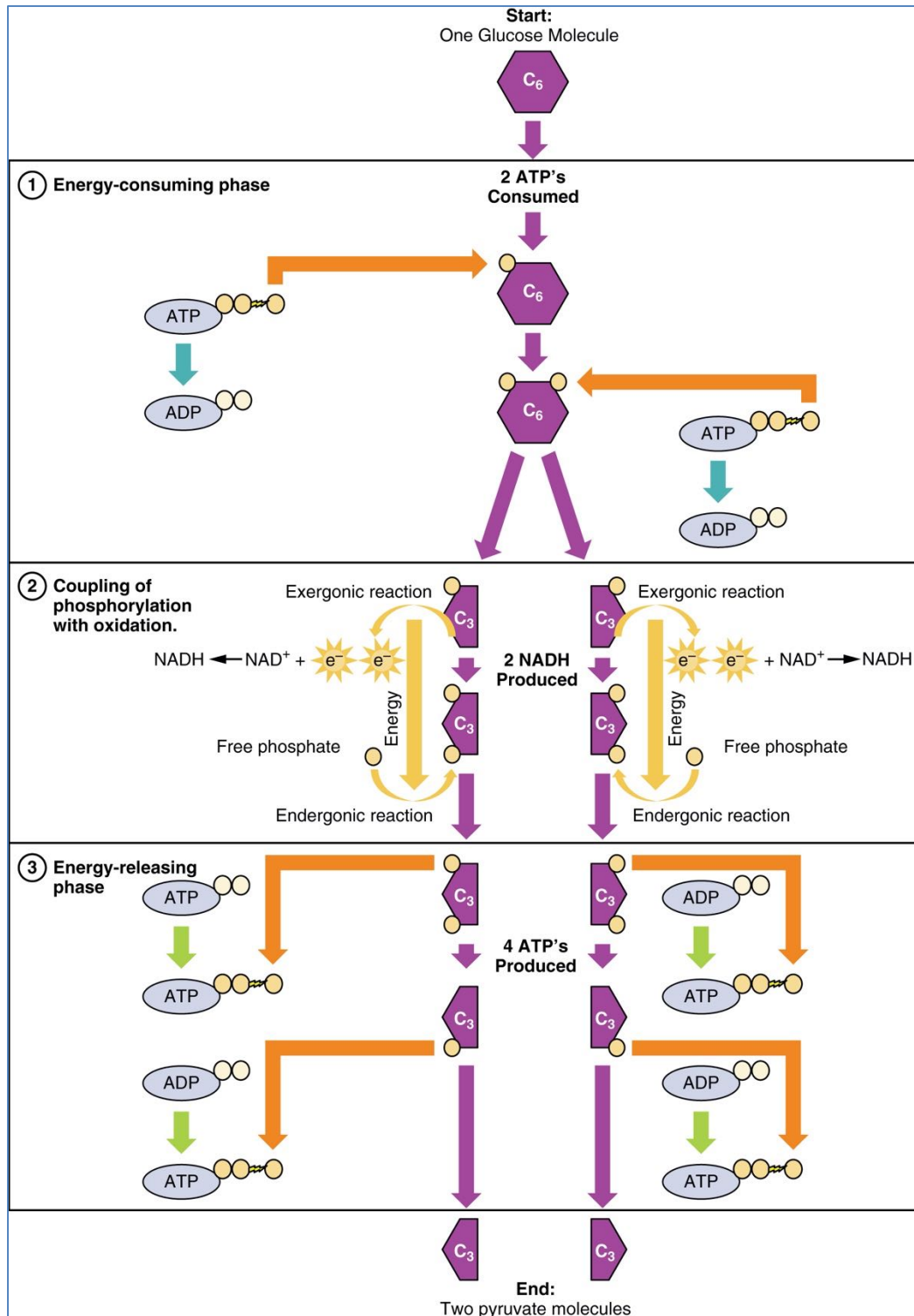
Overview of Carbohydrate Metabolism:

- Energy is stored in the chemical bonds of carbohydrates
- Energy is released as these chemical bonds are broken down and oxidized to CO_2 and H_2O
- This energy is transferred to **activated carrier molecules** which serve as portable energy sources
 - o Eg: ATP
 - o Eg: NADP/NADPH/ NADH_2 / NADH^+
- **Regulated by Insulin, Glucagon & 'Counter-Regulatory Hormones':**
 - o **Insulin:** Promotes decrease in blood [glucose] by increasing uptake (**glycolysis or glycogenesis**)
 - o **Glucagon:** Promotes increase in blood [glucose] by increasing output (**gluconeogenesis and glycogenolysis**)
 - o ***Adrenaline/Cortisol:** Promotes increase in blood [glucose] (as above)



GLYCOLYSIS:

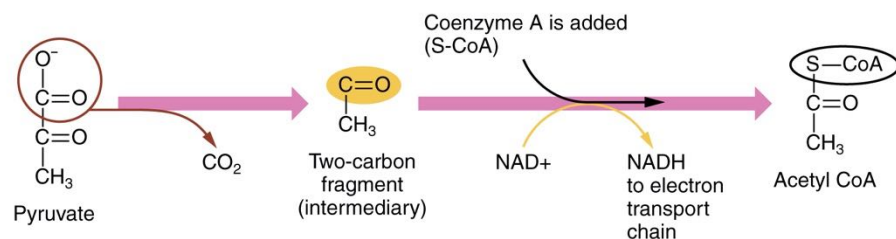
- **What is it?**
 - The energy-producing breakdown of Glucose into Pyruvate
 - The beginning point of cellular carbohydrate metabolism
 - Note: Other non-glucose sugars must first be converted to one of the glycolytic intermediates
- **Where does it occur?**
 - In the Cytoplasm of all cells
 - Therefore, first requires the uptake of extracellular glucose INTO the cell via the GLUT transporter
 - Uptake of extracellular glucose is regulated by Insulin
 - Some cells (**Red Blood Cells**) **rely exclusively on glycolysis** for energy (no mitochondria)
- **Summary: Glycolysis converts 1x Glucose molecule into...**
 - **2x Pyruvates** (Which then pass into the mitochondria → TCA/Krebs Cycle)
 - **4x ATP** – Net gain = 2ATP's (2 spent, 4 produced)
 - **2x NADH** – Net gain = 2NADH's (0 spent, 2 produced)



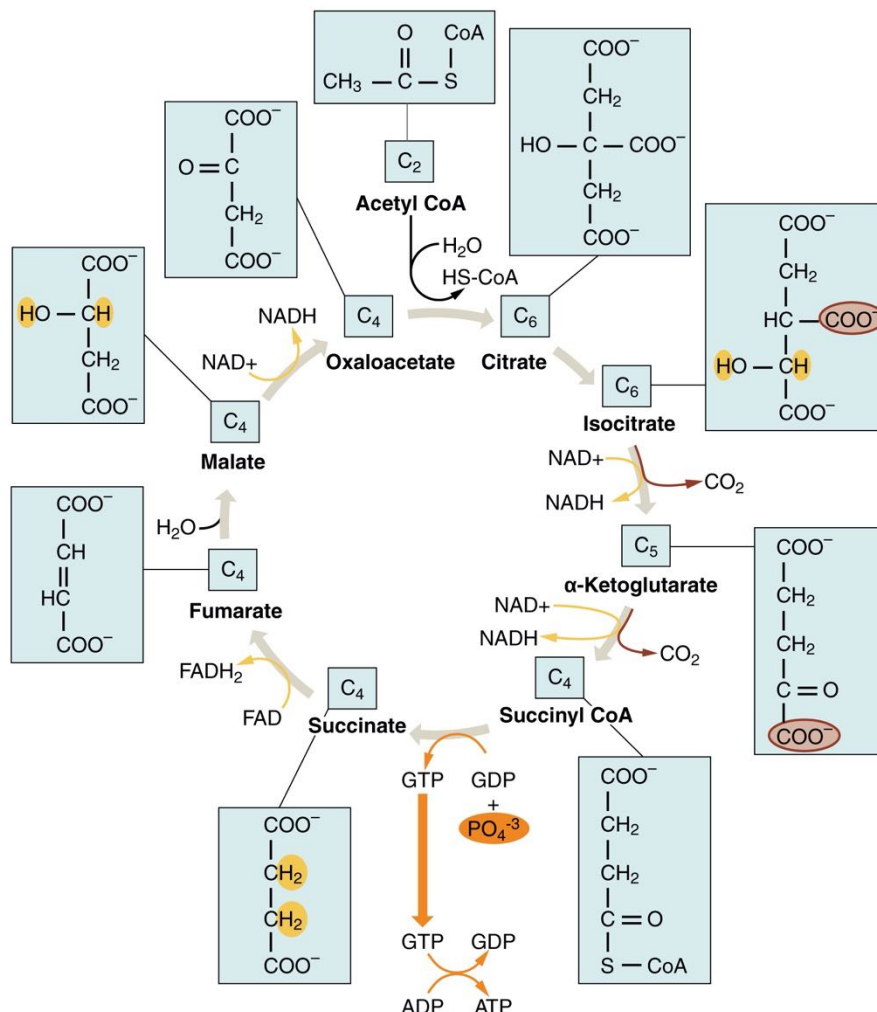
THE CITRIC ACID CYCLE (TCA) / 'KREBS CYCLE':

- **What is it?**
 - o The intermediate step between Glycolysis & Oxidative Phosphorylation (Electron Transport Chain)
 - o Glycolysis supplies the TCA cycle with Pyruvate
 - o TCA Cycle supplies the Electron Transport Chain with NADH, FADH,
- **Where does it occur?**
 - o Occurs in the Mitochondria of All Cells (Except RBCs)
 - o Requires sufficient glucose concentration in the cytoplasm to maintain constant supply of Pyruvate
 - o **Note:** Pyruvate is converted to Acetyl-CoA upon entry into the Mitochondria:
 - Produces 1x NADH (worth ~3ATP)
 - Consumes 1x Coenzyme A
 - Liberates 1x CO₂ molecule
- **Summary: TCA Cycle Converts 1x Pyruvate molecule into...**
 - o **3x NADH** (Which later makes 9 ATP)
 - o **1x FADH** (Which later makes 2 ATP)
 - o **1x GTP** (Which later makes 1 ATP)

Transformation of pyruvate into acetyl CoA

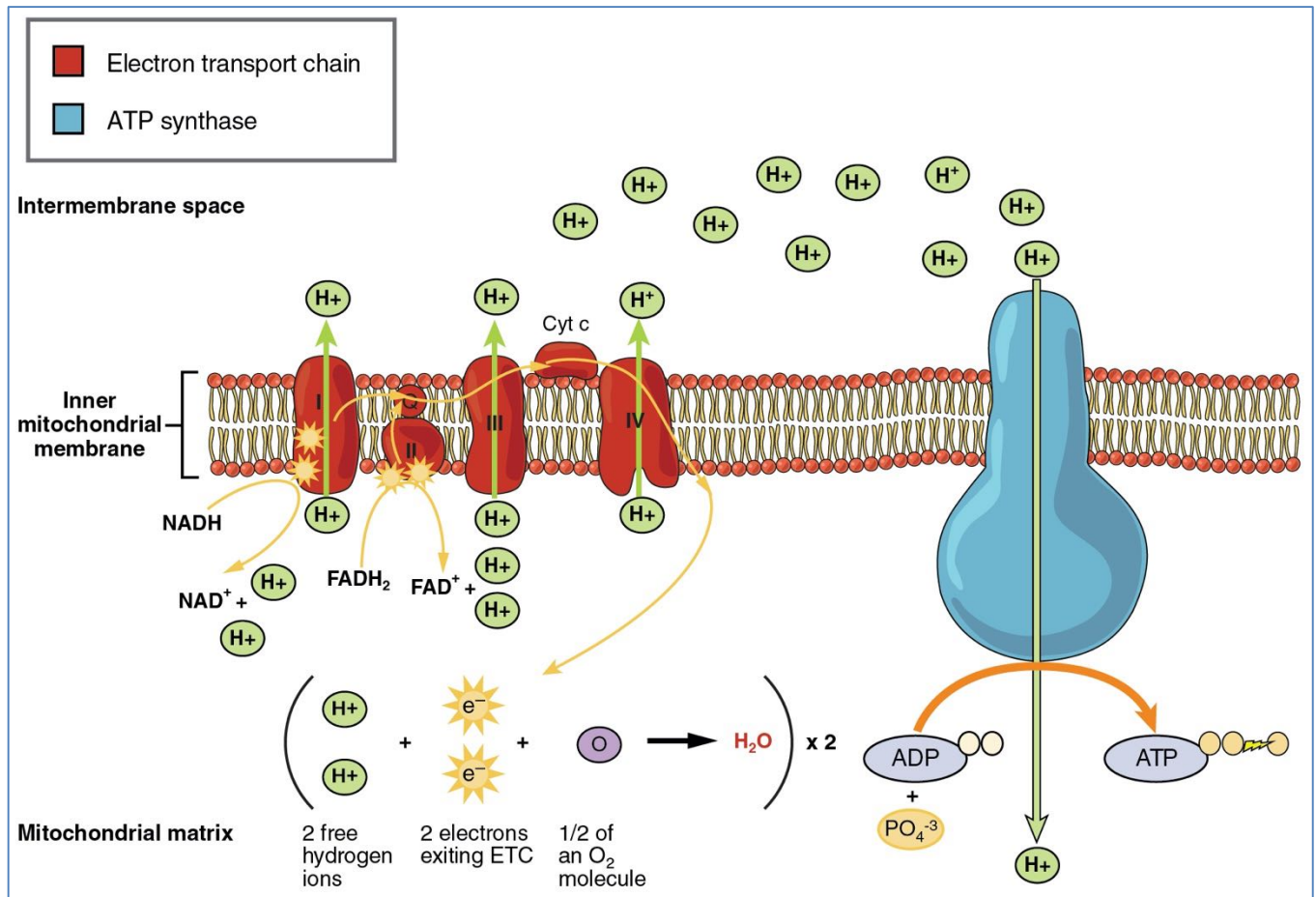


The Krebs/citric acid cycle



ELECTRON TRANSPORT CHAINS / OXIDATIVE PHOSPHORYLATION:

- **What is it?**
 - o A series of proteins, lipids & metals that facilitate electron movement
 - o Electron movement creates a Proton gradient within the Outer & Inner Mitochondrial Membrane
 - o Harnessing the flow of Protons is used by **ATP-Synthase** to generate ATP
- **Where does it occur?**
 - o Occurs in the Inner Mitochondrial Membrane & Inter-Mitochondrial Membrane Space
- **Summary: Oxidative Phosphorylation Converts...**
 - o **NADH** (produced in TCA) $\rightarrow$ **NAD** + 3x **ATP**
 - o **FADH** (produced in TCA) $\rightarrow$ **FAD** + 2x **ATP**
 - o **Large amounts of Oxygen** $\rightarrow$ **CO₂** + **H₂O**
 - o **Yields approximately 30-34 ATP total** (depending on initial fuel)



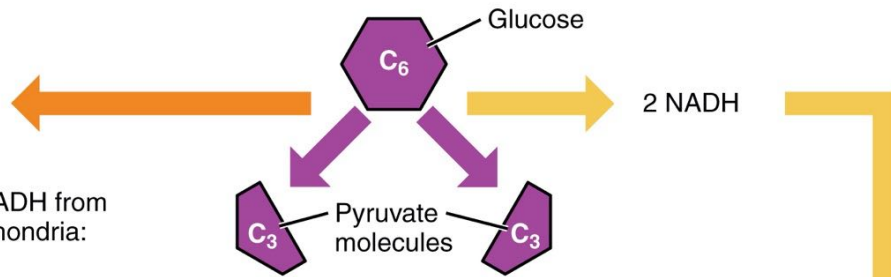
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Summary:

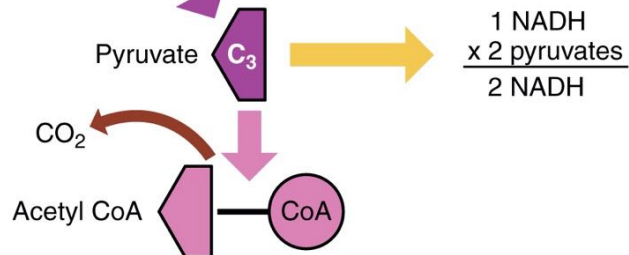
Glycolysis

4 ATP
- 2 ATP used
2 ATP

Active transport of NADH from
cytoplasm into mitochondria:
- 2 ATP

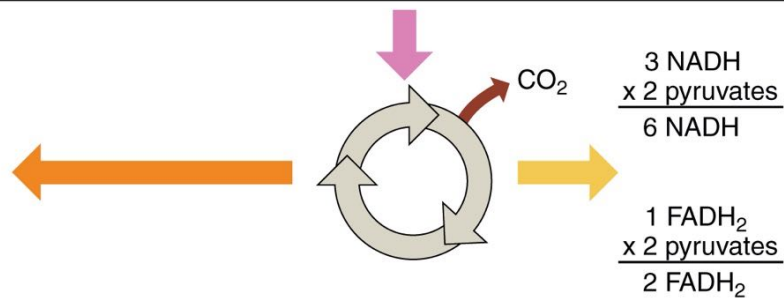


Transformation of pyruvate to acetyl CoA



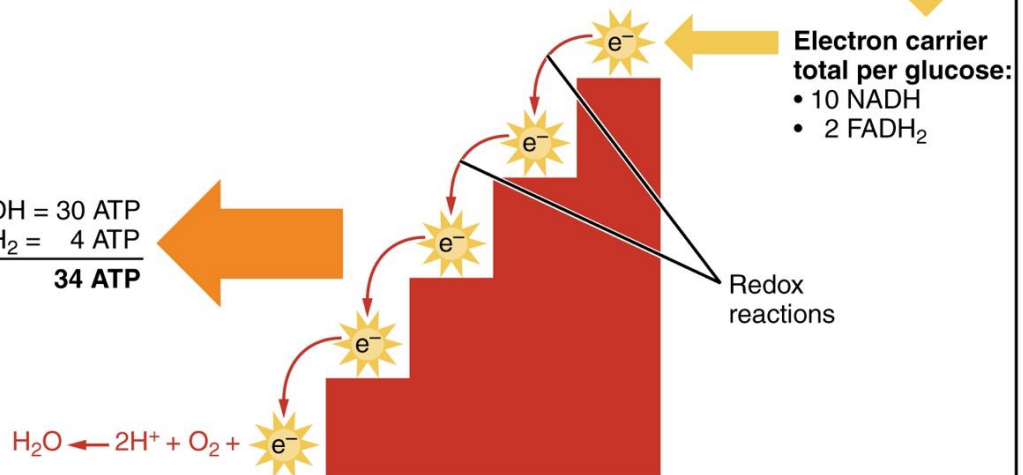
Krebs cycle

1 ATP
x 2 pyruvates
2 ATP



Electron transport chain

3 ATP x 10 NADH = 30 ATP
+ 2 ATP x 2 FADH₂ = 4 ATP
34 ATP



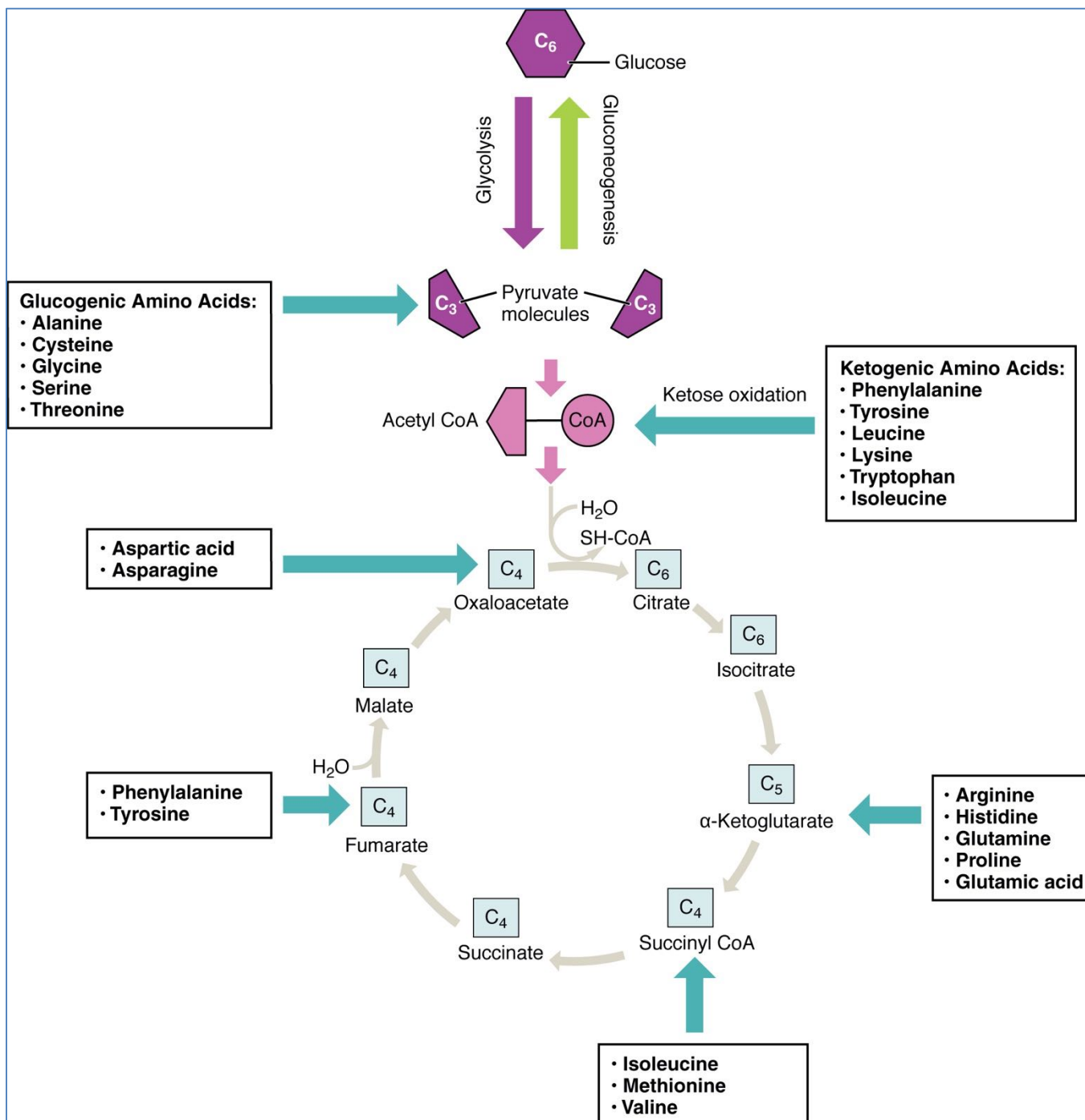
Total ATP produced:

Glycolysis =	2 ATP
NADH transport cost =	-2 ATP
Pyruvate into acetyl CoA =	0 ATP
Krebs cycle =	2 ATP
+ ETC =	34 ATP
Total	36 ATP per glucose

AMINO ACID METABOLISM & THE UREA CYCLE

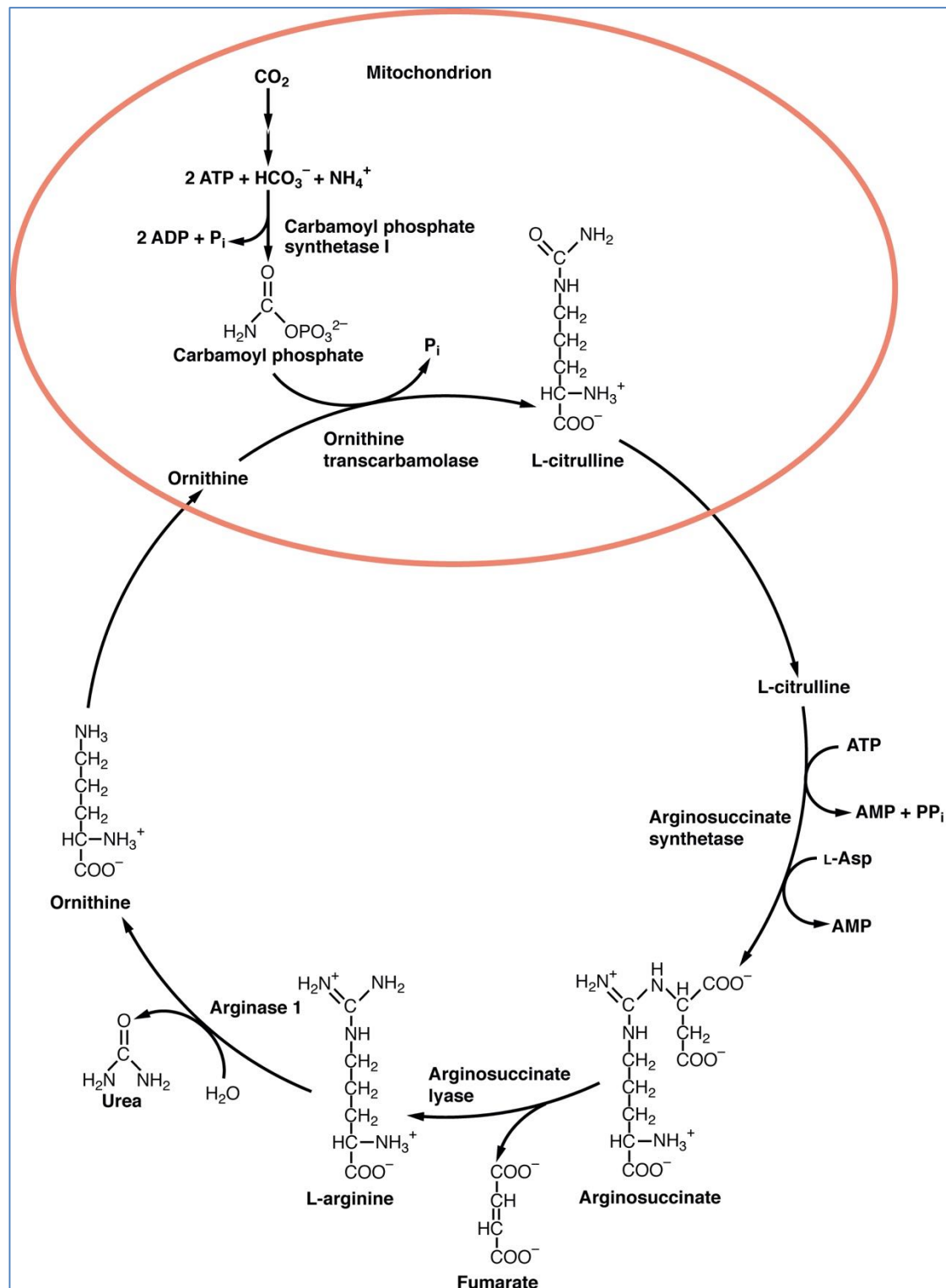
Overview of Amino Acid Metabolism:

- **Amino acids** = Nitrogenous Organic compounds with **-NH₂ & -COOH groups**
 - Humans can only make some of the 20 Amino Acids required by our physiology
 - Plants & microbes can make all 20 amino acids (via transamination reactions)
- **Humans gain the remaining 'essential amino acids' through diet**
- **Dietary proteins must be broken down into their constituent Amino Acids in order to be metabolized:**
 - Once broken down by digestive enzymes, Amino Acids are absorbed in the intestines
 - Intestinal absorption → Portal vein delivers absorbed amino acids → Liver
 - Liver → Synthesizes non-essential amino acids
 - All amino acids → delivered to body cells via blood → Uptake into cells via active transport
- **Some Amino Acids are 'Glucogenic'; some are 'Ketogenic':**
 - **Glucogenic Amino Acids:**
 - Amino acids that can be converted into Glucose through Gluconeogenesis
 - (See diagram)
 - **Ketogenic Amino Acids:**
 - Amino acids that can be converted directly into Acetyl-CoA (the precursor to Ketone Bodies)
 - (See diagram)
 - Note: Leucine & Lysine are *exclusively* ketogenic
 - **Note: Some amino acids are *both* Glucogenic & Ketogenic:**
 - Phenylalanine
 - Isoleucine
 - Threonine
 - Tryptophan
 - Tyrosine
- **Amino Acid Metabolism Produces Ammonia (NH₄):**
 - Ammonia (NH₄) is TOXIC & therefore must be detoxified to Urea (Non-toxic)
 - **Urea Cycle** is responsible for this detoxification

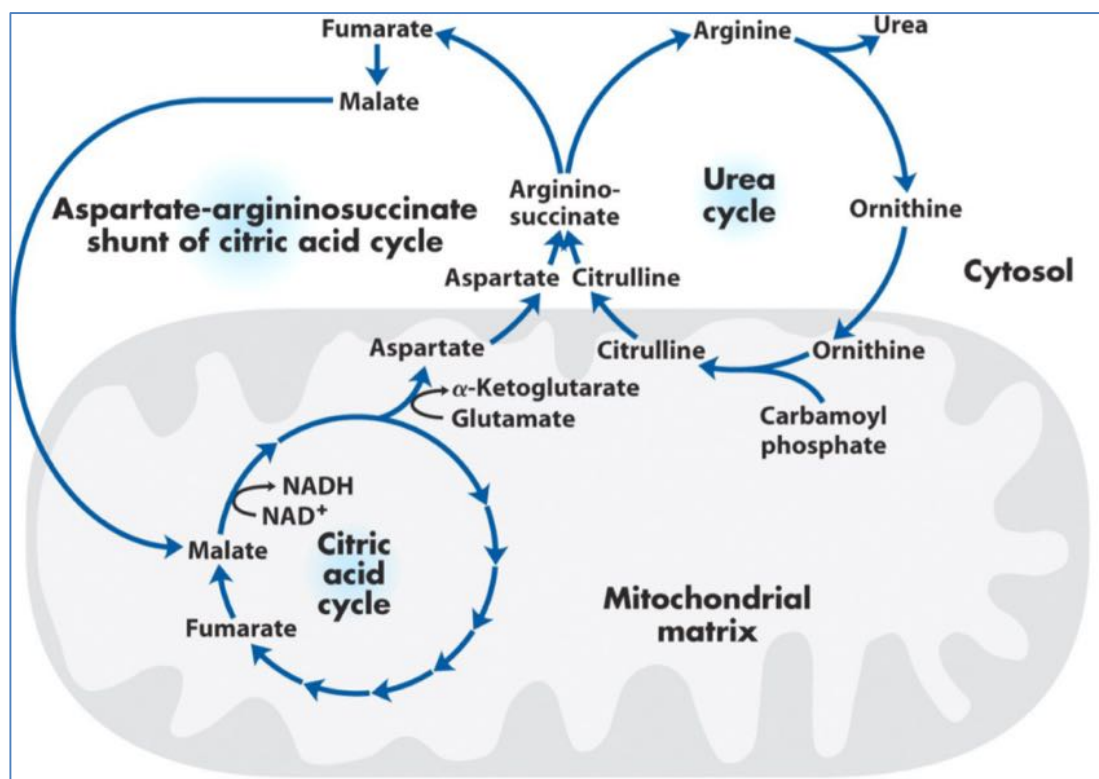
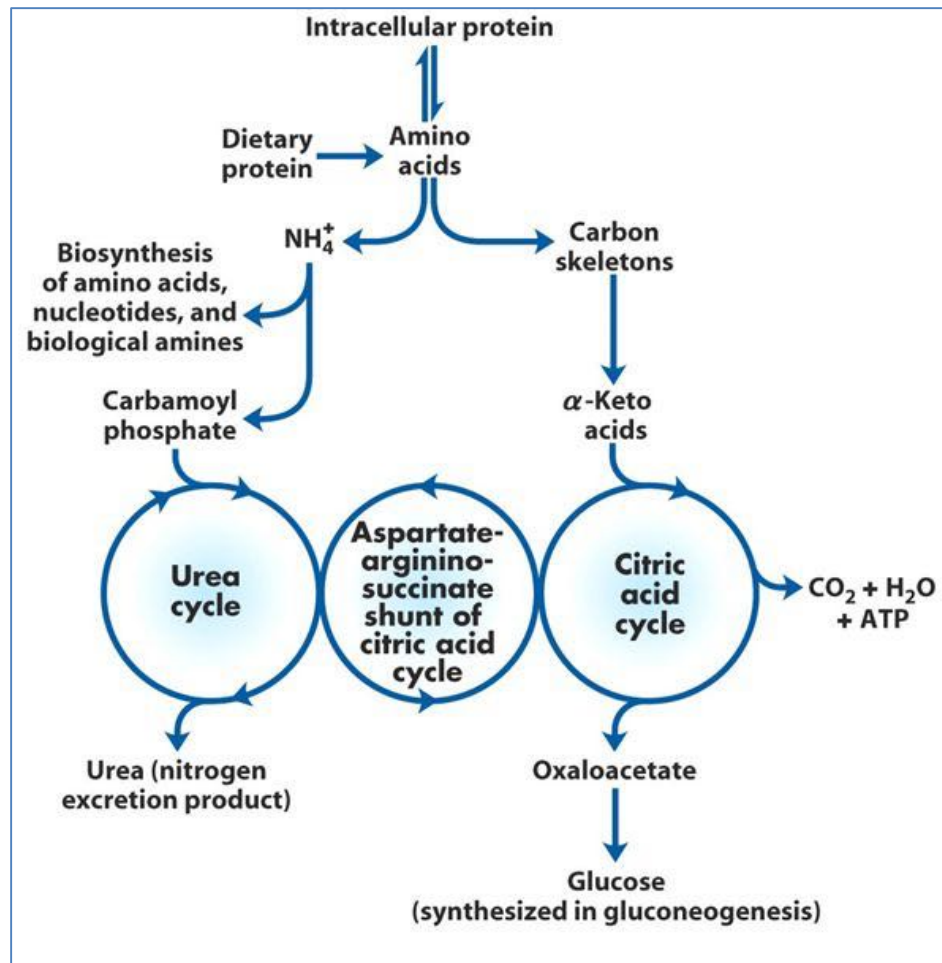


THE UREA CYCLE:

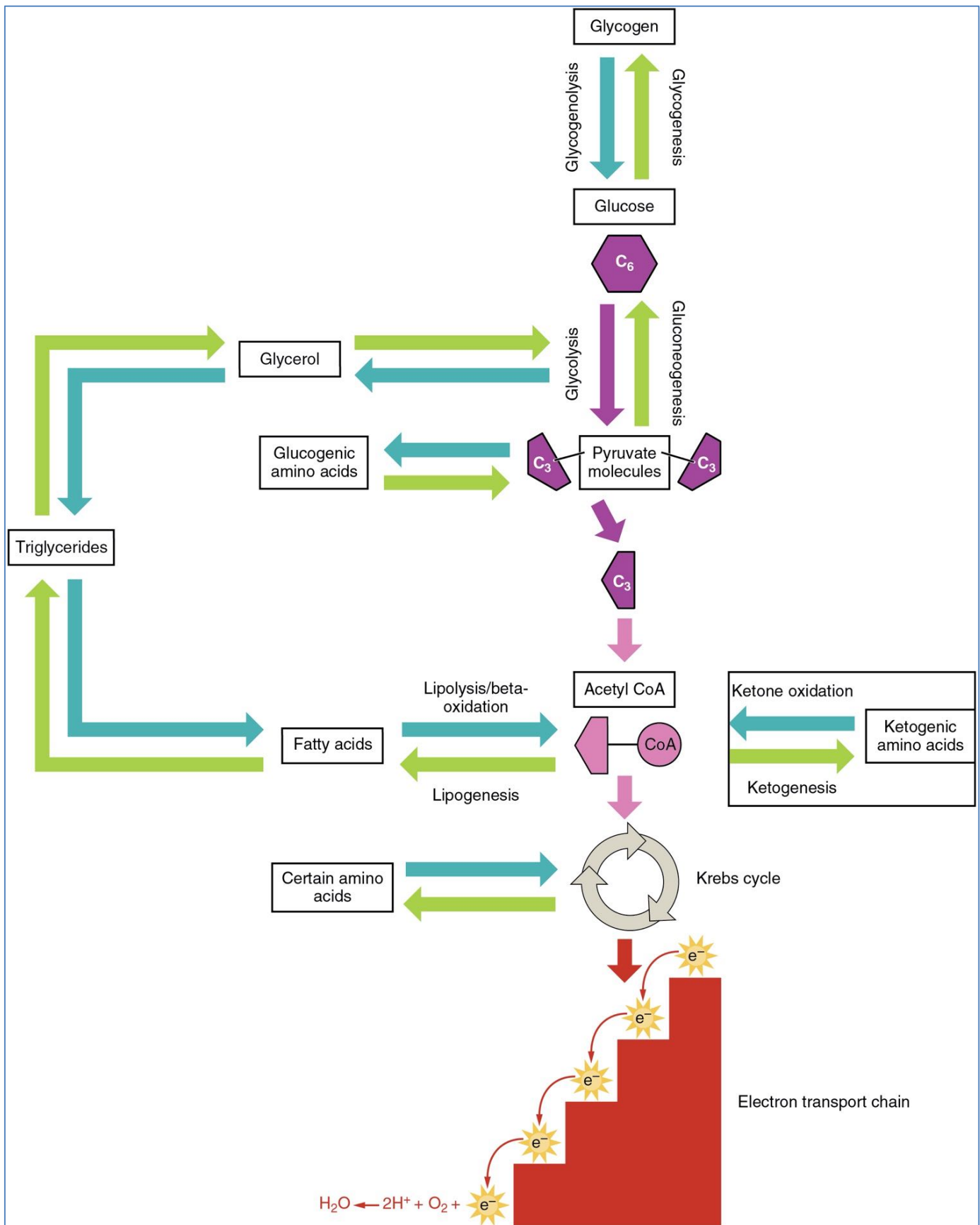
- **What is it?**
 - o The critical detoxification pathway of Ammonia (NH_4) $\rightarrow$ Urea ($(\text{NH}_2)_2\text{CO}$)
 - o Consists of 4x enzymatic reactions (1x mitochondrial reaction & 3x cytosolic reactions)
- **Where in the body does it occur?**
 - o Primarily occurs in the Liver $\rightarrow$ Urea released into bloodstream $\rightarrow$ Excreted into urine by kidneys
 - o Also in kidneys to lesser extent
- **Where in the cell does it occur?**
 - o Mitochondria (1 of the 4 reactions)
 - o Cytosol (3 of the 4 reactions)
- **Summary:**
 - o Costs 3x ATP $\rightarrow$ 2x ADP + 1x AMP



- **'The Aspartate-Arginino-succinate Shunt' (Urea Cycle's Relationship with the TCA Cycle):**
 - o The Urea Cycle & TCA cycle are independent but can feed into each other
 - o Transamination of a TCA-produced Oxaloacetate → supplies Urea Cycle with Aspartate
 - o Fumarate produced by Urea Cycle → Malate → Enters the TCA cycle



Unattributable



Catabolic and Anabolic Pathways Nutrients follow a complex pathway from ingestion through anabolism and catabolism to energy production

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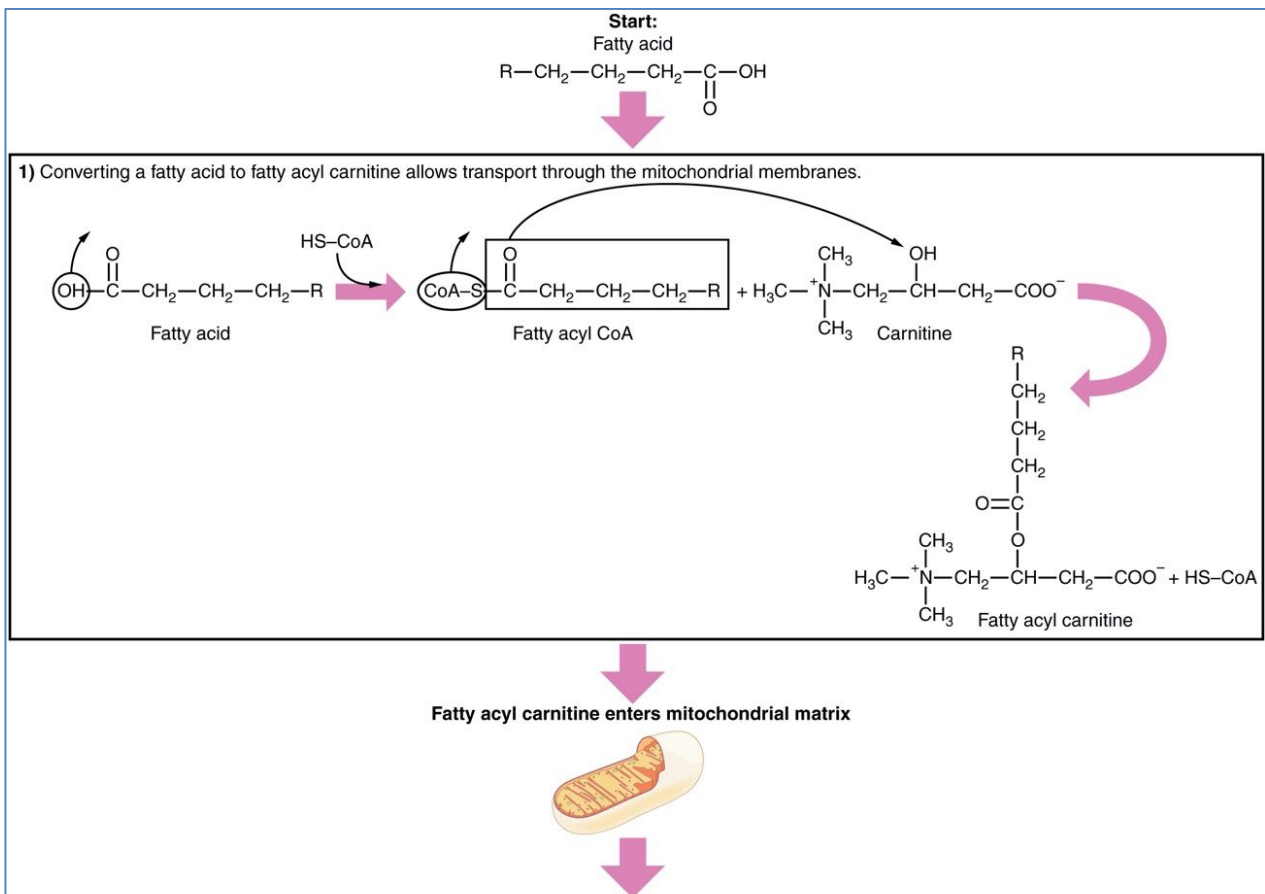
FATTY ACID METABOLISM

Overview of Fatty Acid Metabolism:

- **Fatty Acids** = The Simplest Lipid form
 - o Consist of a Carboxylic Acid
 - o + a Long Carbon+Hydrogen chain
- **Chain Length Varies:**
 - o **Dietary Fats:** Short-Medium Chain Fatty Acids
 - o **In-Vivo Fats (Synthesized by the body; Liver & Adipose):** Long Chain Fatty Acids
- **Functions:**
 - o **Fuel:** Fatty acids are metabolized to produce energy (ATP) via 'Beta-Oxidation'
 - o **Energy Storage:** The highest energy-density of all nutrient classes (most ATP per gram)
 - o **Precursors:** Fatty acids are precursors for Triglycerides, Phospholipids, Hormones & Ketones

FATTY ACID OXIDATION:

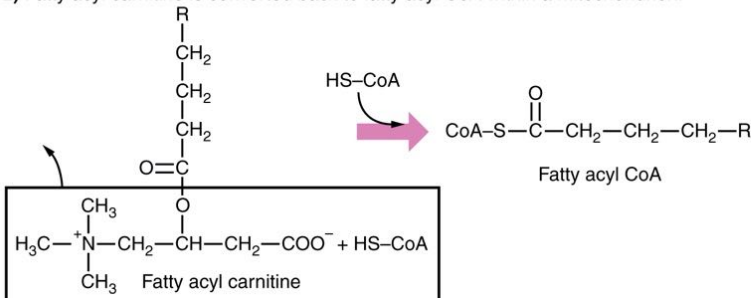
- **1: Lipolysis (Adipose tissue):**
 - o Lipase → Removes the fatty acid chain from the Glycerol on a triglyceride
 - o Stimulated by Glucagon & Epinephrine in response to declining blood glucose levels
- **2: Fatty Acids enter Blood stream:**
 - o Fatty acids are not water-soluble, so are transported by plasma **Albumin**
- **3: Free Fatty Acids enter Metabolizing Cells:**
 - o Enter via Specific transport proteins (eg: SLC27)
- **4: Fatty Acid is reacted with Coenzyme A to give Fatty-Acyl-CoA**
- **5: Fatty-Acyl-CoA enters Mitochondrion via the 'Carnitine Shuttle'**
- **6: Beta Oxidation → TCA Cycle:**
 - o Beta-Oxidation cuts long carbon chains of the fatty acids into multiple 2-carbon (Acetate) units
 - o Each Acetate unit combine with Co-Enzyme-A to form **Acetyl-CoA**
 - o **Acetyl-CoA** → combines with Oxaloacetate → **Citrate**
 - o **Citrate** → Feeds directly into the TCA Cycle



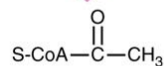
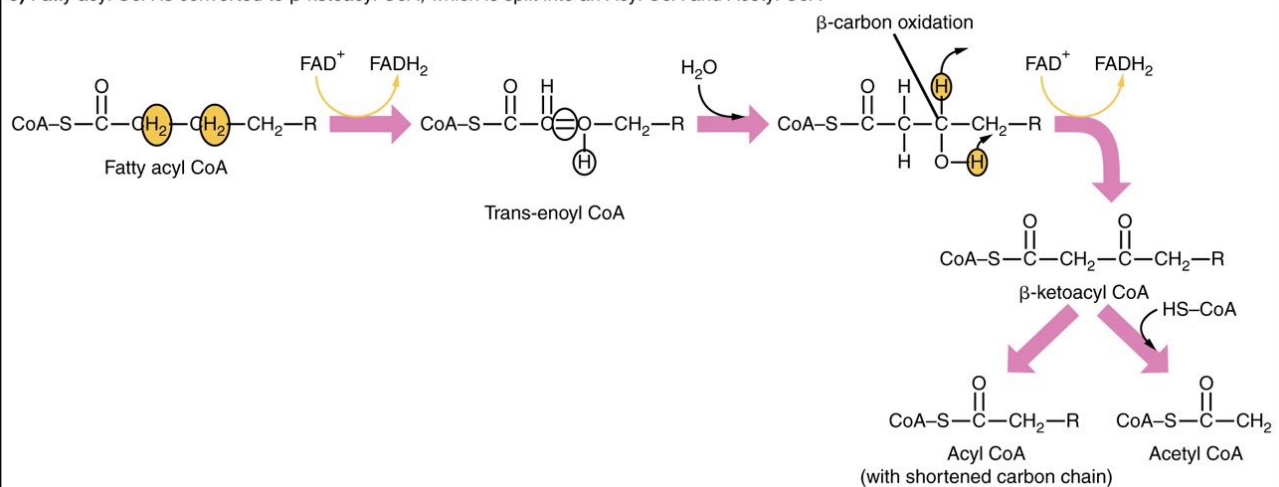
Fatty acyl carnitine enters mitochondrial matrix



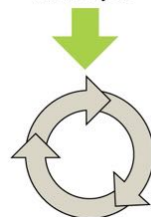
2) Fatty acyl carnitine is converted back to fatty acyl CoA within a mitochondrion.



3) Fatty acyl CoA is converted to β -ketoacyl CoA, which is split into an Acyl CoA and Acetyl CoA

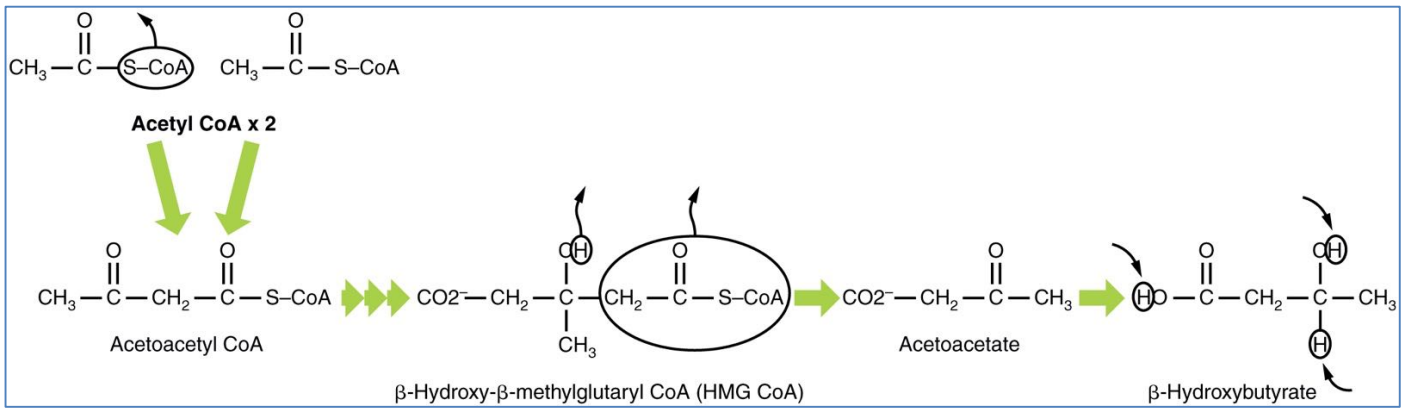


End:
Acetyl CoA enters
Krebs cycle



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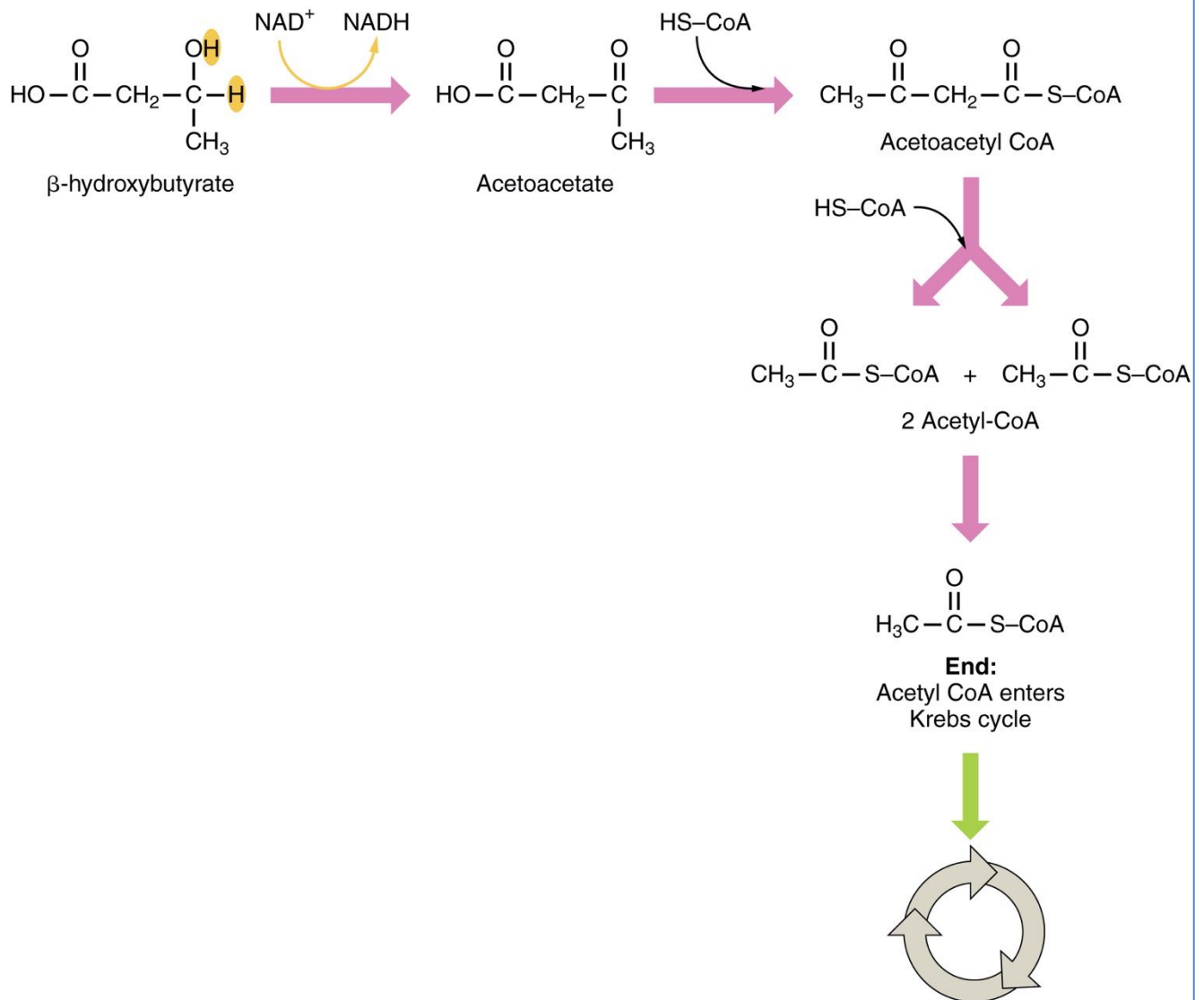
Ketogenesis Excess acetyl CoA is diverted from the Krebs cycle to the ketogenesis pathway. This reaction occurs in the mitochondria of liver cells. The result is the production of β -hydroxybutyrate, the primary ketone body found in the blood.



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Ketone Oxidation When glucose is limited, ketone bodies can be oxidized to produce acetyl CoA to be used in the Krebs cycle to generate energy

Ketone oxidation



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CELLULAR SIGNALLING

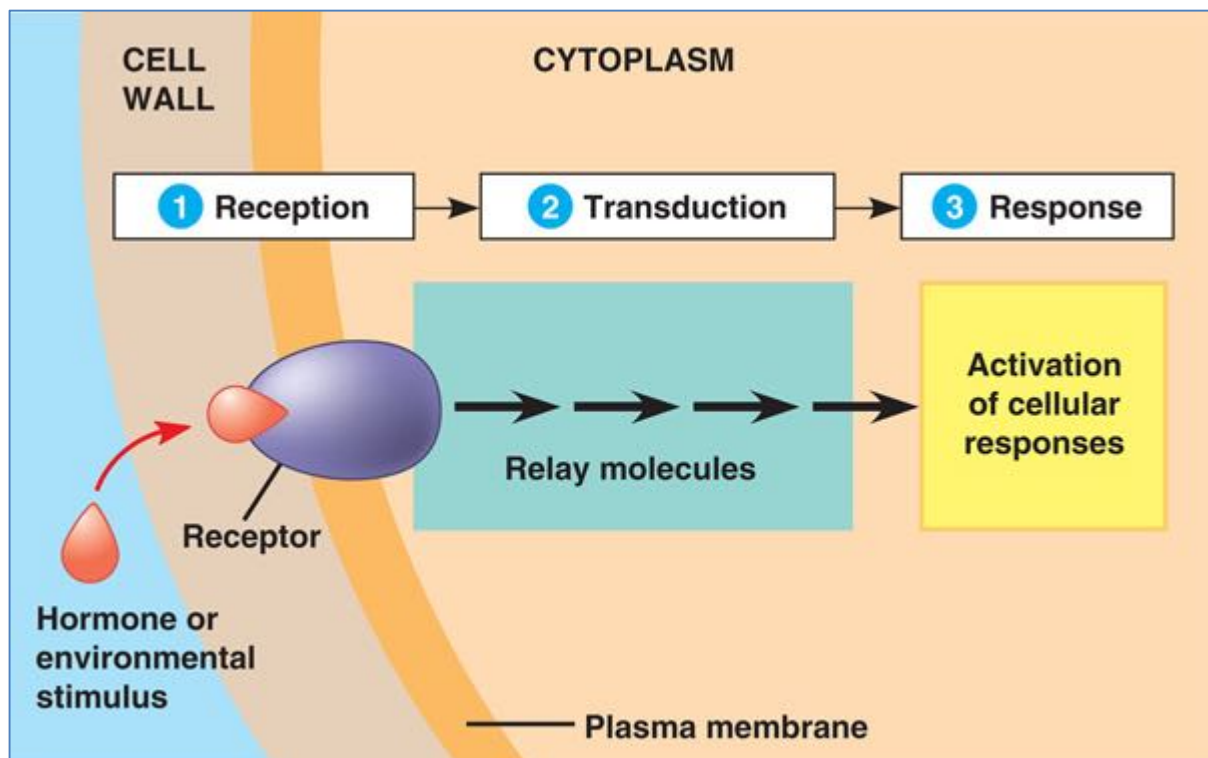
CELLULAR SIGNALLING

Purpose of intercellular signalling:

- To aid cells in coordinating their functions towards the common good of a **multicellular** organism
- Cells must interpret the multitude of signals from other cells to help coordinate their functions
- Effects of coordinated functions include:
 - Movement
 - Growth
 - Reproduction
 - Digestion
 - Metabolism
 - Circulation
 - Respiration
 - Senses
 - Temperature
 - Balance
 - Immune system
 - Differentiation
 - Death (apoptosis)
- Note: different cells may respond to the same signal in different ways

Signal Transduction

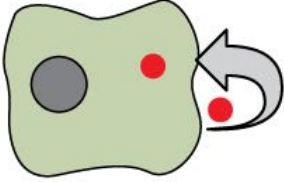
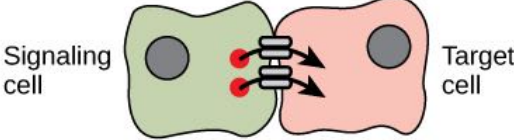
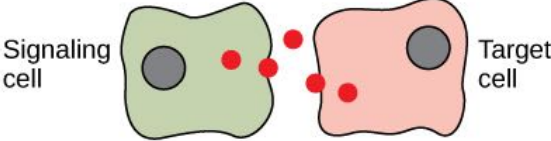
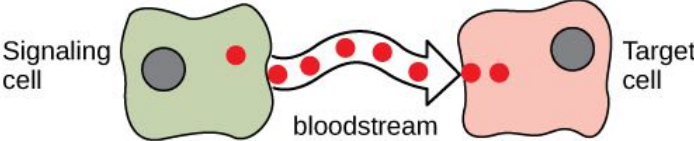
- **Communication frequently involves converting signals from one form to another**
 - Signalling cell produces a signal molecule (proteins /peptides /amino acids /nucleotides /steroids /fatty-acid derivatives /gasses)
 - Signal molecule is detected by target cell
 - Receptor protein receives the signal & transduces it to an intracellular signal
 - Intracellular signal relayed, amplified, & diverged along a signalling 'cascade'
 - Intracellular signals received by target proteins inside cell, altering cell behaviour



Source: http://www.bio.miami.edu/dana/226/226F08_20print.html

Long or Short Range?

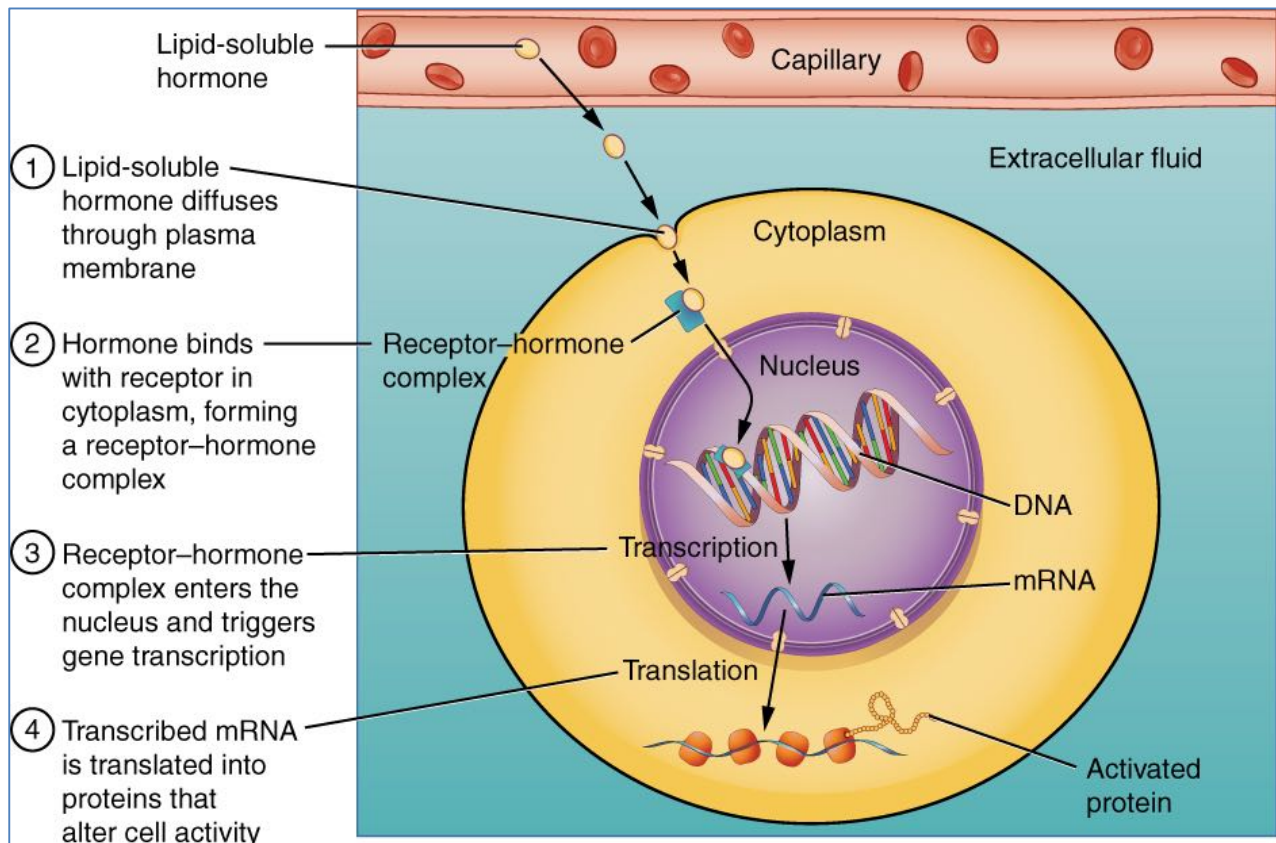
- **Endocrine Signalling:** Some signals are “broadcasted” throughout the entire body via bloodstream → **Hormones** (produced by endocrine cells) [TV]
- **Autocrine:** Signals that affect only cells of the same cell type as the emitting cell [doctor conference]
- **Paracrine:** Signals (aka local mediators) that act on cells in the vicinity of the emitting cell but on different cell types than the emitting cell [Lecture]
- **Neuronal:** Specific messages are delivered across long distances to specific target cells [phone call]
- **Contact dependant:** Does not require secretion of signal molecule. Instead, cells make direct contact through signalling molecules and receptors lodged in their plasma membranes

Forms of Chemical Signaling	
Autocrine	A cell targets itself.
	
Signaling across gap junctions	A cell targets a cell connected by gap junctions.
	
Paracrine	A cell targets a nearby cell.
	
Endocrine	A cell targets a distant cell through the bloodstream.
	

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2 Main Receptor Types: (Intracellular & Membrane-bound Receptors)

- Determined by how the hormone receptor binding is relayed to the cytoplasm
- Mechanism depends on chemical nature of the hormone & the cellular location of receptor
- **Intracellular Receptors:**
 - o Lipid-soluble hormones (steroid/thyroid hormones) & even gasses (nitric oxide-blood vessel dilation)
 - Can diffuse straight through the membrane
 - **Steroid hormones** bind to receptor proteins in the cytosol or the nucleus that regulate gene expression
 - **Other signal molecules** activate intracellular enzymes
 - Once bound, the receptor protein undergoes a large conformational change and 'activates', allowing it to promote/inhibit transcription of a select set of genes



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- **Plasma-Membrane-Bound-Receptors:**

○ **Ion-Channel-Linked Receptors:**

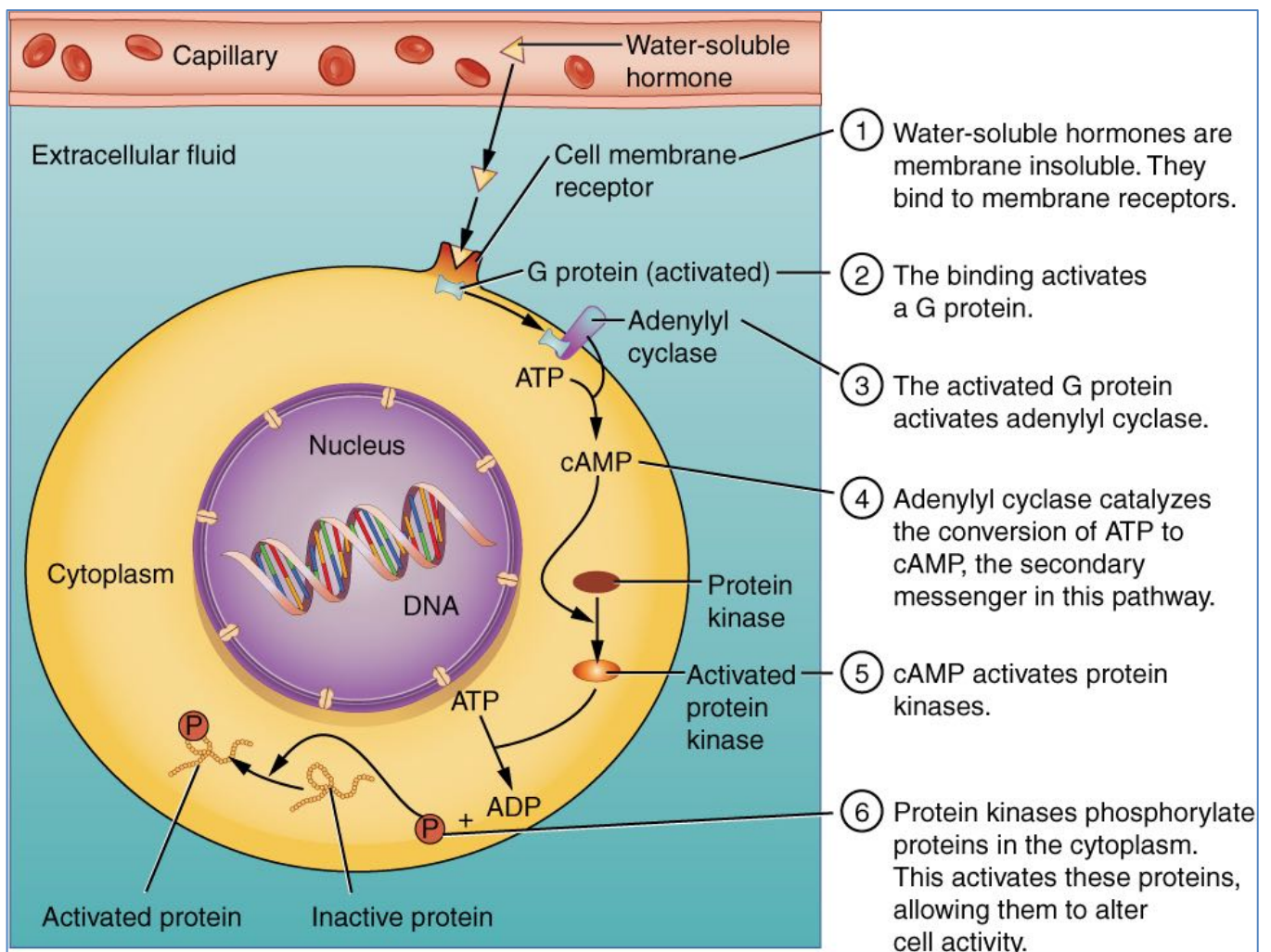
- Resulting signal is a flow of ions across the membrane – produces an electric current
- Signal molecules are often neurotransmitters

○ **Enzyme-Linked Receptors:**

- When activated → receptor acts as an enzyme (or is associated with intracellular enzymes) → Initiates a cascade of other effects → Signal transduction into the cytosol

○ **G-Protein-Linked Receptors (more common):**

- The largest family of cell-surface receptors
- Signal molecule binds to **G-Protein** → G-Protein becomes activated
- **Activated G-Protein** → initiates a cascade of other effects
 - Some activate ion channels
 - Others activate membrane-bound enzymes (eg: Adenylyl-Cyclase → Cyclic AMP)
- G-Proteins automatically switch themselves off by hydrolysing their bound GTP to GDP



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THE 'CELL CYCLE' & CELLULAR REPLICATION

THE 'CELL CYCLE' & CELLULAR REPLICATION

The Cell Cycle:

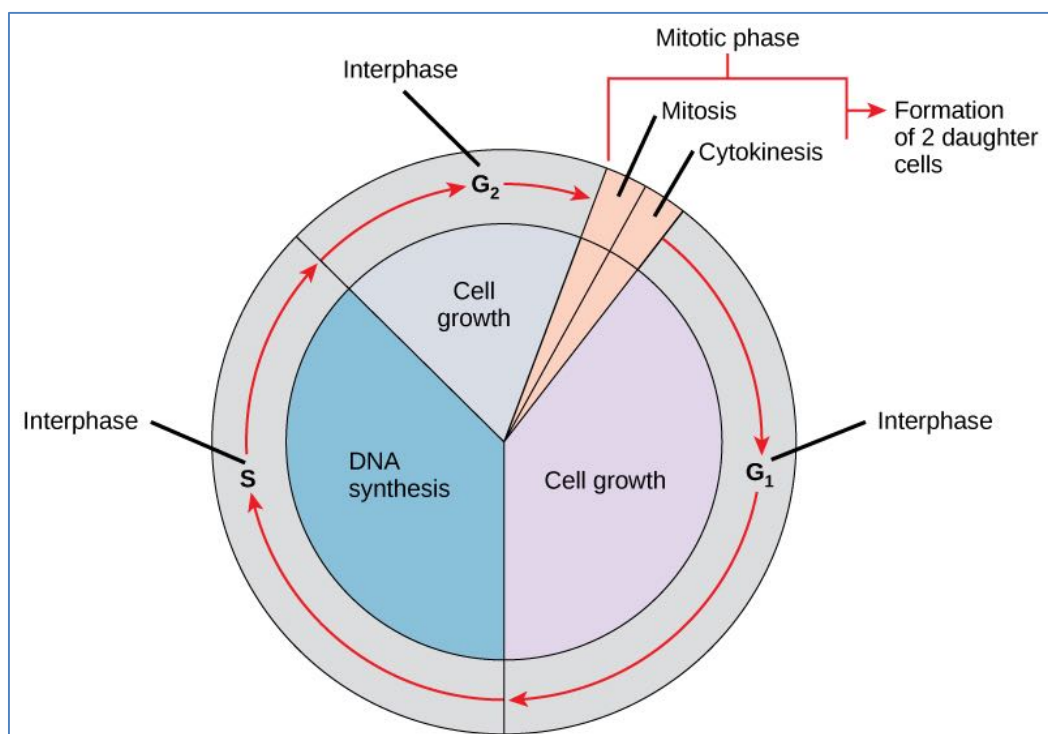
- The *cell doctrine*: "where a cell arises there must be a previous cell"
- For a cell to reproduce, it must duplicate its contents & divide cyclically (**the cell cycle**)

Fundamental tasks of the cell cycle:

- Copy and pass on genetic info to next generation
- Produce 2 genetically identical daughter cells
- DNA in each chromosome must be accurately replicated
- Replicated chromosomes must be equally distributed between daughter cells
- Coordinate growth with division to maintain size & contents

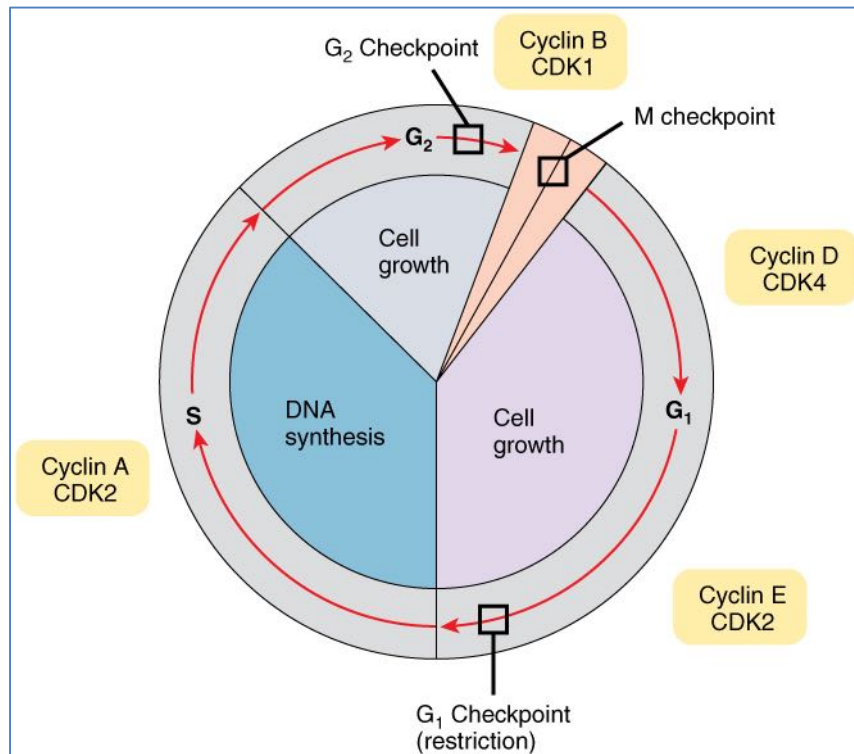
4 Phases of the Cell Cycle:

- **Interphase:**
 - Cells spend most of their time in 'Interphase'
 - During this time, the Cell continues to transcribe genes, synthesize proteins, & grow
 1. **G₁ Phase (gap)**
 - Provide additional time for cell to grow & duplicate Cytoplasmic organelles
 2. **S Phase (synthesis)**
 - DNA Replication
 - Restriction/Commitment Point – no turning back
 3. **G₂ Phase (gap)**
 - Provide additional time for cell to grow & duplicate Cytoplasmic organelles
 - Replicated chromosomes condense
 - **Mitosis & Cytokinesis:**
 4. **M Phase (mitosis & cytokinesis)**
 - **Prophase** – nuclear membrane breaks down
 - **Prometaphase** – replicated chromosomes are lined up
 - **Metaphase** – chromosomes separated into 2 sister chromatids
 - **Anaphase** – chromatids arrive at ends of cell, decondense & form separate nuclear membranes
 - **Telophase** – Plasma membrane pinches cell into 2
- (**G₀ Phase** = Some cells (neurons) that don't divide, just stop at G₀)



The Cell-Cycle Control System:

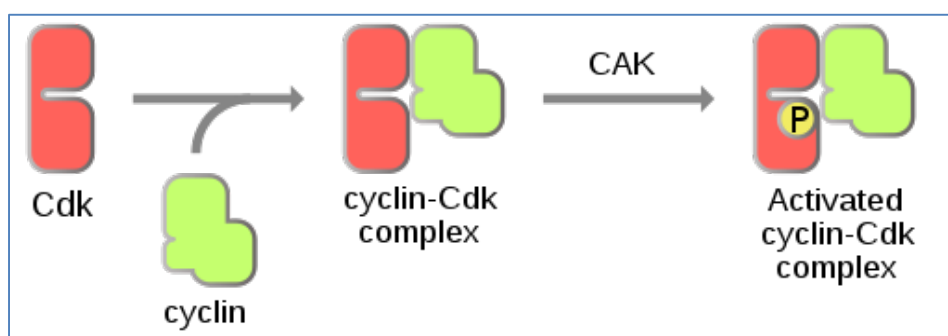
- A complex network of regulatory proteins that ensures a cell replicates properly
- The cell checks that all critical earlier events have occurred:
 - o Eg: DNA replication
 - o Eg: Segregation of duplicated chromosomes
- Responds to various signals from outside & inside the cell
- Critical to regulation of cell numbers
- **Malfunction can lead to cancer** (Abnormal number of chromosomes/Mutated DNA)
- Can stop the cycle using **molecular 'brakes'** at 2 important **checkpoints**



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'Molecular Brakes' of the Cell-Cycle:

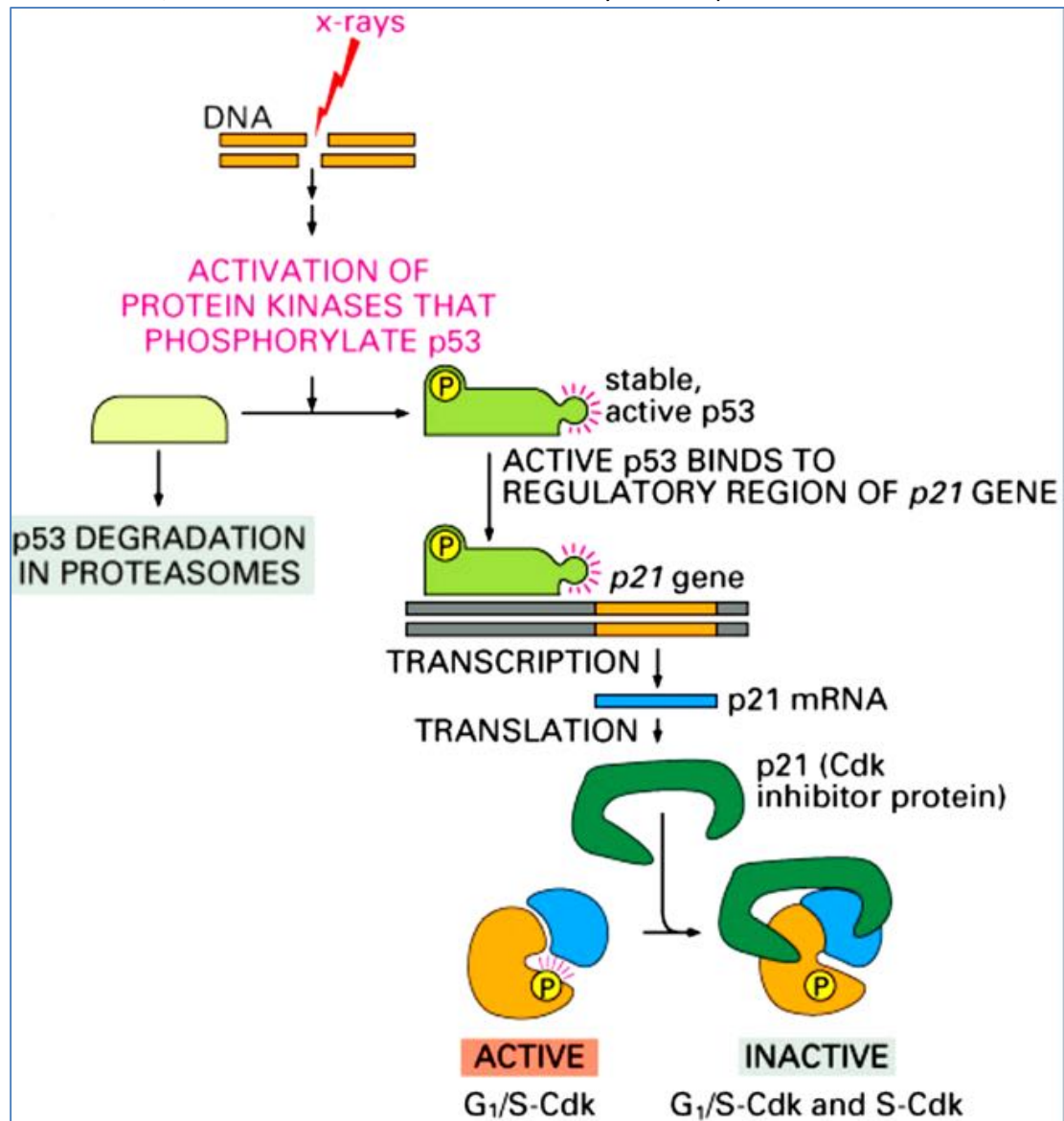
- Govern cell-cycle by activating/inactivating proteins for DNA replication, mitosis & cytokinesis
 - o **Activation** via phosphorylation → by **protein kinases** (consume ATP + phosphorylate substrate)
 - o **Deactivation** via dephosphorylation → by **protein phosphatases**
- Protein kinases must first become partially active by **binding to a cyclin**
- Cyclins – have no enzymatic activity
- Cyclin Dependant Kinases (Cdk's) – partly activated by cyclin
 - o Cyclin + Cyclin Dependant Kinase = **Cyclin-Cdk**
- Cdk-Activating kinase (CAK) – **phosphorylates** the partly activated cyclin-Cdk, fully activating it
- Activation of Cyclin-Cdk complexes can trigger cell-cycle events (Entry into S-phase or M-phase)
 - o 'M'-Cyclin + Cdk = M-Cdk → drives cells into **M-phase**
 - o 'S'-Cyclin + Cdk = S-Cdk → drives cells into **S-phase**
- Cdk Inactivation: Once the relevant phase is complete, **ubiquitin-dependant proteolysis** destroys cyclins



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G₁ Checkpoint: DNA damage prevents S-Phase Entry:

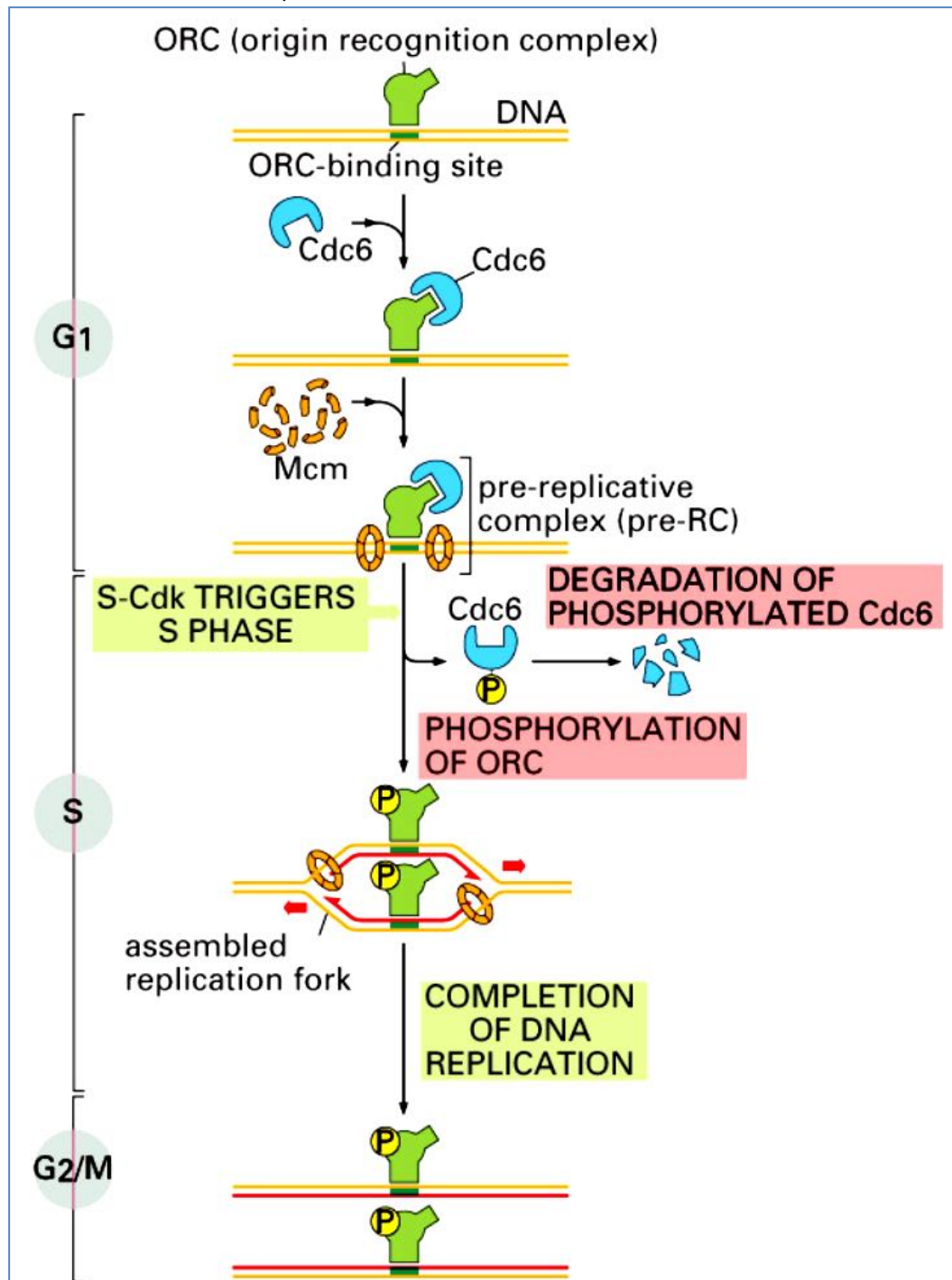
- Ensures **environment is favourable** for cell proliferation
- Ensures **DNA is intact** before committing it to **replication (S-Phase)**
- **S-Cdk:**
 - o Damaged DNA causes expression of a **Cdk Inhibitor Protein** that inactivates the **S-Cdk**
 - o S-Cdk is necessary for DNA replication in S-Phase
 - o Therefore, inactivation of S-Cdk arrests the cell cycle in G₁ phase



https://digitalcommons.olivet.edu/cgi/viewcontent.cgi?article=1102&context=honr_proj

Cdc6:

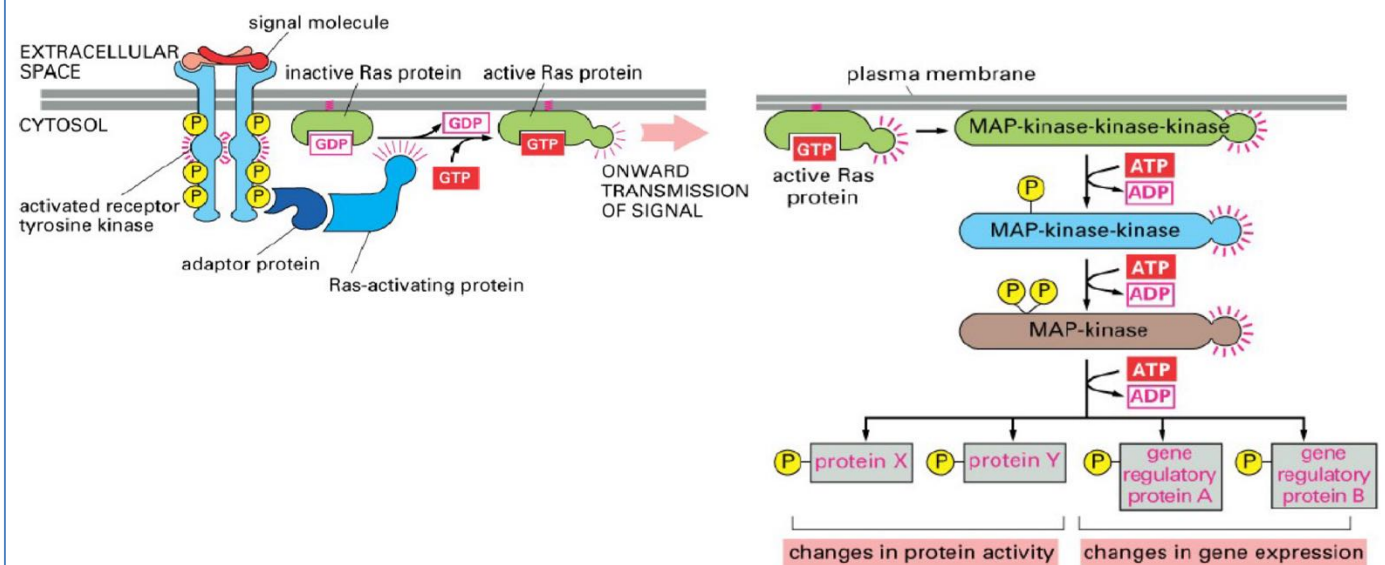
- **Cdc6 is a regulatory protein that binds to ORC's (Origin Recognition Complexes) on DNA**
 - When **Cdc6** is associated with the ORC it puts a safety switch on DNA replication
 - Prevents DNA Replication until G₁ checkpoint is satisfied
- **Activated S-Cdk removes Cdc6 via phosphorylation → allows DNA replication to continue**
 - Initiates DNA replication



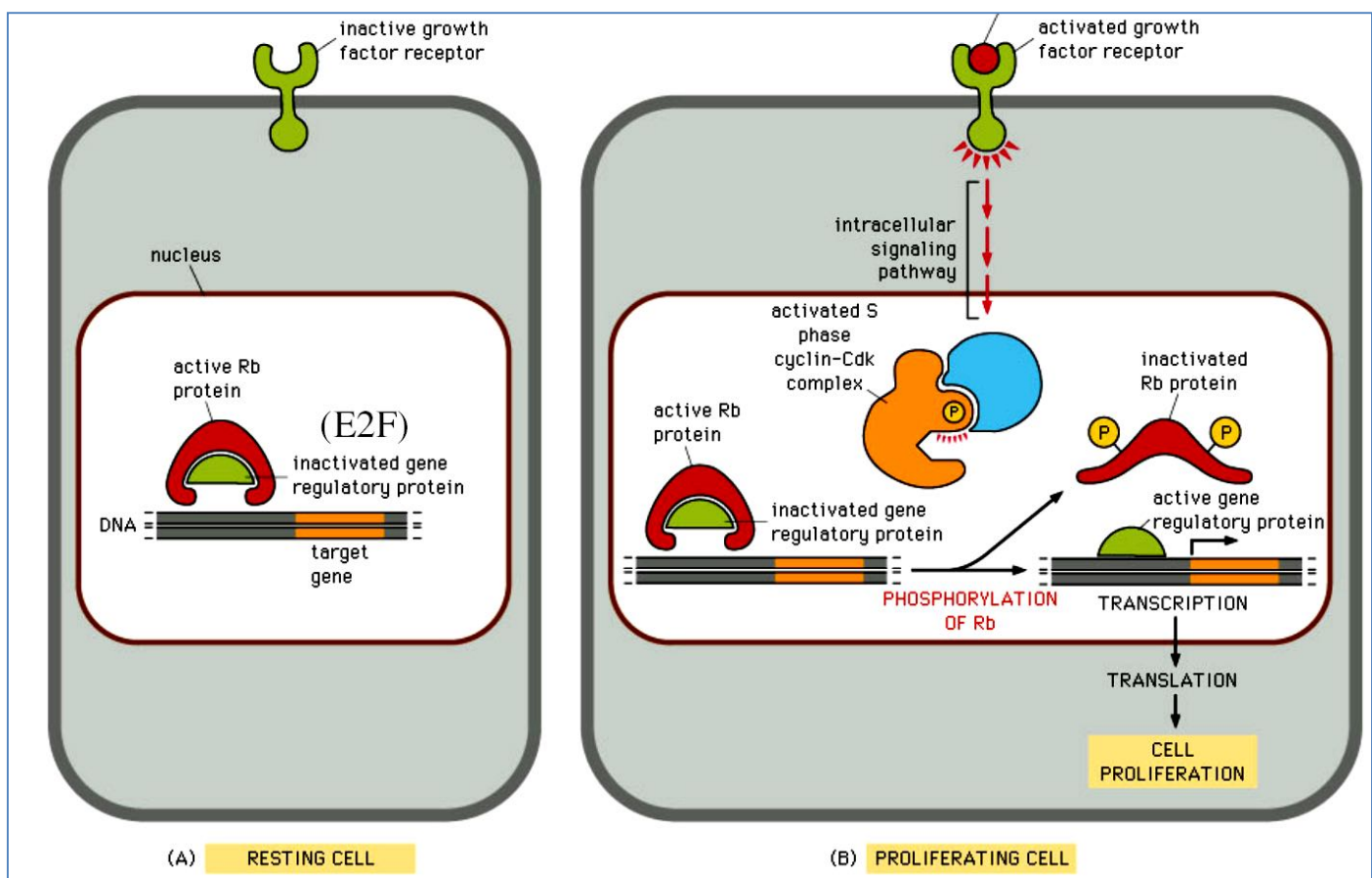
Source: <https://slideplayer.com/amp/7695798/>

- **Rb-protein (Gene Regulatory Protein-B):**

- **Rb-Protein is another regulatory protein that prevents gene transcription of S-Phase proteins**
- Requires the presence of **external growth-factors** → intracellular cascade
 - Enzyme-linked receptor → Ras activated
 - Ras → MAP-Kinase cascade → removes Rb
- Removal of Rb activates allows gene expression → protein synthesis can continue



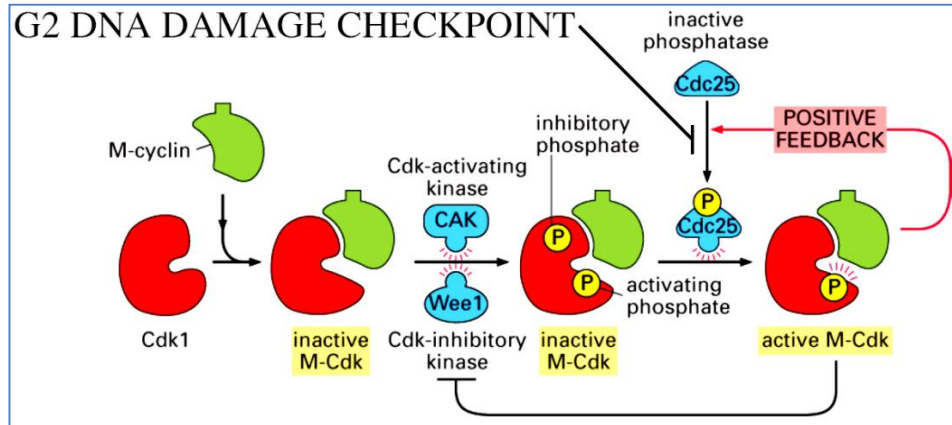
Source: <https://slidetodoc.com/growth-factors-and-enzymelinked-receptor-signal-systems-lecture/>



Source: <https://slidetodoc.com/growth-factors-and-enzymelinked-receptor-signal-systems-lecture/>

G₂ Checkpoint: DNA damage prevents M-Phase Entry

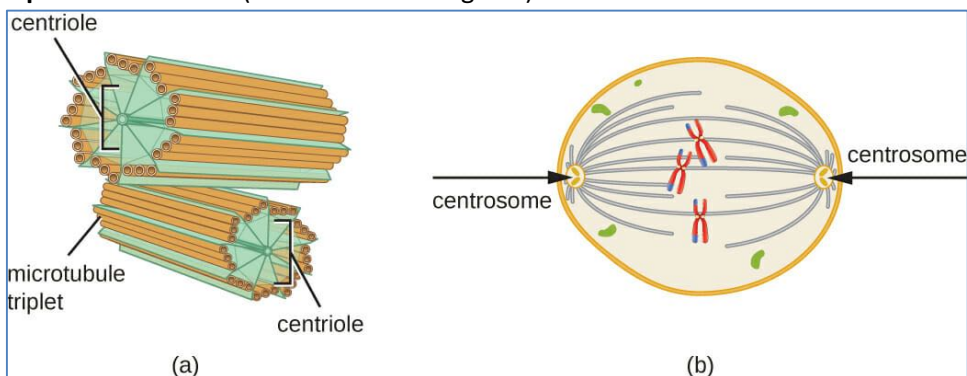
- Ensures that cells don't enter **mitosis** until **DNA replication** is completed correctly
- **M-Cdk:**
 - o Similar function to S-Cdk in the G1 Checkpoint
 - o S-Cdk is inactivated by phosphorylation & activated by dephosphorylation
 - o If DNA is good, **Phosphatase Cdc25 removes the inhibitory phosphor**, instantly activating the M-Cdk
 - o If DNA is bad, phosphatase Cdc25 is inactive, & therefore so is M-Cdk
- **Active M-Cdk further activates phosphatase Cdc25 & also stops the Cdk-inhibitory kinase** from adding the inhibitory phosphor → **triggers MITOSIS**



Source: <https://cdn.technologynetworks.com/ep/pdfs/cell-cycle-regulation-1.pdf>

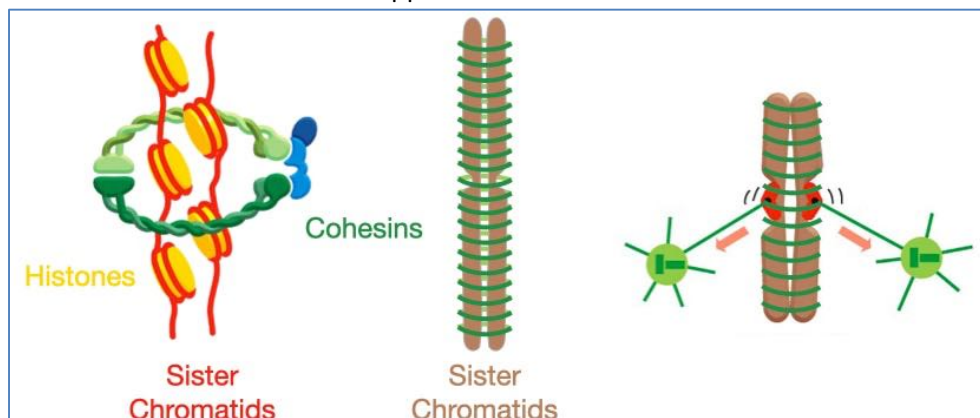
The Centrosomes:

- **Form mitotic spindles** to ensure even distribution of genetic material during mitosis
- Consist of a **pair of Centrioles** (1 mother & 1 daughter)

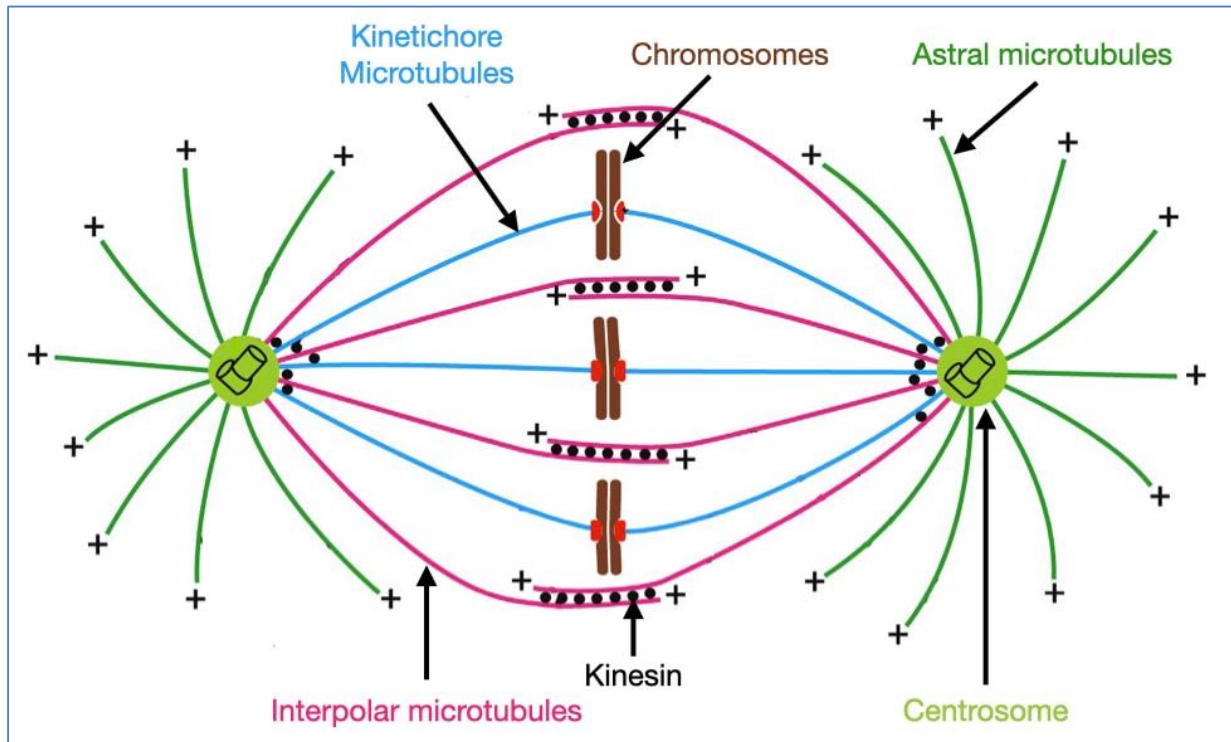


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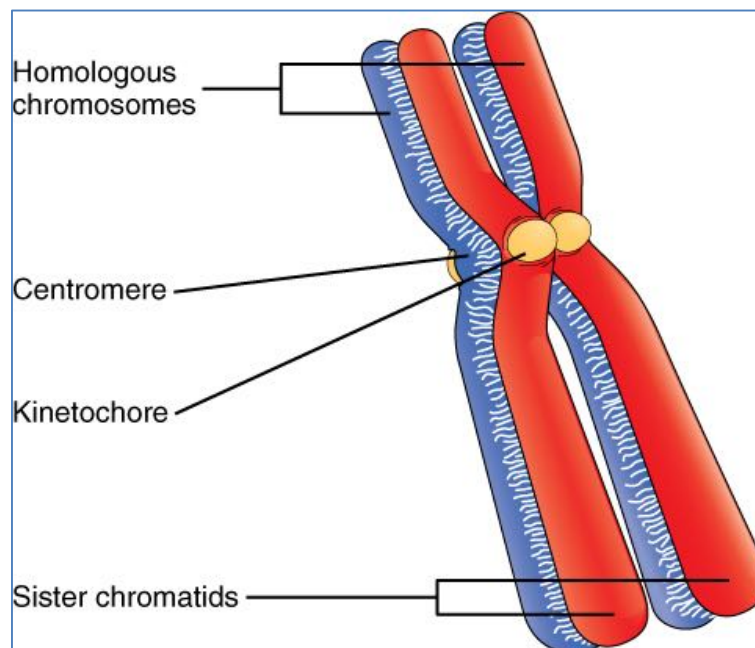
- **Are responsible for the 'pulling apart' of the chromosomes during anaphase**
 - o Sister chromatids are held together by '**cohesin**'
 - o Mitotic **spindles grab the chromosomes** on either side at the **kinetichore**
 - o Spindles take up tension
 - o When this 'spindle checkpoint' is satisfied & all chromosomes are bound to spindles, the cohesin is cleaved & chromatids retract to opposite sides of the cell



Source: http://tbl.med.yale.edu/cell_growth_control/reading.php



Source: http://tbl.med.yale.edu/cell_growth_control/reading.php

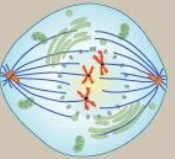
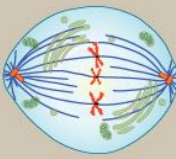
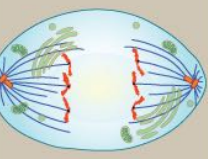
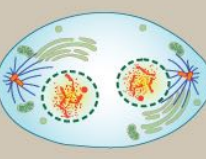
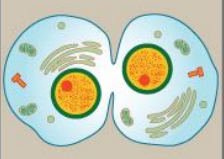
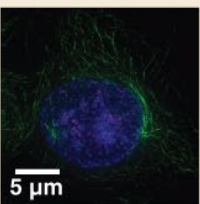
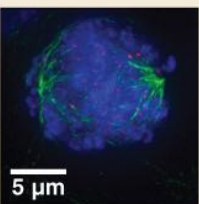
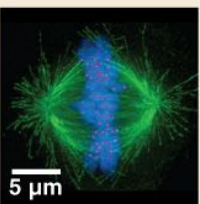
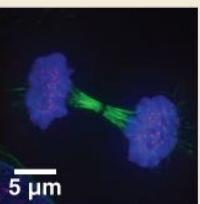
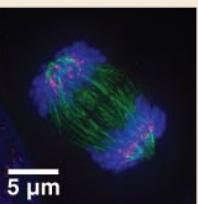
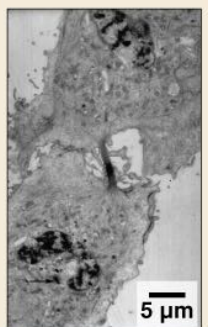


A Homologous Pair of Chromosomes with their Attached Sister Chromatids The red and blue colours correspond to a homologous pair of chromosomes. Each member of the pair was separately inherited from one parent. Each chromosome in the homologous pair is also bound to an identical sister chromatid, which is produced by DNA replication, and results in the familiar "X" shape.

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Mitosis – Cell Division:

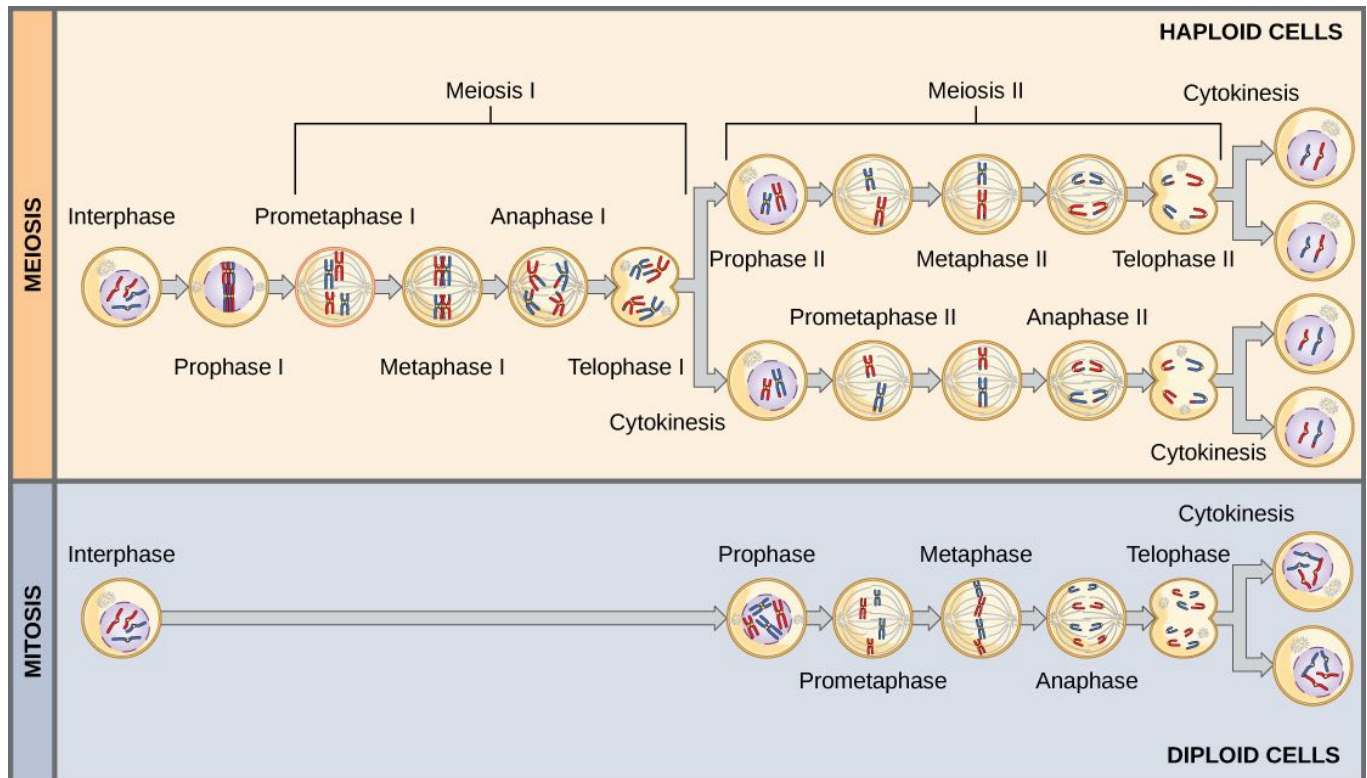
- The division of the nucleus of a eukaryotic cell, involving condensation of the DNA into visible chromosomes
- **5 Phases:**
 - **1: Prophase:** Replicated chromosomes condense each with 2 sister chromatids
 - **Prometaphase:** The nuclear envelope breaks down allowing the chromosomes to attach to the spindle microtubules
 - **2: Metaphase:** The stage at which chromosomes are firmly attached to the mitotic spindle and align at the cell's equator but have not yet split
 - **3: Anaphase:** Where the paired chromatids separate to form pairs of 2 daughter chromosomes & each is pulled slowly toward the spindle pole it is attached to
 - **4: Telophase:** Final stage where the 2 sets of separated chromosomes arrive at the spindles. Chromosomes decondense and become enclosed by new nuclear envelopes
 - **5: Cytokinesis:** The division of the cytoplasm of the cell into two (the division of the entire cell into 2 cells)

Prophase	Prometaphase	Metaphase	Anaphase	Telophase	Cytokinesis
					
• Chromosomes condense and become visible • Spindle fibers emerge from the centrosomes • Nuclear envelope breaks down • Centrosomes move toward opposite poles	• Chromosomes continue to condense • Kinetochores appear at the centromeres • Mitotic spindle microtubules attach to kinetochores	• Chromosomes are lined up at the metaphase plate • Each sister chromatid is attached to a spindle fiber originating from opposite poles	• Centromeres split in two • Sister chromatids (now called chromosomes) are pulled toward opposite poles • Certain spindle fibers begin to elongate the cell	• Chromosomes arrive at opposite poles and begin to decondense • Nuclear envelope material surrounds each set of chromosomes • The mitotic spindle breaks down • Spindle fibers continue to push poles apart	• Animal cells: a cleavage furrow separates the daughter cells • Plant cells: a cell plate, the precursor to a new cell wall, separates the daughter cells
					
MITOSIS					

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Sidenote on Meiosis:

- Similar to Mitosis, but different purpose → To produce gametes
- (Covered in our Sexual Health / Reproductive Subject)



						OUTCOME
PROCESS	DNA synthesis	Synapsis of homologous chromosomes	Crossover	Homologous chromosomes line up at metaphase plate	Sister chromatids line up at metaphase plate	Number and genetic composition of daughter cells
MEIOSIS	Occurs in S phase of interphase	During prophase I	During prophase I	During metaphase I	During metaphase II	Four haploid cells at the end of meiosis II
MITOSIS	Occurs in S phase of interphase	Does not occur in mitosis	Does not occur in mitosis	Does not occur in mitosis	During metaphase	Two diploid cells at the end of mitosis

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REGULATION OF CELL FATE

REGULATION OF CELL FATE

Proliferation/Division

- **Regulated process** of cytoplasmic duplication, followed by mitosis
- Regulated by nutrients, secreted chemical messengers & environmental/local signals
- Essential in development, growth, maintenance & repair

Differentiation

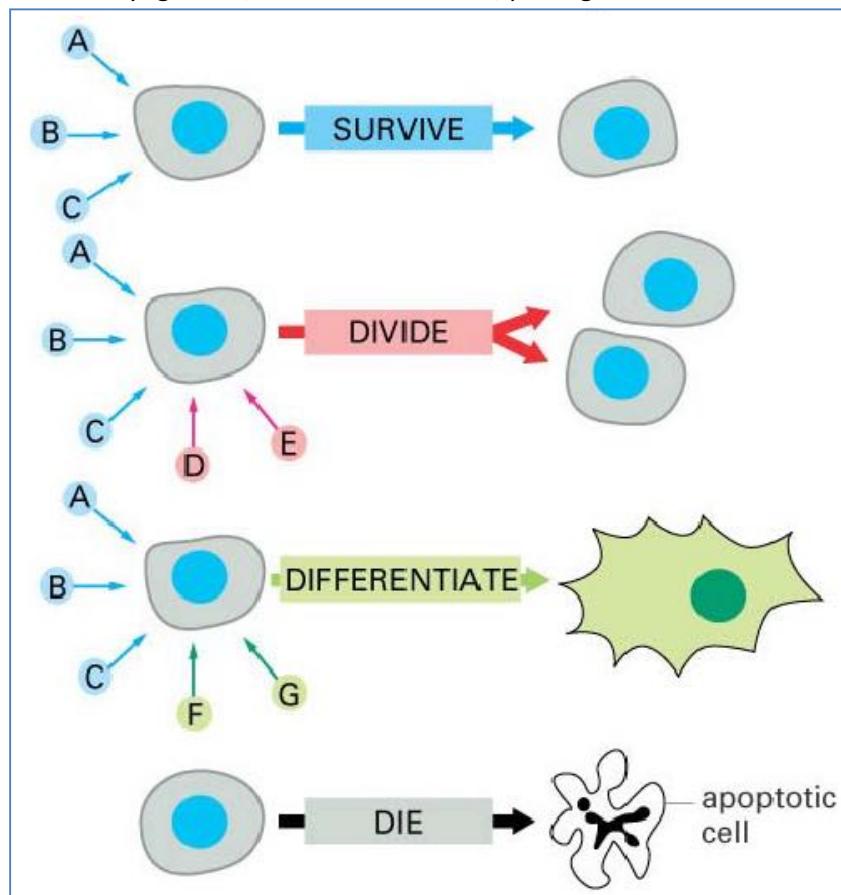
- **Regulated step-wise process** where cells **gain/lose specialised characteristics** (morphological or biochemical)
- Regulated by nutrients, secreted chemical messengers & environmental/local signals
- Essential in the formation & maintenance of specialised tissues/organs

Terminal Differentiation

- **Regulated process** where cells differentiate but can no longer proliferate
- Regulated by nutrients, secreted chemical messengers & environmental/local signals

Apoptosis

- **Regulated process** where cells die for the benefit of the organism
- Regulated by nutrients, secreted chemical messengers & environmental/local signals
- Essential process in embryogenesis, ovulation & menses, pathogenesis

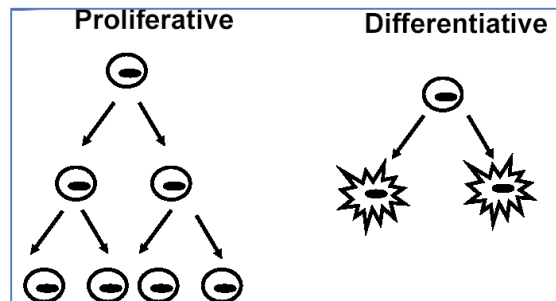


****Imbalance of these 4 processes can result in necrosis and/or cancer****

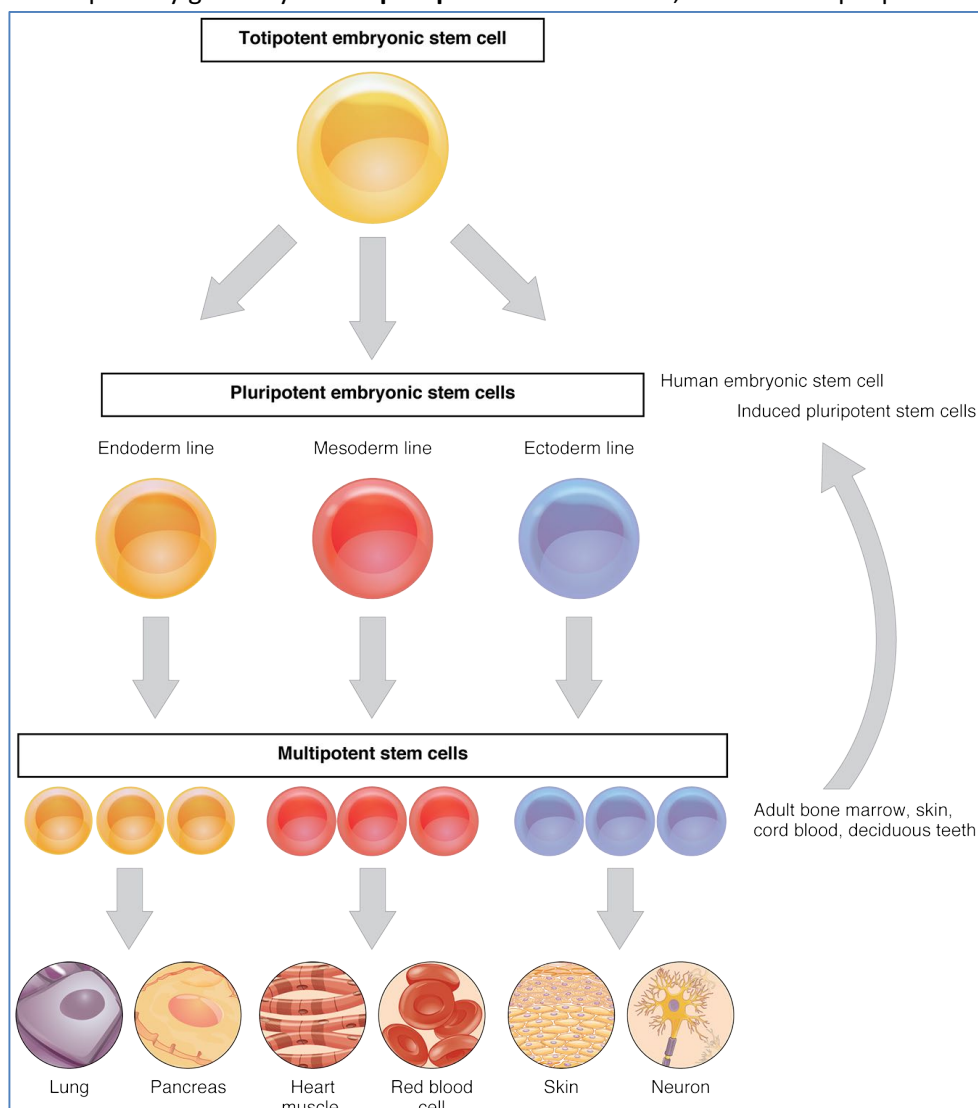
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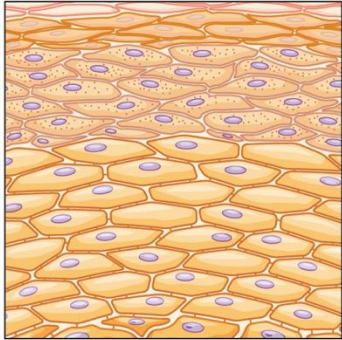
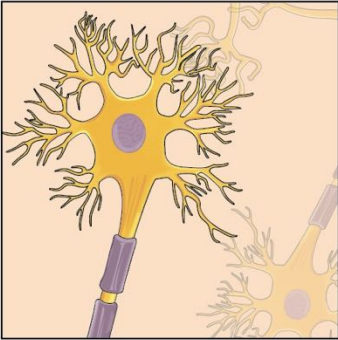
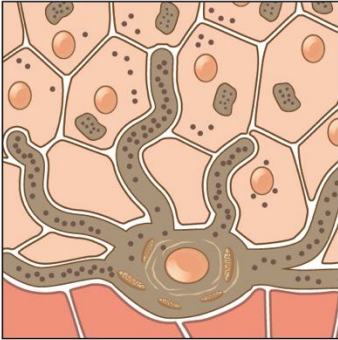
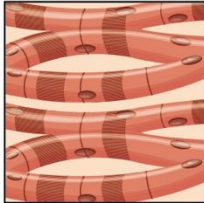
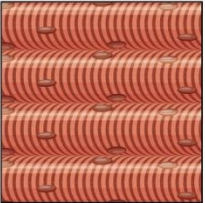
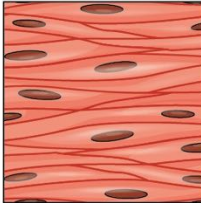
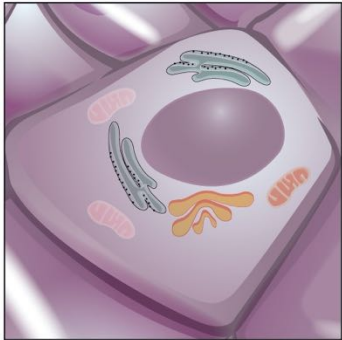
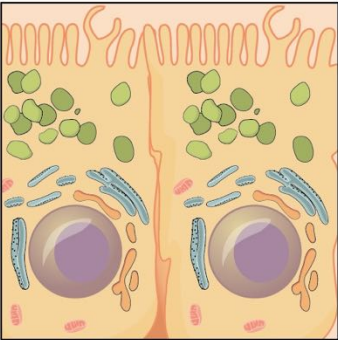
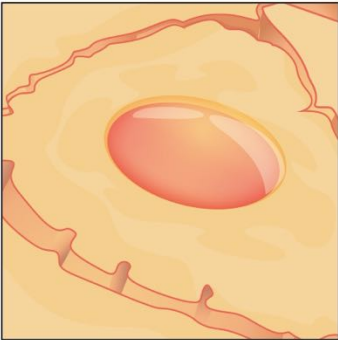
Stem Cells

- Can proliferate
- Can be determined
- Can differentiate into any type of cell
- **Proliferation**
 - Proliferative - eg: Early embryogenesis
 - Differentiative – eg: Oogenesis (stem cells run out)
 - Both



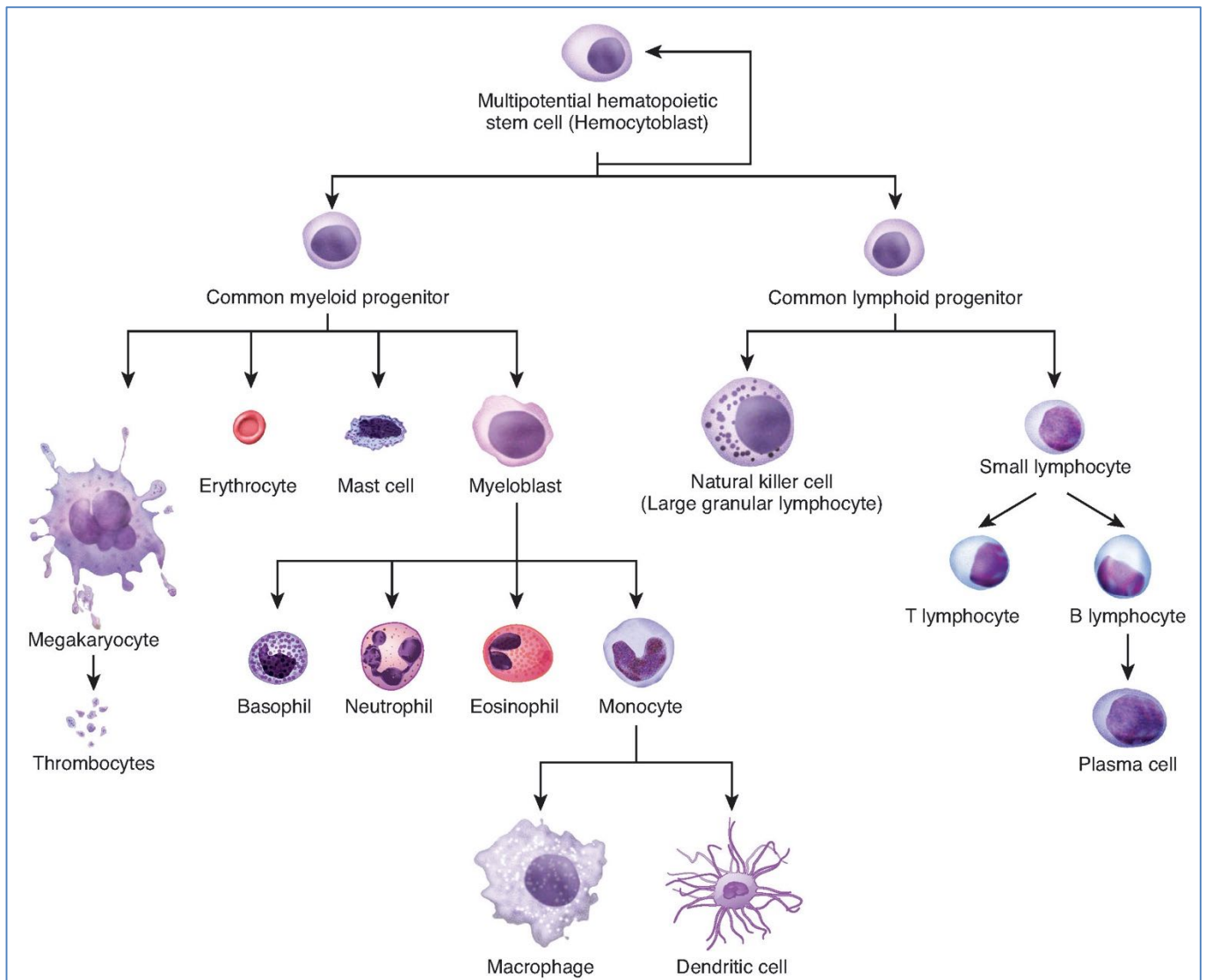
- **Determination**
 - Where the cell is pre-set to a specific phenotype but has not yet differentiated
 - Determined cells are morphologically indistinguishable from stem cells
 - **Eg: Early embryogenesis**
 - **before morula stage**, cells can become any human cell → **totipotent**
 - **at the morula stage**, the cells are determined to be either a primary germ layer or trophoblasts; but have not yet differentiated
 - Also the primary germ layers are **pluripotent**=determined, but for multiple possible pathways



Germ Layer	Gives rise to:
Ectoderm	Epidermis, glands on skin, some cranial bones, pituitary and adrenal medulla, the nervous system, the mouth between cheek and gums, the anus  Skin cells  Neurons  Pigment cell
Mesoderm	Connective tissues proper, bone, cartilage, blood, endothelium of blood vessels, muscle, synovial membranes, serous membranes lining body cavities, kidneys, lining of gonads  Cardiac muscle  Skeletal muscle  Tubule cell of kidney  Red blood cells  Smooth muscle
Endoderm	Lining of airways and digestive system except the mouth and distal part of digestive system (rectum and anal canal); glands (digestive glands, endocrine glands, adrenal cortex)  Lung cell  Thyroid cell  Pancreatic cell

- **Differentiation**

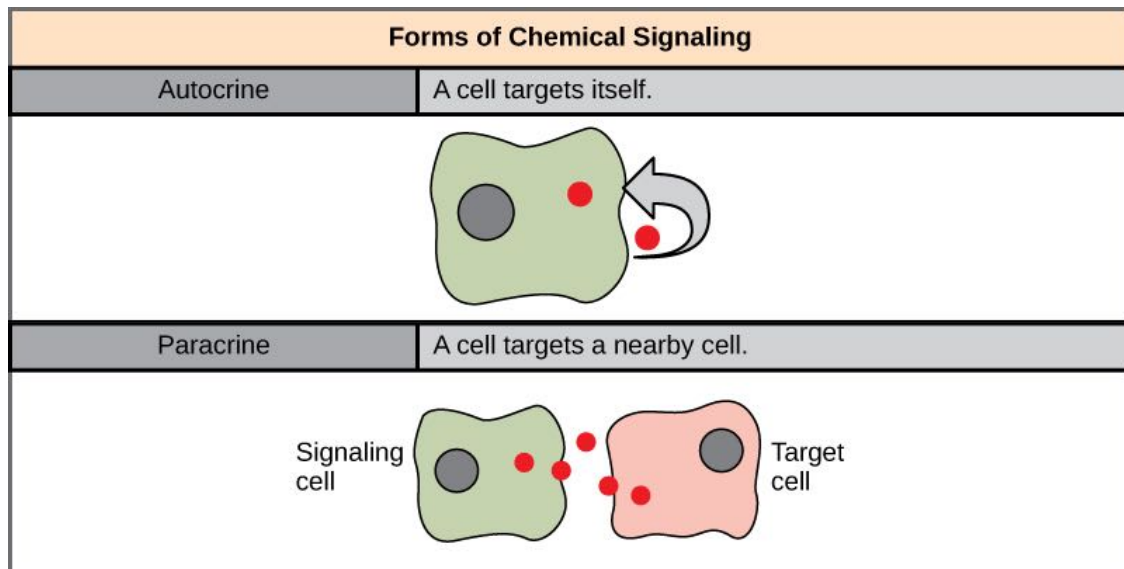
- Once determined, stem cells will differentiate into that cell
- **Eg: Adult immune system – hemopoietic stem cells** in bone marrow can differentiate into any of the immune-system cells



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Regulation of Cell Fate

- **Cell Memory**
 - Gene expression is limited (determined) so can't differentiate
- **Chemical Messengers – “Growth Factors”**
 - **Autocrine or Paracrine**

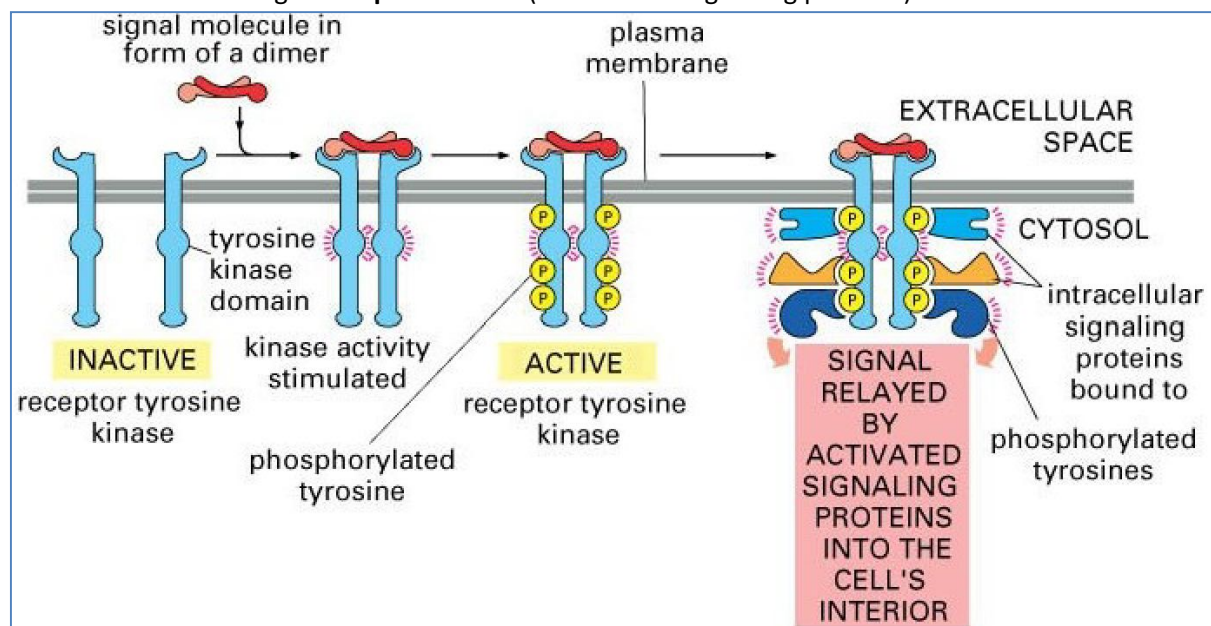


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****note: Mitogens** → Stimulate proliferation only;

Growth Factors → Stimulate proliferation & other constructive processes**

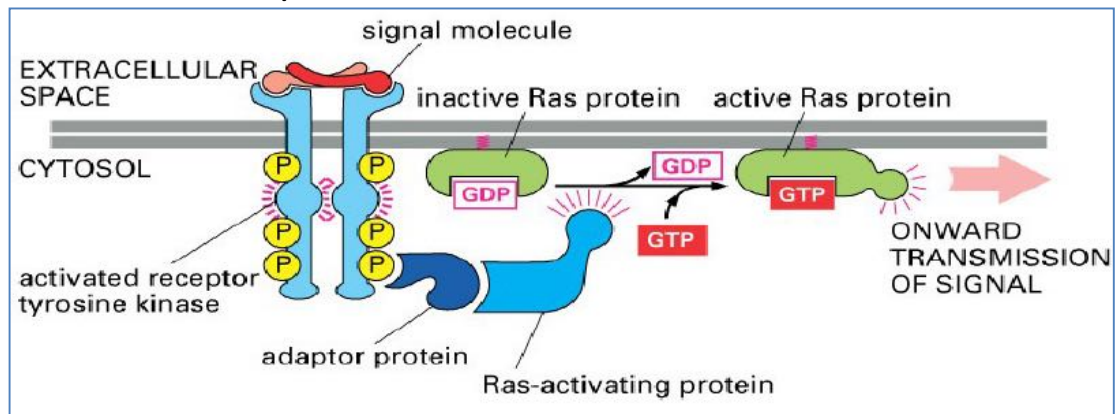
- **Enzyme-Linked Receptors**
 - Binding of **growth factor**
 - **Dimerisation** of tyrosine kinase receptor
 - **Autophosphorylation** of dimer – monomers phosphorylate each other
 - Binding of **Adapter Protein** (intracellular signalling proteins)



Source: <http://home.ku.edu.tr/~okeskin/Biol200/chapter16.pdf>

○ **Activation of Ras**

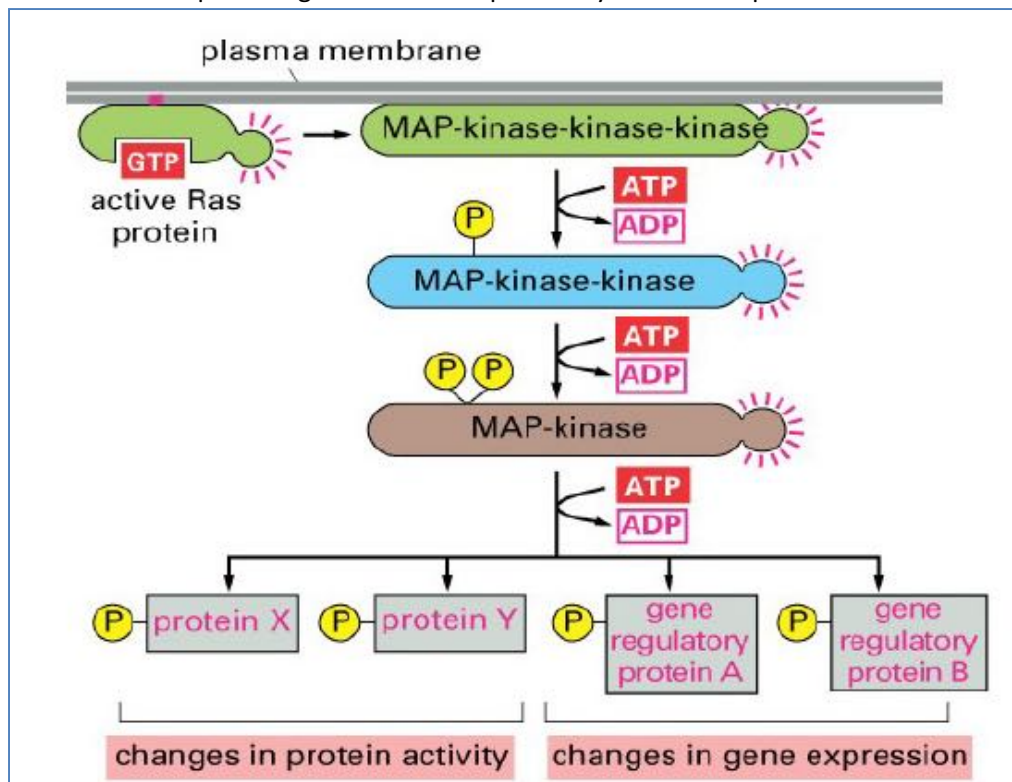
- Binding of **Ras-activating protein** to adapter protein
- **Ras loses GDP and binds GTP** → becoming **active**
- **Active Ras protein** initiates the **MAP-Kinase cascade**



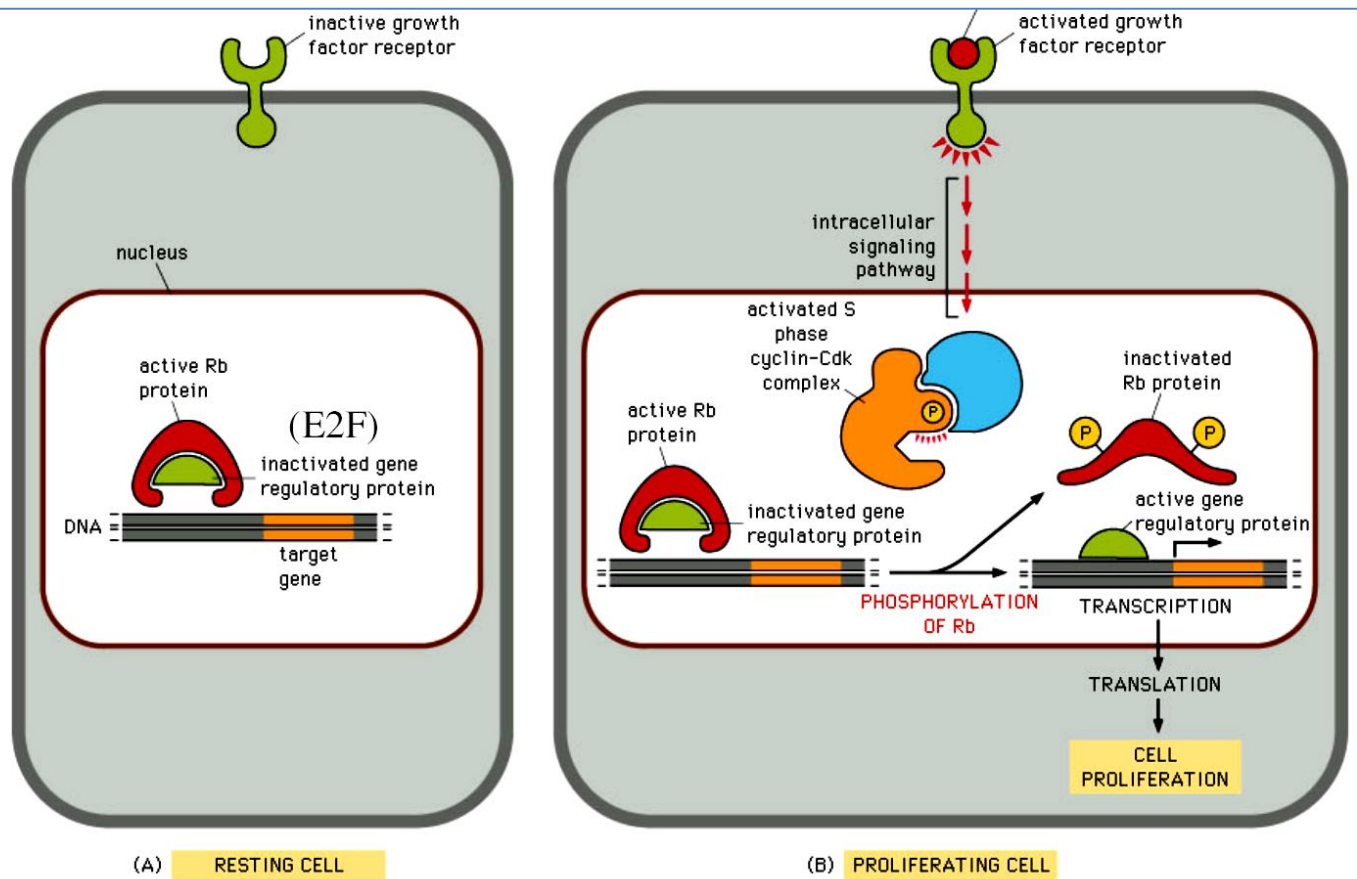
Source: <http://home.ku.edu.tr/~okeskin/Biol200/chapter16.pdf>

○ **MAP-Kinase cascade**

- Active **Ras** protein **initiates** Mitogen-Activating-Protein-Kinase **cascade**
- **MAP-Kinase-Kinase-Kinase** phosphorylates **MAP-Kinase-Kinase**
- **MAP-Kinase-Kinase** phosphorylates **MAP-Kinase**
- **MAP-Kinase** **inactivates** gene-inhibitory **Rb-Protein** via phosphorylation
- **Rb-Protein** releases **E2F-Transcription factor**, initiating transcription
- Transcription of genes results in protein synthesis and proliferation



Source: <http://home.ku.edu.tr/~okeskin/Biol200/chapter16.pdf>

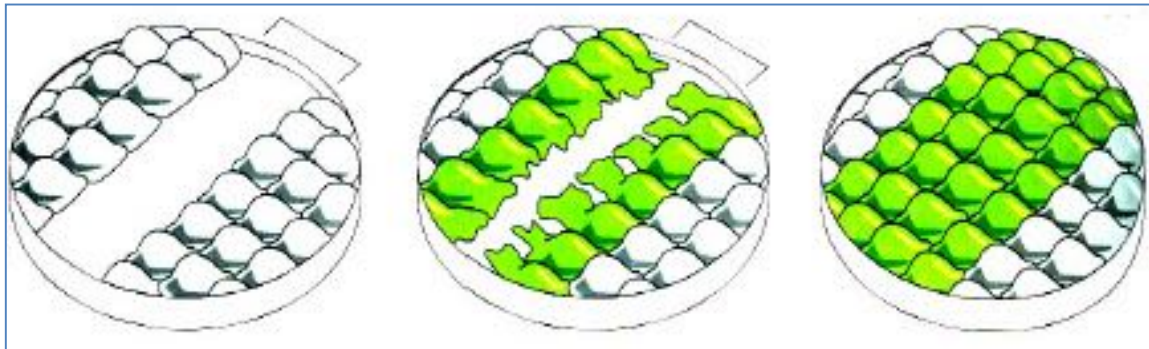


Source: <https://slidetodoc.com/growth-factors-and-enzymelinked-receptor-signal-systems-lecture/>

- **Positional Information**

- **Cell/Cell Contact**

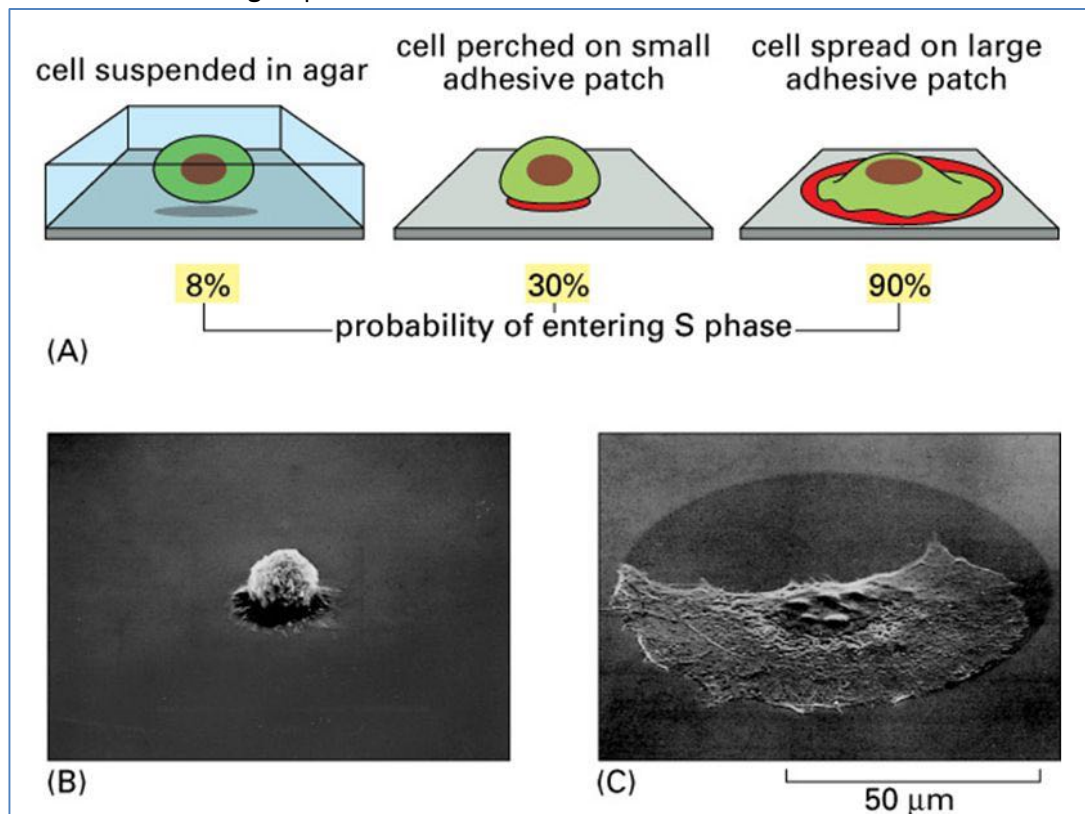
- Signalling through gap junctions for gene transcription & proliferation
 - Also **contact inhibition** stops over proliferation
 - Eg: Experiment:



Cells scraped away → Cells grow back → Cells stop dividing once all-round contact is made

- **Cell-Matrix Contact**

- Fluid, nutrients, waste diffusion medium, fibres guide direction of cell proliferation
 - **Anchorage dependence** of cell division:
 - Eg: Experiment:



Source: <https://slideplayer.com/slide/4663410/>

CANCER, CELL DEATH & CELLULAR AGEING

CANCER, CELL DEATH & CELLULAR AGEING

Growth Characteristics of Normal Cells:

- Subject to **contact inhibition**
- Limited **lifespan**
- **Anchorage dependant**
- **Growth-factor** dependant
- **Able to apoptose**

Growth Characteristics of Tumour & Cancer Cells:

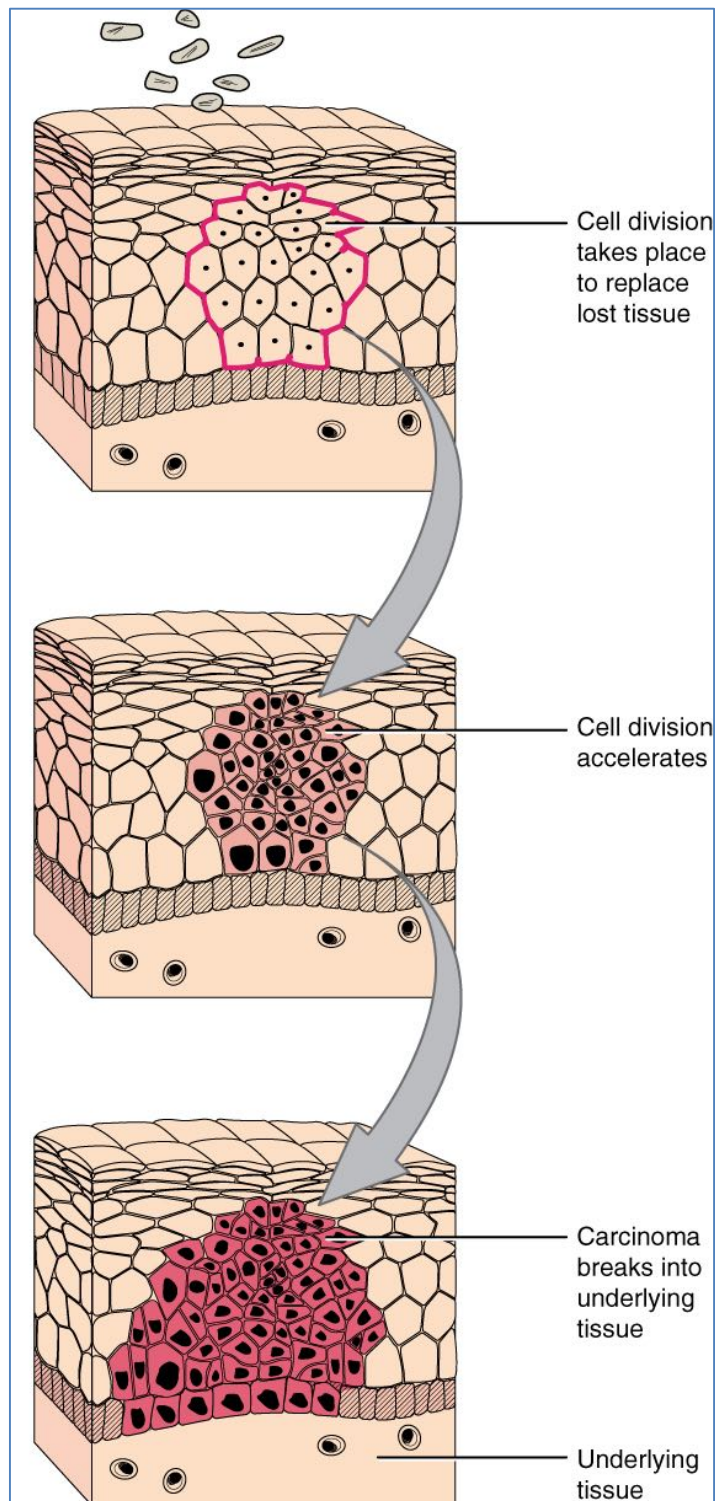
- **Not** subject to **contact inhibition**
- Unlimited **lifespan**
- **Anchorage Independent**
- **Unresponsive** to **growth-inhibitors**
- **Unable** to **apoptose**
- **Differentiate** independently
 - Differentiated tumours = **teratomas**
 - May form teeth, hair, bone, nails, toes, brain matter etc

General Characteristics of BENIGN TUMOUR CELLS:

- **Lower mitotic index than cancerous tissue**
- **Well-defined capsule**
- **NOT INVASIVE**
- **Well differentiated** – still exhibit characteristics of their normal cells of origin
- **NOT METASTATIC**

General Characteristics of METASTATIC CANCERS:

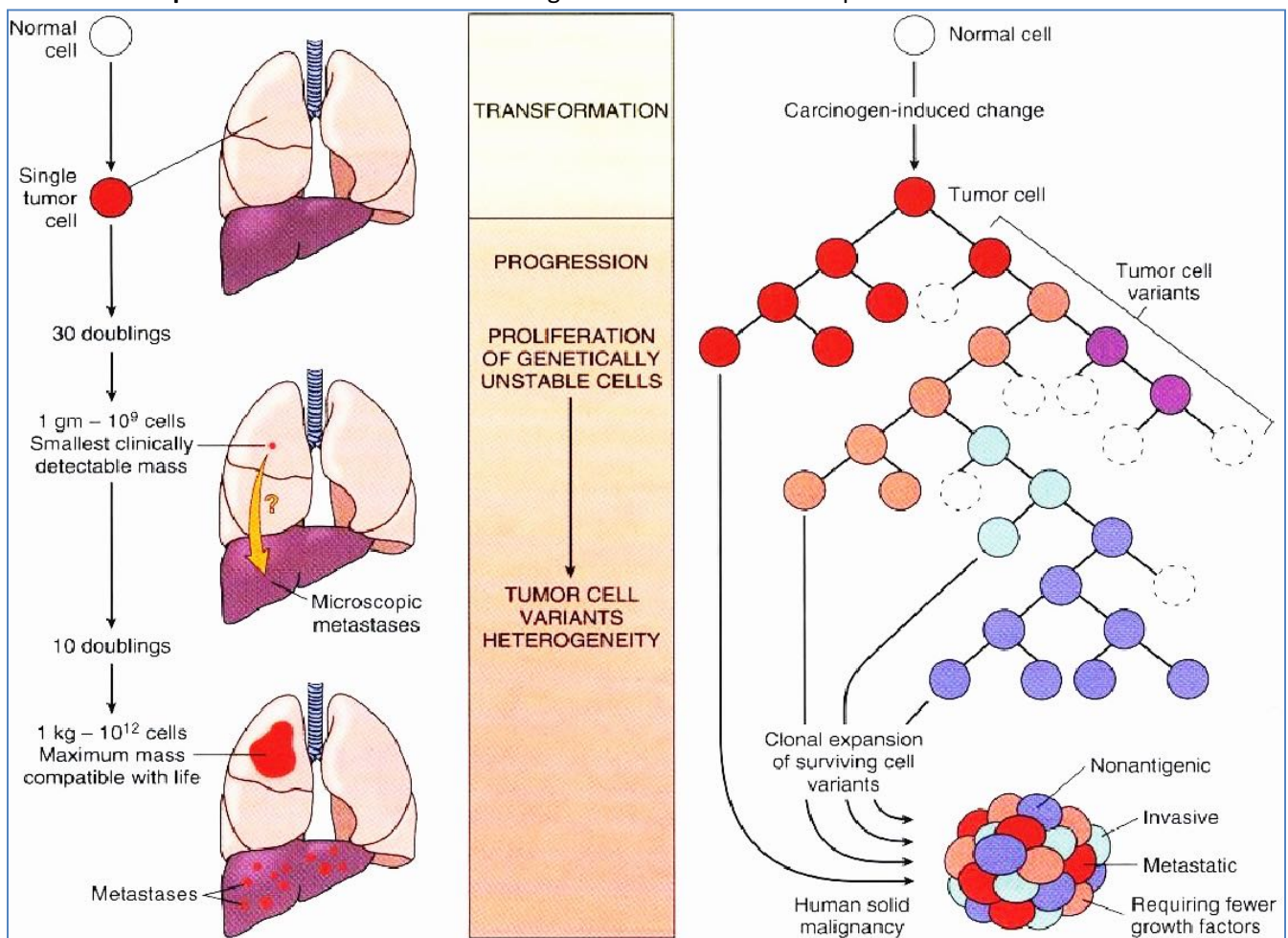
- **Abnormally high mitotic rate**
- **Show signs of de-differentiation**
 - have features of **primordial stem-cells**
- **Disordered growth patterns**
 - Grow as a chaotic mass in all directions
- **Can be Metastatic (Colonise distant tissues)**
 - Cells can break away from primary tumour & travel through blood/lymph
 - Establish new tumours (secondaries) called **metastases**
- **Show gross genetic abnormalities**
 - Aberration in chromosome number
 - Deletions, translocations in genome
- **Grow in the absence of growth factors**
- **Are Immortal**
 - Escape cellular ageing (senescence)
 - Many also don't apoptose
- **Malignant phenotype is heritable**
 - Cancer cells propagate through many mitotic divisions without losing cancerous features



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Cause of Cancer

- Genetic **mutations** that are **non-lethal** to the cell
- Results from **mutagens**:
 - Chemicals
 - Radiation
 - Carcinogens
 - Free-Radicals
 - Microbes (viruses)
 - Inherited
- **DNA damage** = dysregulated growth patterns → uncontrolled proliferation
 - **Damage to regulatory genes**:
 - Results in **loss / gain of function** of:
 - **DNA repair Genes**
 - **Cell Ageing Genes**
 - **Protooncogenes**
 - **Growth-inhibiting (anticancer) Genes**
 - **Apoptosis Genes**
- **Clonal expansion**: cancers arise from a single cell with uncontrolled proliferation



Source: <http://www1.lasalle.edu/~frizzell/tumorgrowth.htm>

Defects in DNA Repair Genes:

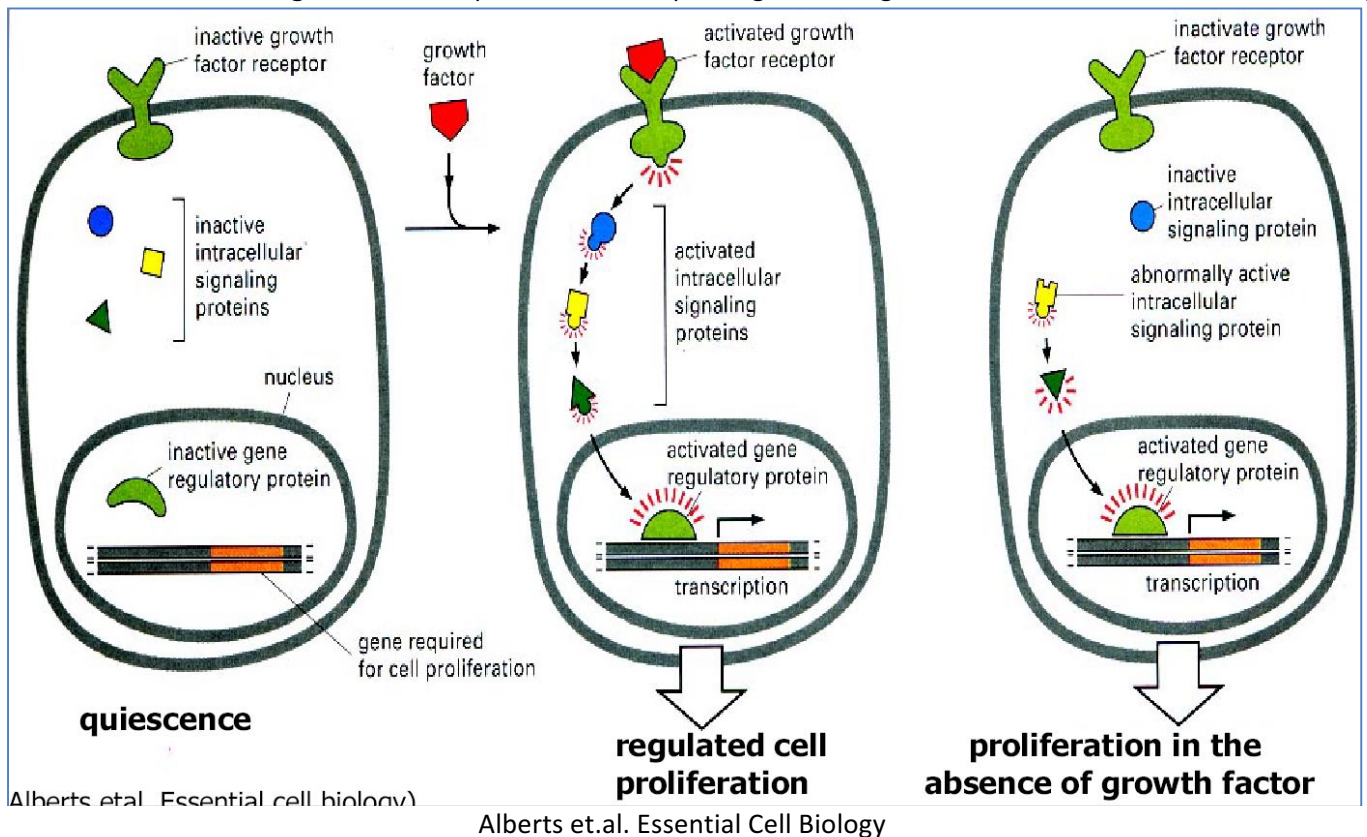
- Genetic mutations happen all the time
- However, DNA is repaired by the cell
- Defective DNA repair can lead to uncorrected mutations → cancer

Defects in Cell Ageing Genes:

- Cellular age is determined by the number of divisions
- When cells age, they enter **senescence**: a terminal non-dividing state
- Mutations that enable the cell to avoid senescence → cancer

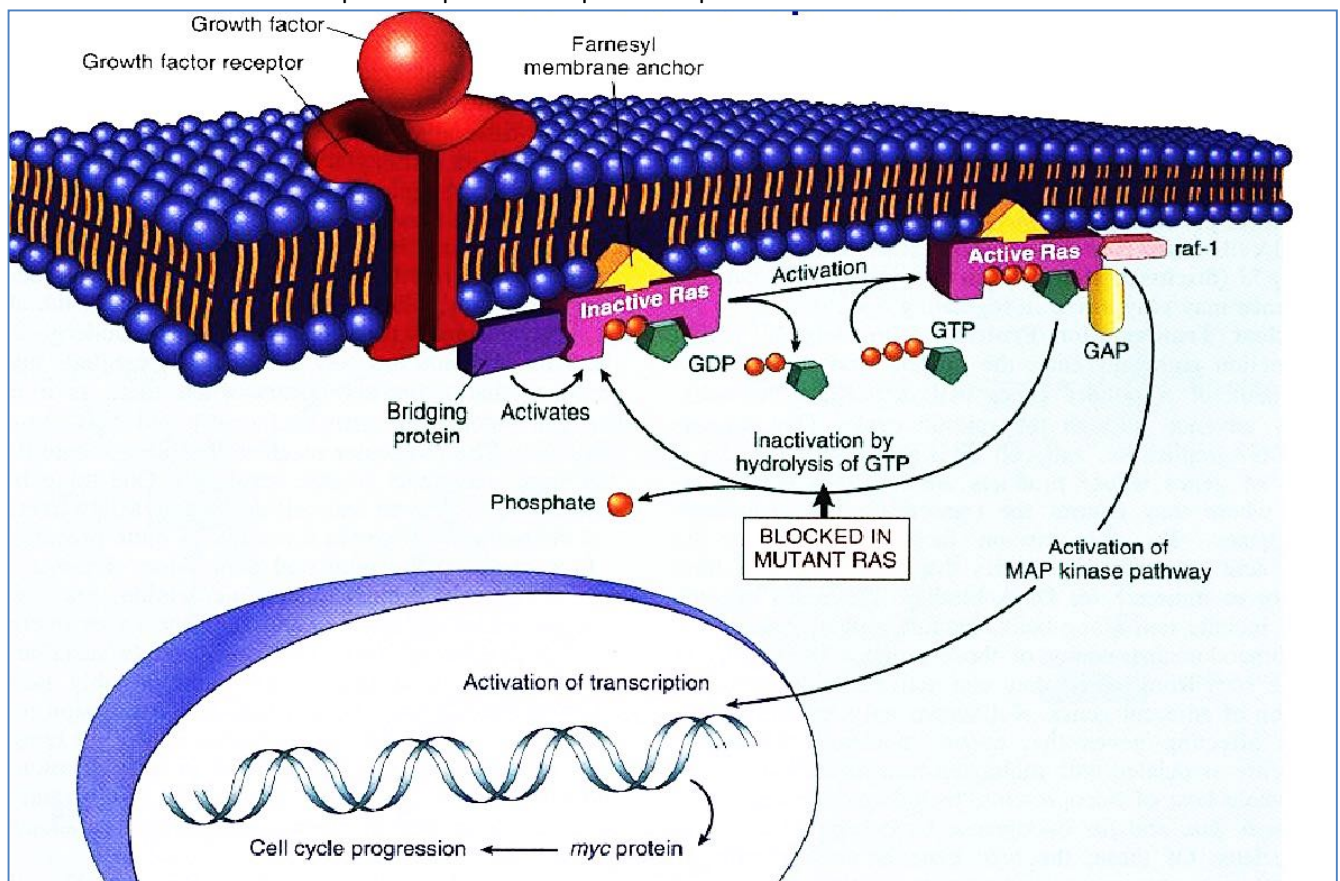
Gain of Function in a Proto-oncogene:

(**Proto-oncogene**: a normal gene that can become an **oncogene** due to mutations or increased expression. An **oncogene** is a protein encoding gene, which — when deregulated — participates in the onset and development of cancer. **Proto-oncogenes** code for proteins that help to regulate cell growth and differentiation → such as Ras)



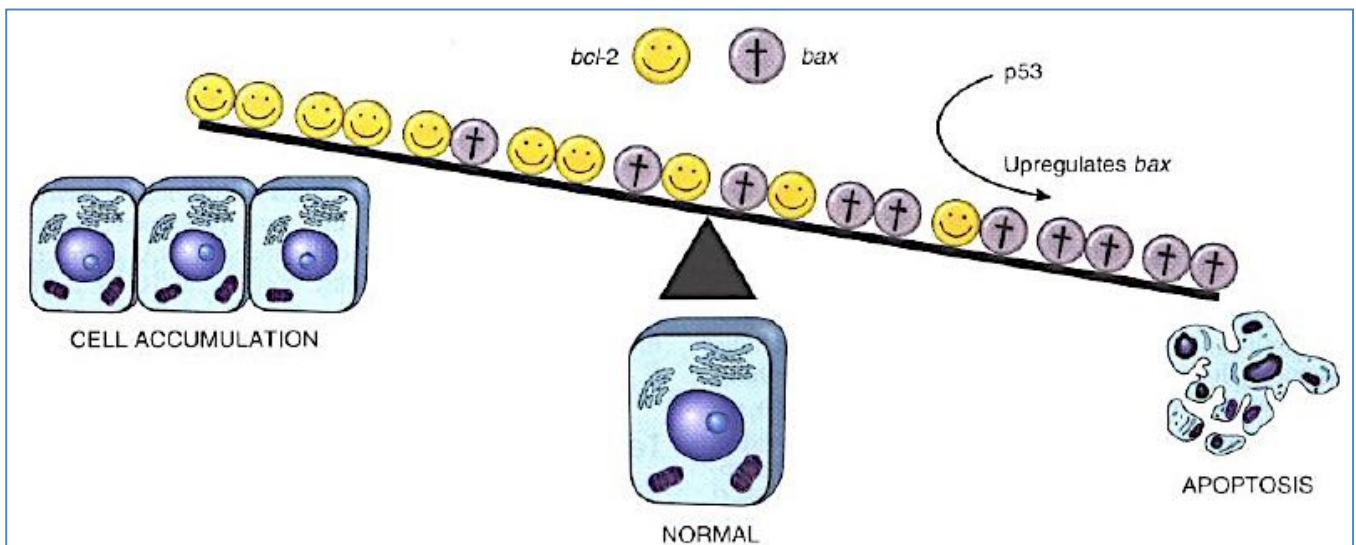
Oncogenic activation of Ras:

- Active Ras protein loses its hydrolysing ability and cannot be turned off
- Results in over-transcription of proteins required for proliferation



Abnormal Expression of Apoptosis-Regulating Genes: (Bcl-2)

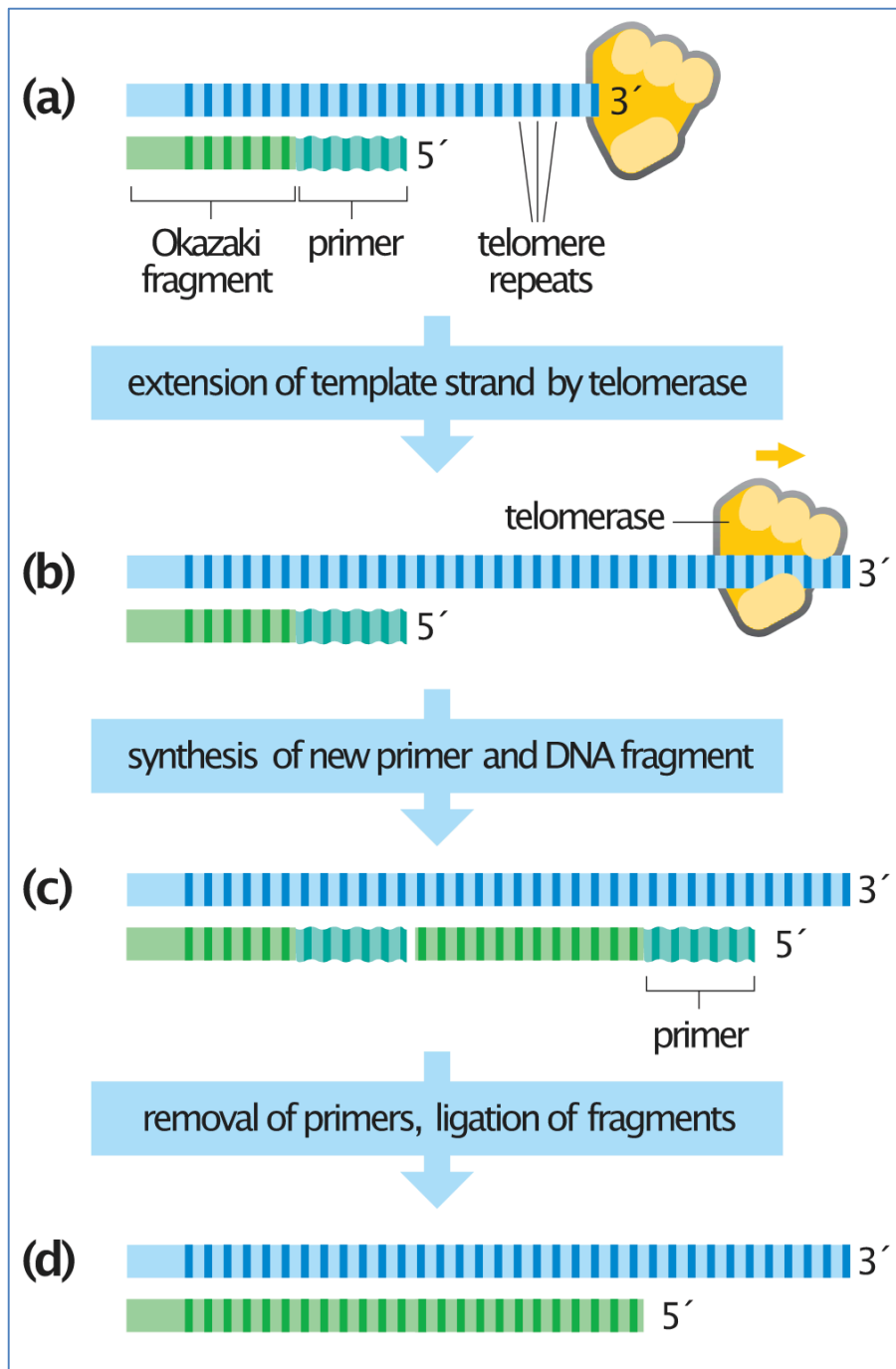
- Over-expression of **anti-apoptotic regulators**
- Under-expression of **pro-apoptotic regulators (Bax & Bak)** → initiates the **CASPASE CASCADE**
 - Leads to abnormalities in regulation of cell proliferation



Source: Unattributable

Cell Ageing

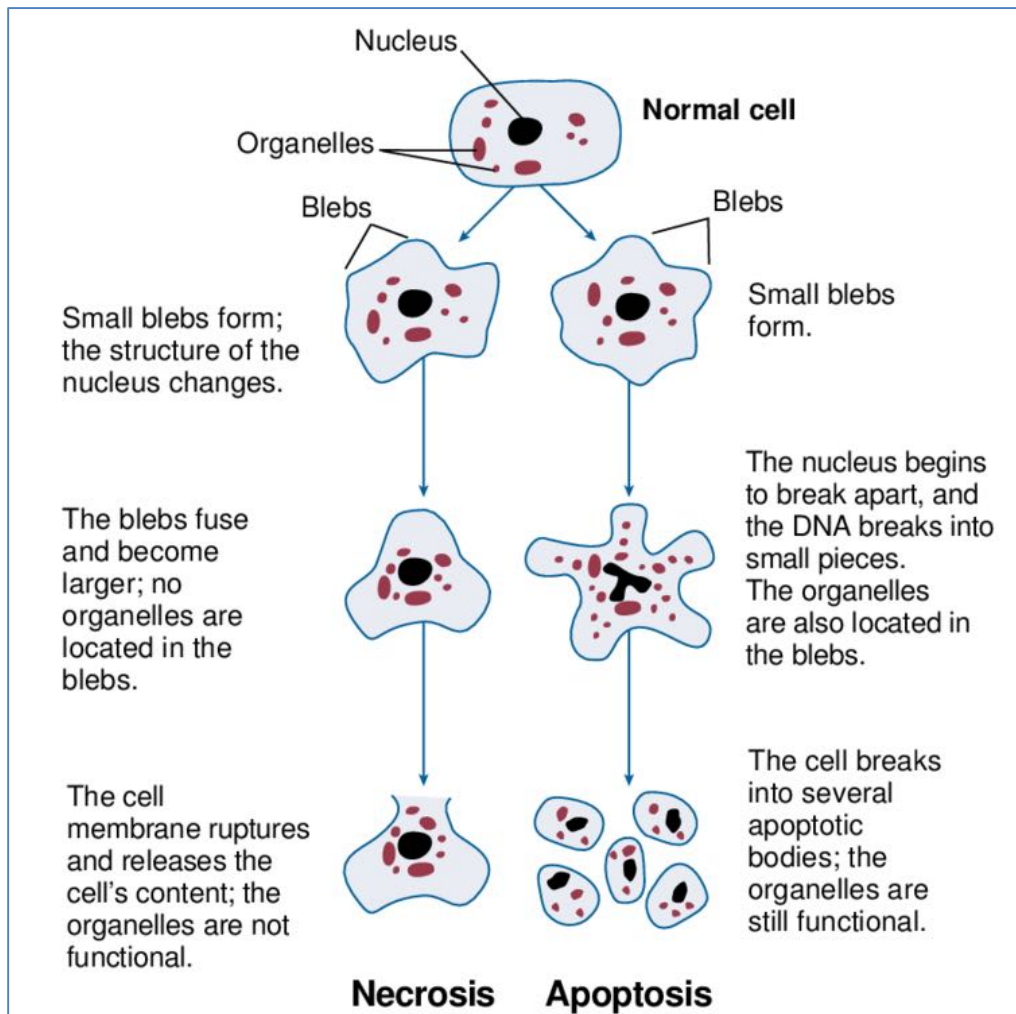
- Progressive alterations in structure → loss of functional capacity (senescence) ending in death
- **On a Cellular Basis:**
 - **Changes in Structure & Function:**
 - **Decrease in:**
 - rate of mitochondrial oxidative phosphorylation
 - nucleic acid synthesis
 - synthesis of proteins (structural/enzymes/receptors/transcription factors)
 - effectiveness of DNA repair mechanisms
 - **Increase in:**
 - incorrectly folded proteins
 - irregularly shaped nuclei
 - **Changes in organelle structure & function**
- **Senescence:**
 - Irreversible arrest of cell division in G1 phase
 - Non-responsive to mitogens
 - Abnormalities in morphology, metabolism & functions
 - Increased resistance to apoptosis
 - Correlation between # of divisions & senescence
 - Suggests that # of divisions is limited & decreases with age
- **Cellular Clocks (the cause of senescence??):**
 - **Telomere Replication:**
 - **Telomere:** *The non-coding end-region of a chromosome-protects the start of the coding sequence from shortening during successive replications*
 - However, each time DNA replication takes place, the telomere itself gets shorter
 - **Telomerase** counters this by adding repeats to the template strand, allowing telomere elongation on the new DNA strand
 - Unfortunately, **telomerase activity decreases with # of cell divisions**



David O Morgan, Attribution, via Wikimedia Commons

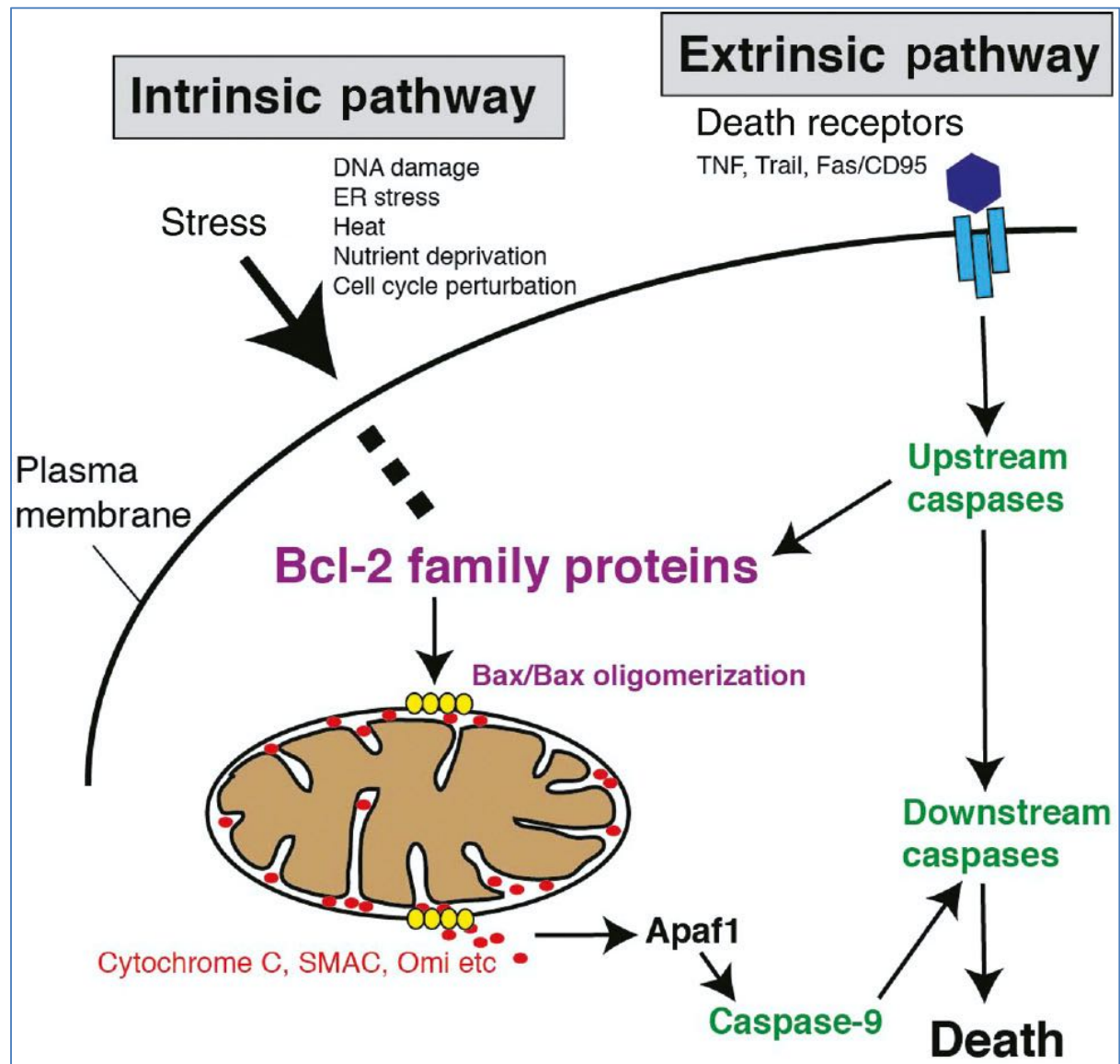
Cell Death / 'Apoptosis':

- **Cells that undergo apoptosis shrink & condense, dying neatly without damaging neighbours**
 - A normal, ordered, regulated, intentional & active cellular process of disassembly
 - Causes Minimal inflammation or scarring
 - Apoptotic bodies attract phagocytes (eg: Macrophages) and are engulfed/phagocytosed
- Distinct from Necrosis:



National institute on alcohol abuse and alcoholism (NIAAA), Public domain, via Wikimedia Commons

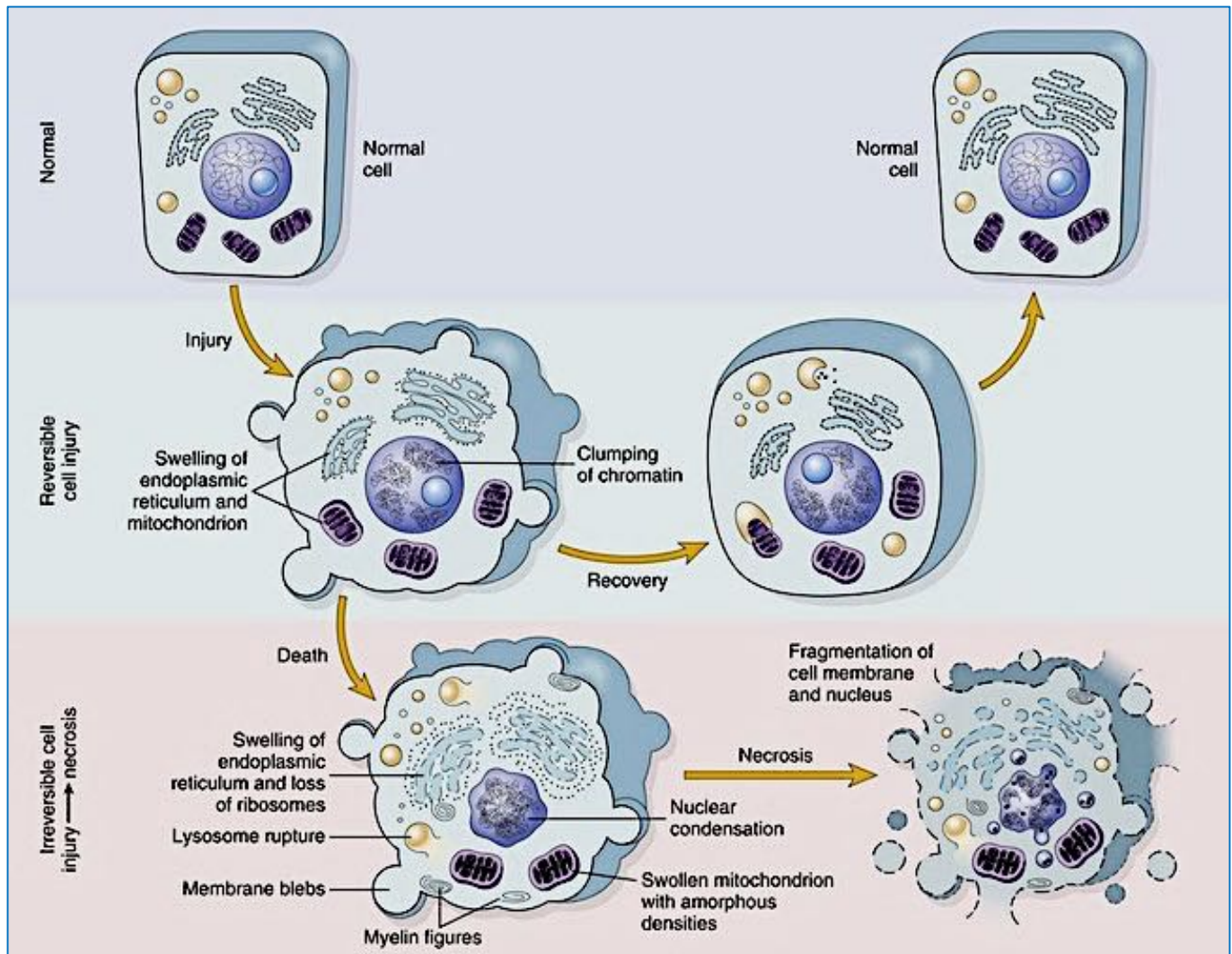
- **Apoptosis is Regulated by CASPASES:**
 - **Caspases** = Proteolytic enzymes that cut up proteins & nuclear laminin (nuclear envelope)
 - Initially caspases are created as inactive enzymes called **procaspases**
 - The activation of procaspases is regulated by the **Bcl-2** protein family (**Bax & Bak**)
 - → **Bax & Bak** increase outer-mitochondrial-membrane permeability
 - → releases **cytochrome-C** into cytosol
 - → **Cytochrome-C** then binds to **adapter proteins**
 - → **activates procaspases** → **active caspases**
 - → **Active caspases** → **Activate other procaspases**
 - → causes an explosive chain reaction (Caspase cascade)
 - **Caspase cascade** → is destructive, self-amplifying, and irreversible



From the journal Nanotechnology Reviews; <https://doi.org/10.1515/ntrev-2016-0070>

Necrosis:

- Cells that die from acute injury typically swell & burst, spilling their contents
 - o Abnormal process (pathophysiological)
 - o Unregulated
 - o Energy independent
 - o Enzyme independent (doesn't require gene expression)
 - o Chaotic destruction
 - o Activates inflammatory response
 - o Results in scarring



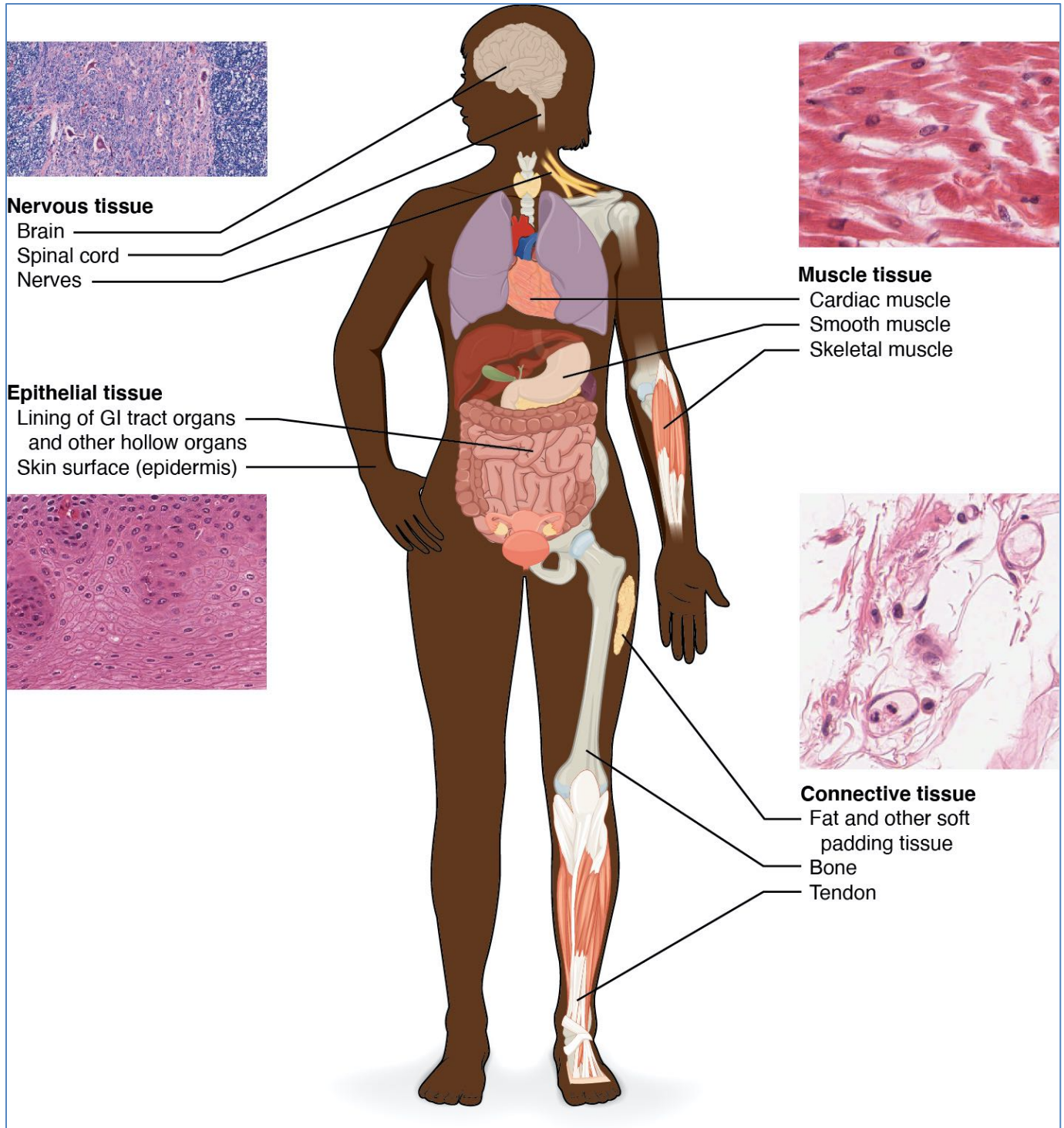
<https://medicoapps.org/m-irreversible-cell-injury/reversible-and-irreversible-cell-injury/?amp>

EPITHELIAL TISSUES (MEMBRANES & GLANDS)

EPITHELIAL TISSUES (MEMBRANES & GLANDS)

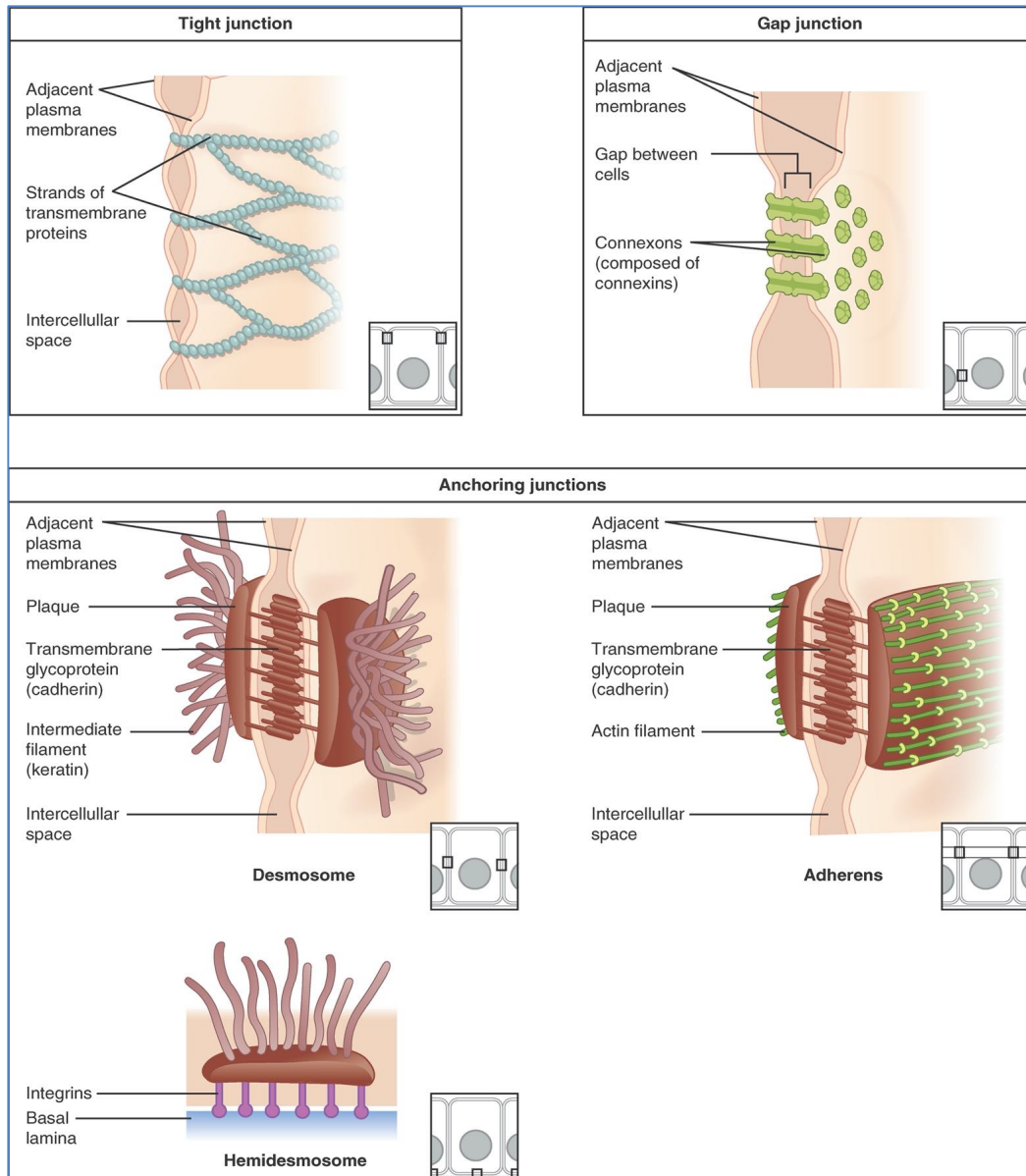
Epithelial Tissues (epithelium):

- **One of the 4 basic tissue types** (nerve/muscle/connective/epithelial)
- **Epithelial tissues form *either* Membranes or Glands**
- ***Nearly all substances received or given off by the body must pass through an epithelium**
- **General Functions:**
 - Protection
 - Absorption
 - Filtration
 - Excretion
 - Secretion
 - Sensory reception



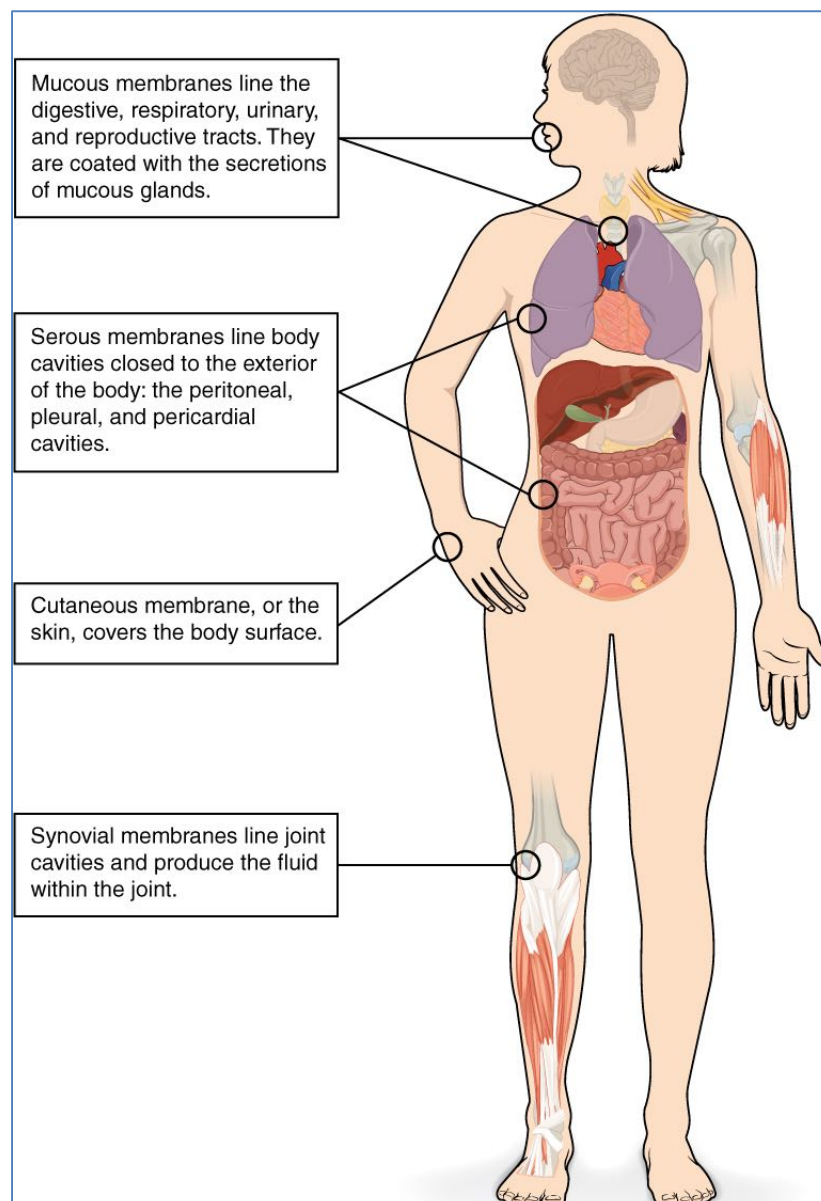
Characteristics of Epithelial Cells:

- **Exhibit Polarity:** (apical-basal polarity)
 - o **Have an Apical Surface**
 - Surface **exposed to the inside** of the lined cavity
 - Most have **microvilli** (finger-like extensions of the plasma membrane)
 - Some have **cilia** (arms that propel substances in 1 direction – eg: trachea)
 - o **Have a Basal Surface**
 - Surface facing connective tissue **on the outside** of the lined cavity
 - Supported by a **Basement Membrane**:
 - Lining the basal surface is a thin supporting sheet called the **basal lamina** - determines which molecules can diffuse through the basal membrane
 - Below the basal lamina is the **reticular lamina** – fine network of collagen fibres belonging to the underlying connective tissue
- **Held Together By Specialised Contacts:**
 - o **Tight junctions** – maintain polarity – protect basal side from apical environment
 - o **Desmosomes** – resist mechanical forces – holds cells together
 - o **Gap junctions** – Allows intercellular transfers & communication
- **Supported by Connective Tissue:**
 - o Provide support
 - o Reinforces the epithelial sheet
 - o Resists stretching/tearing forces
 - o Defines epithelial boundary



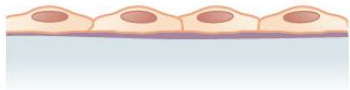
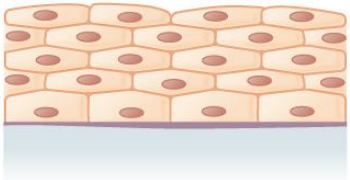
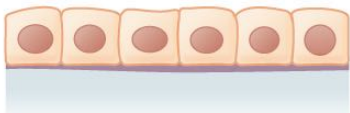
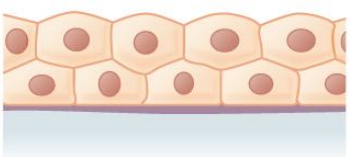
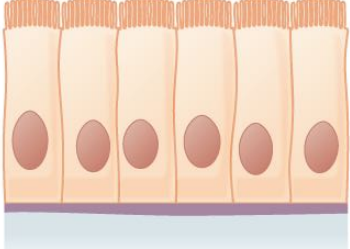
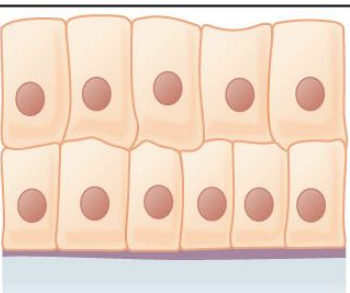
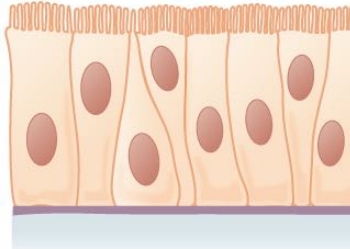
Epithelial Membranes:

- **Continuous multicellular sheets composed of an epithelial layer bound to underlying connective tissue**
- **Cutaneous Epithelial Membranes:**
 - Keratinized stratified Squamous epithelium attached to a thick layer of dense irregular connective tissue
 - Is a dry membrane – exposed to the air
- **Mucous Epithelial Membranes:**
 - Either stratified Squamous or simple columnar epithelial underlain by the lamina propria (loose connective tissue)
 - Is a wet membrane – line body cavities that open to the exterior (digestive, respiratory, urogenital)
 - Bathed by secretions of copious amounts of mucus (except urinary tract)
 - Adapted for absorption and secretion
- **Serous Epithelial Membranes:**
 - Simple Squamous epithelium (a mesothelium) resting on a thin layer of areolar tissue (loose connective tissue)
 - Is a moist membrane – found in closed ventral body cavities
 - **2 layers:**
 - **Visceral:** Tightly encases organs
 - **Parietal:** Loosely encases organs
 - Mesothelial cells create a thin, clear lubricating fluid (Serous Fluid) which fills the gap **between the visceral and parietal** layers

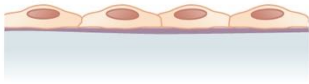
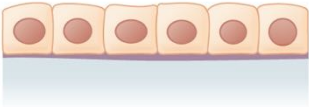
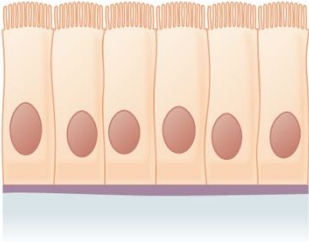
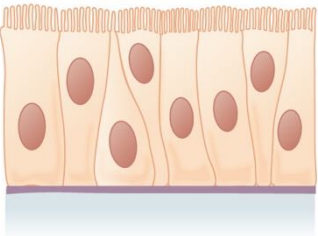
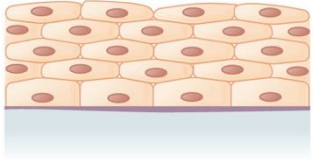
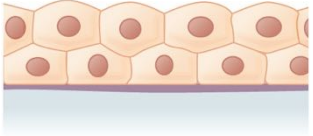
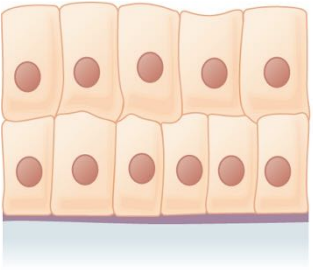
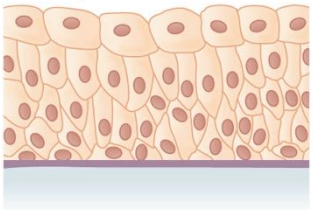


Epithelial Histology:

- **Classed on a combination of the number of cell-layers present and the shape of the cells:**
 - **Shape:**
 - **Squamous**
 - Flat, tile-like (fried egg)
 - Flat central nucleus
 - **Cuboidal**
 - Box-like shape
 - Spherical central nucleus
 - **Columnar**
 - Column/prism shaped
 - Elongated basal nucleus
 - **Layering:**
 - **Simple:**
 - Single layer
 - Common in high-secretion/absorption/filtration areas (requiring a thin epithelial barrier for high diffusion)
 - Not usually for protection
 - **Stratified:**
 - Multiple layers
 - Common in high-abrasion areas for protection (eg: skin/mouth)
 - Named by the shape of the cells in the apical layer
 - **Pseudostratified:**
 - Seemingly layered but is still simple (all cells touch the basal lamina)
 - Generally columnar

	Simple	Stratified	
Squamous	 Simple squamous epithelium	 Stratified squamous epithelium	
Cuboidal	 Simple cuboidal epithelium	 Stratified cuboidal epithelium	
Columnar	 Simple columnar epithelium	 Stratified columnar epithelium	Pseudostratified  Pseudostratified columnar epithelium

- **Simple Squamous:**
 - **Diffusion / Filtration / Friction-reducing** lining in lymphatic & cardiovascular systems
 - Present in **kidneys, lining of heart, blood vessels, alveoli, lymphatic vessels & serosae**
- **Simple Cuboidal:**
 - **Secretion / absorption**
 - Present in **kidneys, ducts, secretory portions of small glands, ovary surface**
- **Simple Columnar:**
 - Many contain **cilia**
 - **Secretion / absorption**
 - **Ciliated:**
 - Line small **bronchi, uterine tubes**, regions of the **uterus**
 - **Non-Ciliated:**
 - Line the **digestive tract** and the **gallbladder**
- **Pseudostratified Columnar:**
 - Cells with different heights (but all touch basal lamina)
 - Nuclei are seen at different levels
 - Secretion & propulsion of mucus
 - **Ciliated:**
 - **Trachea**
 - **Non-Ciliated:**
 - **Sperm-Carrying Ducts**
- **Stratified Squamous:**
 - Thick membrane (multiple layers)
 - **Protects areas subject to abrasion**
 - External part of **skin's epidermis / oesophagus lining / mouth / vagina**
- **Transitional Epithelia:**
 - Multiple layers
 - **Basal** cells = Cuboidal
 - **Apical** cells = Dome shaped
 - **Stretches** to permit expansions / contractions
 - Lines the **urinary bladder / ureters / part of urethra**

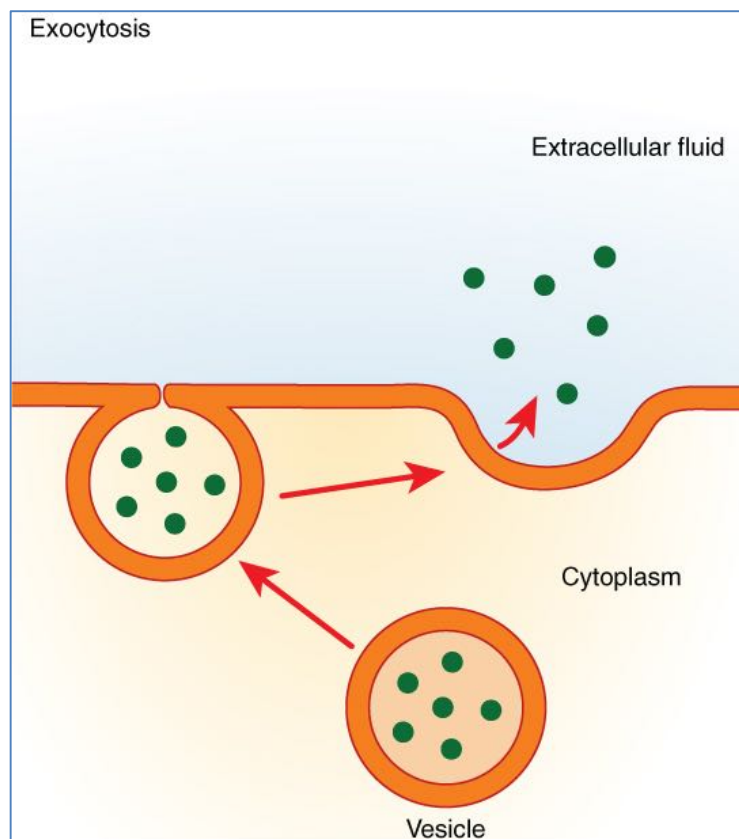
Cells	Location	Function
Simple squamous epithelium 	Air sacs of lungs and the lining of the heart, blood vessels, and lymphatic vessels	Allows materials to pass through by diffusion and filtration, and secretes lubricating substance
Simple cuboidal epithelium 	In ducts and secretory portions of small glands and in kidney tubules	Secretes and absorbs
Simple columnar epithelium 	Ciliated tissues are in larger bronchioles, uterine tubes, and uterus; smooth (nonciliated tissues) are in the digestive tract, bladder	Absorbs; it also secretes mucus and enzymes
Pseudostratified columnar epithelium 	Ciliated tissue lines the bronchi, trachea, and much of the upper respiratory tract	Secretes mucus; ciliated tissue moves mucus
Stratified squamous epithelium 	Lines the esophagus, mouth, and vagina	Protects against abrasion
Stratified cuboidal epithelium 	Sweat glands, salivary glands, and the mammary glands	Protective tissue
Stratified columnar epithelium 	The male and female urethrae and the ducts of some glands	Secretes and protects
Transitional epithelium 	Lines the bladder, urethra, and the ureters	Allows the urinary organs to expand and stretch

Glandular Epithelia:

- **One or more cells that make & secrete an aqueous fluid** – usually contains proteins / steroids / lipids
- **Classified by:**
 - **Site of product release:**
 - Endocrine
 - Exocrine
 - **Number of cells forming the gland:**
 - Unicellular – scattered within epithelial sheets – ductless
 - Multicellular – form by invagination / evagination from an epithelial sheet – have ducts (tube-like connections to epithelial sheets)

Endocrine (diffuse) Glands:

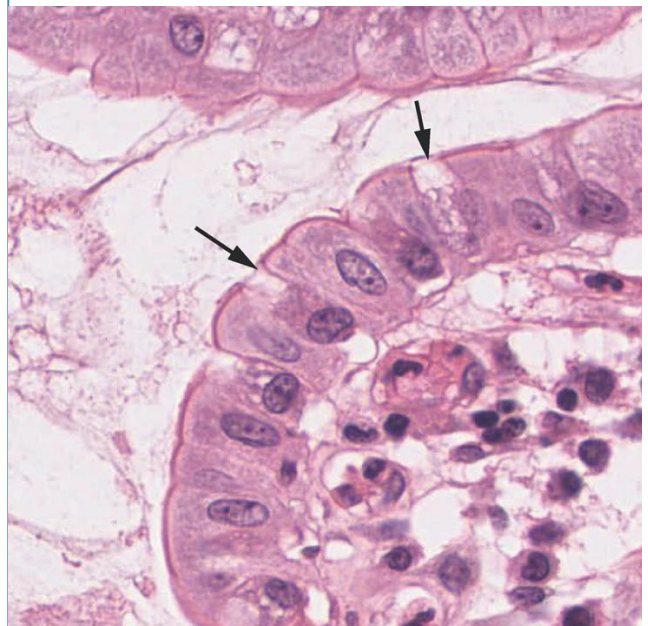
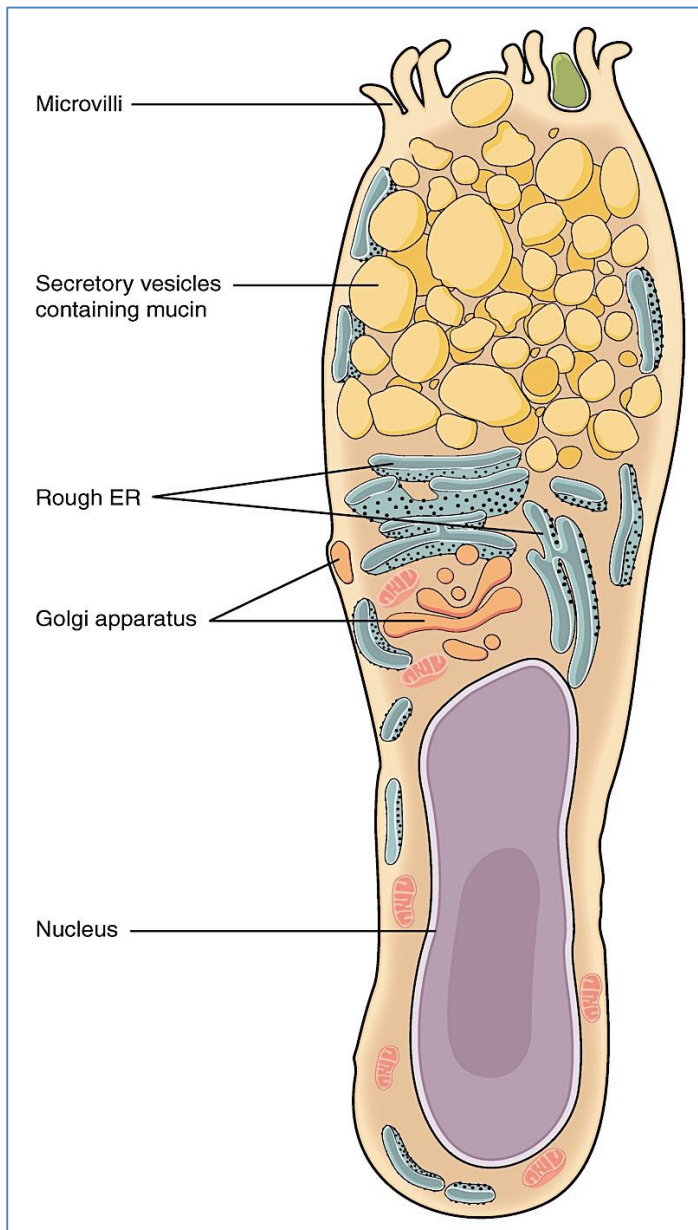
- **Ductless glands that produce mainly hormones**
 - Prompts target organs to respond in some way
- Secretions also include amino acids, proteins, glycoproteins
- Secrete by **exocytosis** directly into the extracellular space – hence **diffuse endocrine system**
- Most are compact Multicellular organs (eg: Thyroid, ovaries, testes, pancreas)
- Situated close to blood/lymphatic vessels



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Exocrine Glands:

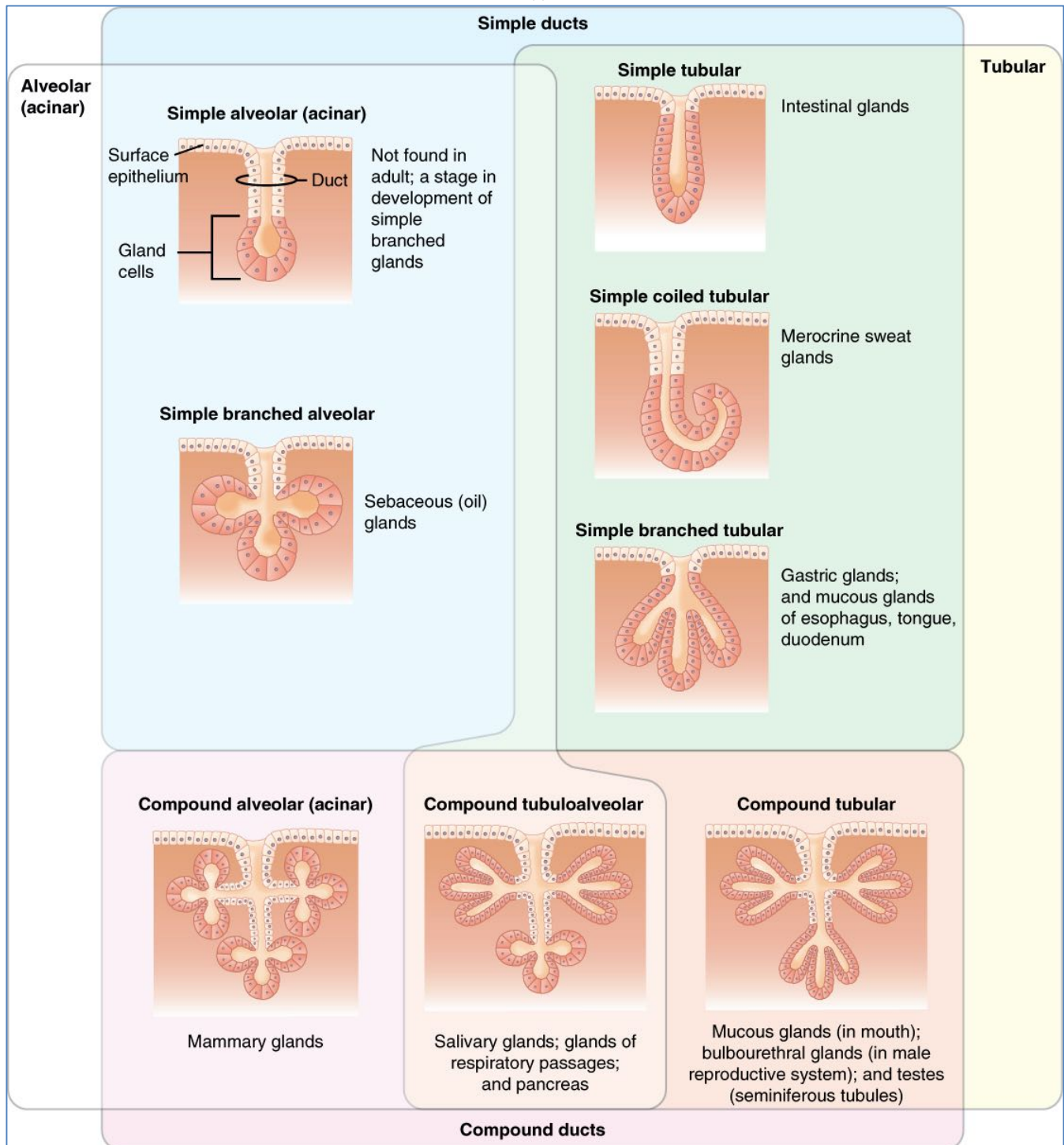
- More numerous than endocrine glands
- All secrete products onto body surfaces/into body cavities
- (mucous glands, sweat glands, oil glands, saliva glands, liver, pancreas-exocrine part)
- **Unicellular Exocrine Glands:**
 - o Goblet Cells
 - o Sprinkled among columnar epithelial linings of intestinal & respiratory tracts
 - o Secrete mucin directly by exocytosis into lumen
 - (mucin + water = mucous) slimy coating that protects & lubricates



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- **Multicellular Exocrine Glands:**

- **2 Parts:**
 - Epithelium-derived duct
 - Secretory unit (acinus) at base of duct – secretory cells
- **Ducted Type Glands: Either -**
 - **Simple** (DUCT is unbranched)
 - **Compound** (DUCT is branched)
- **Structure of secretory units:**
 - **Tubular**
 - **Alveolar**
 - **Tubuloalveolar** – mixture of both types



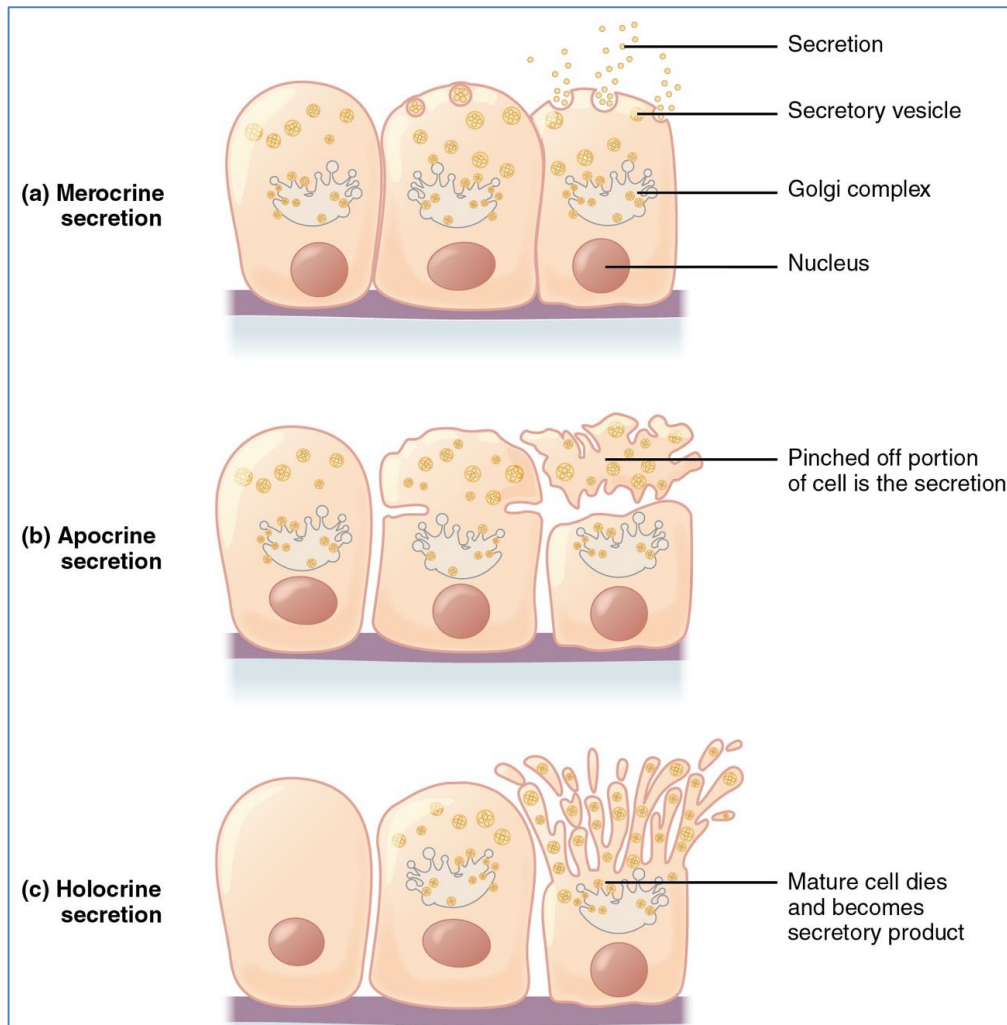
- **Modes of Secretion:**

- **Merocrine Glands:**

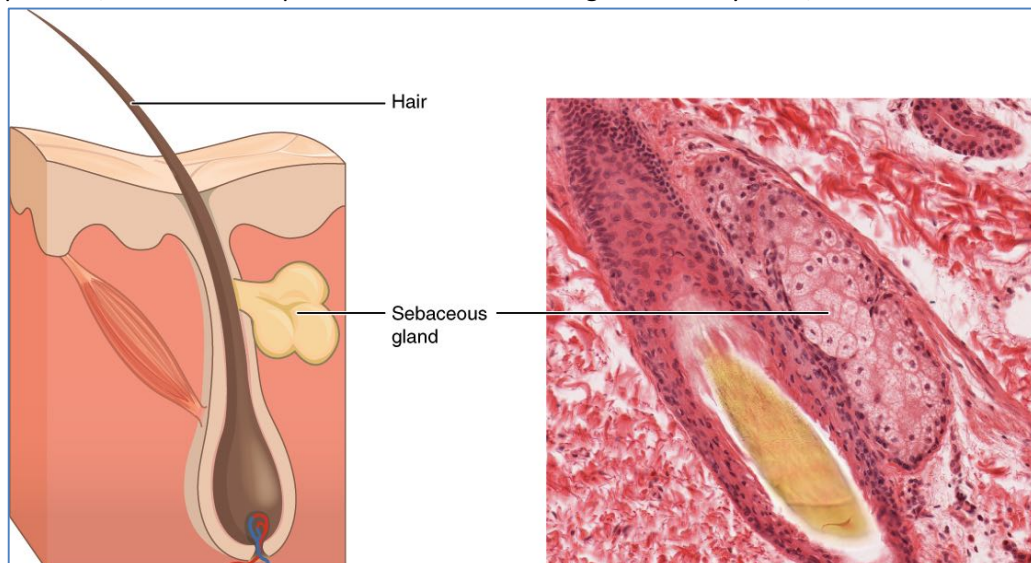
- Products secreted by exocytosis (pancreas, sweat, salivary glands)

- **Holocrine Glands:**

- Products secreted by the rupture of the apical cells of the gland
 - Underlying cells replace ruptured cells and repeat process
 - (Only human example: Sebaceous glands – oil glands of the skin)



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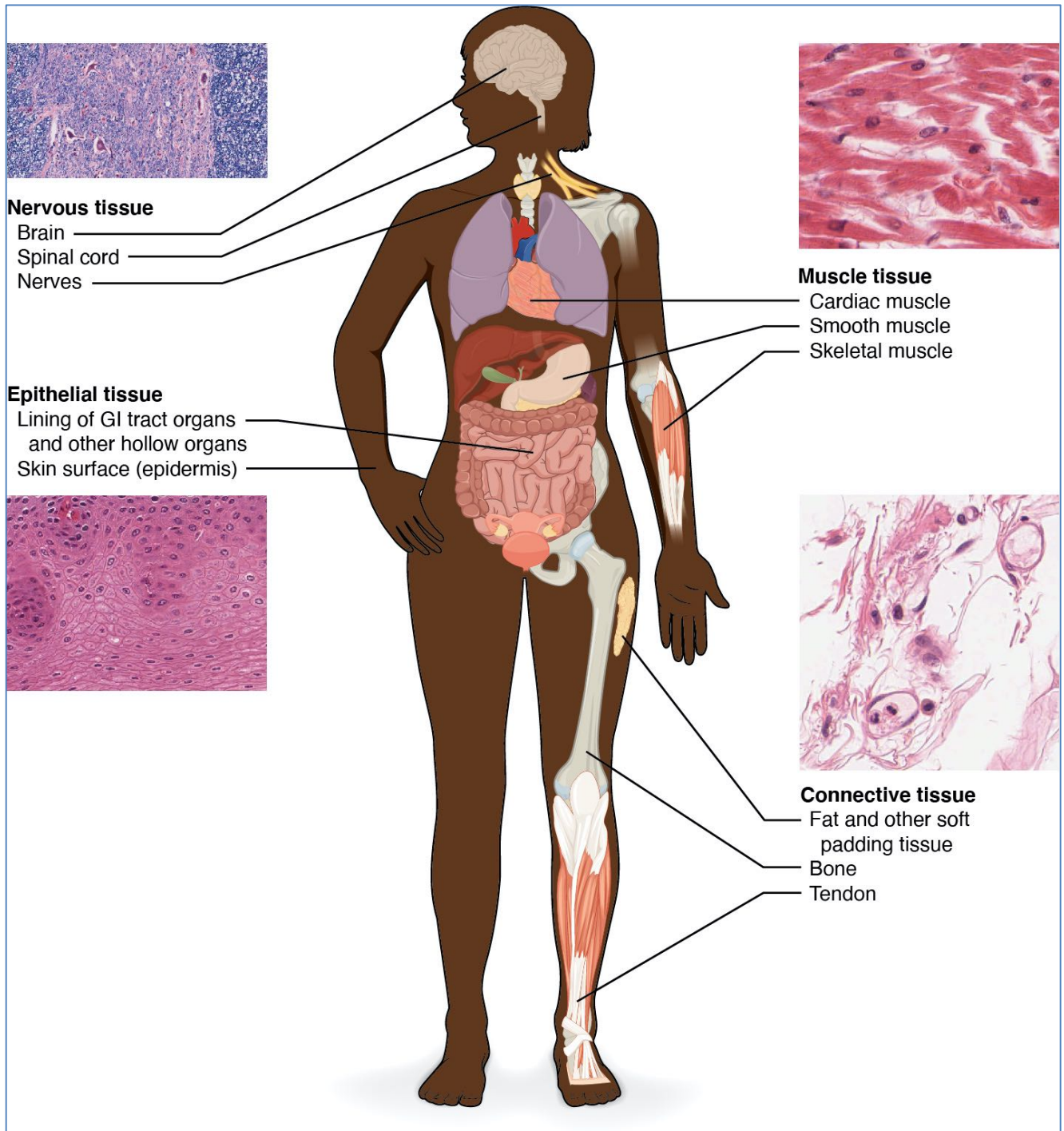


Sebaceous Glands These glands secrete oils that lubricate and protect the skin. They are holocrine glands and they are destroyed after releasing their contents. New glandular cells form to replace the cells that are lost.

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CONNECTIVE TISSUE:

CONNECTIVE TISSUE:



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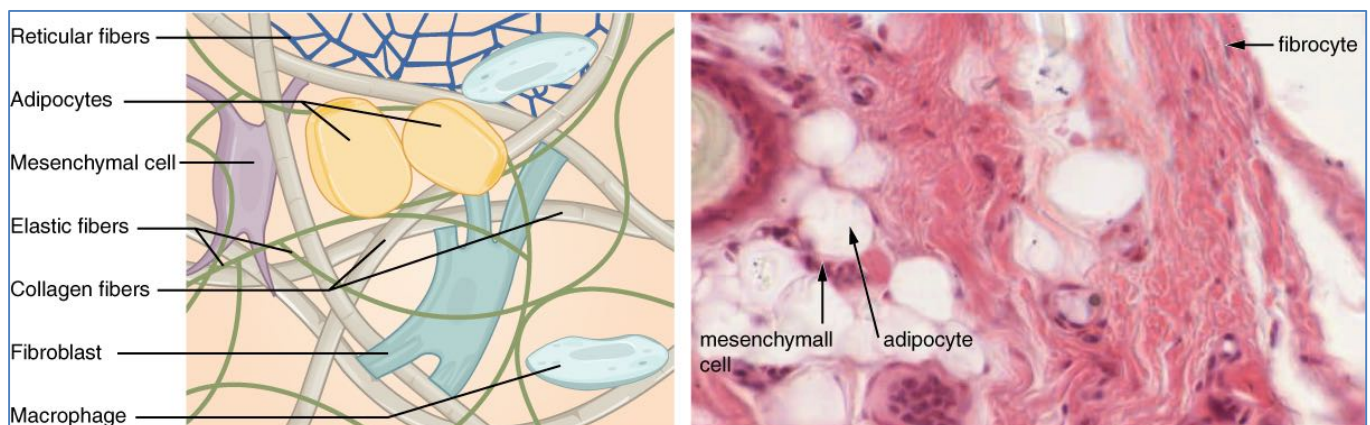
Connective Tissue:

- Found everywhere in the body
- Most abundant primary tissue
- **4 Classes of Connective Tissue:** (see latter pages)
 - o Connective tissue proper
 - o Cartilage
 - o Bone
 - o Blood

Connective Tissue Examples

Connective tissue proper	Supportive connective tissue	Fluid connective tissue
Loose connective tissue • Areolar • Adipose • Reticular	Cartilage • Hyaline • Fibrocartilage • Elastic	Blood
Dense connective tissue • Regular elastic • Irregular elastic	Bones • Compact bone • Cancellous bone	Lymph

- **4 Major Functions:**
 - o Binding / Support
 - o Protection
 - o Insulation
 - o Transportation (blood)
- **Common Characteristics:**
 - o **Common Origin**
 - All arise from *mesenchyme* (embryonic tissue)
 - o **Degrees of vascularity**
 - Avascular (cartilage)
 - Poorly vascular (dense connective tissue)
 - Rich blood supply
 - o **Largely non-living:**
 - Connective tissue is made up largely of non-living extracellular matrix
 - Some sparse living cells
 - Can therefore bear weight, withstand tension, trauma & abrasion



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- **Structural Elements:**

○ **Extracellular matrix**

▪ **Ground Substance**

- Unstructured interstitial (tissue) fluid
- Cell adhesion proteins (laminin & fibronectin) – allows connective tissue cells to attach to matrix elements
- Proteoglycans – trap water, making it more viscous (varying degrees)
- Functions as a molecular sieve – nutrients diffuse between blood capillaries and cells

▪ **Fibres**

• **Collagen Fibres (white fibres)**

- Constructed of the fibrous **protein** “Collagen”
- Found in all connective tissues
- Small cross-linked fibrils bundle together into thick collagen fibres
- Are extremely tough – high tensile strength

• **Elastic**

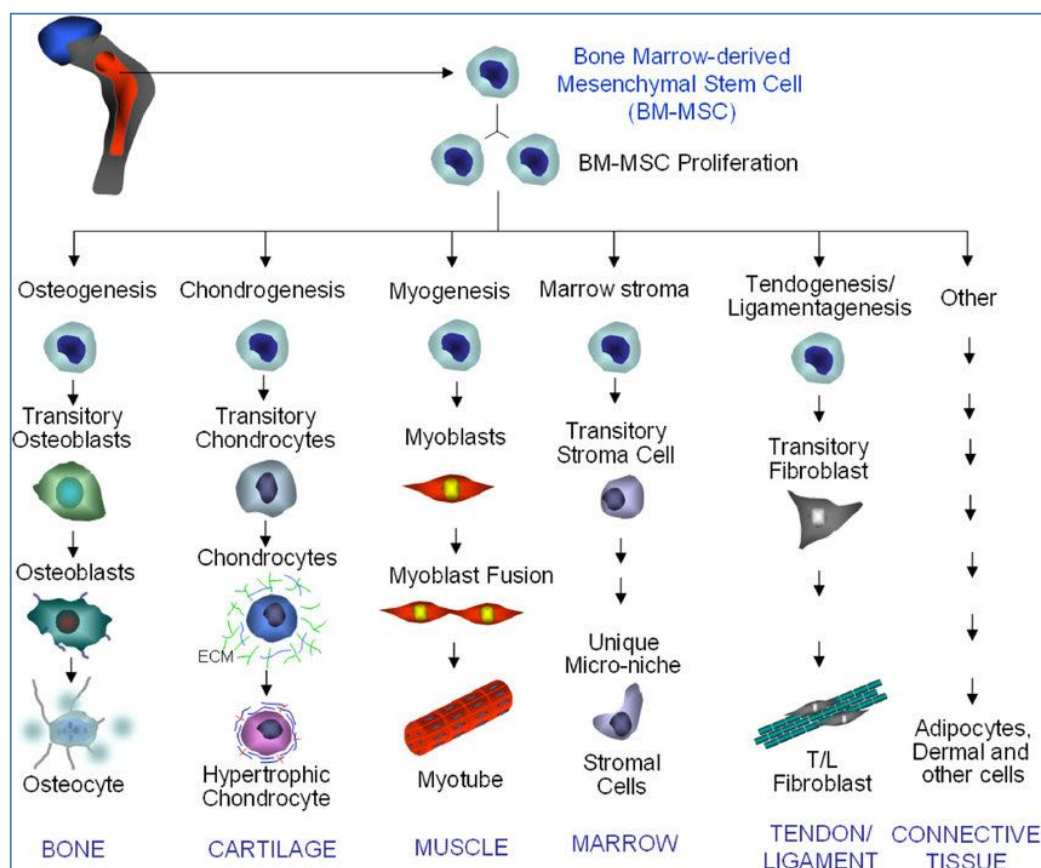
- Long, thin fibres forming branching networks
 - Contain a rubberlike protein, *elastin*
- Allow for stretch and recoil (skin, lungs, blood-vessels)

• **Reticular**

- Collagenous fibres that form delicate networks
- Surround blood vessels & support soft tissue of organs
- Particularly common where connective tissues meet other tissue types (basement membrane of epithelial cells)

○ **Cells**

- Make up connective tissue once matured (blasts = immature, cytes = mature)
- Fibroblasts – Connective tissue proper
- Chondroblasts – Cartilage tissue
- Osteoblast – Bone
- Hematopoietic Stem cells – Blood
- White blood cells, plasma cells, macrophages, mast cells



Stem cells in bone tissue engineering - Scientific Figure on ResearchGate. Available from:

https://www.researchgate.net/figure/BM-MSC-differentiation-strategy_fig3_47349594 [accessed 21 Feb, 2022]

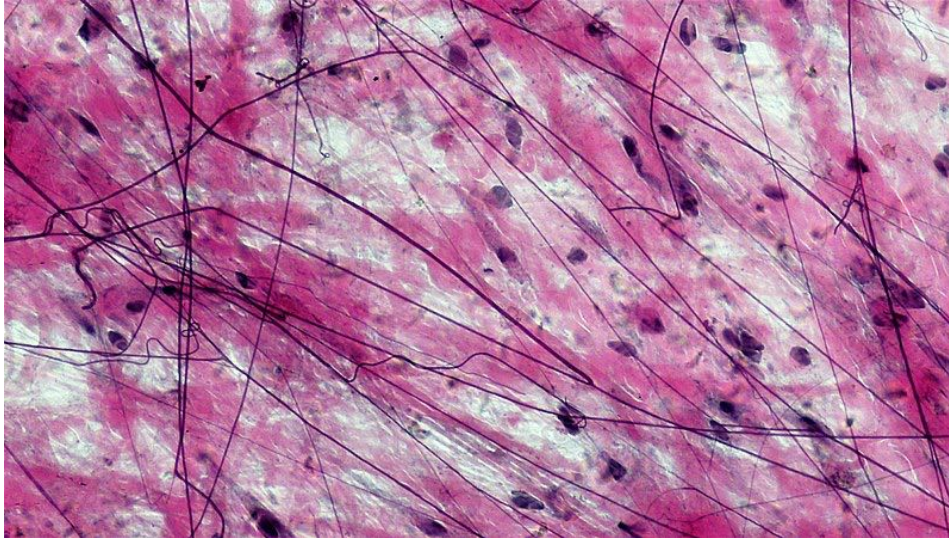
4 Classes of Connective Tissue: (see latter pages)

1: Connective Tissue Proper:

○ Loose Connective Tissue:

▪ Areola

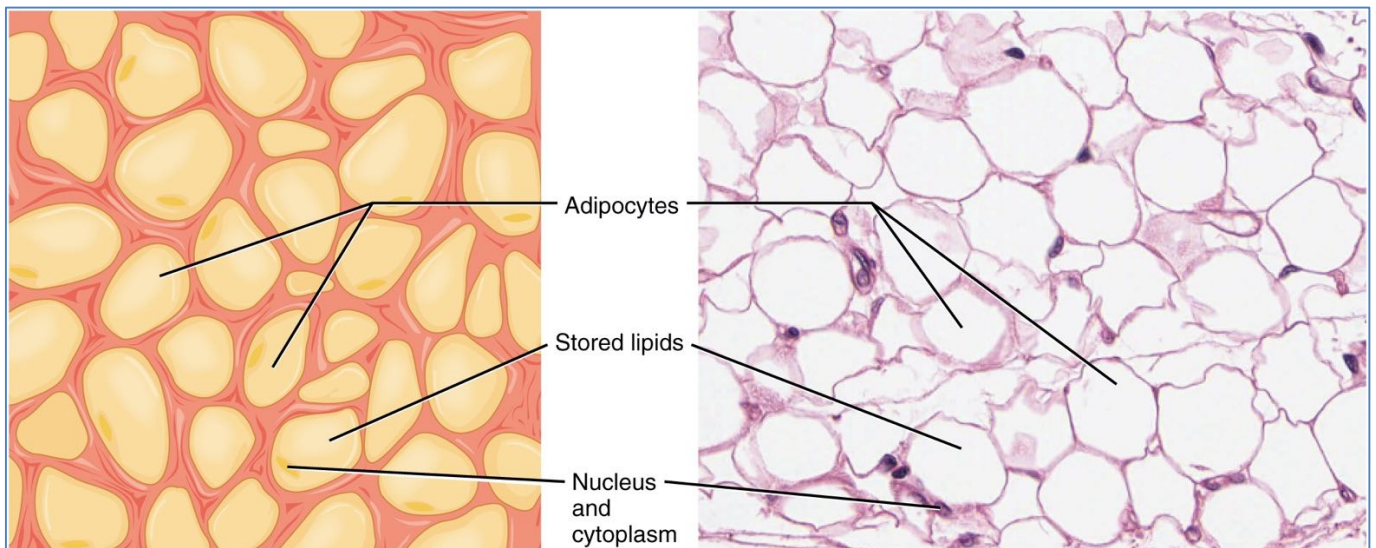
- Most widely distributed connective tissue in the body
- Loose gel-like matrix
- All 3 connective fibres
- Contains fibroblasts, macrophages, mast cells & white blood cells



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▪ Adipose

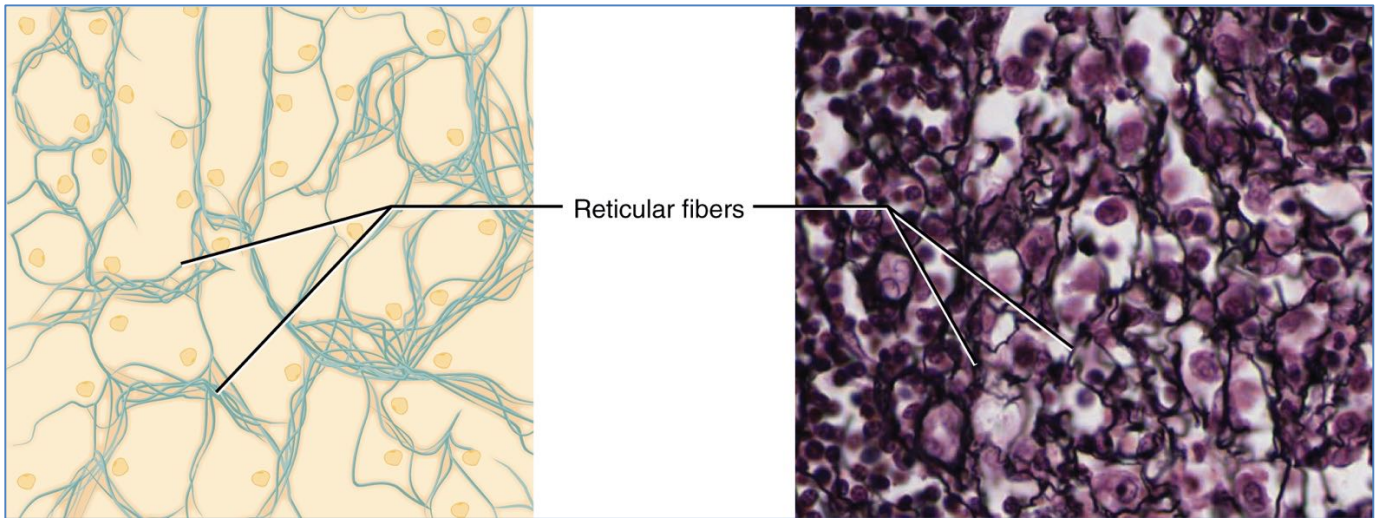
- High nutrient-storing ability
- Closely packed adipocytes (fat cells) predominate – 90% of tissue mass
- Insulates, supports & protects
- (under skin, around kidneys, breasts)
- Richly vascularised



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▪ **Reticular**

- Reticular fibres only in matrix
- Cells lie on a **reticular fibre** (mattress) **network**
- Function: forms a soft internal skeleton (stroma) that supports other cells (Ie: White blood cells, mast cells, & macrophages)
- Found in lymph nodes, bone marrow and the spleen



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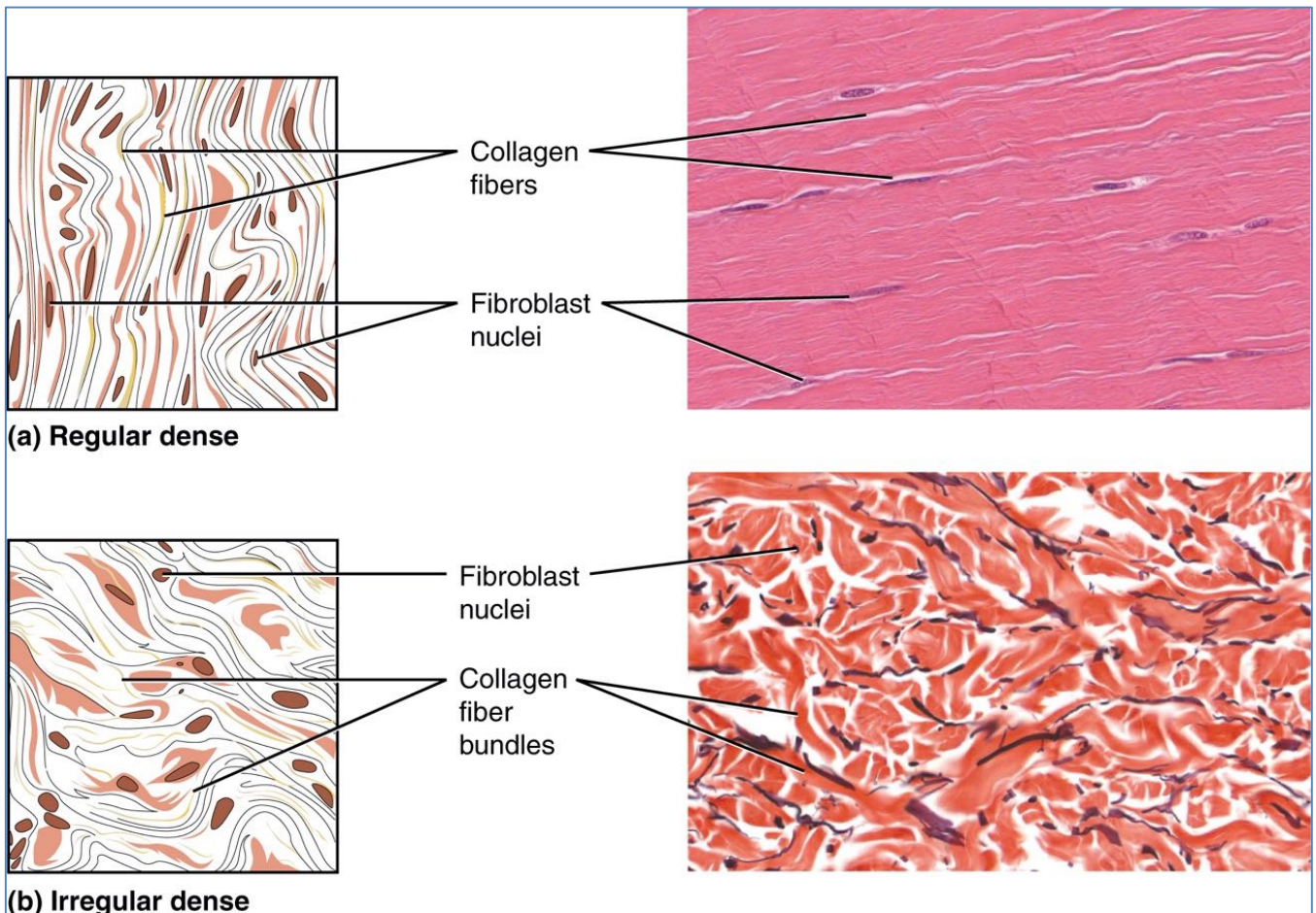
- **Dense Connective Tissue (fibrous connective tissues)**

- **Dense Regular**

- **Fibres** form the **predominant** constituent (**mainly collagen** – few elastic fibres)
 - Contains closely packed bundles of **parallel collagen fibres**
 - Collagen fibres are wavy to stretch a little, but once straight, they have no more give
 - Major cell type: **Fibroblasts**
 - High tensile strength in a specific direction. Still flexible.
 - Poorly vascularised
 - Forms tendons – attach muscles to bones & muscle to muscle
 - Forms Fascia (cling wrap) – fibrous membrane – wraps around muscles, blood vessels & nerves
 - Form ligaments – attach bones to bones @ joints

- **Dense Irregular**

- Same structural elements as Dense Regular, however the bundles of collagen fibres are much thicker and are arranged irregularly
 - Forms sheets in body areas where multidirectional tension is exerted
 - Skin (dermis)
 - Forms fibrous joint capsules and fibrous coverings surrounding some organs:
 - Kidneys, bones, cartilages, muscles, nerves



2: Cartilage

○ Hyaline Cartilage:

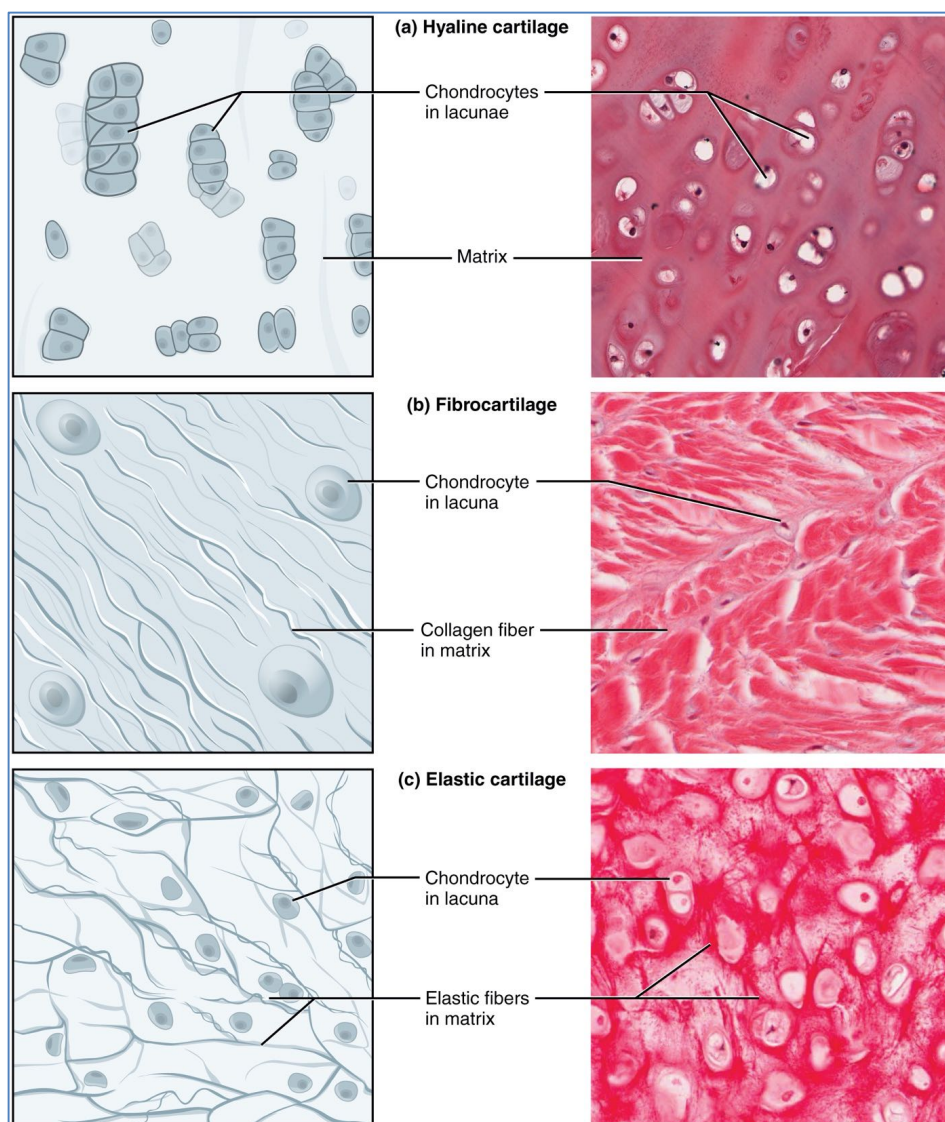
- Amorphous but firm matrix
- Chondroblasts produce matrix Chondrocytes = mature cartilage cells
- Cartilage matrix is approx 80% water – enables it to rebound after compression
- Firmly bound collagen fibres (some elastic)
- Has qualities intermediate between dense connective and bone tissue
- Withstands **tension** and **compression**
- Flexible but rigid
- Lacks nerve fibres
- Is **avascular**
- Receives nutrients by diffusion from blood vessels in the connective tissue membrane (perichondrium) surrounding it
- **Embryonic skeleton, synovial joints, respiratory tubes, nose, sterna-rib joint**

○ Elastic Cartilage:

- Nearly identical to hyaline cartilage - but has more elastin fibres
- Found where strength & stretchability are needed
- **External ear, epiglottis**

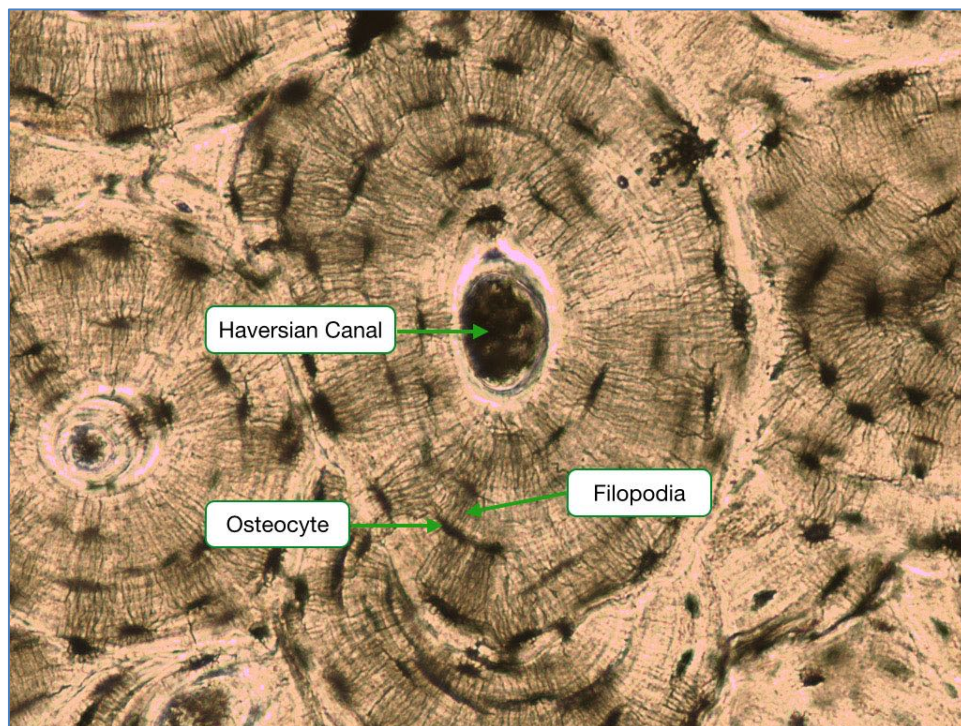
○ Fibrocartilage

- Perfect structural intermediate between hyaline cartilage and dense regular connective tissue
- Compressible and resists tension
- Found where strong support and heavy pressure resistance is needed
 - **Intervertebral discs, knee, etc**



- **3: Bone (osseous tissue)**

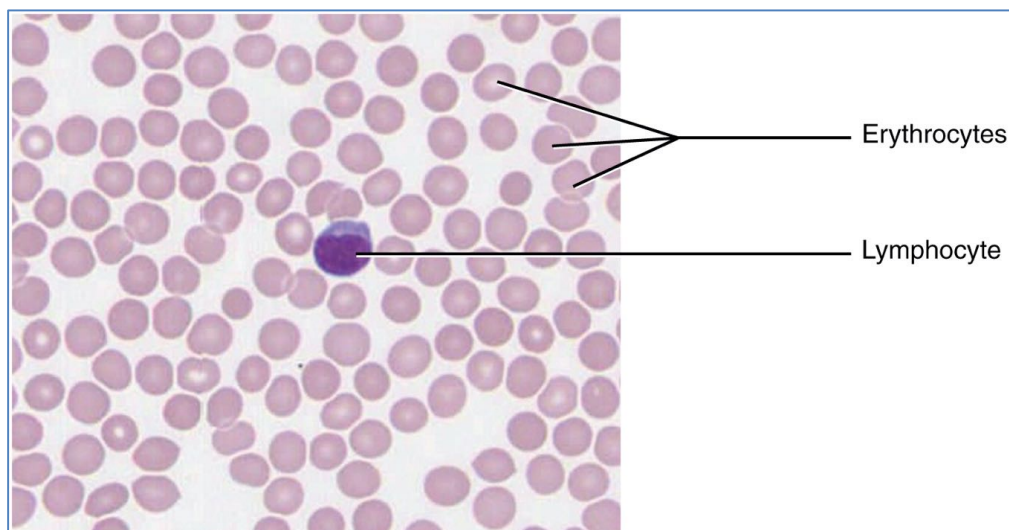
- Similar to that of cartilage but harder and more rigid
- Highly abundant collagen fibres and inorganic calcium make it very hard and rigid
- Osteoblasts produce the organic material in the matrix, and bone salts (calcium) deposits between the fibres
- Mature bone cells have osteocytes
- Provides cavities for fat/mineral storage & synthesis of blood cells
- Cross-section – closely packed structural units called osteons
 - Concentric rings of bony matrix surrounding central canals of blood vessels and nerves
- Well vascularised



http://histology.med.yale.edu/bone/bone_reading.php

- **4: Blood**

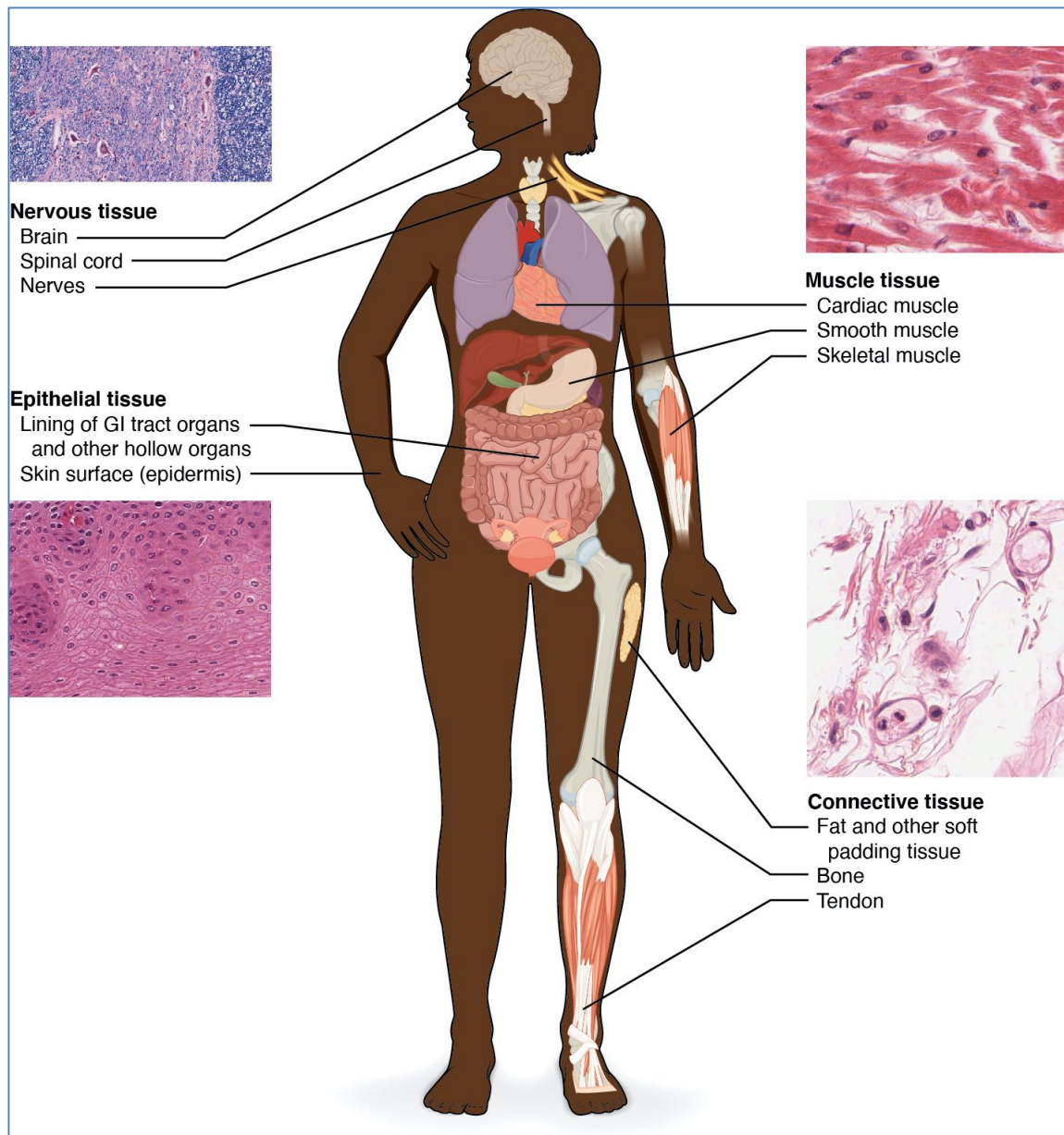
- Classed as a “connective tissue” because cells develop from mesenchyme and consists of cells surrounded by a non-living matrix
- Red (erythrocytes) and white cells in a fluid matrix (plasma)
- Contained within blood vessels
- Transports respiratory gases, nutrients and wastes



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MUSCLE TISSUE:

MUSCLE TISSUE:



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Comparison of Structure and Properties of Muscle Tissue Types

Tissue	Histology	Function	Location
Skeletal	Long cylindrical fiber, striated, many peripherally located nuclei	Voluntary movement, produces heat, protects organs	Attached to bones and around entrance points to body (e.g., mouth, anus)
Cardiac	Short, branched, striated, single central nucleus	Contracts to pump blood	Heart
Smooth	Short, spindle-shaped, no evident striation, single nucleus in each fiber	Involuntary movement, moves food, involuntary control of respiration, moves secretions, regulates flow of blood in arteries by contraction	Walls of major organs and passageways

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- **3 Types of Muscle:**

- **a) Skeletal**

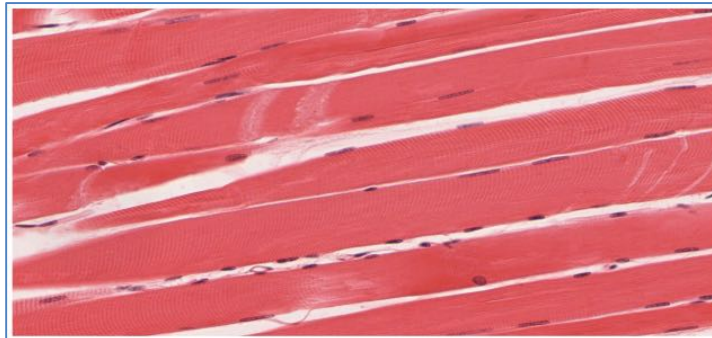
- Attaches to bone for movement (voluntary)
 - Long, Cylindrical
 - Multinucleated
 - Obvious striations → sarcomeres

- **b) Smooth**

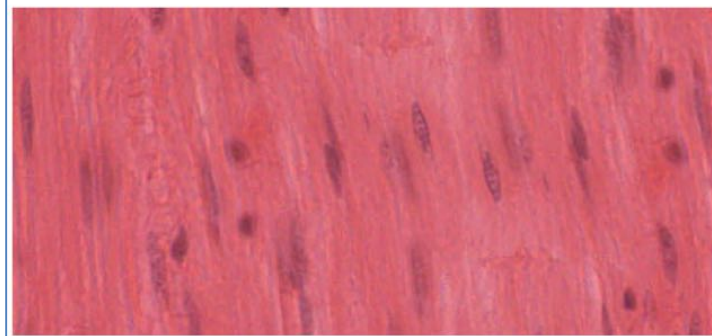
- In the walls of visceral organs – eg: GI tract/urinary tract/birth canal
 - Spindle-shaped cells
 - Central nuclei
 - No striations → no sarcomeres
 - Cells arranged closely to form sheets (often opposing-laterally perpendicular)
 - Usually involuntary – Controlled by the autonomic nervous system

- **c) Cardiac**

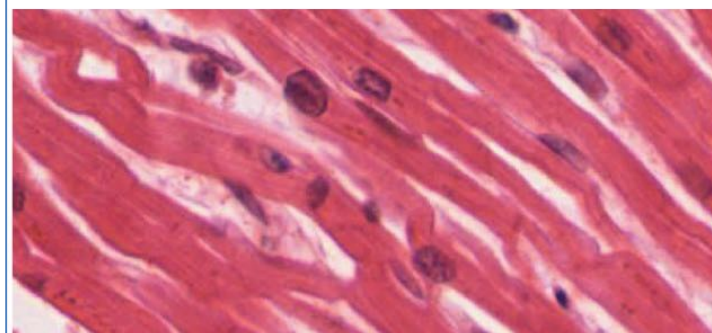
- Makes up the heart
 - Long, **Branched**, Cylindrical
 - Striations → sarcomeres
 - Usually single-nucleated
 - Intercalated discs – cell membranes of 2 adjacent cells bound mechanically (desmosomes), chemically & electrically (gap junctions). Essentially makes the entire heart one single muscle
 - Involuntary – controlled by autonomic nervous system



(a)



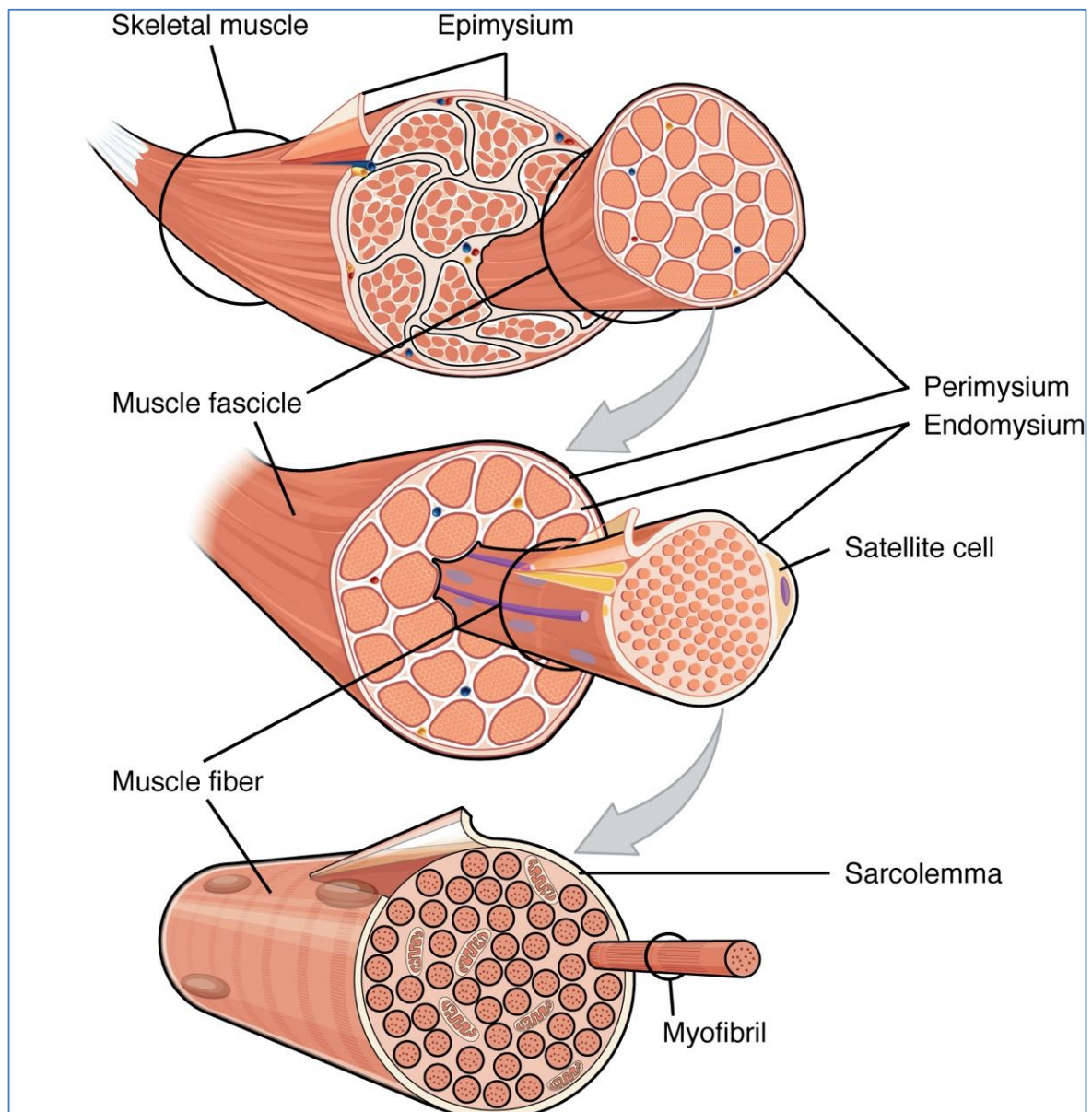
(b)



(c)

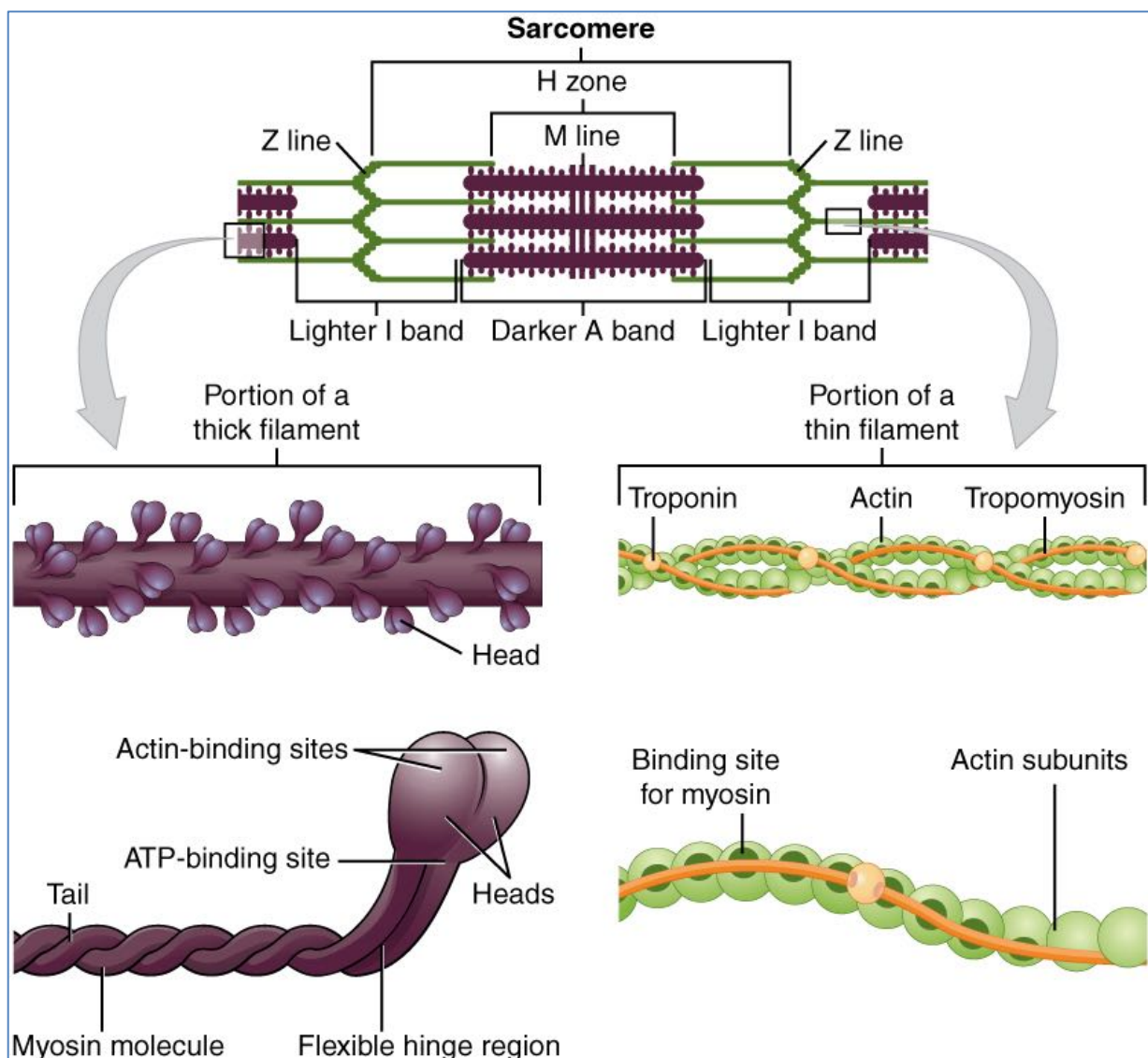
Organisation of Muscle Tissue:

- **Individual Muscle Fibres**
 - Each Muscle Fibre Contains many **Myofibrils (a Muscle Cell's Contractile *Organelles*)**
 - Each Myofibril contains many **Myo-Filaments** (Actin & Myosin) – Contractile *Proteins*
- **Endomysium**
 - Connective Tissue
 - Wraps single muscle fibres (cells)
- **Muscle Fascicles**
 - Bundles of muscle fibres (cells)
- **Perimysium**
 - Connective Tissue
 - Wraps Fascicles
- **Single Muscle**
 - Muscle as a whole – eg: The bicep
- **Epimysium**
 - Connective Tissue
 - Wraps whole muscle
- **Tendons**
 - A fusing together of all connective tissue layers
 - Connects muscle to bone

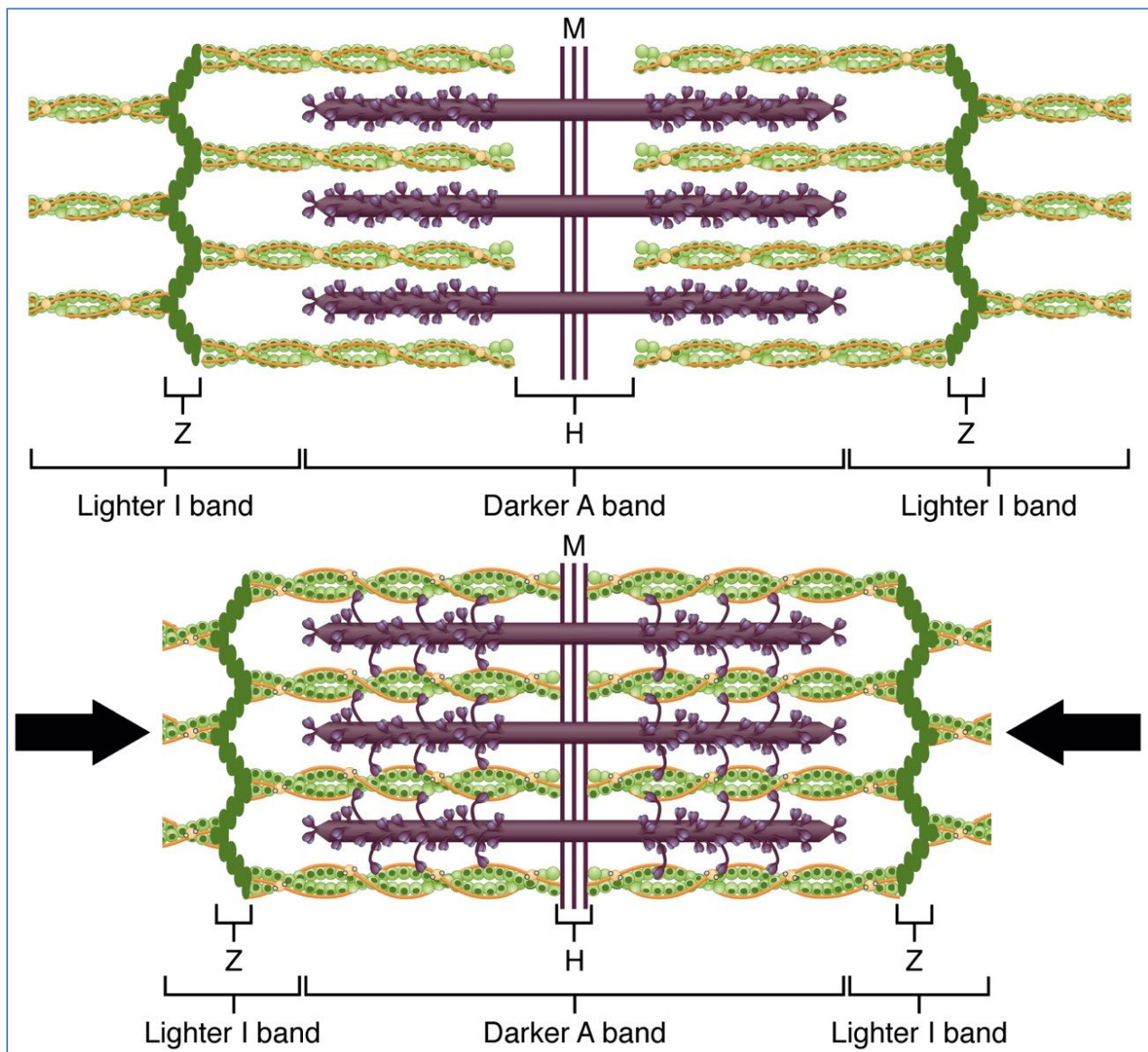


Internal Machinery of Skeletal Muscle Cells:

- **The Sarcomere** is the functional unit of muscles
- **Each Sarcomere contains Myo-Filaments** (Actin & Myosin) that slide past each other during contraction
 - **Actin** (Thin Myofilaments)
 - **Globular Actin:**
 - Kidney-shaped polypeptide subunits intertwined → double helix
 - Bear the active sites → myosin heads attach to during contraction
 - **Tropomyosin** – 2 strands that spiral along the actin
 - Stiffens the actin filament
 - Blocks myosin binding sites in relaxed muscle so myosin heads can't bind to the actin
 - **Troponin:**
 - 3 polypeptide complex
 - Binds to tropomyosin
 - Binds Ca^{+}
 - **Myosin** (Thick Myofilaments)
 - **Tails:**
 - Rodlike & helical
 - Start at the 'M-line'
 - Each ends with a 2 flexible hinges supporting a pair of globular **heads**
 - **Heads:**
 - Form '**Cross bridges**' – link thin & thick filaments during contraction
 - Contain ATPases to generate energy for contraction



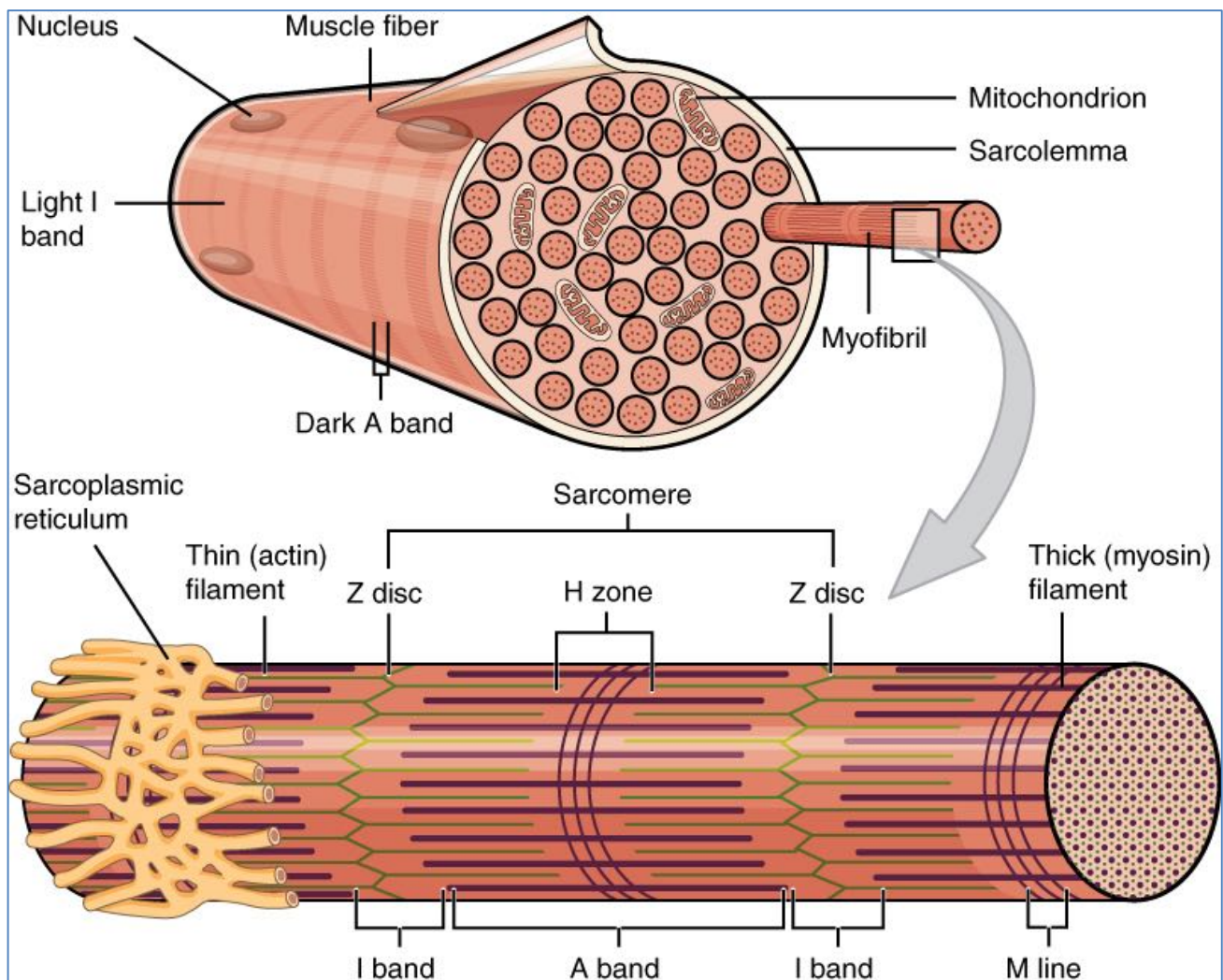
- **When Muscle is Relaxed:**
 - Thick & thin filaments only overlap at the ends
- **When Muscle is Stimulated:**
 - (by nervous system)
 - Myosin heads latch to myosin binding sites on actin
 - Form **cross bridges**
 - Formed & broken many times in a contraction
 - Act like tiny ratchets
 - Generate tension
 - Propel thin filaments toward centre of sarcomere
- **Z-Disc** – anchors sarcomeres together
 - Ensures whole cell contraction



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- **Muscle Fibres (Contractile Cells)**

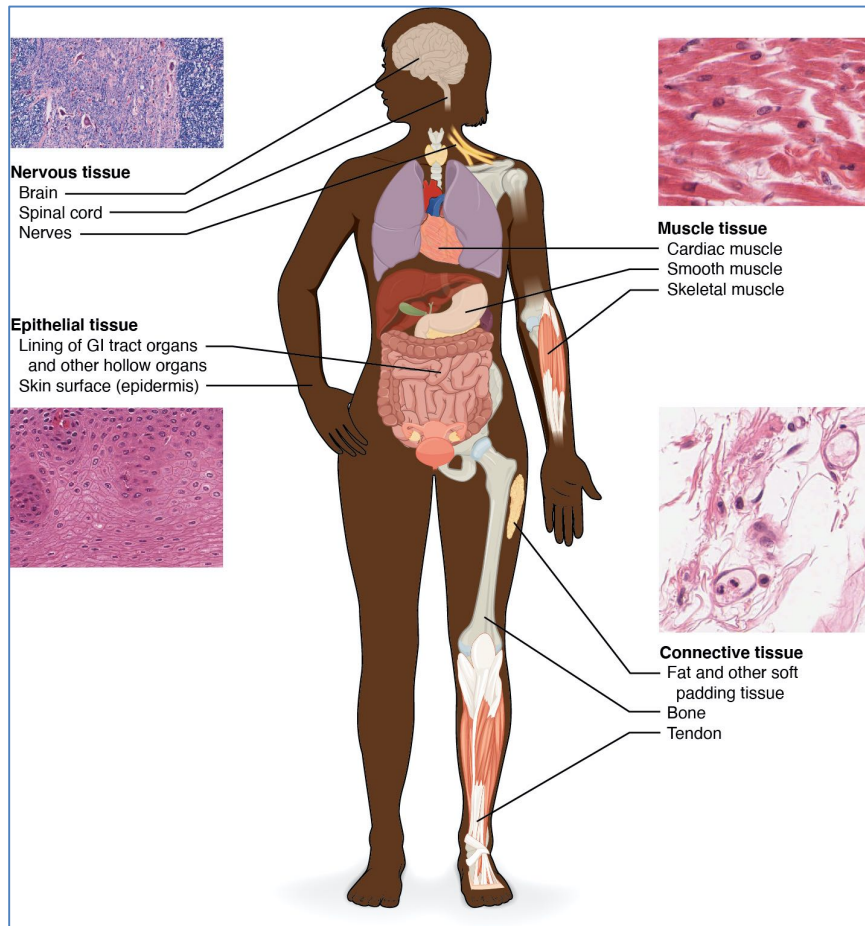
- **Sarcolemma** (plasma membrane)
 - **Transverse ('T') Tubules**
 - Perpendicular Invaginations of the sarcolemma (PM)
 - Runs between paired terminal cisterna of Sarcoplasmic Reticulum
 - Conducts impulses from sarcolemma deep into cell for mass myofibril contraction
- **Sarcoplasm** (cytoplasm – large glucose stores + myoglobin – oxygen supply)
- **Sarcoplasmic Reticulum**
 - Tubular network
 - **Stores & Regulates intracellular Ca^+** levels necessary for contractions
 - Surrounds each myofibril (contractile organelle)
 - **Terminal Cisternae** of the SR butt up on either side of the T-Tubules → forms a '**Triad**'
 - Triads occur at every I_Band–A_Band junction
- Abundant **Mitochondrion** – energy



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NERVOUS TISSUE

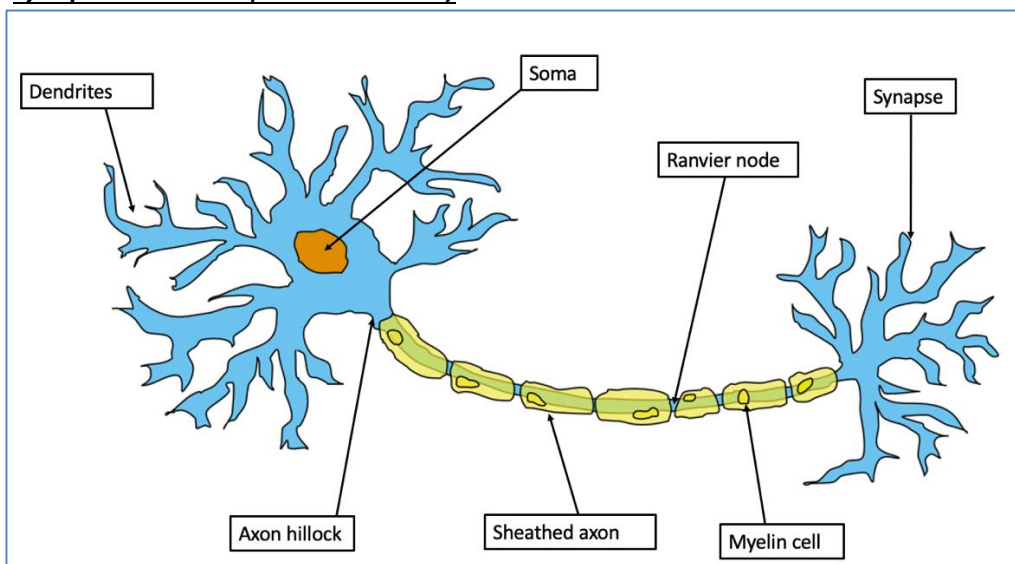
NERVOUS TISSUE



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The Neuron - Structural Features:

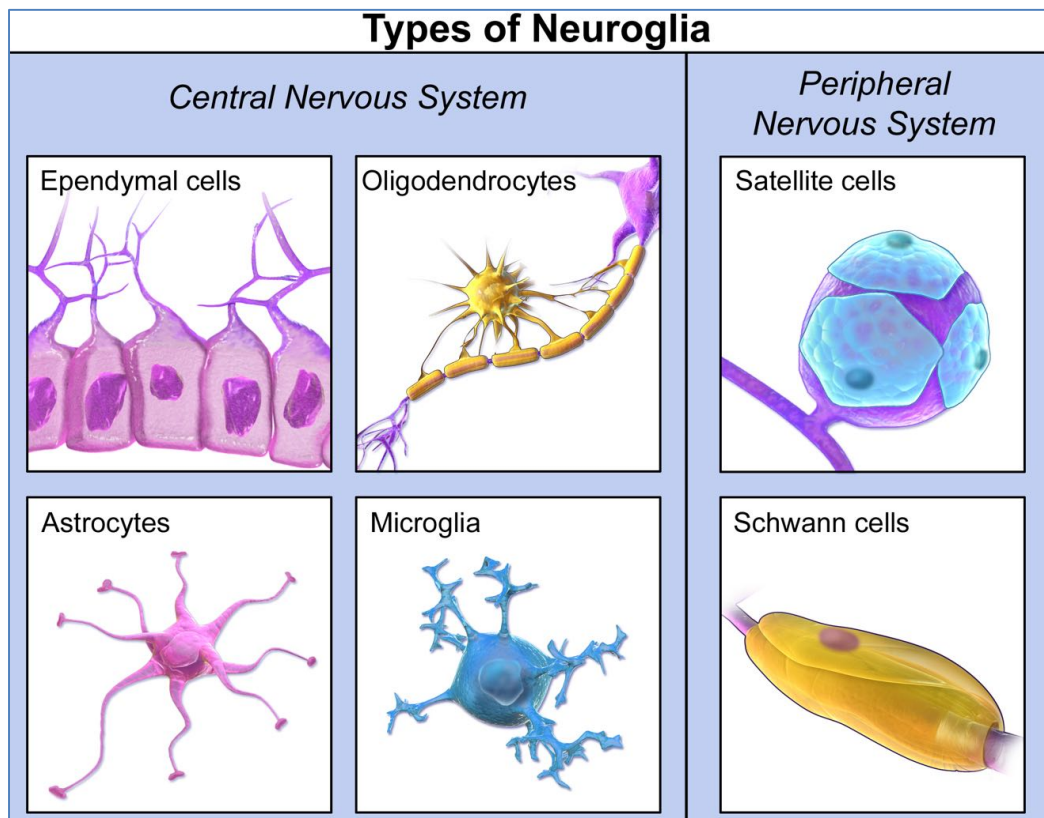
- a) **Receptive Field: Dendrites**
 - Stimulated by inputs
- b) **Cell Body: Soma**
 - Responds to graded inputs
- c) **Efferent Projection: Axon (and Axon Hillock)**
 - Conducts nerve impulses to target
 - Myelinated and unmyelinated
- d) **Efferent Projection: Myelin Sheath**
- e) **Efferent Projection: "Nodes of Ranvier"**
- f) **Output: Synaptic Terminals (Axon Terminals)**



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Supporting Cells: "Neuroglia" (Glia)

- Smaller **support cells** of NS
- Outnumber neurons 10:1
- Structural & mechanical support
- Roles in maintaining homeostasis & Myelination
- Immune responses via phagocytosis
- **Neuroglia of the Central Nervous System (CNS):**
 - **Astrocytes**
 - Nutrient bridge between neuron & capillaries
 - Guide migrating young neurons
 - Synapse formation
 - Mop up excess K^+ ions + neurotransmitters
 - **Microglia**
 - Long thorny processes
 - Monitors neuron health
 - Senses damaged neurons
 - Migrates to damaged neuron
 - Phagocytoses microbes & debris (immune cells are denied access to CNS)
 - **Oligodendrocytes**
 - Myelin formation in CNS
 - **Ependymal Cells**
 - Lines central cavities of brain + spinal chord
 - Blood-brain barrier
 - Beating cilia circulates cerebrospinal fluid
- **Neuroglia of the Peripheral Nervous System (PNS):**
 - **Schwann Cells**
 - Myelin Formation – wrap around axon
 - Regeneration of damaged neurons
 - **Satellite cells**
 - Surround neuron bodies
 - Structure, nutritional support & protection



MEMBRANE POTENTIAL & EXCITABLE TISSUES

MEMBRANE POTENTIAL & EXCITABLE TISSUES

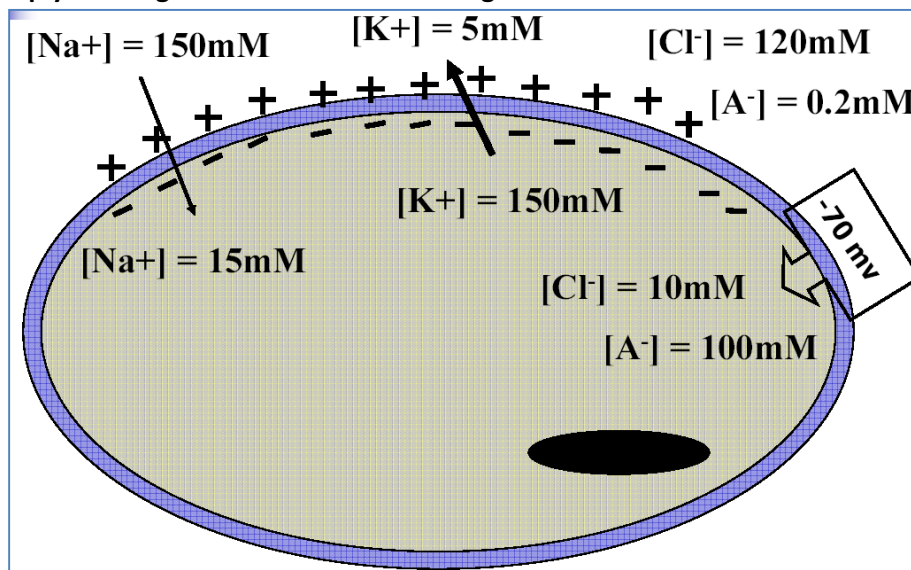
Membrane Potential:

- = Voltage across membrane
- **Cause:** unequal distribution of ions (K^+ & Na^+)
- Result of selective/differential permeability of specific plasma membrane proteins (mainly ion channels)
- Evident in all living cells (ranges between -20 & -200mV)
- **In excitable tissues – stimulation causes change in Membrane Potential**
 - Results in activation of the cell
 - Nervous & Muscle Tissues = Excitable Tissues
- **Membrane Potential Depends on:**
 - **Relative permeability** of PM to ions
 - Each ion's **concentration gradient**
 - **Electrochemical gradient**
- **Resting Membrane Potential:**
 - Stable membrane potential of cells when unstimulated
 - **For Nerve Cells** – approx -70mV

Ion Distribution Across Plasma Membrane

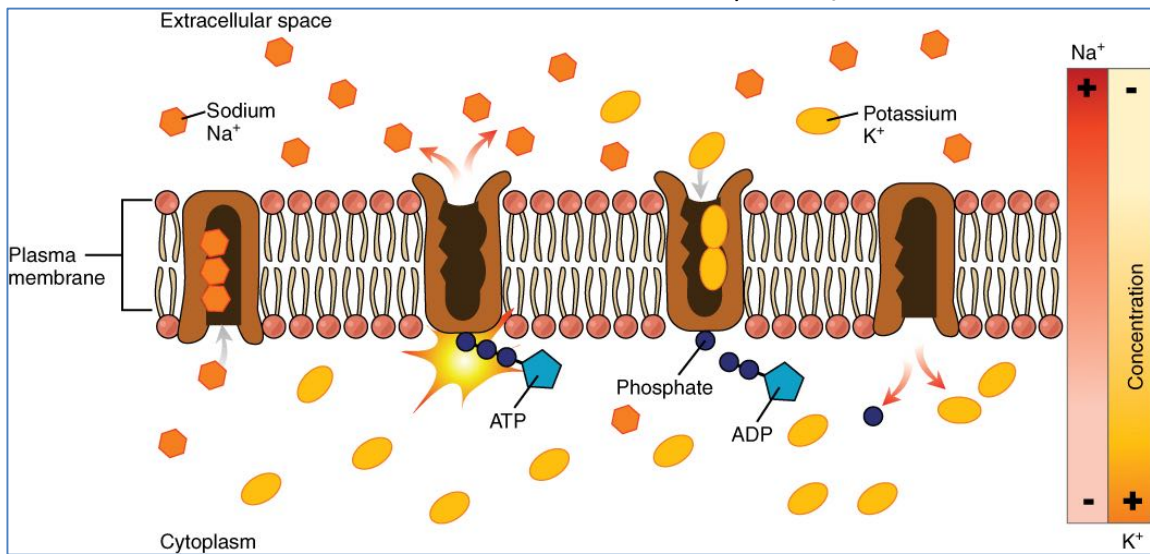
- **K^+ : [greater inside cell]**
 - At rest, membrane is **much** more permeable to K^+ than to Na^+
 - I.e: K^+ diffuses out of the cell through **leakage channels** down its concentration gradient
 - Therefore, loss of positive charge from cell makes:
 - Inside the cell negative
 - Outside the cell positive
 - Eventually the negativity of the inner membrane face attracts K^+ back into the cell
 - Therefore, concentration gradient drives K^+ out and is equally opposed by electrical gradient (**equilibrium potential has been reached**)
- **Na^+ : [greater outside cell]**
 - Membrane has much lower permeability to Na^+
 - Negative inner membrane-face attracts Na^+ into cell, but is opposed by low permeability
 - Therefore low diffusion of Na^+ into cell

****Simply: Resting MP is established due to greater diffusion of K^+ out than Na^+ in****



The Na/K ATPase: Maintaining the Resting Membrane Potential

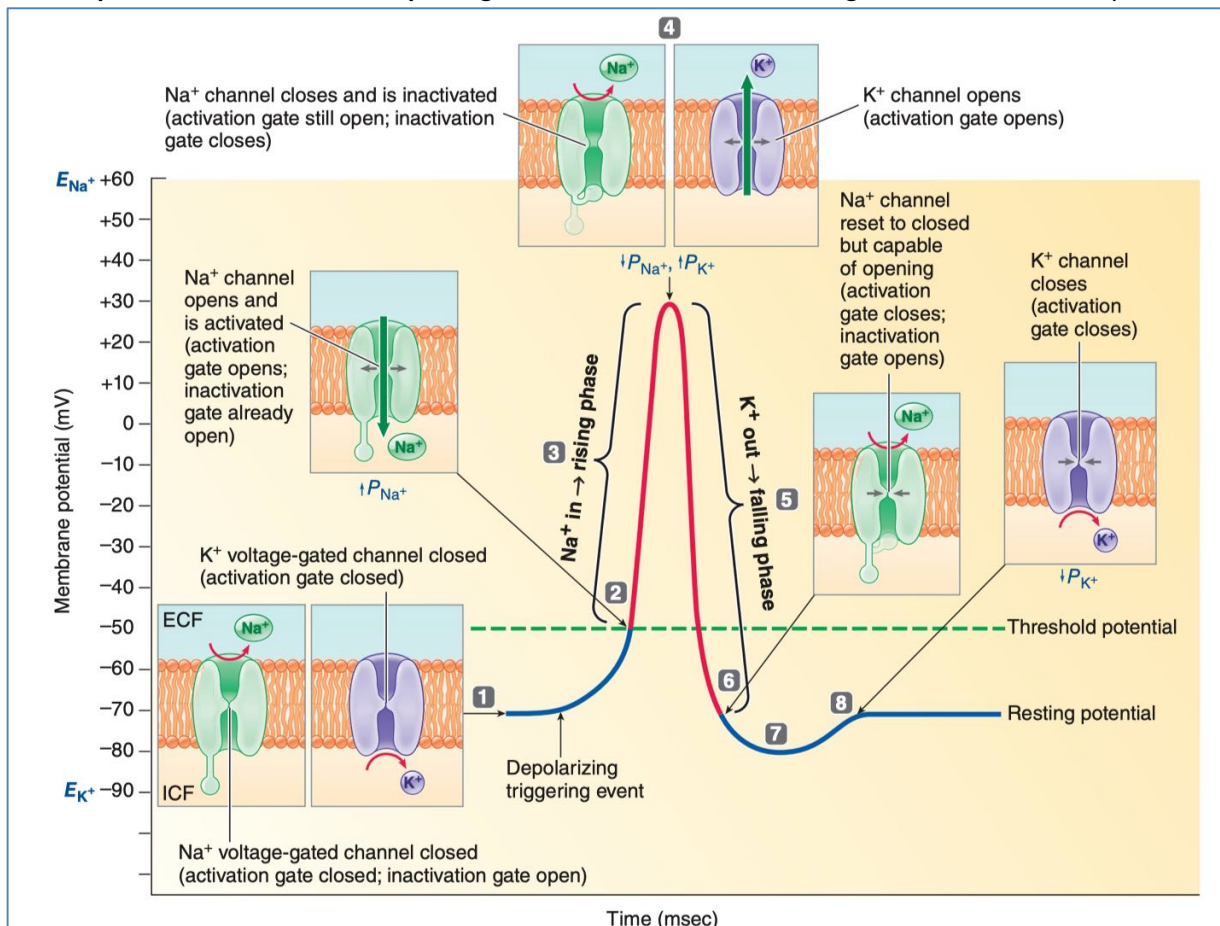
- Na passively diffuses into cell and K passively diffuses out
- So Why doesn't the chemical & electrochemical gradients dissipate?
- The **Concentration Gradients** for both Na & K are **maintained** by the **Na/K ATPase**



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Excitable Tissues (Nerves/Muscle)

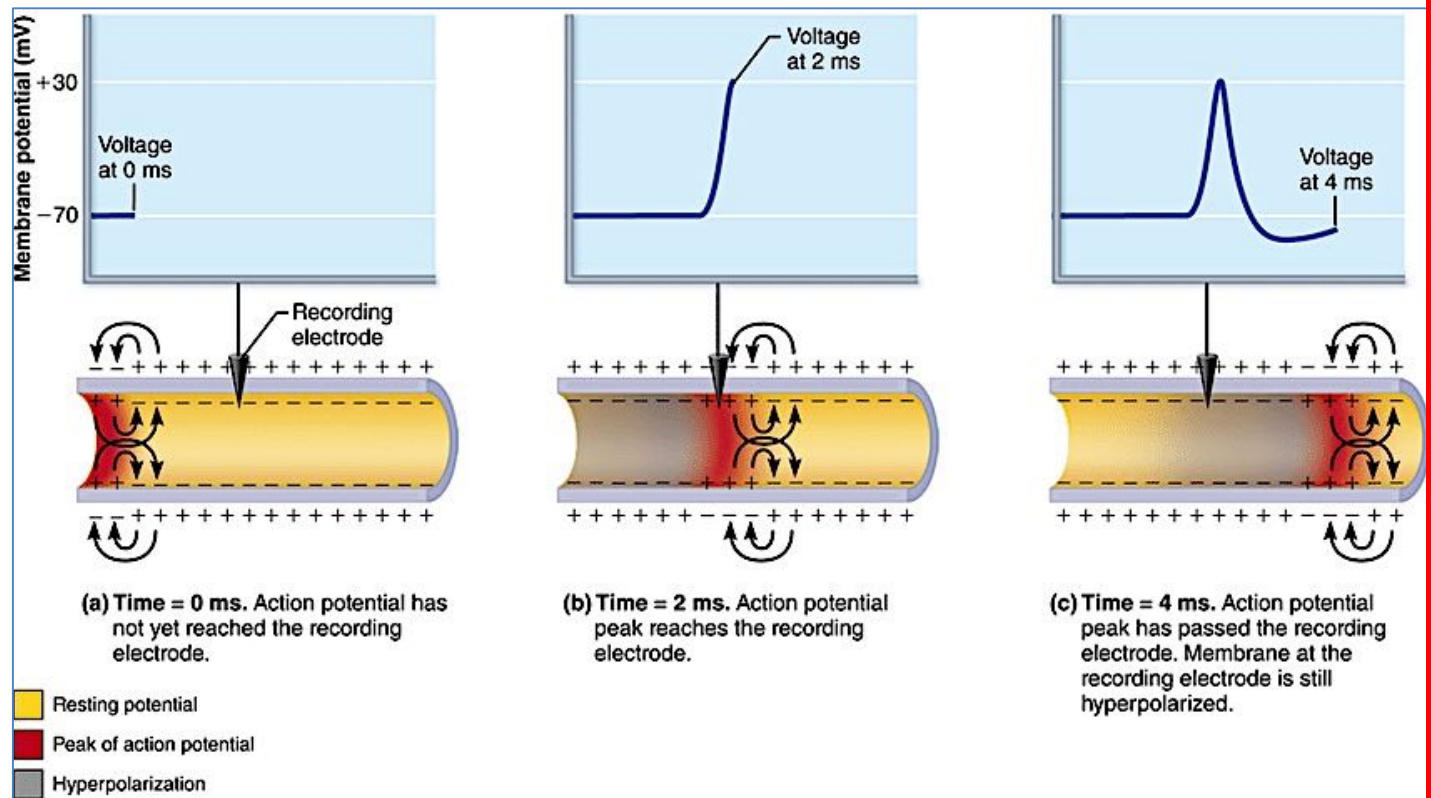
- In excitable cells, **stimuli** can **alter the permeability** of the membrane **to K^+ and/or Na^+**
 - Via opening/closing **gated ion channels** (**ligand/chemically gated**, **voltage gated**, mechanically gated, vibration gated, temperature gated)
- This changes the membrane potential
- If the membrane potential is sufficiently altered, an action potential is initiated
- **Action potential:** an **electrical impulse** generated and conducted along a **nerve's** axon in response to stimuli



Güler and Linaro et al Model in an Investigation of the Neuronal Dynamics using noise Comparative Study - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Phases-of-action-potential-12_fig2_334605708 [accessed 17 Jan, 2022]

Impulses: are conducted along the length of the axon

- A wave of action potentials – opening and closing of voltage gated ion channels
- **Action potential:** an impulse frozen in time
- Depolarisation, repolarisation and hyperpolarisation of membrane



Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Action-potential-propagation-along-an-axon-12_fig5_308369018 [accessed 21 Feb, 2022]

Neuronal Action Potentials:

Phase 1 – Resting Phase:

- Membrane is **much** more permeable to K^+ than to Na^+
- Greater diffusion of K out than Na in
- Therefore inside is negative/Outside is positive
- **Both Na & K voltage gated channels are CLOSED**

Phase 2 – Depolarisation Phase:

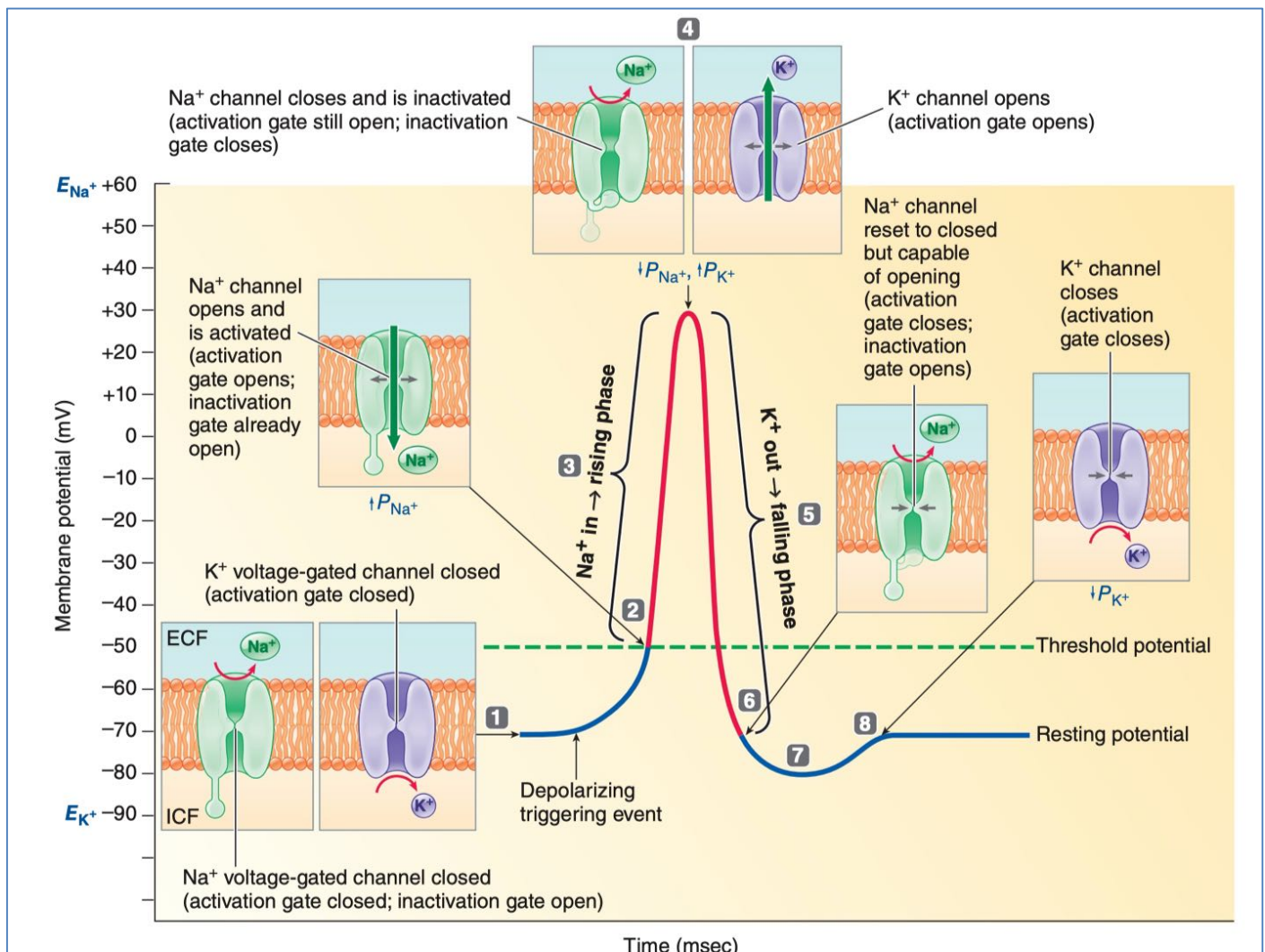
- Mechanical/chemical/vibratory/other stimulus opens some Na^+ channels
- → Na^+ flows into the cell
- Therefore membrane potential becomes less negative (i.e. It depolarises)
- If the MP reaches approximately **-55mV (threshold)**, the **voltage gated Na^+ channels open**
- → Na^+ influx increases dramatically – until MP reaches approximately **+30mV** where the voltage-gated Na^+ channels close

Phase 3 – Repolarisation Phase:

- @ approximately +30mV K^+ voltage gated channels open (permeability of K increases & Na decreases)
- Large outflow of K^+ → membrane potential becomes more negative (repolarises) and returns to -70mV

Phase 4 – Hyperpolarisation (undershoot) Phase:

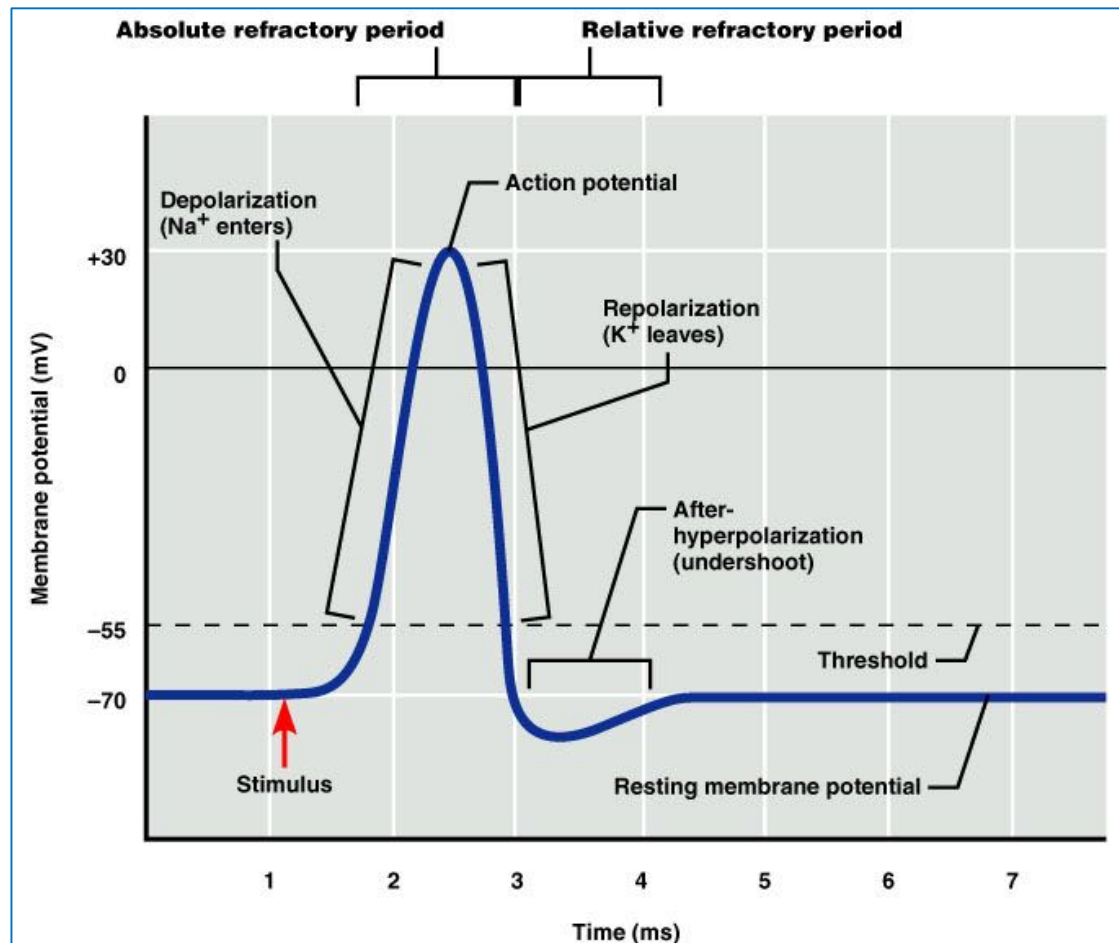
- K^+ channels remain open past -70mV and MP becomes more negative than at rest
- K^+ channels close and Na/K ATPase returns the MP to normal (-70mV)



Güler and Linaro et al Model in an Investigation of the Neuronal Dynamics using noise Comparative Study - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Phases-of-action-potential-12_fig2_334605708 [accessed 17 Jan, 2022]

Refractory Periods During the Action Potential:

- Basically the total time between a stimulus creating an action potential and the MP returning to rest
 - Usually 3-4ms
 - Determines how soon a neuron can respond to another stimulus
- Divided into 2 sub-periods:
 - **Absolute Refractory Period** – no additional stimulus (no matter how large) can initiate a further action potential
 - Stimulus → Depolarisation
 - Repolarisation
 - **Relative Refractory Period** – If an additional stimulus is to initiate another action potential during this time, it must be larger in order to reach threshold
 - Hyperpolarisation → Rest



Unattributable

TISSUE INJURY & CELLULAR ADAPTATIONS

TISSUE INJURY & CELLULAR ADAPTATIONS

Cellular Responses to Stress & Noxious Stimuli:

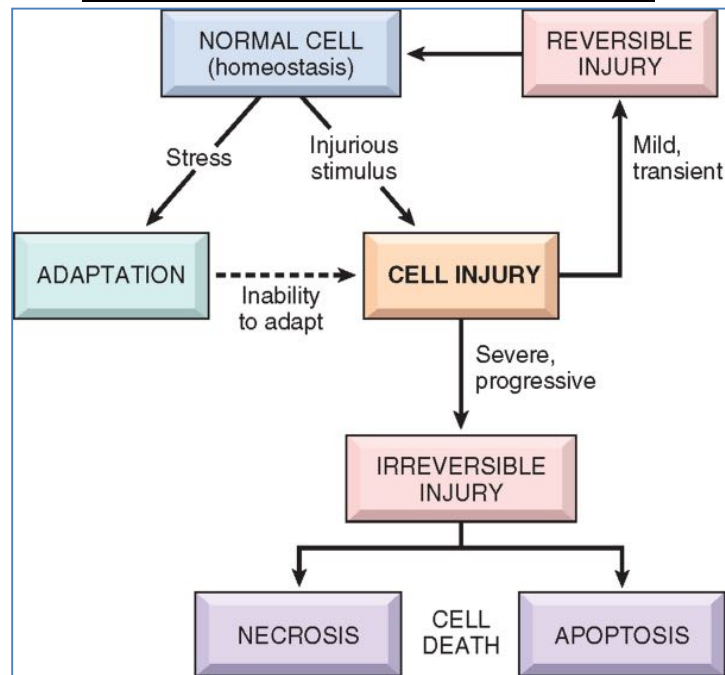


TABLE 1-1 Cellular Responses to Injury

Nature of Injurious Stimulus	Cellular Response
ALTERED PHYSIOLOGICAL STIMULI; SOME NONLETHAL INJURIOUS STIMULI - Increased demand, increased stimulation (e.g., by growth factors, hormones) - Decreased nutrients, decreased stimulation - Chronic irritation (physical or chemical)	CELLULAR ADAPTATIONS - Hyperplasia, hypertrophy - Atrophy - Metaplasia
REDUCED OXYGEN SUPPLY; CHEMICAL INJURY; MICROBIAL INFECTION - Acute and transient - Progressive and severe (including DNA damage)	CELL INJURY - Acute reversible injury Cellular swelling fatty change - Irreversible injury → cell death Necrosis Apoptosis
METABOLIC ALTERATIONS, GENETIC OR ACQUIRED; CHRONIC INJURY	INTRACELLULAR ACCUMULATIONS; CALCIFICATION
CUMULATIVE SUBLETHAL INJURY OVER LONG LIFE SPAN	CELLULAR AGING

1 Cellular Responses to Stress and Toxic Insults: Adaptation , Injury , and Death.

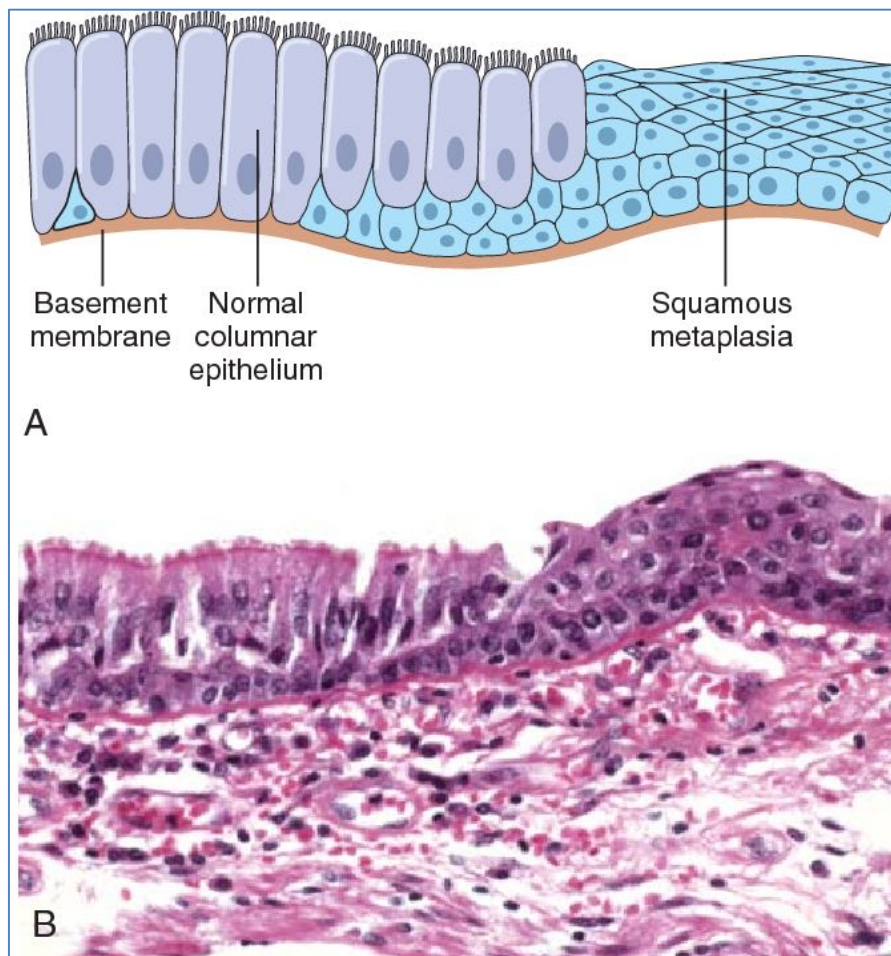
<https://www.semanticscholar.org/paper/1-Cellular-Responses-to-Stress-and-Toxic-Insults-%3A/fbffc46199b0539126b45c3f23933708f7dc3948>

Cellular Adaptations - in Response to Stress:

- (Cells may *Adapt & Change Their Size/Number/Structure/Function* in response to Changing Demands/↑Physiological Stress/Pathological Stimuli)
- **Hypertrophy:** (↑Size of Cells)
 - Hypertrophied organs have NO *New Cells*, just *Larger Cells*
 - Often a response in cells that are unable to divide – Eg: Muscles Cells
 - The Most Common Stimulus = ↑Workload
 - **Physiological Hypertrophy:**
 - Eg: Growth of Myometrium (Uterine Muscle) during Pregnancy
 - Eg: ↑Muscle Mass following Exercise (Both Cardiac & Skeletal)
 - **Pathological Hypertrophy:**
 - Eg: Cardiac Hypertrophy as a compensatory mechanism for Heart Failure
 - **Mechanisms of Hypertrophy:**
 - ↑ Workload → Triggers Mechanical Sensors/Growth Factors/Vasoactive Agents → ↑Synthesis of Cellular Proteins
 - Hypertrophy is the result of **Increased Production of Cellular Proteins**

- **Hyperplasia:** (↑ Number of Cells)
 - Same stimulus as Hypertrophy (↑ Workload), however cells are capable of dividing → ↑ in Number
 - **Physiologic Hyperplasia:**
 - **'Hormonal':**
 - Increases the Functional Capacity of a Tissue *When Needed*
 - Eg: Mammary Gland Hyperplasia during Pregnancy
 - **'Compensatory':**
 - Increases Tissue Mass after Damage or Partial Resection
 - Eg: Hyperplasia after removing part of the Liver
 - **Pathologic Hyperplasia:**
 - Mostly caused by *Excesses of Hormones/Growth-Factors* acting on target cells
 - Distinct from 'Cancer' in 2 ways:
 - **1)** There are NO MUTATIONS in genes regulating cell division, &
 - **2)** The Hyperplasia regresses if the Hormonal Stimuli is removed
 - Eg: 'Endometrial Hyperplasia' – an example of *Abnormal* hormone-induced hyperplasia
 - Eg: 'Benign Prostatic Hyperplasia' – induced by Androgens
 - Eg: Skin warts due to Papillomavirus
 - **Mechanisms of Hyperplasia:**
 - Hyperplasia is the result of *Growth-Factor-Driven Proliferation* of Mature Cells & sometimes Stem Cells
- **Atrophy:** (↓ Size & ↓ Cell Number)
 - Can be Due to:
 - ↓ Workload
 - Loss of Innervation
 - Diminished Blood Supply
 - Loss of Endocrine Stimulation (Eg: Ovaries during menopause)
 - Inadequate Nutrition
 - **Physiologic Atrophy:**
 - Eg: Common during normal foetal development
 - Eg: Atrophy of Uterus following Parturition
 - **Pathologic Atrophy:**
 - Depends on the underlying cause; Can be general or localized:
 - Disuse Atrophy – (↓ Workload)
 - Denervation Atrophy – (Loss of innervations)
 - Diminished Blood Supply – (Ischaemic)
 - Inadequate Nutrition
 - Loss of Endocrine Stimulation:
 - Tissue Compression
 - **Mechanisms of Atrophy:**
 - Initial Response – Cell decreases in size & organelles → ↓ Metabolic Demands
 - This results from ↓ Protein Synthesis & ↑ Protein Degradation in cells
 - Cells may also resort to **Autophagy** ("Self-Eating"), eating its own components for nutrients

- **Metaplasia:** (Reversible Change in Phenotype of Cells)
 - A reversible change in which one differentiated cell type is replaced by another cell type
 - It is an adaptive substitution of vulnerable cells for cells types better able to withstand the adverse environment
 - **Pathologic Metaplasia:**
 - Eg: Gastro-Oesophageal Reflux Disease → Oesophagus changes from *Squamous* to *Columnar* Epithelium in lower Oesophagus (Gives better protection against acid)
 - Eg: Chronically Irritated Mucous Membranes of Respiratory Tract change from *Columnar* to *Squamous* from Smoking (This affects the Mucociliary Escalator since there are no cilia)
 - Eg: Connective Tissue Metaplasia – The formation of cartilage/bone/adipose tissue in tissues that normally don't contain these elements (Eg: Bone formation in muscle)
 - **Mechanisms of Metaplasia:**
 - It is the result of a *Reprogramming of Stem Cells* beneath the stressed cells, which change their potential phenotype
 - This is due to Cytokines/Growth Factors/Extracellular Matrix-interactions with environment

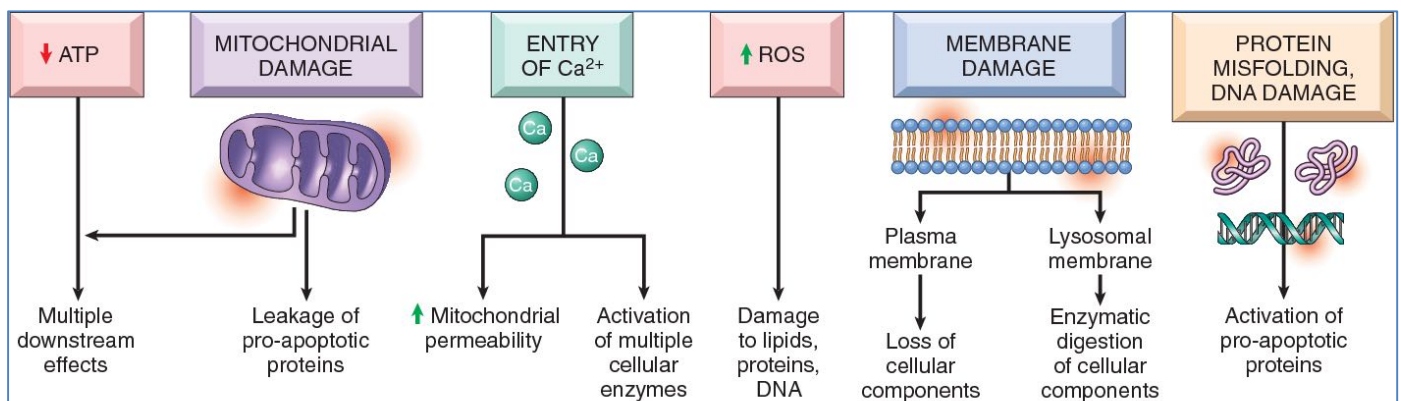


1 Cellular Responses to Stress and Toxic Insults: Adaptation , Injury , and Death.

<https://www.semanticscholar.org/paper/1-Cellular-Responses-to-Stress-and-Toxic-Insults-%3A/fbffc46199b0539126b45c3f23933708f7dc3948>

Cell Injury & Cell Death – in Response to **Noxious Stimuli**:

- When cells are stressed so severely that they can no longer adapt
- Injury may progress through a reversible stage and may culminate in cell death
- **Note:** Cell injury is Reversible up to a point, but if the stimulus persists, or is severe enough, it becomes irreversible → Cell Death
- **Causes of Cell Injury:**
 - **#1 Oxygen Deprivation → Ischaemic/Hypoxic Injury:**
 - Deficiency of oxygen (Hypoxia) → ↓ Aerobic Oxidative Respiration
 - The most common form of injury in clinical practice
 - **Causes include:**
 - Reduced Blood Flow (Ischaemia)
 - Inadequate oxygenation of the blood (Systemic Hypoxia)
 - Decreased O₂ carrying capacity of blood (eg: Anaemia/CO-Poisoning)
 - **Physical Agents:**
 - Eg: Mechanical Trauma
 - Eg: Extreme Temperatures (Hot & Cold)
 - Eg: Sudden changes in pressure
 - Eg: Radiation
 - Eg: Electric Shock
 - **Chemicals & Drugs (Acid/Basic/Toxic/etc)**
 - **Infectious Agents (Directly or by Toxins)**
 - **Immunological Reactions:**
 - Eg: Immune reactions to self-antigens → Autoimmune Diseases
 - **Genetic Derangements (Mutations affect essential cellular constituents)**
 - **Nutritional Imbalances:**
 - Eg: Vitamin Deficiencies
 - Eg: Excess cholesterol → Atherosclerosis



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- **Q – Why do the Centro-Lobular Hepatocytes always seem to be worse off during Toxic/Ischaemic Injury?**

- **A – 2 Reasons:**
 - Centro-lobular areas receive second-hand blood from the outer-lobular areas, which is:
 - a) Low in Oxygen,
 - b) Low in Nutrients
 - & c) Full of cell-waste
 - Blood tends to pool in the centro-lobular region due to the slow-draining single central vein → The cells are exposed to the toxins for slightly longer

Biochemical Mechanisms of Cell Injury:

- Depletion of ATP:

o Major Causes:

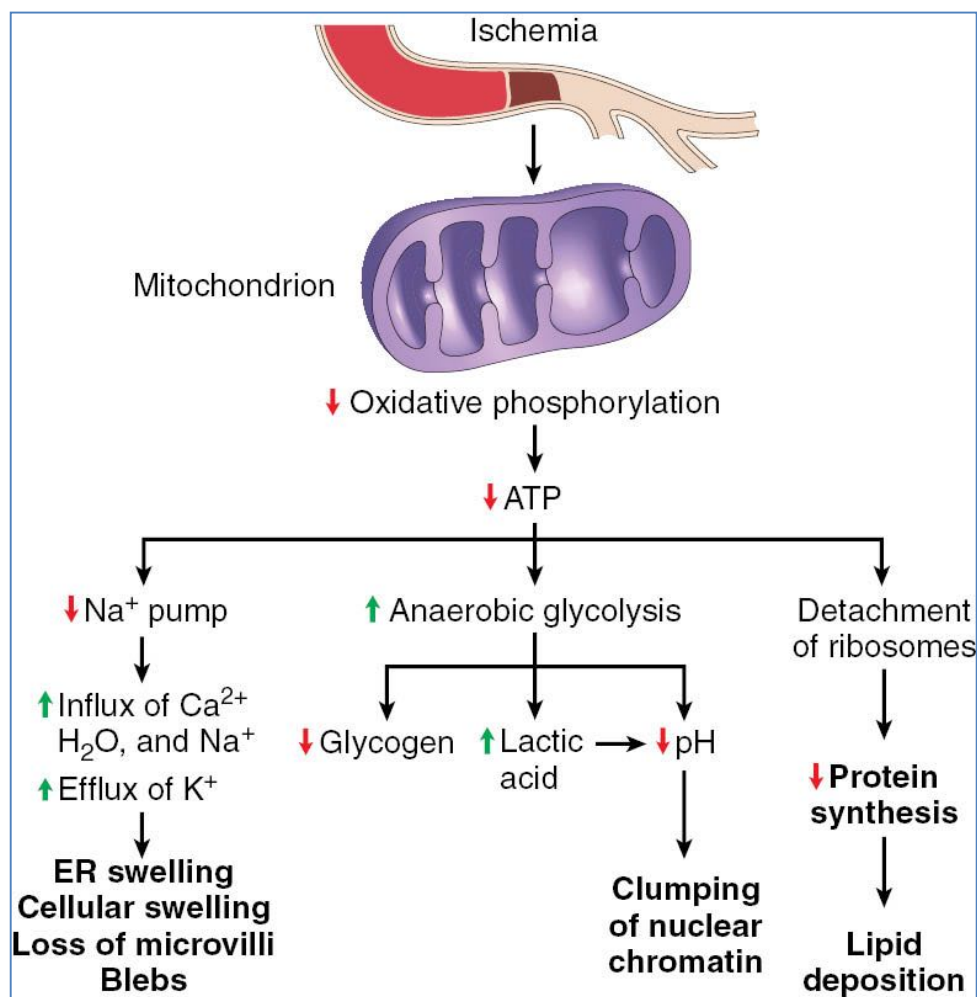
- **Ischaemia** → ↓O₂ & ↓Nutrients
 - Hypoxia → ↓Oxidative Phosphorylation
 - Nutrient Depletion → ↓Metabolism (including Glycolytic pathway)
 - Note: Tissues with greater glycolytic capacity (eg: Liver) last longer than those without (eg: Brain)
- **Certain Toxins** – Affecting Electron Transport Chain - (Eg: Cyanide, Oligomycin, Rotenone, Antimycin, Carbon-Monoxide) → Prevents Oxidative Phosphorylation
- **Mitochondrial Damage** → Leakage of Pro-Apoptotic Proteins (Eg: Cytochrome-C)

o Major Metabolic Pathways that are Vulnerable:

- *Glycolytic Pathway (Some anaerobic capacity)
- *Oxidative Phosphorylation of ADP→ATP (Electron Transport Chain - Mitochondria)

o Consequences → ATP is required for all processes within the cell. Hence, deficiency →

- ↑Anaerobic Glycolysis:
 - →Glycogen Depletion
 - →Lactic Acid Buildup → ↓pH → ↓Activity of essential Enzymes
- ↓Active Membrane Transport:
 - Failure of Na/K-ATPase → Intracellular Na⁺ Accumulation →Cell Swelling
 - Failure of Ca pump → Ca Influx → widespread damage (see diagram)
- ↓Protein Synthesis
- ↓Lipogenesis
- Ultimately leads to Irreversible Mitochondrial/Lysosomal Membrane Damage → Cell Death.

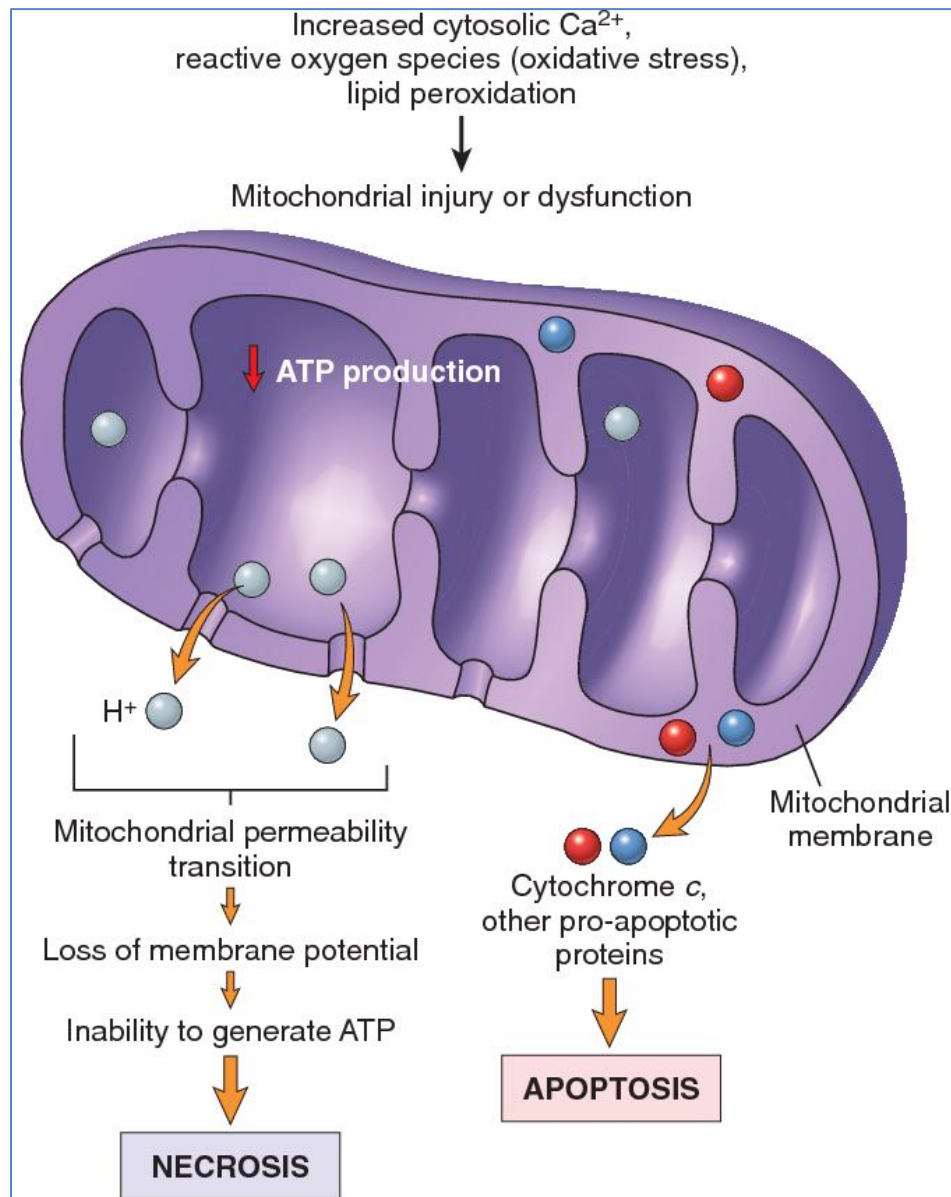


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- **Mitochondrial Damage:**

- **Note:** Irreparable mitochondrial damage *kills cells* – due to reliance on oxidative metabolism
- **Major Causes:**
 - Hypoxia
 - Free Radicals
 - High Cytosolic Ca^{2+}
- **Consequences →**
 - High-Conductance channels, called **Mitochondrial Permeability Transition Pores**, form in mitochondrial membrane → Loss of Mitochondrial Membrane Potential → Failure to Oxidative Phosphorylation → ATP Depletion → Necrosis of cell
 - Leakage of Inter-mitochondrial-membrane substances (Eg: Cytochrome-C) can activate Apoptotic Pathways (Ie: The Caspase Cascade)

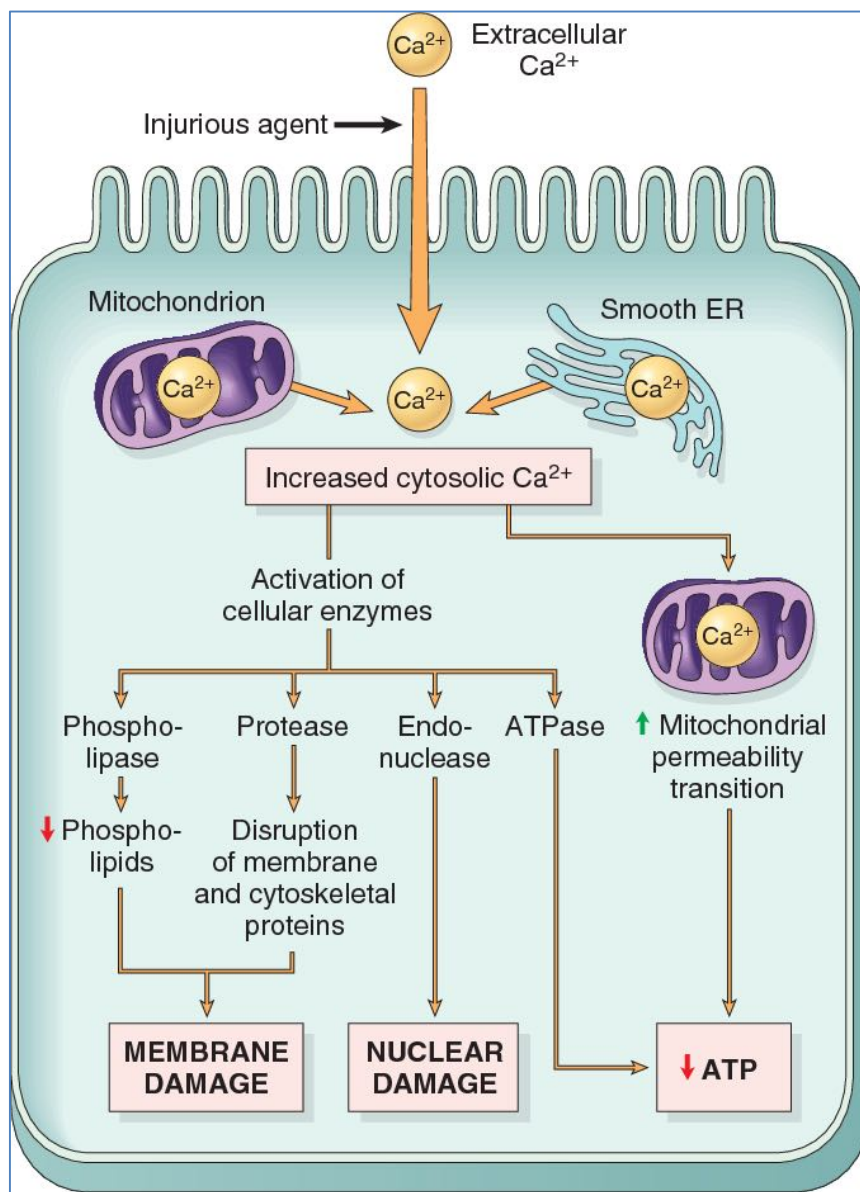


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- **Loss of Calcium Homeostasis & Ca^{2+} Influx:**

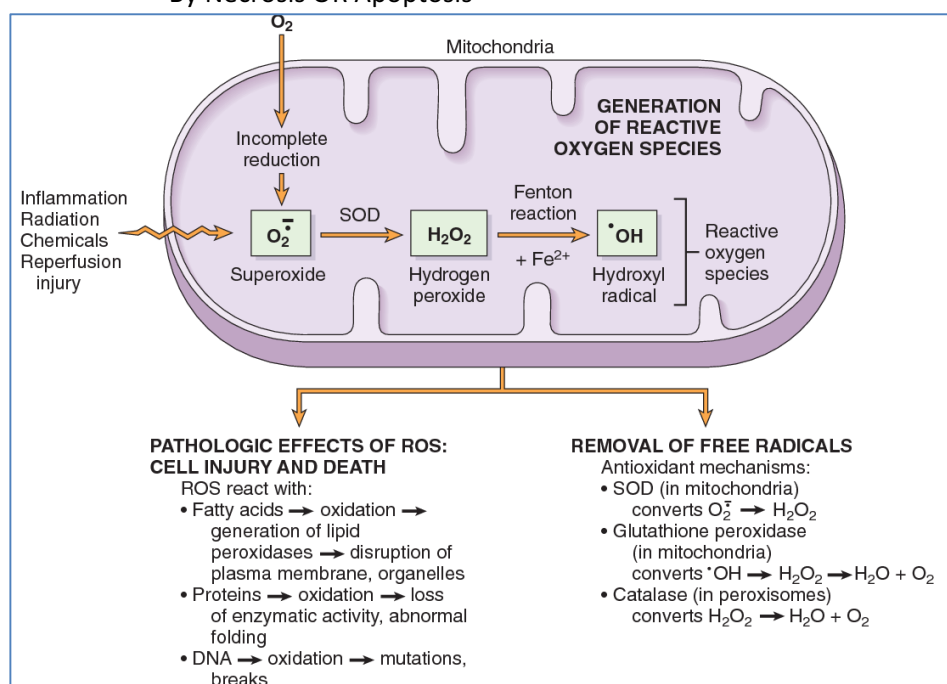
- Ca^{2+} is normally maintained at very low concentrations in the Cytosol by *Energy-Dependent Systems*
- **Major Causes:**
 - **Toxins** → Causing Ca^{2+} release from intracellular stores (Mitochondria & Endoplasmic Reticulum)
 - **Hypoxia** → ATP Depletion → Can't feed Active Ca^{2+} Export Systems → Net Ca^{2+} influx across the Plasma Membrane
- **Consequences** →
 - $\uparrow \text{Ca}^{2+}$ → Opens *Mitochondrial Permeability Transition Pores* in mitochondrial membrane → Loss of Mitochondrial Membrane Potential → Failure to Oxidative Phosphorylation → ATP Depletion → Necrosis of Cell
 - $\uparrow \text{Ca}^{2+}$ → Activates destructive enzymes (ATPases, Phospholipases, Proteases & Endonucleases) – See Diagram
 - $\uparrow \text{Ca}^{2+}$ → Direct Activation of Caspases & Leakage of Cytochrome-C → Induces Apoptosis by Direct Activation of Caspases & release of pro-apoptotic substances (Including Cytochrome-C)



1 Cellular Responses to Stress and Toxic Insults: Adaptation , Injury , and Death.

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- **Oxygen-Derived Free-Radicals (Oxidative Stress):**
 - **What are Free Radicals?**
 - AKA: Reactive Oxygen Species
 - Are normal by-products of mitochondrial respiration
 - Are chemicals with a **Single Un-Paired Valent Electron** → Oxidising Potential
 - **Major Causes: (Generation of ROS):**
 - **Normal Metabolism** - Normal Redox reactions during normal metabolic processes
 - **Radiation** - Absorption of radiant energy (Eg: UV/Xray/Microwave)
 - **Inflammation** - Produced by Phagocytes during Inflammation
 - **Chemicals** - Metabolism of some exogenous Chemicals (eg: some Drugs)
 - **Re-Perfusion Injury** – Exacerbation of Injury due to Restoration of blood flow to Ischaemic Tissues
 - **Removal of ROS:**
 - **Spontaneous Decay** - in the presence of H_2O
 - **Radical-Scavenging Systems** - Enzymatic mechanisms that remove Free Radicals
 - **Antioxidants** - (eg: Vits A/E/C) Remove/Mop Up/Prevent/Inactivate Free Radicals
 - **Proteins** - Reactive metals are bound to storage/transport proteins in blood
 - **What is “Oxidative Stress”?:**
 - Imbalance occurs between Free-Radical production & Radical-Scavenging-Systems
 - Ie: ↑↑Free-Radicals
 - **Consequences of ↑ROS→**
 - **Membrane-Lipid Peroxidation:**
 - Oxidative damage of Lipids within Plasma/organelle Membranes
 - → Damages membranes
 - **Oxidative Modification of Proteins:**
 - → Damage Active Sites on Enzymes
 - → Disrupt Conformation of Structural Proteins
 - Enhance action of Proteases → Continued Protein Degradation
 - **DNA Damage:**
 - Oxidative damage → breaks in the DNA strands/Mutations/Cross-Linking/etc – Ie: Stuff that isn’t supposed to happen
 - → Cellular Ageing
 - → Cancer
 - **Cell Death:**
 - By Necrosis OR Apoptosis



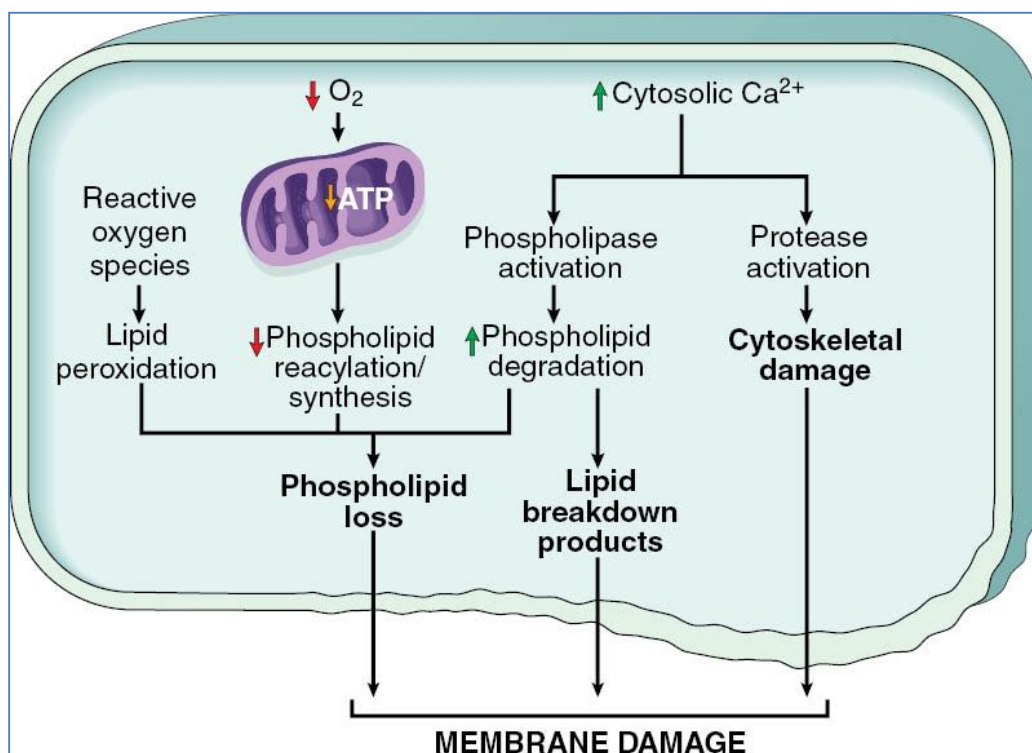
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<https://www.semanticscholar.org/paper/1-Cellular-Responses-to-Stress-and-Toxic-Insults-%3A/fbffc46199b0539126b45c3f23933708f7dc3948>

- **Defects in Membrane Permeability:**

- **Causes: Occurs in all forms of Cell Injury – (EXCEPT Apoptosis):**
 - **Eg: Ischaemia →**
 - ATP Depletion → ↓Phospholipid Synthesis
 - **Eg: Free-Radicals →**
 - Membrane-Lipid Peroxidation
 - Oxidative Modification of Proteins (structural/enzymes/Cytoskeleton)
 - **Eg: Ca^{2+} - Mediated activation of Phospholipases →**
 - ↑Phospholipid Degradation
 - **Eg: Bacterial Toxins (Endotoxins)**
 - **Eg: Viral Proteins**
 - **Eg: Lytic Complement Components (Eg: The “Membrane Attack Complex”)**
 - **Eg: Perforins – from cytolytic lymphocytes (Cytotoxic-T & NK cells)**
 - **Eg: Physical Trauma**
 - **Eg: Chemical Agents**
- **Consequences →**
 - **Mitochondrial Membrane Damage (↑Permeability) →**
 - ↓ATP Production &
 - Release of Pro-Apoptotic proteins (Cytochrome-C)
 - **Plasma Membrane Damage →**
 - Loss of Osmotic balance
 - Influx of Ions
 - Influx of fluids
 - Loss of Cellular Contents & Essential Metabolic Substrates
 - **Lysosomal Membrane Damage →**
 - Leakage of Destructive Enzymes into Cytoplasm:
 - RNases
 - DNases
 - Proteases
 - Phosphatases
 - Glucosidases
 - Cathepsins

→ Cells Die by Necrosis

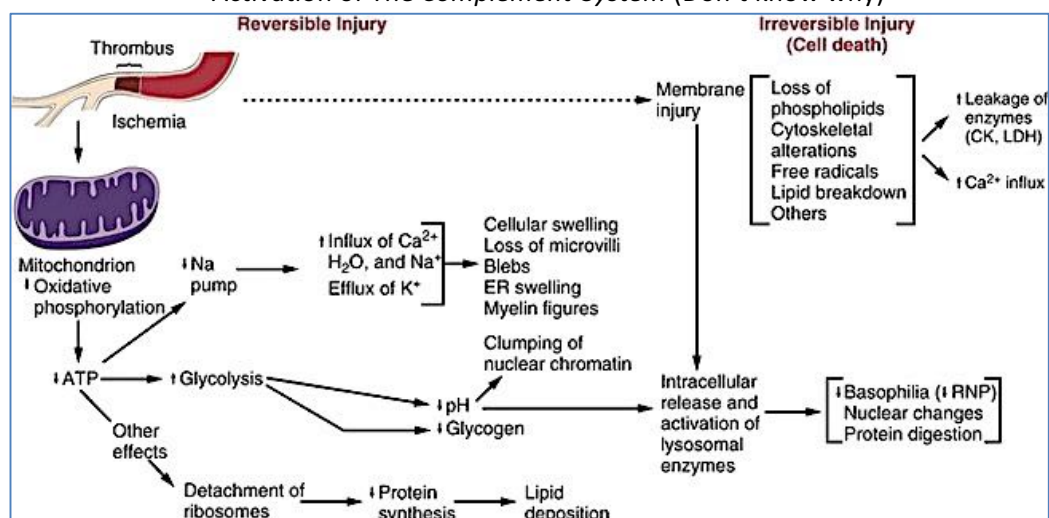


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- **Ischaemic & Hypoxic Injury:**

- Most common type of cell injury
- **Ischaemia Vs Hypoxia:**
 - **Ischaemia** = ↓Supply of O₂ & Nutrients due to ↓Blood Flow
 - Note: Anaerobic Glycolysis stops after glycolytic substrates are exhausted
 - **Hypoxia** = ↓Supply of O₂
 - Note: Anaerobic Glycolysis can still continue
- **Consequences →**
 - **Reversible Consequences:**
 - ↓Oxidative Phosphorylation → ↓ATP
 - Failure of Na/K-ATPase → Na Influx → Fluid Influx → Cell Swelling
 - Anaerobic Metabolism → ↓Glycogen, ↑Lactic Acid, ↓pH
 - ↓Protein Synthesis
 - Ca⁺ Influx
 - Cytoskeleton Disperses → Loss of Ultrastructural Features → Formation of “Blebs” on cell surface
 - Membrane Damage
 - Mitochondrial Damage
 - **Note: If O₂ is restored, all of the above are reversible**
 - **Irreversible Consequences:**
 - MASSIVE Ca⁺ Influx (Particularly if the ischaemic zone is re-perfused)
 - →Leakage & Activation of Self-Digestive Enzymes
 - Severe Swelling of Mitochondria
 - Extensive Plasma-Membrane Damage:
 - → Continued Loss of: Proteins/Enzymes/Coenzymes/RNA
 - Swelling of Lysosomes
 - **Note: Even If O₂ is restored, the above are Irreversible**
 - **Death by Necrosis →**
 - Cell Components degraded
 - Leakage of cellular enzymes (Eg: Troponin I & Creatinine Kinase)
 - Entry of Extracellular molecules into dying cell
 - Dead cells become ‘Myelin Figures’ (composed of phospholipids) →
 - →Phagocytosed
 - →Degraded further to FFA’s
 - →Calcified
- **What is Ischaemia-Reperfusion Injury?:**
 - Phenomenon where Restoration of blood flow to **Irreversibly-Injured** Ischaemic Tissues →Exacerbation & Acceleration of Injury AS WELL AS Further Injury
 - **Theories as to Why:**
 - Re-oxygenation → ↑Generation of Free Radicals
 - Activation of The Complement-System (Don’t know why)



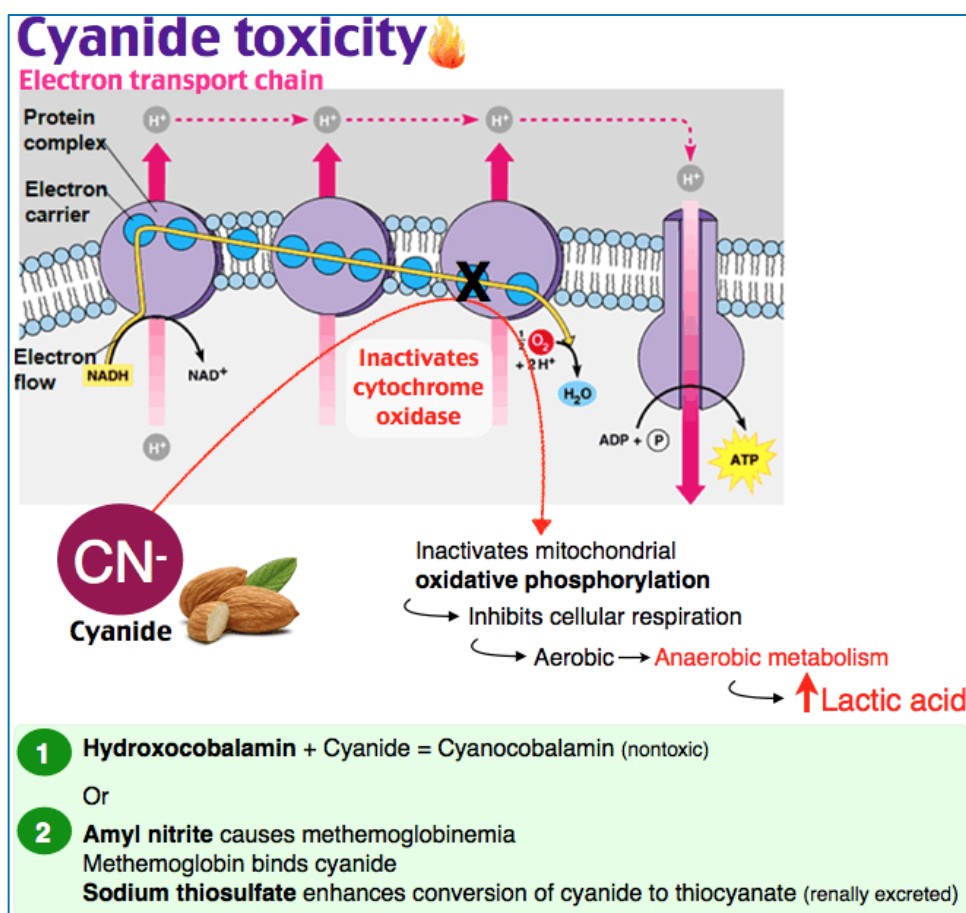
- **Chemical Injury:**

○ **Major Causes:**

- Direct damage

○ → **Direct injury by combining with critical molecular components:**

- Eg: Mercury - binds to cell-membrane proteins → ↑ Permeability & ↓ Ion Transport
- Eg: Cyanide – binds with Cytochrome-Oxidase → Inhibits Oxidative Phosphorylation



<https://richkosh.blogspot.com/2019/05/zyklon-b-hydrogen-cyanide.html?m=1>

Morphological Alterations in Cell Injury:

- **Reversible:**

- Early stages where the functional/morphological changes are reversible if damaging stimulus is removed

○ **2 Common Histological Features:**

▪ **1: Cellular Swelling (Hydropic Change)**

• **Mechanism:**

- Failure of energy-dependent ion pumps in the PM → Cells are incapable of maintaining Ionic & Fluid Homeostasis

▪ **2: Fatty Change**

- Frequently seen in injured Hepatocytes (Toxic injury) & Myocardial Cells (Hypoxic Injury)

- Typically seen in Toxic or Hypoxic Injury

• **Mechanism:**

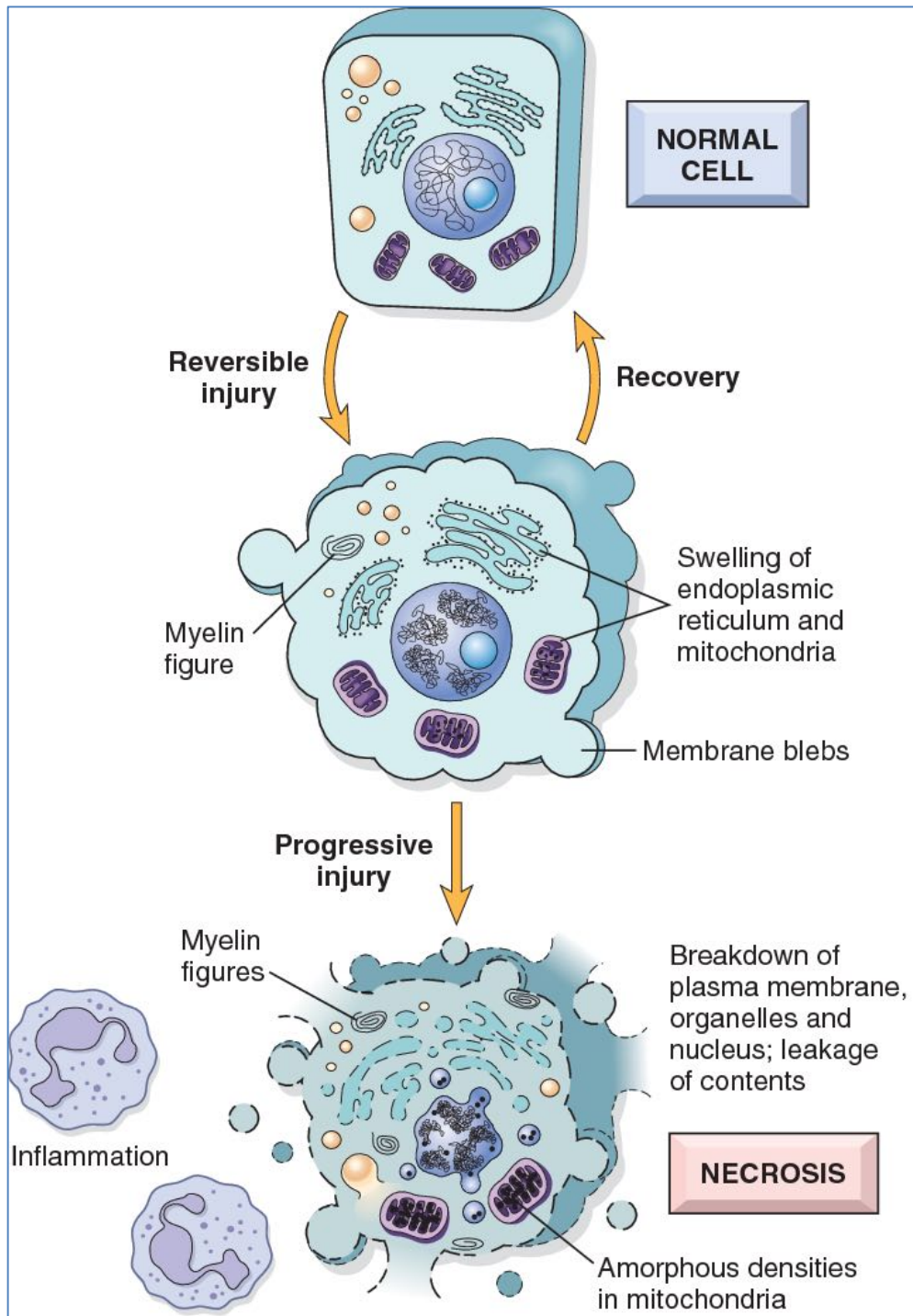
- Interferes with the enzymes that package fat into Lipoproteins & allow fat export from the liver. Decreased function of these enzymes leads to lipid accumulation in Hepatocytes

○ **Other Features:**

- **Blebbing of the Plasma Membrane**
- **Detachment of Ribosomes from the ER**
- **Clumping of Nuclear Chromatin**

- **Irreversible → Necrosis:**

- Result of Denaturation of Intracellular Proteins & Enzymatic Self-Digestion
- Cells Lose Membrane-Integrity → Spill their contents → Local Inflammation
- **Morphological Features of Nuclear Degeneration:**
 - **Pyknosis** – Nuclear Shrinkage, chromatin condenses into a solid, shrunken basophilic mass → ↑Basophilia (Staining with basic dyes)
 - **Karyorrhexis** – The Pyknotic Nucleus fragments → within 1-2 days, the Nucleus totally disappears
 - **Karyolysis** – Basophilia of the chromatin fades due to loss of DNA through enzymatic degradation



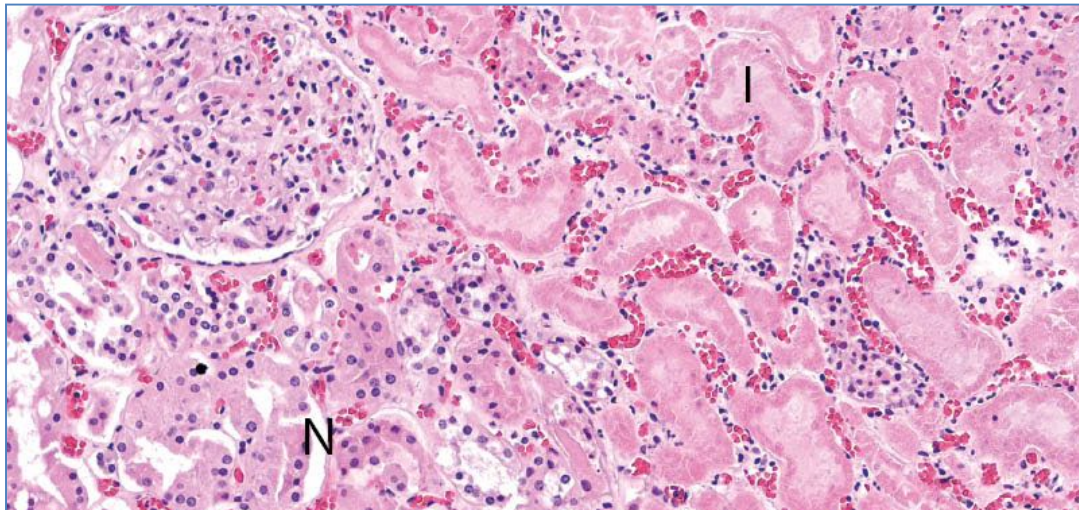
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Patterns of Tissue Necrosis:

- #1 - Coagulative Necrosis:

- **Caused by Hypoxia/Ischaemia/Infarction** → Denaturation of cell Proteins (similar to cooking an egg); also blocks the proteolysis of the dead cells → Cell outline remains for days/weeks
 - **Eg:** Myocardial Infarction
 - **Eg:** Gangrene – Usually Appendages that have lost blood-supply
- **The basic cell-outline remains** for several days. This is because the Lysosomal Enzymes usually responsible for structural breakdown are denatured. Hence, affected tissues have a firm texture
- **Mechanism of Cell Death:**
 - **Ischaemia** (except in the brain) leads to Coagulative Necrosis (Infarct)

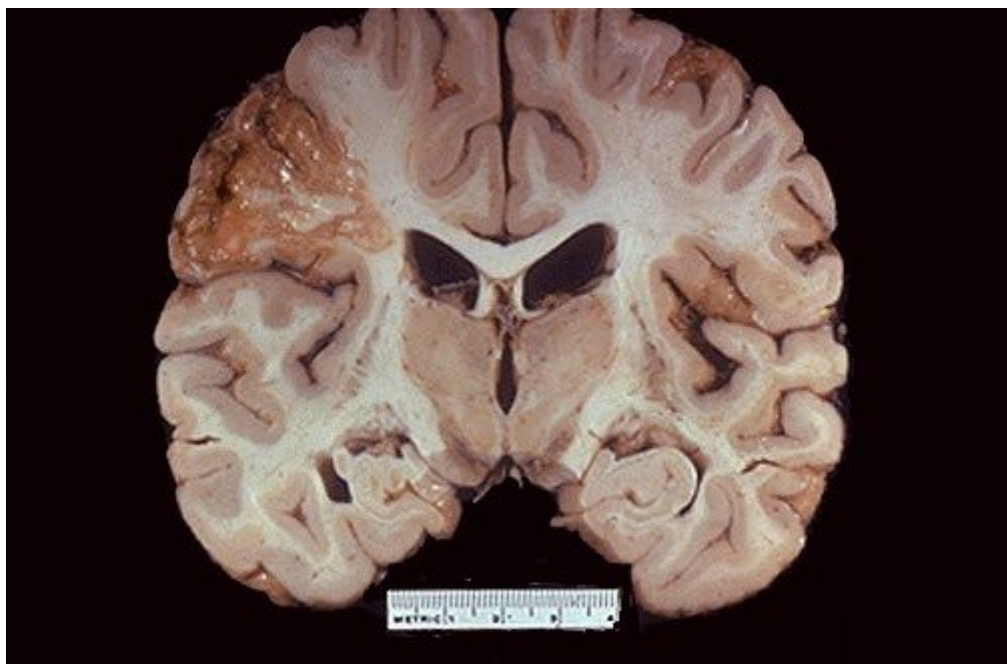


Above: Kidney Infarct (Macro & Micro) "N" = Normal Tissue; "I" = Infarct

Calicut Medical College, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons

- Liquefactive Necrosis:

- **Caused by Bacterial (sometimes fungal) Infections & Subsequent Inflammation** → Dead bacteria & dead Neutrophils form *Pus*, the liquid viscous mass → Abscess
 - **Note: Exception** – *Hypoxic Death* of CNS Neurons leads to Liquefactive Necrosis
- **Eg:** Abscess in Lymph Node



<https://webpath.med.utah.edu/CINJHTML/CINJ024.html>

- **Caseous ("Cheese-Like") Necrosis:**

- **Caused by Tuberculosis, Syphilis & Certain Fungi;** It can be considered a combination of Coagulative & Liquefactive Necrosis
- **Microscopically – A "Granuloma"** - The necrotic area is a collection of lysed cells & amorphous granular debris, enclosed within a distinctive Inflammatory Border

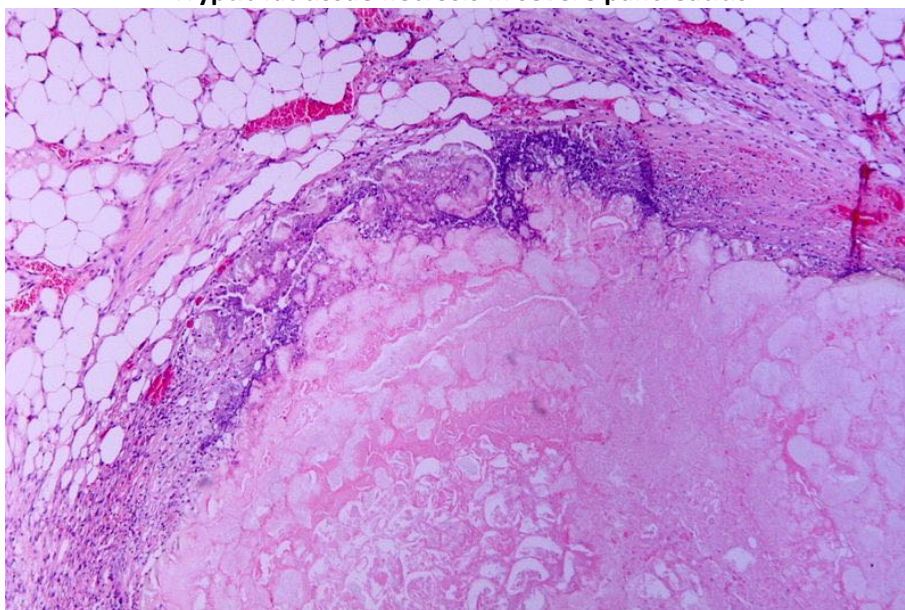


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- **Fat Necrosis:**

- **Local areas of Fat Destruction, Typically Caused by release of Activated Pancreatic Lipases on Adipose Tissues** → (Acute Pancreatitis)
 - **Eg: Acute Pancreatitis** → Release of activated pancreatic lipases from damaged pancreas into peritoneal cavity → acts on fat on mesenteries
 - **Eg: Breast-Tissue Necrosis**
- Looks like – Chalky-white areas surrounded by an Inflammatory Reaction

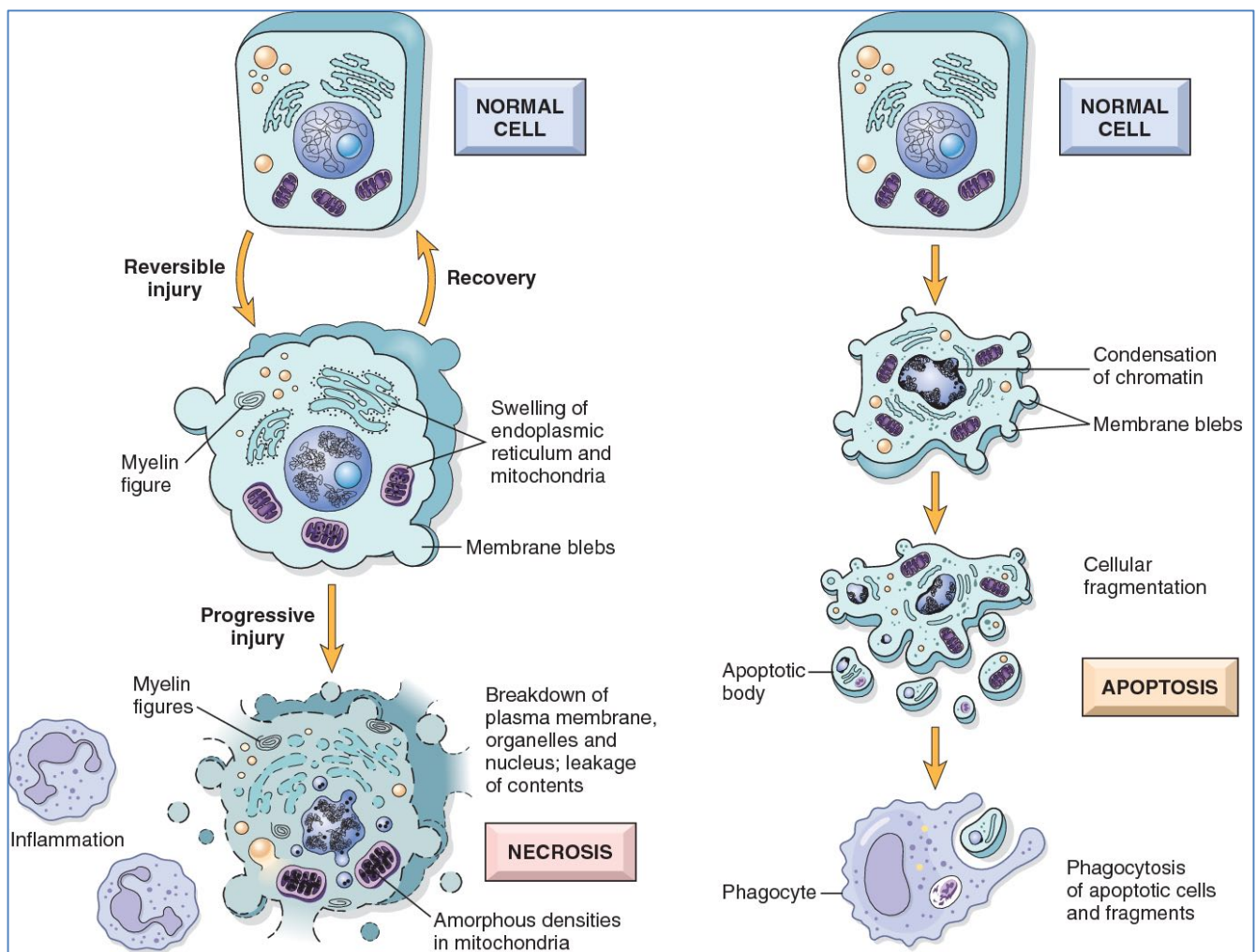
Tryptic fat tissue necrosis in severe pancreatitis



Patho, CC BY-SA 3.0 <<https://creativecommons.org/licenses/by-sa/3.0>>, via Wikimedia Commons

Apoptosis (NOT Necrosis):

- **What is it?**
 - A pathway of cell death induced by tightly-regulated 'Suicide' program → Activate enzymes that degrade the cell's own Nuclear DNA & Cytoplasmic Proteins
 - The Cell then breaks up into fragments (**Apoptotic Bodies**), which are phagocytosed
 - By not spilling cell contents, Apoptosis doesn't elicit inflammation (Unlike Necrosis)
- **Causes of Apoptosis:**
 - **Apoptosis in Physiologic Situations:**
 - **Elimination of Unwanted/Aged/Harmful cells**
 - **Eg:** Embryogenesis
 - **Eg:** B/T-Cell Negative-Selection
 - **Eg:** Endometrial Breakdown during menstrual cycle
 - **Apoptosis in Pathologic Conditions:**
 - **Elimination of Cells that are Irreversibly Injured, without Collateral Damage**
 - **Eg:** DNA damage (Radiation/Chemotherapy/Hypoxia)
 - **Eg:** Accumulation of Misfolded Proteins → Endoplasmic-Reticulum Stress (or ER-Stress)
 - **Eg:** T-Cell Mediated Apoptosis of Virally-Infected Cell



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MICROBIAL DIVERSITY

MICROBIAL DIVERSITY

(Only an introduction; we cover microbiology in more detail in our Population Health/Infectious Disease subject)

- Some bugs are good (even essential) and some bugs are bad
- Organisms capable of causing disease are **pathogens**

Normal Flora (commensals)

- Heavily colonise skin – armpit, perineum, interdigital areas
 - Nose and oropharynx
 - GI Tract
 - Uro-genital tract
- Are normal at certain places where they are not harmful
 - However when they colonise an area where they shouldn't, they cause disease (nosocomial infection)

Pathogenesis

- The biochemical mechanisms whereby microbes (bacteria, fungi, parasites & viruses) causes disease
- **Virulence:** the propensity of a microbe to cause infection → disease

Steps to disease:

- I. **Entry**
 - Oral
 - Skin
 - Trans-placental
 - Inhalation
 - Inoculation (wound/skin penetration)
 - Sexual
- II. **Colonisation**
 - Breach of skin/epithelia/conjunctiva
 - Attachment
- III. **Persistence + avoiding host defences**
 - Beat natural barriers – flushing, mucous + cilia, stomach pH, Lysosomes in saliva, etc
- IV. **Replication**
 - Mucosal (GI tract)/systemic (blood)/nerves (viruses)/cerebrospinal fluid (meningitis)
- V. **Dissemination – (Host-Host)**
 - Faecal-oral (diarrhoea), Aerosols (sneezing), Sexual (intercourse)
 - Depends on:
 - Organism size
 - Ability to survive in external environment
- VI. **Cause Disease**
 - Can release toxins – either local effects / or systemic
 - Can cause unusual cellular activity
 - Can cause tissue damage

Host-Parasite Interactions:

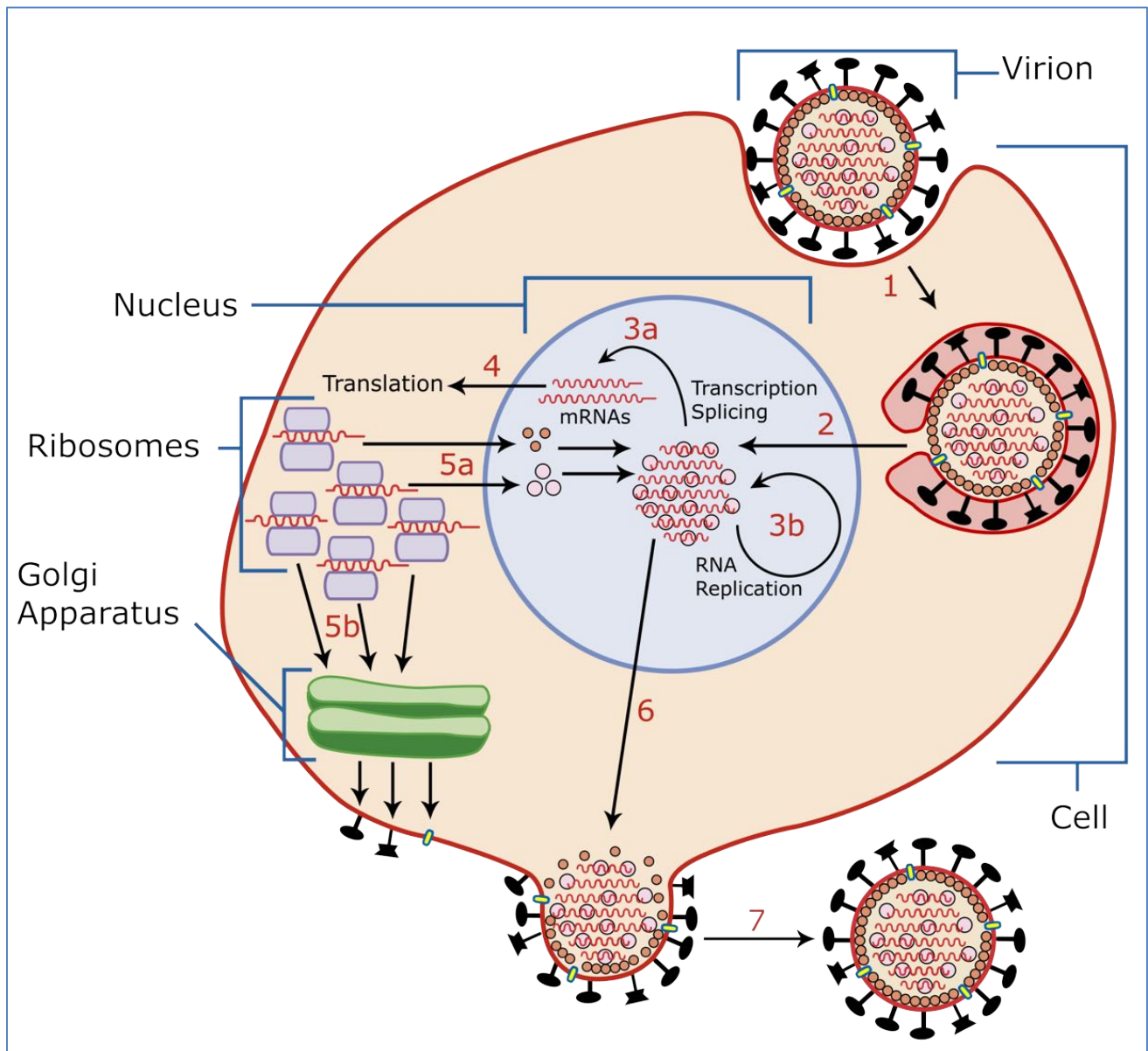
- 1) Colonised, no disease, no illness (asymptomatic)
Eg: Helicobacter – in stomach
- 2) Colonised, disease, no illness (asymptomatic)
Eg: Chlamydia & other genital tract infections
- 3) Colonised, disease, illness (symptomatic)

The Organism Classifications:

- **Prokaryotes:**

- **Viruses**

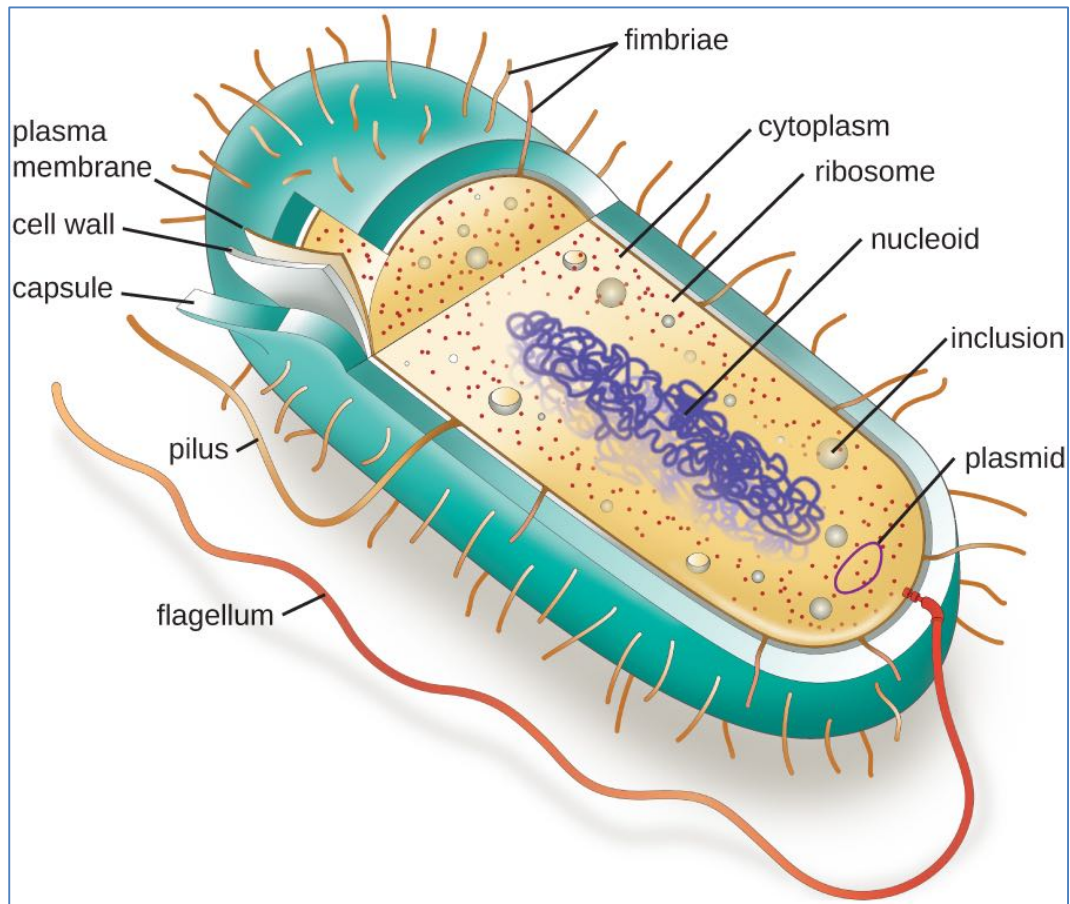
- Very small
 - Nucleic acid inside protein coat (DNA or RNA)(ss or ds)
 - Complete parasitic dependency
 - Replicates inside cell - but metabolically inert in external environment
 - Need close/direct contact
 - Need a moist environment
 - Lyses host cells and then infects more
 - Respiratory route / oral / inoculation / sexual transmission



User:YK Times, CC BY-SA 3.0 <<http://creativecommons.org/licenses/by-sa/3.0/>>, via Wikimedia Commons

- **Bacteria**

- Larger than viruses
- Visible under light microscope
- Living → replicate by binary fission
 - Can be killed
- Intracellular or extracellular
- Motile
- Can produce toxins
- Contain DNA, Ribosomes + Inclusions – no true nucleus
- Resulting disease often more severe

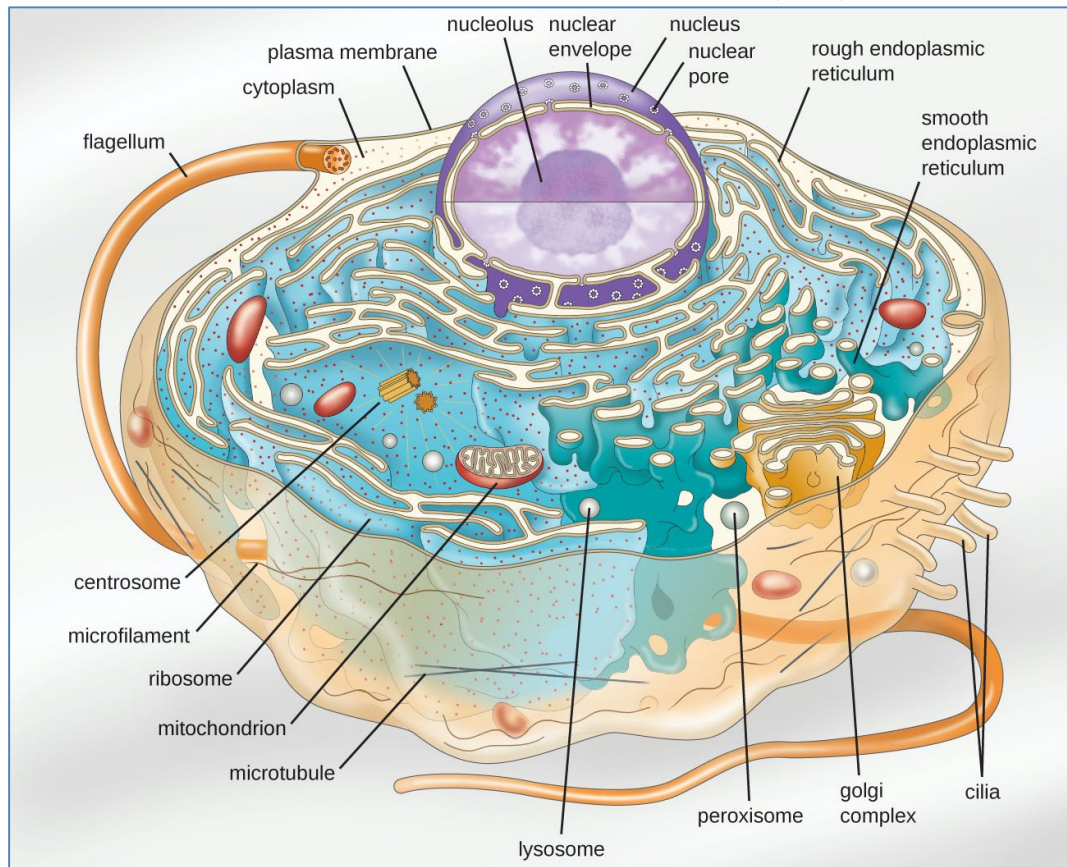


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- **Eukaryotes:**

- **Protozoa**

- **Single-Celled Animals**
 - Larger than bacteria – still small enough to live intracellularly
 - Can also live extracellularly
 - Vectors / faecal-oral route → most infections occur tropically



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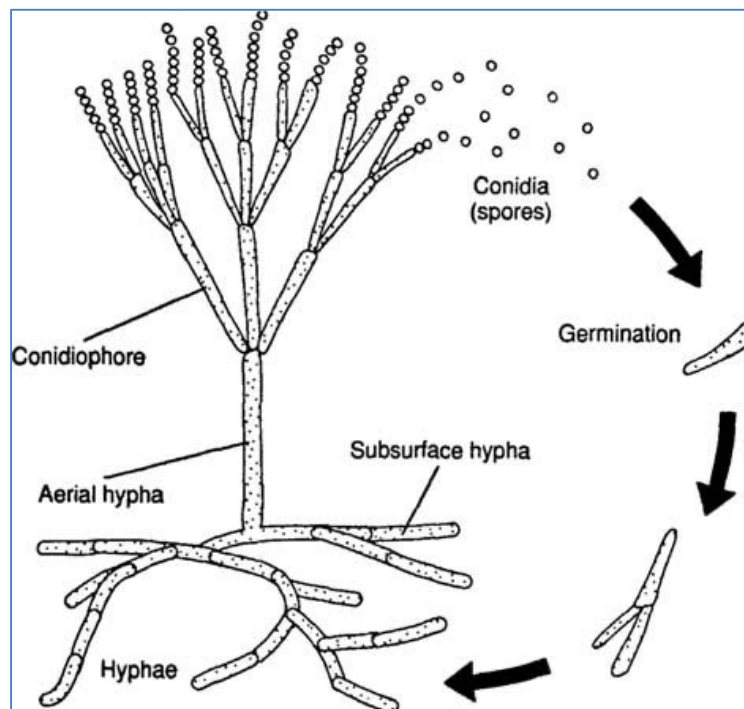
- **Helminths – Eg: Whipworm, Ascaris, Pinworm, Tapeworm, etc**

- Multi-celled, often macroscopic organisms
 - Complex body organisation and reproduction (some have sexual dimorphism)
 - Difficult for immune system to destroy – too big
 - Cause inflammation
 - Are often never eliminated



<https://upload.wikimedia.org/wikipedia/commons/b/bf/Hookworms.JPG>

- **Fungi**
 - Thousands of species
 - Few are pathogenic to humans
 - 20ish are fatal
 - Resulting Mycoses (disease) either:
 - Superficial
 - Cutaneous
 - Subcutaneous
 - Systemic
 - Opportunistic – seen in compromised hosts
- Depending on site of infection
- Exist as branched filamentous forms, or yeasts
- Asexual spores (conidia)
- Spores commonly inhaled & cause infection



<https://aspergillusproject11.wordpress.com/2013/04/18/life-cycle/amp/>



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